NiCE

Nexidia Data Exchange Framework

User Guide

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NiCE invites you to join the **NiCE User Group (NUG)**.

Visit the NUG Website at www.niceusergroup.org and follow the instructions.

All queries, comments, and suggestions are welcome! Please email: nicebooks@nice.com

For more information about NiCE, visit www.nice.com

CONTENTS

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1

Introduction

WELCOME TO THE NEXIDIA DATA EXCHANGE FRAMEWORK

The Nexidia Data Exchange Framework extracts media and metadata from call recording platforms on a scheduled basis. You can quickly configure workflows for a wide range of recording platforms and metadata sources, and routinely add or modify workflows to accommodate new platforms and sources as needed.

The application consists of an Administration server (NEAT) and a Worker server (NEST). These components can be deployed in a distributed-server deployment model in either a hosted or dedicated environment. Most deployments have one Administration server with many associated Worker servers. The Worker servers report state and status information to the Administration server via a heartbeat message.

The system constantly monitors itself for issues or failures, and alerts administrators immediately based on the configuration options.

Building Processing Workflows

The Home view provides a tree view of the Data Exchange Framework components. From here, you work with the hierarchy of companies, sites, servers, processes and resources to create processing workflows. The tree view displays the full hierarchy (Companies, Sites, Servers, Processes, and Resources) by default. You can filter the navigation tree as needed.

- Display only the component that you wish to work with by entering the component name (such as Process) in the **Filter navigation tree...**.
- Specify a different view by selecting one of the following options from the **Show All** drop-down list.

- ***Show All*** - Displays the full environment (default)
- ***Show Problems*** - Displays only the components that have a critical or warning status.
- ***Show Online*** - Displays all the components with an online status.
- ***Show Running Process*** - Displays all the components with an online status.
- ***Show Scheduled Process*** - Displays all the components with an online state.
- ***Expand All Nodes*** - Expands all of the nodes in the tree view.
- ***Collapse All Nodes*** - Collapses all of the nodes in the tree view.

Components

You must create and configure these components before you can begin building your processing workflows. When you are working with these components you have the option to right-click and select items from the tree or to click the action buttons in toolbar.

- Companies
- Sites
- Servers
- Processes
- Resources

When you click a defined company, site, server, process, or resource under its main component in the tree, details about it are populated in the Properties area. You can optionally edit, copy and delete these items. The **Deactivate** option is used for server maintenance and is only enabled for Administrators and Super Administrators.

Keep in Mind

- Activation and deactivation cascades down the hierarchy from company to site to server.
- Super Administrators can deactivate all companies at one time using this url: localhost:8080/NEAT/CustomerController?EVENT_ID=DEACTIVATE_ALL_CUSTOMERS
- Once a server has been deactivated, running processes are allowed to complete and no new processes launch.
- Once all processes are quiet, Worker servers shut down after a 10 second pause. In a single-server configuration, a 5 minute shutdown process is triggered, during which time logged in users receive a shutdown notice.
- On start up, all servers take themselves out of maintenance mode.

- Administrative servers in distributed configurations never shut down. They can be stopped using the stop.sh/bat command-line scripts.

Scope

Who should use this help?

- Administrators, analysts, supervisors.

What is included?

- Building Process Workflows
- Configuring Actions
- Administration Settings
- Deployment View
- Other Tools

What is not included?

- Installing and Configuring the Data Exchange Framework, see the *Nexidia Data Exchange Framework Installation Guide*.
- Instructions for using the Eclipse SDK project to create new actions that are fully integrated with the Data Exchange Framework, and that cover any gaps in functionality required for a Data Exchange Framework project. See the *Nexidia Data Exchange Framework SDK Guide*.
- Integration Templates. The Data Exchange Framework templates have been removed from the user interface and are now available on the Data Exchange Framework Confluence site for those who have access (Process+Library). You can also download them as a complete package via the NiCE Software Download Center: (https://nice.subscribenet.com/control/nice/login?nextURL=%2Fcontrol%2Fnice%2Fpurchases).

  The integration templates are pre-configured process definitions used to simplify the DEF process design. When you add a process to a server from the Processes window, you can import a template. The actions in the templates are pre-configured, but you can modify them. You will configure the resources for your environment. There are placeholders for the resources in the templates.

Document Revision History

SOFTWARE VERSION	DATE	DESCRIPTION
12.5.2.1	March 2026	Documented Google API Media and Metadata Retrieval Action. Documented Import JSON Action. Documented Generate Desktop Tags Action. Documented how to update ENL CSAT Index Thresholds in the DEF Database. Documented the parameters for the Workflow Enlighten, Workflow Request, and Call Summarization Response actions related to complaints-extraction data. Documented the language code parameter in the Workflow Request to improve the v11 transcript quality. Changed the default value of Include Sentiment Scores parameter in Workflow Enlighten actions to true. Combined AutoSummary LLM and AutoSummary JSON Creator into a single AutoSummary LLM V2 action.

SOFTWARE VERSION	DATE	DESCRIPTION
12.5.2	December 2025	• Added a new action to parse ElevateAI Real-Time metadata and transcripts and send the parsed data to NIA. • Convert Transcript to ASR action, added a new Transcript Type config option - Analytics Hub. • . • Added a new Analytics Hub Response action to retrieve a call recording transcript from Analytics Hub. • Added a new Analytics Hub Request action to provide a call recording for transcript generation by Analytics Hub. • Documented Topic AI Request action to retrieve the correct IPM model from Topic AI Editor. • Documented Topic AI Response action to retrieve the response from the Voice Grid and send it to NIA. • Added support for using a Key Management Service key when writing to an AWS S3 resource.

SOFTWARE VERSION	DATE	DESCRIPTION
12.5.1	July 2025	• Added an option to collect some additional metrics for the chat media for new ingest path. • Added an ability to support SOCKS5 protocol during SFTP transfer. • Updated the CXoneMediaRetrieverV3Action to provide the option to retrieve IVR media. • Added new action CXone Storage Media RetrieverV1Action to retrieve media using the Cxone Storage Services API. • Added new resource type - Artemis. • Added new action - CXone Snowflake Query. • Added new action - CXone Snowflake Extractor.
		• Added warning feature in Inbox Publisher for unmapped metadata fields in Working Table. • Added Process Type feature to differentiate monitoring of Production and Testing DEF processes. • The DEF calculates the values of two metrics, OneHalfSentimentScore & ThreeForthsSentimentScore and passes them to the NIA. • Added new Skip bad CSV Files and continue option to CSV to DB action for improved error handling and continuous processing.

SOFTWARE VERSION	DATE	DESCRIPTION
		• Added temporary file naming feature for DB-to-CSV and Metadata Mapping and Export actions to prevent access to incomplete files. • Added process-level retention period setting to override server-level retention and simplify working table cleanup management. • Added Empty folder strategy option to the Cleanup action to specify how to handle empty folders in input resources after file cleanup. • Added the Output Format option to the Parse General Text action to generate output in the TextGrid JSON, VoiceGrid ASR JSON format, and both formats.
		• Added the Output Format option to the Parse Email to Text action to generate output in the TextGrid JSON, VoiceGrid ASR JSON format, and both formats. • Added a new Resource: Snowflake. • Added configuration options for automatic S3 credential rotation to enhance security and meet PCI compliance requirements. • Added ability to exclude mono calls. • Added ability to redact the email content in Parse Email Text to JSON action. • Added ability to redact the email content in Parse General Text to JSON action.

SOFTWARE VERSION	DATE	DESCRIPTION
		• Added ability to decrypt stored passwords for privileged users. • Added ability to get the sentiment transition value of interaction from Voice Grid and send it to NIA for further analysis. • Added ability to send failed interactions information to the NIA that can't be retried. • Added XO Billing - New action to feed to the CXone billing mechanism. • Updated import and export process to capture additional information. • Added how to add an annotation in Templates/exported processes. • Updated Call Summarization Response v2 action to include offsets.
		• Updated EML Parse Text and Parse Email Text actions to capture Email media type as NIA first-class field and send to NIA.
12.5.0.2	February 2025	• Added new configuration parameter to Inbox Publisher - Stylesheet Path. • Added configuration options to CXoneMediaRetrieverV3 action to add prefix to additional and custom chat metadata.

SOFTWARE VERSION	DATE	DESCRIPTION
12.5.0.1	January 2025	• Added new action - Media URL Download action. • Added new parameters for Twilio Media Retriever action. • Added new action CXone Completed Contacts Metadata Retriever V2 Action to extract DFO mails for completed contacts and CXone Media Retriever V3 Action. • Added enhancements to DFO chat integration to strip bot and split agent conversations. • Added enhancements to Salesforce chat integration to remove chatbot data and split agents in Salesforce Metadata Parser action. • Now the entire conversation transcript will be sent to LLM to summarize the call. • Added an option to create a new test mode option for Inbox Publisher and delete fake inbox once the migration test completes.

SOFTWARE VERSION	DATE	DESCRIPTION
12.5	December 2024	• Documented new action - Convert 3rd party transcript to VoiceGrid ASR format. • Documented the feature - Rule-based summary should support non-English language in Auto Summary JSON Creator action. • Documented resizing windows in the DEF UI using mouse. • Documented Rule-based summary support for non-English language. • The DEF monitoring data supports UTC format. • CXone media retriever: Default value for "Download voice and screen media" set to false.

SOFTWARE VERSION	DATE	DESCRIPTION
		• You can now enter negative values in quantum field. • Support a different queue name for each text analysis IPM model. • Added feature allowing users to select media files based on metadata, improving performance. • Documented AppLink SAP Generic Upload action. • The new filtering option can be used to route traffic to/around AutoSummarization, as appropriate. • Removed scoring model metadata mapping field. • Documented Calabrio User Metadata Retriever action. • Updated Calabrio Metadata Retriever action.

SOFTWARE VERSION	DATE	DESCRIPTION
		• Append DEF Certificate field added to add a database resource section. • For CXone Media Retriever V2 Action, in the Download voice and screen media drop-down, changed the default setting to false to download the voice only media file. • Added Check signature config option to verify file signatures before decryption, ensuring only files signed by the user are accepted. • Updated DEF monitoring data date/time fields to support UTC format for consistency in DW and improved customer benefit, as requested by NIA team. • Documented - Updated CXOne Metadata Retriever action to organize business data. • Documented - Update CXone Metadata Retrieve action to retrieve data only for specific hours in a day. • Added 'Include Formatted Transcript' configuration option to Call Summarization Response for GPT integration. • To improve efficiency, the DEF team will no longer require manual updates for newly added first-class fields from NIA. Instead, DEF will optionally retrieve available fields from a configurable XML file. • Documented new action - Custom SQL v2 that allows only single SQL statements, preventing invalid SQL entries. The existing Custom SQL action will be deprecated.

SOFTWARE VERSION	DATE	DESCRIPTION
		• After copying a process, publish it manually.
		• Documented Salesforce Metadata Retriever action. • Documented new action - AutoSummary LLM. • Enhanced DEF to add configurable milliseconds to single-turn chat durations to ensure accurate evaluation, with a default of 5ms for hits and session end time. • Added multi-agents Salesforce chats can be split into “one agent per” chat. • Redesigned action pop-up screen with searchable groups and expandable accordion view to improve user experience and ease of finding specific actions.
		• Documented new action Convert Transcript to ASR, which converts the Nexidia Standard transcript format into the VoiceGrid ASR. • Documented Salesforce Metadata Parser Action. • Enhanced Call Summarization Request action. • Added an ability to choose either interval schedule or fixed schedule in Quantum. • Added Workflow Enlighten action calculates the index scores for the autosummarization tag. • Documented Enhance Enlighten scores functionality to be compatible with ElevateAI integration.

SOFTWARE VERSION	DATE	DESCRIPTION
12.4.2 FP1	April 2024	• Documented - Intent Models - Configure / Process specific Intent Models for specific Interactions. • Documented - Add Formatted Transcipt to Call Summarization Response JSON Output. • Documented - Make a DEF/Grid Response Queue if needed. • Documented - Update the CXone Metadata Retriever action based on the CXone API changes. • Documented - Workflow Request: adjust media URI based on file filter.

SOFTWARE VERSION	DATE	DESCRIPTION

12.4.1	August 2023	• Added following topics for Twilio Integration.
		  • Twilio Calls Metadata Retriever Action
		  • Twilio Media Retriever Action
		  • Twilio Recordings Metadata Retriever Action
		• Added Desktop Tags in Call Summarization Response.
		• Updated the Call Summarization Response topic so that it can be mapped into UDFs for NIA.
		• Added the transcript type and the option to Flip roles (Channel) in Call Summarization Response.
		• Added Transcript Type as an option to flip channels in CSV in Call Summarization Request.
		• Added an option to provide output file name in Metadata Mapping and Export Action.
		• Added an option for Engage validation in Convert Engage Hierarchy V2.
		• Added configurations to push reconciliation report data to NIA in Configure Processes.
		• Updated the Configure Process topic to configure the retry interval after failure at the process-level.
		• Added the Channel-Flipping Option to the Worfklow Response Action.
		• Updated the AppLink SAP Upload to include extract appropriate keys from Engage, and stored them inside of a Java SealedObject(s).
		• Removed the Agent and User Import Action V2 Action as it has been deprecated as a result of CXF dependencies.

SOFTWARE VERSION	DATE	DESCRIPTION
		• Updated the following topics for CXone integration authentication mechanism for AppLink: • AppLink SAP Business Data Upload • AppLink SAP User Upload • AppLink SAP Upload
5.7 Continued		• In the Compare Process History Preview mode, differences in the components of the selected versions are highlighted in light blue. • The DEF Internal log retention days configuration was added in the Add a Server window. • The ESM File Mover action was removed.

DATA ACCESS REQUIREMENTS

The Data Exchange Framework must have access to the following databases. Control over the NiceKeyCertificate is also required.

- nice_admin
- nice_rule
- nice_storage_center
- nice_interactions
- nice_cti_analysis

Execute permissions are required on these stored procedures:

- [nice_interactions].[dbo].[GetSetNumber]
- [nice_interactions].[dbo].[dbo.spPlaybackGetSegmentByListInteractionID]
- [nice_interactions].[dbo].[udf_ScGetUnitPath]
- [nice_crypto].[dbo].[spDecryptData]

THE EXTRACTION PROCESS

An extraction process moves files for interactions and metadata from a company location to a location recognized by the Interaction Analytics ingestion process. The files are formatted and processed according to a data structure expected by the ingestion process. An extraction process contains both actions and resources. Actions are specific tasks a process performs, and resources are the entities that the actions act upon.

For example, using the File Mover action, you can specify to move a particular file type from location A to location B. Location A and B are the resources that specify the original and destination directories. A process definition includes one or more actions and resources connected as a workflow. You define processes, and then execute them.

Use the Company, Site, and Server windows to organize the processes. In the Company window, you identify the owner of a process, the Site window provides location information, and the Server window contains the computing environment hosting the processes. You must define these before you can define a process.

The application provides a visual process designer where you connect multiple actions and resources to create a workflow. A process workflow contains two key components: resources and actions. A resource is a logical entity upon which an action can act upon. You link these together to form an extraction process.

There can be a single process running in a company dedicated environment or multiple processes running for various companies simultaneously in a hosted environment. Once the workflow is defined, the tool provides visual feedback on any problems encountered. If problems were encountered, you can make corrections and continue running the workflow from the point of the last error.

Depending on the requirements of a deployment, you can create one comprehensive workflow or several smaller workflows that work together. Each workflow you create will include processes and resources. Processes are tasks such as moving files or filtering files. Processes use resources to expedite actions and can be used by multiple processes.

Communication Options

You can set up your environment to run in the following installation modes. An Administration server can be configured to use either SFTP or File System communications. The Worker server can be a mix of either.

- Administration server in SFTP mode—The administration servers communicate over SFTP. The SFTP connection information is entered into the sites, and the communications subdirectories are entered into the servers.
- Worker server in SFTP mode—The processing server communicates with its administration server over SFTP. The SFTP connection information is entered during installation.
- Administration server in File System mode—The administration server communicates with its servers through the file system (either local or a CIFS share). Folder information is entered into the sites, and the communications subdirectories are entered into the servers.
- Worker server in File System mode—The processing server communicates with its administration server through a file system folder. Folder information is entered during installation.
- Compute Node—Used to horizontally scale a Worker server quickly and easily in response to increased workload or backlog.

DATA COMPLIANCE

The Data Exchange Framework is one of the Nexidia Data Compliance participants that support delete requests. The following actions were created to perform the Data Compliance operations:

- GDPR ID Creator action
- GDPR Requests action

See the Data Compliance chapter In the *Interaction Analytics Installation and Maintenance Guide* to learn more about Data Compliance configuration and the flow of Data Compliance requests in the Nexidia applications. Refer to the *Interaction Analytics Troubleshooting Guide* for help with issues you may encounter.

GDPR ID Creator Action

This action creates GDPRID number columns based on configured incoming metadata columns. If the selected metadata column is a phone number, then a phone number normalization function can be applied to the phone number to remove all non-numeric characters. This component can be configured to update the NX Compliance service when OPTOUT GDPR ID values are blocked/deleted from a working table or roll over table.

- Users can create configure one or more incoming metadata columns as GDPR ID columns. These columns are created in the Data Exchange Framework working table if configured:
  - GdprId 1
  - GdprId 2
  - GdprId 3
  - GdprId 4
  - GdprId 5
- Example mapping: Map metadata Email column to GdprId 1.
- You can apply an optional normalization function to a GdprId column. The available normalization function is Normalize Phone Number. It removes all non-numeric characters from a phone number metadata field.
- If the configured GdprId column does not exist in the target DEF working table, the column is created as follows: nvarchar(1000) COLLATE Latin1_General_100_CI_AS_KS_WS_SC type column.

- Following is working table example with a created GdprId column:

Results Messages

	quence	MediaFile	InteractionID	languageId	interactionType	ExternalMediaId	Starttime	Email	GdprId1
1		gdpr_media_0.wav	gdpr_media_0	0	call	C:\Temp\gdpr\gdpr-demo-media-out\gdpr_media_0.wav	2019-10-04 07:48:44	bsmith@gmail.com	bsmith@gmail.com
2		gdpr_media_1.wav	gdpr_media_1	0	call	C:\Temp\gdpr\gdpr-demo-media-out\gdpr_media_1.wav	2019-10-04 08:52:23	ejones@gmail.com	ejones@gmail.com
3		gdpr_media_2.wav	gdpr_media_2	0	call	C:\Temp\gdpr\gdpr-demo-media-out\gdpr_media_2.wav	2019-10-04 09:36:34	dmarchman@gmail.com	dmarchman@gmail.com
4		gdpr_media_3.wav	gdpr_media_3	0	call	C:\Temp\gdpr\gdpr-demo-media-out\gdpr_media_3.wav	2019-10-04 10:25:44	dabbott@gmail.com	dabbott@gmail.com
5		gdpr_media_4.wav	gdpr_media_4	0	call	C:\Temp\gdpr\gdpr-demo-media-out\gdpr_media_4.wav	2019-10-04 11:05:01	skelly@gmail.com	skelly@gmail.com
6	JLL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL
7		gdpr_media_5.wav	gdpr_media_5	0	call	C:\Temp\gdpr\gdpr-demo-media-out\gdpr_media_5.wav	2019-10-04 11:34:21	pfarmer@gmail.com	pfarmer@gmail.com
8		gdpr_media_7.wav	gdpr_media_7	0	call	C:\Temp\gdpr\gdpr-demo-media-out\gdpr_media_7.wav	2019-10-04 07:34:324	rthompson@gmail.com	rthompson@gmail.com
9	JLL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Block Requests Processing

- The GDPR ID Creator action will delete matching blocked metadata as follows:
- The Data Exchange Framework deletes metadata and associated media based on matching block requests.
- The block requests are stored in the nx_gdpr_block_requests table.
- The action inserts the block requests (OPTOUT) into the nx_gdpr_block_requests table.
- The action deletes matching blocked metadata and associated media from the target working table.
- The action deletes matching blocked metadata from the target roll over table:
  - This occurs for historical purposes when it is the first time through after the columns have been created in the target Data Exchange Framework working table.
  - All matching metadata that has already been processed is deleted.

Configuration Options

PROPERTY	DESCRIPTION	INPUT SOURCE
Request Source	The Data Compliance request input source, either CSV or NIA.	
Participant ID	The participant ID used to identify the client application to the Management Console. The Compliance Manifest tab in the Management Console is used to define which environments, or Participants, must be tracked for Data Compliance.	NIA
Host Name	The host name of the Data Compliance requests service.	NIA
Port	The port number of the Data Compliance requests service. The default value is 47306.	NIA

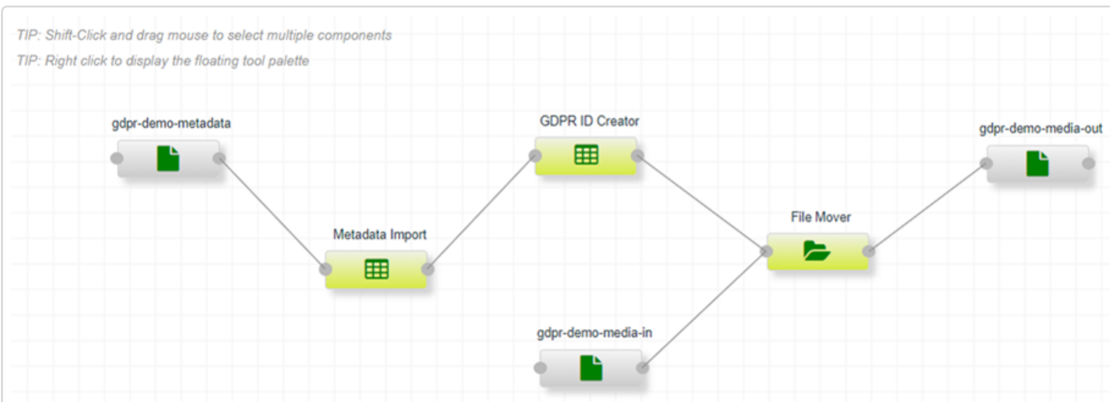
PROPERTY	DESCRIPTION	INPUT SOURCE
Gdprid 1-5 Field (s)	The name of a metadata column that will be used to uniquely identify a metadata row.	
Gdprid 1-5 Normalization Function	The normalization function to be applied to the metadata value. Normalize Phone Number is the available option. It removes all non-numeric characters from a phone number metadata field. By default, no normalization function is applied.	
Service Name	The name of the Data Compliance requests service. This defaults to nx-datacompliance-service and should not be changed.	NIA
Terminal Action	Specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This options is provided for instances where the default behavior is not desirable.	
Disable Action	Specify if you want to prevent this action from processing any files during its run; leave "false" to allow processing as usual. This options is useful for troubleshooting the process.	

Audit Logging

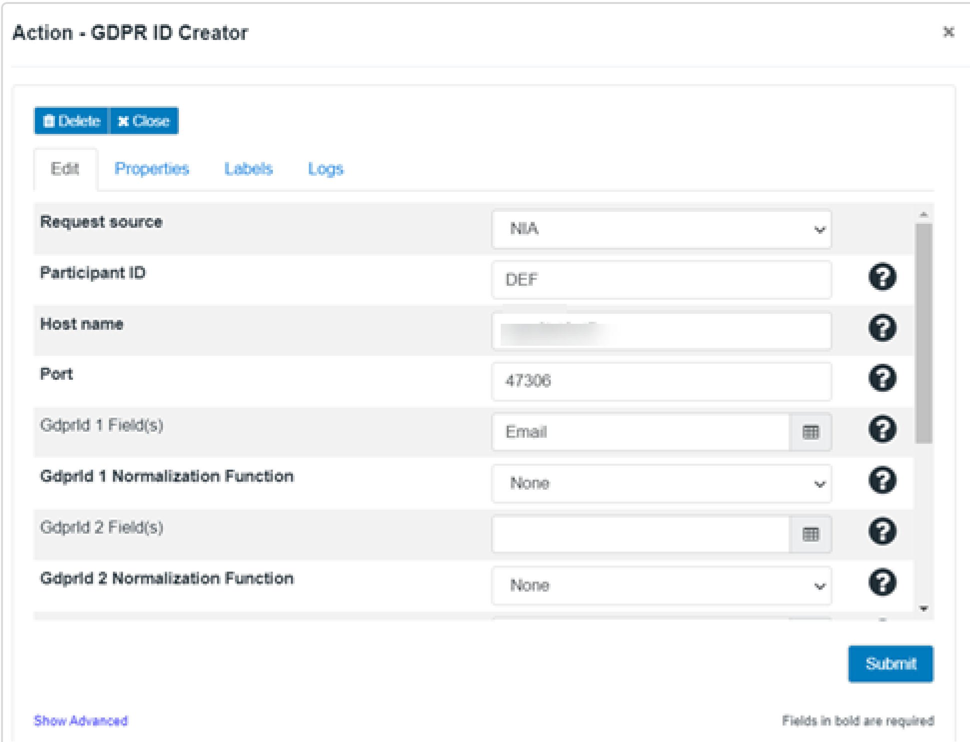
The GDPR ID Creator audit log file is created in the <DEF Data Folder>/audit_
logs/gdpr/GDPRIdCreatorAction_<nnnnn>.log. The following information is logged:

- Each blocked metadata row is logged. This includes GDPR ID column name, GDPR ID value, working table name, request type and request ID.
- Each deleted associated media file is logged. The includes the full media file path.

Process Example



Configuration Example



GDPR Requests Action

Honors GDPR and other data protection requirements by allowing either direct interaction with *Interaction Analytics*, or using file system resources to process the same requests out of delivered CSV files. Only one GDPR Requests action is allowed per DEF server. It scans all working tables contained below the parent server. Any working table that contains one or more GDPR ID columns is eligible for GDPR metadata deletion/removal.

This action processes the following request types:

- DELETE - Deletes matching metadata and associated media from DEF working tables.
  - Retrieves a list of all working tables for the current server that contain one or more GDPRID columns, and looks for these columns: GDPRID1, GDPRID2, GDPRID3, GDPRID4, GDPRID5.
  - For each working table, attempts to find a matching metadata row for the specified GDPRID column.

  - If a match is found that metadata row is deleted (from either the working table or roll over table).
  - If the metadata row to be deleted is a working table row, then any associated media files based on that metadata row are deleted.
  - If the metadata row to be deleted is a roll over table row, then associated media is not deleted.
- OPTOUT - Inserts the OPTOUT request into the new DEF nx_gdpr_block_requests table so that the specified metadata is not ingested in the future. OPTOUT sends only incremental results.
- OPTIN - Deletes the OPTOUT request from the new DEF nx_gdpr_block_requests table.
- EXPORT/EXTRACT - Extracts matching metadata and associated media from DEF working tables.
  - Retrieves a list of all working tables for the current server that contain one or more GDPRID columns, and looks for the following columns: GDPRID1, GDPRID2, GDPRID3, GDPRID4, GDPRID5.
  - For each working table, attempts to find a matching metadata row for the specified GDPRID column.
  - If a match is found that metadata row is extracted to a csv file.
  - If the metadata row to be extracted is a working table row, then any associated media files based on that metadata row will also be extracted.
  - If the metadata row to be extracted is a roll over table row, then associated media is not extracted.
  - Extracted/exported output is in the form of CSV files.
  - CSV files are created on a request ID and process group basis.
  - Extracted associated media is extracted to the same location as the CSV file containing the corresponding media.
  - Exported metadata and media files are written to the GDPR Requests action output file resource.

Configuration Options

PROPERTY	DESCRIPTION	INPUT SOURCE
Request Source	The Data Compliance request input source, either CSV or NIA.	
Force File Encoding	Leave blank in most cases to auto-detect. If auto-detect is not working, enter a valid Java InputStream encoding.	
Participant ID	The participant ID used to identify the client application to the Management Console. The Compliance Manifest tab in the Management Console is used to define which environments, or Participants, must be tracked for Data Compliance.	NIA
Host Name	The host name of the Data Compliance requests service	NIA
Port	The port number of the Data Compliance requests service. The default value is 47306.	NIA
Age Out Temporary Block Requests	Specify how long to keep temporary block requests. The default value is 24-hours.	
Service Name	The name of the Data Compliance requests service. This defaults to nx-datacompliance-service and should not be changed.	NIA
File Filter	If you selected CSV in the Request Source drop-down, .csv is populated in the File Filter field.	CSV
Delete GDPR Requests After Processing	If you selected CSV in the Request Source drop-down, specify if you want to delete Data Compliance requests after processing.	CSV

PROPERTY	DESCRIPTION	INPUT SOURCE
Default Separator Character	If you selected CSV in the Request Source drop-down, enter the character used to separate delimited column values in the output CSV (for example ,).	CSV
Default Quote Character	If you selected CSV in the Request Source drop-down, enter the character used to quote column values in the output CSV (for example ").	CSV
Default Escape Character	If you selected CSV in the Request Source drop-down, enter the character used to escape column values in the output CSV (for example \).	CSV
Default Line End Character	If you selected CSV in the Request Source drop-down, select the character used to end/terminate a line in the output CSV (for example \n or \r\n)	CSV
Delete Associated media Files?	Specify if you want to delete the associated media files.	
Terminal Action	Specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This options is provided for instances where the default behavior is not desirable.	
Disable Action	Specify if you want to prevent this action from processing any files during its run; leave "false" to allow processing as usual. This option is useful for troubleshooting the process.	

Audit Logging

The GDPR ID Creator audit log file is created in the <DEF Data Folder>/audit_ logs/gdpr/GDPRRequestsAction_<nnnnn>.log. The following information is logged:

- Start processing request CSV file (CSV source only)
- Stop processing request CSV file (CSV source only)
- GDPR block request created (OPTOUT) (Request ID and block request criteria)
- GDPR block request removed (OPTIN) (Request ID and block request criteria)
- Rows deleted from working table (DELETE) (Request ID, # of rows deleted, table name, selection criteria)
- Rows deleted from roll over table (DELETE) (Request ID, # of rows deleted, table name, selection criteria)
- Deleted associated media file (DELETE) (Deleted media file full path)
- Exported/extracted associated media file (Exportedmedia file full path)

Detail Logging

The GDPR Requests action generates detailed logs for every request processed. They are written to the GDPR Requests action output file resource in these formats:

- CSV format: gdpr-response-YYYYMMDD-HHHMMSS.csv
- JSON format: gdpr-response-YYYYMMDD-HHMMSS.json

Example CSV Log File Output

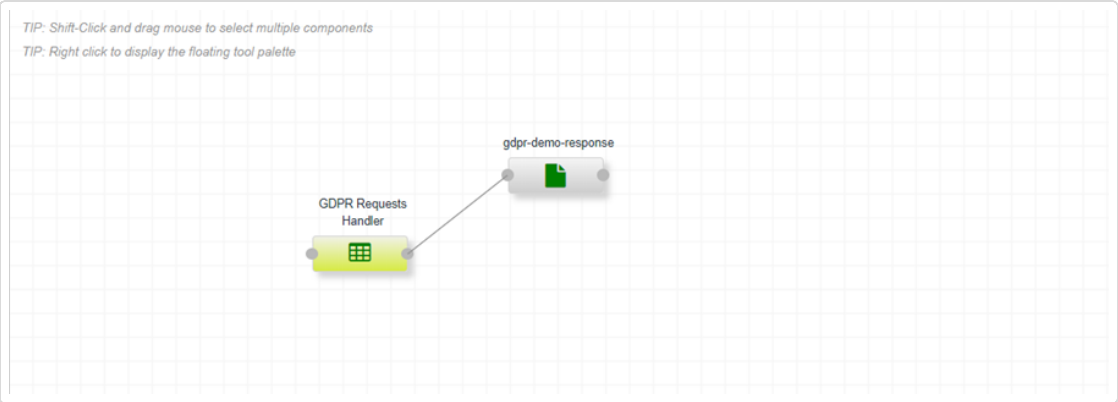
"GDPRRequestID","NumberOfRowsDeleted","NumberOfRowsDeletedBlocked"

"64ade1ad-cdec-488a-88b6-e5b73dcf4785","2"","0"

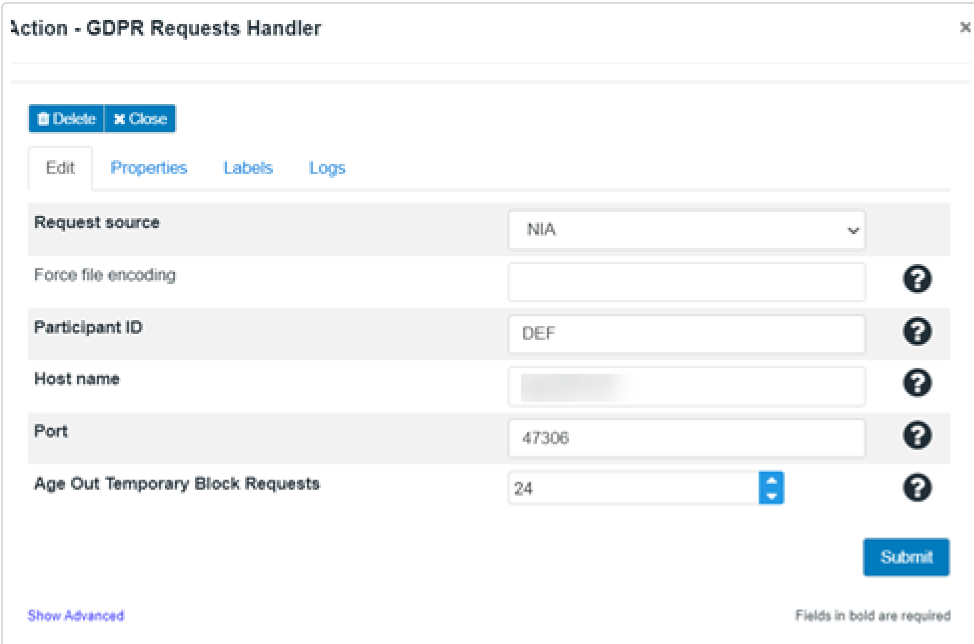
Example JSON Log File Output


```
{
  "gdprRequests": [
    {
      "requestId": "64ade1ad-cdec-488a-88b6-e5b73dcf4785",
      "requestDate": "May 27, 2020 1:12:51 PM",
      "requestType": "DELETE",
      "numberDeletedWorkingTable": 0,
      "numberDeletedRollOverTable": 2,
      "numberOptOut": 0,
      "numberOptIn": 0,
      "numberExported": 0,
      "requestItems": [
        {
          "gdprId1": "[redacted].com",
          "deletedFilesList": [],
          "deletedMetadataList": [
            {
              "fileName": "gdpr_media_0.wav",
              "createDate": "2020-07-07 10:07:17.610000",
              "deleted": false,
              "exported": false,
              "rolloverTable": false,
              "processGroup": "gdpr_demo",
              "workingEntryId": 1,
              "criteria": "GdprId1 EQUALS [redacted].com"
            }
          ]
        },
        {
          "gdprId1": "[redacted]om",
          "deletedFilesList": [],
          "deletedMetadataList": [
            {
              "fileName": "gdpr_media_1.wav",
              "createDate": "2020-07-07 10:07:17.620000",
              "deleted": false,
              "exported": false,
              "rolloverTable": false,
              "processGroup": "gdpr_demo",
              "workingEntryId": 2,
              "criteria": "GdprId1 EQUALS [redacted].com"
            }
          ]
        }
      ]
    }
  ]
}
```


Process Example



Configuration Example



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2

Settings

ADMIN SERVER CONFIGURATION

Use the Admin Server Configuration window to specify information about the Admin and Worker servers. The heartbeat is a communication between the Worker and Administration servers and indicates the latest status of a server.

To define the heartbeat interval:

1. Click **Settings** from the sidebar navigation options, and then select the **Admin Server Configuration** option. The Configure tab is selected by default.
2. Enter the e-mail address where notifications regarding the administration server should be sent. This contact will get notifications only if no administrator e-mail addresses were provided at the Company, Site, or Server levels.
3. Enter the e-mail address where notification emails from the system should be sent from.
4. Enter the **SMTP server**.
5. Optionally enter the **SMTP user**.
6. Optionally enter the **SMTP password**.
7. Select the heartbeat interval seconds from the associated drop-down list. This value indicates how often to send a message to remote sites.
8. Select the heartbeat error threshold from the associated drop-down list. This value indicates the number of seconds that there has been no response from a remote site before an error condition is created.
9. Specify the SFTP file check interval (in seconds) from the associated drop-down list. The check verifies the FTP connectivity.

10. Select the Administration server log retention days. This value indicates how many days of admin server logs to retain. Anything beyond this range is deleted during midnight cleanup monitor runs. The default value is 240. This only applies to standalone NEAT servers in distributed environments. For Worker server cleanup, server-level retention settings will take precedence.
11. Specify the Administration server debug level. The possible options are:
    - DEBUG
    - INFO
    - WARN
    - ERROR
    - FATAL
12. Specify if you want the application to show grid activities for the selected item only (no sub-items). The valid options are "true" or "false". You should only change this option if the "Activities" tab performance at higher levels becomes a problem.
13. Specify if you want the application to delete existing action patch files. The valid values are ***true*** or ***false***.
14. Specify if you want the application to delete existing custom actions. The valid values are ***true*** or ***false***.
15. Specify if you want to use the *Interaction Analytics* user interface parameters for the Metadata Mapping and Export action user interface display. The valid values are ***true*** or ***false***. When you select ***true***, the following four additional fields display. When you specify values for these fields, the *Interaction Analytics* mapping fields in the Configure Metadata Mapping and Export action will accept any customized values from *Interaction Analytics*, and change the user interface names of those fields to match the values in *Interaction Analytics* Search and Reporting.
    - NXESIDW JDBC connect string - Enter the value to reach the NXESIDW database where the metadata mapping table is located. Use the format "jdbc:sqlserver://{databaseServer}[\namedInstance];databaseName=NexidiaESIDW".
    - NIA version - Specify the application version. The valid options are 10.x; 11.0; 11.1, or 11.2; 11.3, 12.x.

- Database user name - Enter the database user name.
- Database password - Enter the database password.

16. Click **Submit** to save your Administration server configuration updates.

To upload patches:

1. Click the **Choose** button to navigate to files to upload.
2. Select the Patch tab and then select a file or files and click **Upload**. The following patch types are available:
   - actions.jar files - This file type may be uploaded to distribute customer-related action classes to all worker servers as a hot patch.
   - patches.jar files - This file type is a *Interaction Analytics*-provided hot patch file for overriding core process action components. It is automatically distributed to worker servers.
   - ndef#.#.jar or .war files - This file type is a system patch that triggers automatic server process quiescence, patching, and updates.
   - basic.jar files - This file type is distributed to worker servers, and should be used in support of customer-created actions. These also trigger server process quiescence + restart.

To decrypt stored password in DEF actions:

1. Click the Password tab.
2. Enter the encrypted password, process ID, and Process Property ID to decrypt the password. Only users with SUPER-level privileges can decrypt the password.
3. Click **Decrypt**.

USER MANAGEMENT

You will use the User Management window to create new users, delete users, edit users, change a user's password, add or remove a user to or from a company, and to add or remove a user to or from a role. Access this window by clicking Settings from the sidebar navigation options, and then selecting User Management.

To add a new user:

1. Click the **Add a user** button to open the Add a User window.
2. Enter the **user name** in the User Name field.
3. Enter the **e-mail address** of the user in the E-mail field.
4. Enter the **first name of the user** in the First Name field.
5. Enter the **last name of the user** in the Last Name field.
6. Enter the **password for the user** in the Password field.
7. Click **Submit**. The user is added to the Admin User Management window.

To delete a user:

1. Do one of the following:
   - Click the User options button to the right of the user name that you want to delete, and then select the Delete User option.
   - Select the user names that you want to delete, and then click the Delete selected users button.

To edit a user:

1. Click the **User options** button to the right of the user name that you want to edit, and select the Edit User option.

2. In the Edit User Window, make changes to the user information as needed.
3. Click **Submit**. The user information is updated on the User Management window.

To change a user password:

1. Click the **User options** button to the right of the user name that you want to edit, and select the Change Password User option to open the Change Password window.
2. Change the password as needed.
3. Click **Submit** to save the change and close the window.

To add or remove a user to or from a company:

1. Click the **User options** button to the right of the user name that you want to edit, and select the Assign user to companies option to open the Assign a User to Companies window.
2. Select the **check box** next to the company name that you want to associate this user with. You can select multiple companies for a user. To remove a user from a company, de-select the check box for the company.
3. Click **Submit** to save the update and close the window.

To add or remove a user to or from a role:

1. Click the **User options** button to the right of the user name that you want to edit, and select the Assign user to roles option to open the Assign User to Roles window.
2. Select the **check box** next to the roles that you want to associate this user with. The possible values are ADMIN or SA.
3. Click **Submit** to save the update and close the window. The user will be required to logout of the application, and then log in again for the user role changes to take effect.

USER AUDIT LOG

The User Audit Log provides a list of user activity logs that include add/delete company, site, and user. When you select the User Report option, the User Activity Report window opens with the log information. Access this window by clicking Settings from the sidebar navigation options, and then selecting the User Audit Log option.

The following information is displayed for each user:

- User ID
- Event ID
- Event Time
- Data

You can filter the display based on the User Name or an Event Name to narrow the display of information.

→ **To apply a filter:**

1. Do one of the following:
    - Select a user name from the **User** drop-down list.
    - Select an event name from the **Event** drop-down list.
2. Click the **Refresh** user activity list button. The information displayed is modified to match your filter selection.

ADMIN ROLE PERMISSIONS

Administrative users use this window to define the tasks that are allowed for the different user roles. The application populates the data during starting processes and related activities. Access this window by clicking Settings from the sidebar navigation options, and then selecting the Role Permissions option.

The following information is displayed:

- ID - A system-generated identification number.
- UI Entity - Defines which HTML elements in the user interface are visible to the user.
- Event - Events that the user role is allowed to invoke.
- Access - Specifies the objects that the user role has right-click access to.
- User Roles - The associated user role.

To add a permission:

1. Click **Add a permission**. The Add Permission window opens.
2. Enter a UI Entity.
3. Enter an event.
4. Enter a process.
5. Select the user-role(s) that will have access.
6. Click **Submit**.

To edit a permission:

1. Right-click on the UI Entity and select the **Edit permission** option. The Edit Permission window opens.
2. Modify the current specifications as needed.
3. Click **Submit**.

To add a role to a permission:

1. Right-click on the UI Entity and select the **Add role to a permission** option.
   The Add role to a permission window opens.
2. Add and or remove user roles as needed by.
3. Click **Submit**.

ACTIONS PREFERENCES

Administrative users use this window to specify the availability of actions. The default value for each action is enabled. Access this window by clicking Settings from the sidebar navigation options, and then selecting the Actions Preferences option.

The following information is displayed:

- Friendly Name - The name used for the action.
- Version - The current version of the action.
- Description - A description of the action.
- Disabled - Indicates if the action is disabled.

To disable an action:

1. Select the **Disabled** check box. Your selection is automatically saved.
   When you disable an action, it no longer displays in the process builder. If an action is not in use by any processes, it does not display in the list of actions.

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3

Deployment View

DEPLOYMENT VIEW

The Deployment View provides a graphical representation of all processes for a specific deployment in a single view so that you can easily understand and debug the workflows associated with it. When you access the Deployment View you can:

- Get a list of servers.
- Get a list of processes for each server.
- Get a list of resources for each process.
- Identify the input resources for a rocess.
- Identify the output resources for a process.
- Identify resources that are shared across servers.
- Configure the Graph.

User Interface

When you access the Deployment View, a graphical representation of the selected company displays. Each element is represented as a graphic which indicates the element type. You can zoom in and out of the display, and move the display up or down as needed.

A Legend/Filters/Configurations pallet scroll area resides on the left side of the screen. You can:

- Use the legend as a reference of the chart display information.
- Filter on companies, servers, and process groups.
- Specify the configuration and update the chart display.

Legend

The Legend pallet displays the element types, status indicators, and process relationships.

- Available element types: Server, Process Group, Process, Database Resource, File Resource, and SFTP Resource.
- Status indicators: Critical, Warning, Running, and Online.
- Process Relationships: Changed and Launched.

The running processes display in gray and then turn green or red depending on the status after processing completes. They total number of files display as the process runs.

Filters

The current filters for the selected company are displayed in the Filters pallet. You can change the selections for the companies, servers, and process groups.

Configuration

The current configurations for the selected company are displayed in the Configuration pallet. You change the selected layout and direction, specify resource connections, select level separation and node spacing. Click Update Chart to display the chart with your updates.

→ To use the deployment view:

1. Click on a node to view its properties and most recent activity. At the server level, if there are active compute nodes attached, CN and the number of compute nodes attached is displayed to the right of the server name in the display. When you hover over this display, the IDs of the active compute nodes display.
2. You can make changes as needed by selecting from the available options. The options that are available vary depending on the node type you selected.
    - **Hide button** - The node and associated children are hidden from the view.
    - **Filter button** - Allows you to change the filtering options for the selected item.
    - **Clear Filter button** - Clears the current filtering options for the selected item.
    - **Properties Tab** - Displays the status information for the selected item.
    - **Edit Tab** - Allows you to make changes to the selected item. The appropriate Edit window displays. Refer to the following topics for information about the options that you can modify.

  - Configure Servers
  - Configure Processes
  - Configure Resources
- **Recent Activity** -Displays the recent activity for the selected item.
- **Design** - Displays the Workflow Designer window for the process.
- **Launch** - Launches a process.
- **Start** - Starts a process.
- **Stop** - Stops a process.
- **End** - Ends a process.

3. Optionally make changes to the graph configuration.
   - Change the method for sorting the nodes that are displayed view between Directed and Hubsize. HubSize sorting takes the nodes with the most edges (or lines coming into or out of the node) and puts them at the top. Directed sorting follows the order that the nodes where created starting with the Server nodes, then Process Group nodes, then Process nodes, and finally Resource nodes. Directed sorting is the more useful of the two sorting methods since it sorts the nodes in a hierarchical fashion with server nodes at the top.
   - Connect resources. A dotted line is added between resources.
   - Connect across servers only.
   - Specify node spacing. Node spacing is the minimum distance between nodes on the free axis. The number can be any positive number (for example: 50, 100, 150, 200). The node spacing can be re-configured in cases where you have a really congested chart that is hard to view. Increasing the node spacing typically spreads out the spacing between the nodes.

4

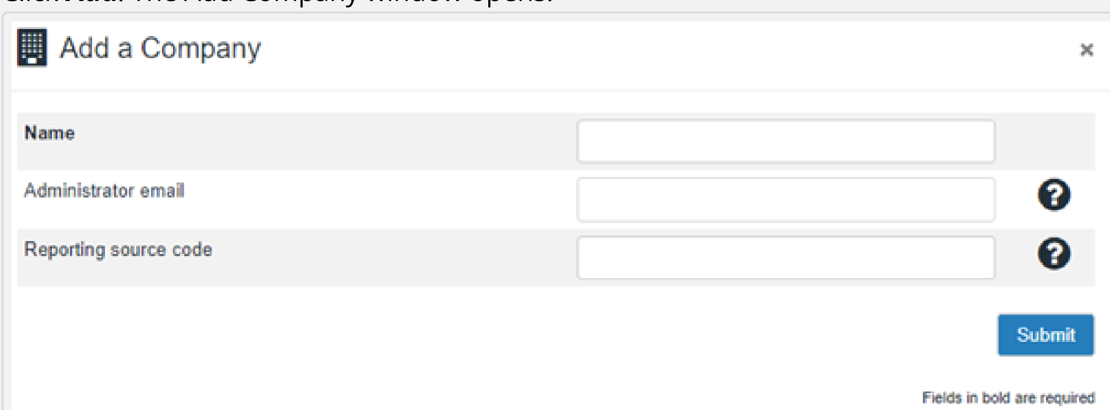
Process Workflows

COMPANIES

Companies are the first component to create in your workflow. Access the Companies window by selecting the Companies node in the tree view. When you select a company name, the configured properties and most recent activities are loaded for it.

To add a company:

1. Click **Add**. The Add Company window opens.



-

2. In the **Name** field, enter the company name.
3. In the **Administrator Email** field, enter the administrator email address for processing notifications. If you do not enter an address, the application administrator account is used.

4. In the **Reporting source code** field, enter a value to override the default company name in generated reports.
5. Click **Submit**. The company is added to the grid with a status of New.

To run and view the Deployment Report:

You can generate a detailed HTML report of a customer deployment that includes company, sites, servers, processes, resources and admin configuration settings.

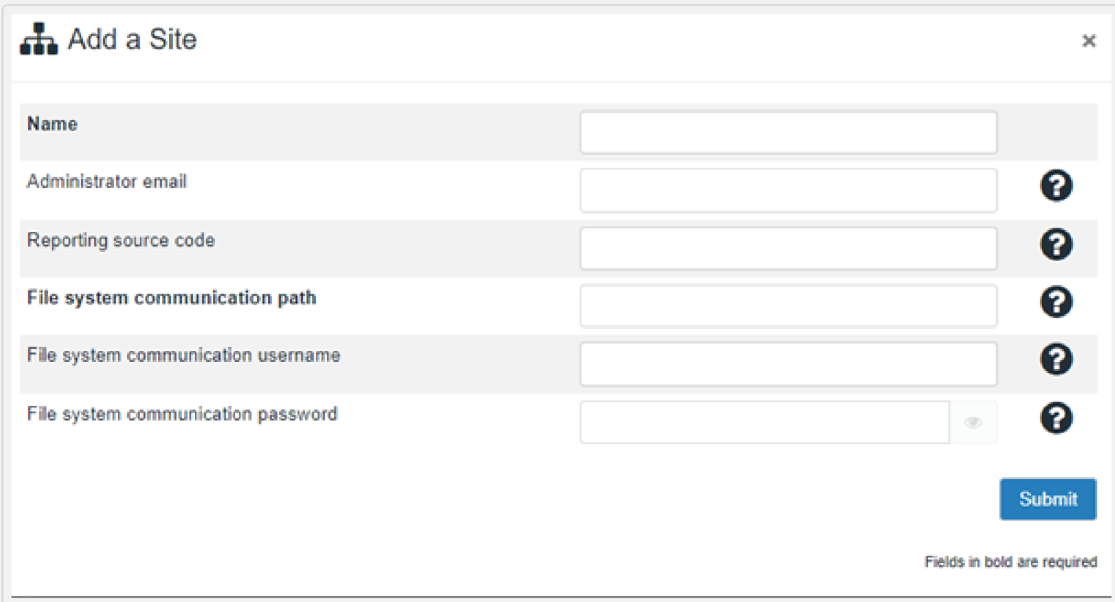
1. Select a company and click **Deployment Report**. The Deployment Report window opens.
2. Click the checkbox to the right of the report options to select them:
   - **Show Quick Start Guide**
   - **Show Process Components**
   - **Show Process Thumbnails**
3. In the **Available Processes** table select the processes to include in the report.
4. Click **Run Report**. A confirmation message displays.
5. Click **OK** to generate the report.

SITES

After you have created a company, you need to create sites for the company. Access the Site window by selecting the Sites node for an existing company. When you select a site name, the configured properties and most recent activity are loaded for the site. You can toggle between these tabs.

To add a site:

1. Click **Add a Site**. The Add a Site window opens.



2. Enter the site name in the **Name** field.
3. Enter the contact e-mail address for the site in the **Administrator E-mail** field.
4. Optionally enter a value to override the default column value that will display in reports in the **Reporting source code** field.
5. Enter the **File System Communication Path**. This is the root path where server communications occur. In most cases, this is "c:\path\to\comms\folder", and servers configured beneath this site will define sub-folders beneath that. For example, a server's "\serverOne" communication path would resolve to "c:\path\to\comms\folder\serverOne".

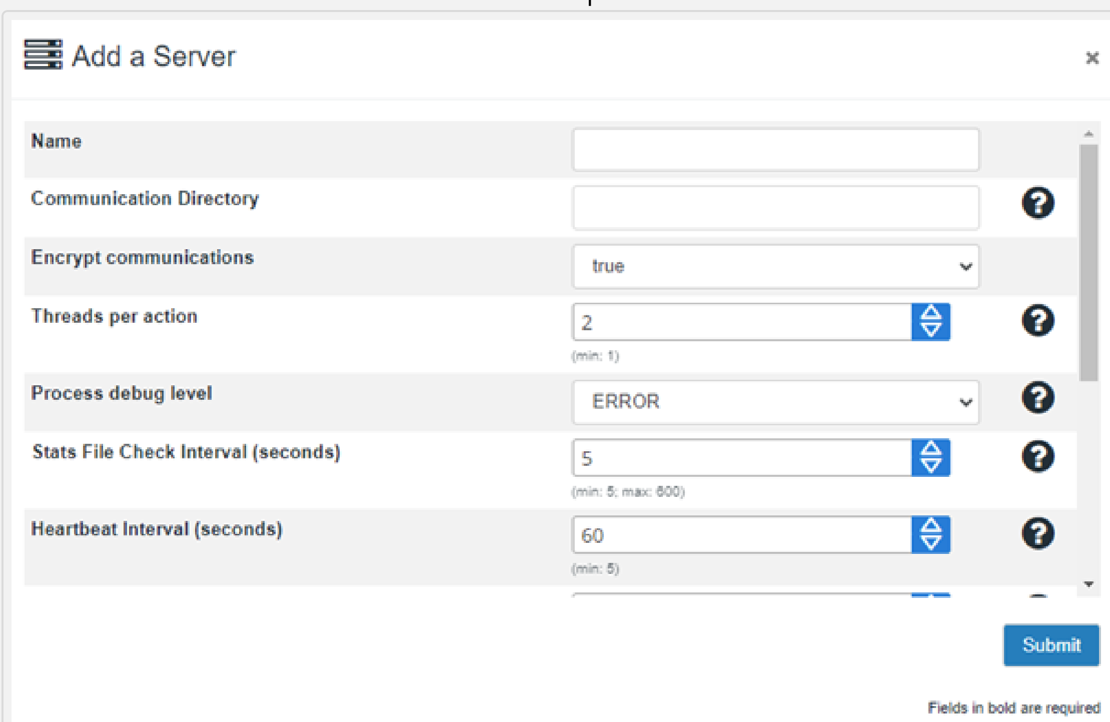
6. Enter the **File System Communication User Name**. For Windows authentication only; "username" or "domain;username".
7. Enter the **File System Communication Password**. For Windows authentication only.
8. Click **Submit**. The site is added to the grid with a status of New.

SERVERS

After you have created sites for your company, you need to add servers. Access the Servers window by selecting the Servers node for an existing site. When you select a server name, the configured properties, most recent activity and compute nodes are loaded for the server. If there are compute nodes attached to the server, their IDs are displayed along with the root node. You can toggle between these tabs.

To add a server to a site:

1. Click **Add a Server**. The Add a Server window opens.



2. Enter the server name in the **Name**.
3. Enter the communication directory to be used for communication with the Administration server in the **Communication Directory** field. This is the folder path used by this server for communication with the administration server. The full path for the SFTP directory is /server name/communication directory. The system verifies the path provided. If it is invalid, this value is left blank.

4. Specify if you want the application to encrypt communications. When you select ***true***, all communications to and from the Administration server are encrypted. This option is only valid for SFTP or File System servers.
5. Modify the threads per action value as needed. This is a global setting for the default number of threads each action on this server is allowed to partition its work into. This can be overridden at the action level. The default value is 1.
6. Modify the process debug level as needed. This is the level of messages from the processing server to the administration server. The valid values are ***ERROR***, ***DEBUG***, ***INFO***, ***WARN***, ***ERROR*** and ***FATAL***. The default value is ***ERROR***.
7. Modify the stats file check interval seconds as needed. This specifies how often processes send the status of their currently running processes to the Administration server for review. The default value is 5.
8. Modify the heartbeat interval seconds as needed. This specifies how often to send tests for heartbeats between the Admin and Processing servers. The default value is 60.
9. Modify the file check interval seconds as needed. This specifies how often to check for synchronization between the Admin and Processing servers. The default value is 30.
10. Modify the log capture threshold as needed. This specifies the maximum number of log records to send back from Processing servers to the Administration server in each process run. The default value is 100.
11. Modify the server retention days as needed. This is the number of days beyond which it is safe to delete DEF internal database and file system logs, the contents of auto-provisioned file system resources, and old rows in all working tables' rollover and quarantine tables. Automated cleanup of these items happens nightly. Note that active working table cleanup is also included in the nightly run, but it uses the maximum self-reconciliation setting at the process level and ages out from there into the corresponding '{working table}$Q' quarantine table. This setting then determines the aging out from there.
12. Modify the DEF internal log retention (days) as needed. This is the number of days beyond which it is safe to delete the DEF internal logs. Automated cleanup of these items happens nightly.

13. Modify the audit log retention days as needed. This specifies the number of days beyond which it is safe to delete old audit log files. Automated cleanup of these items happens nightly. Audit logs are distinct from regular logging, and include only output from those actions that require it (like the NMF Decrypt suite, or the File Mover when it is in auditing mode for specific extensions). The default value is 60.
14. Enter the e-mail address for the server administrator. Notifications and messages are delivered to this e-mail address automatically.
15. Optionally enter a value to override the default column value that will display in reports in the Reporting source code field.
16. Click **Submit**. The server is added to the grid with a status of New. The ID is the database key. This ID is generated when the administration server is created. The same ID also must be used for the processing server and is used for communication between the two. In order for the status information to properly update, the server identification must be shared between the servers.

To review the server process time line:

1. Select a server and click **Process Analysis**. The Process Analysis - <Server Name> window opens with the Server Process Timeline tab displayed.
2. Enter or select the desired date.
3. If there are active compute nodes for the server, select a compute node from the **Root Node** drop-down list.
4. Click **Run**. The window is populated with information about the processes. You can see when process are running together on a specific date, to troubleshoot how processes relate to system performance. You can use the mouse to drag the time line left or right, or use the scroll wheel to zoom the time line in or out. The total number of records processed by all actions in each process is displayed in the colored segment. You can hover over the colored segment to display the total time taken and the total number of records processed. You can also click on segments to see a breakdown by action from the nx_process_block_log table.
5. Click **Close**.

To review the action inventory:

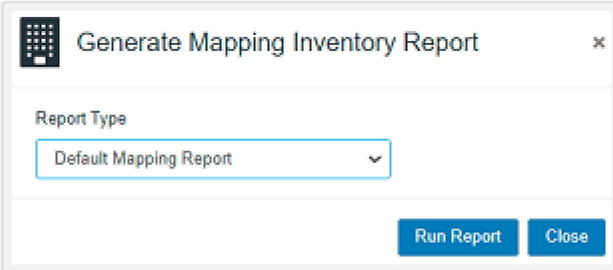
1. Select a server, click **Options**, and select **Process Analysis**. The Process Analysis - <Server Name> window opens with the Server Process Timeline tab displayed.
2. Select the Action Inventory tab. The available actions and the associated counts display in a tree view.
3. Expand the actions to see the processes where the action is used.
4. Click on processes as needed to display a preview of the process.
5. Click **Close**.

To review the resource inventory:

1. Select a server, click **Options**, and select **Process Analysis**. The Process Analysis - <Server Name> window opens with the Server Process Timeline tab displayed.
2. Click the **Resource Inventory** tab. The available resources and the associated counts display in a tree view.
3. Expand the resources to see the processes where the resource is used,
4. Click on **processes** as needed to display a preview of the process.
5. Click **Close**.

To review the default mapping inventory:

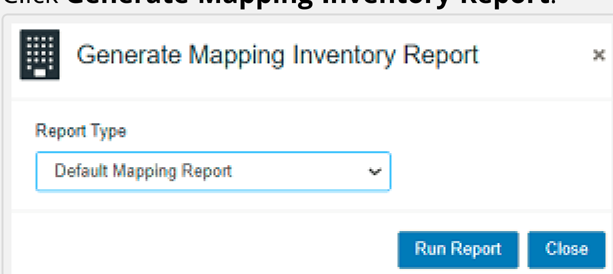
1. Select a server, click **Options**, and select **Process Analysis**. The Process Analysis - <Server Name> window opens with the Server Process Timeline tab displayed.
2. Click the **Inbox Mapping** tab. You can view mapping of NIA fields with metadata fields for listed processes and InboxPublisher action.
3. Click **Generate Mapping Inventory Report**.



4. Select Default Mapping Report, then click **Run Report**. The report is generated in PDF format, and shows detailed mapping such as process name, action name, and NIA field.
5. Click **Close**.

To review the full mapping inventory:

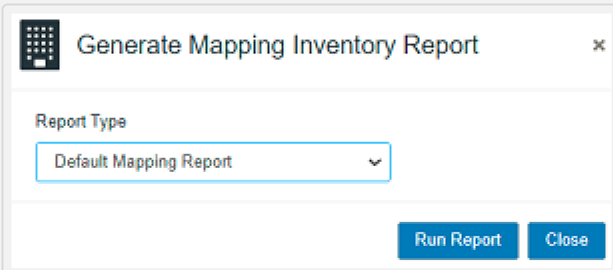
1. Select a server, click **Options**, and select **Process Analysis**. The Process Analysis - <Server Name> window opens with the Server Process Timeline tab displayed.
2. Click the **Inbox Mapping** tab. You can view mapping of NIA fields with the metadata field for listed processes and the InboxPublisher action.
3. Click **Generate Mapping Inventory Report**.



4. Select **Full Mapping Inventory**, then click **Run Report**. The report is generated in PDF format, showing the mapping of various NIA fields.
5. Click **Close**.

To review the full mapping inventory in CSV format:

1. Select a server, click **Options**, and select **Process Analysis**. The Process Analysis - <Server Name> window opens with the Server Process Timeline tab displayed.
2. Click the **Inbox Mapping** tab. You can view mapping of NIA fields with metadata fields for listed processes and InboxPublisher action.
3. Click **Generate Mapping Inventory Report**.

4. Select **Full Mapping Report (CSV)**, then click Run Report. A CSV report is generated with the following header fields:
   - NIA Field
   - Mapped Field
5. Click **Close**.

To review the NEST (worker server) properties:

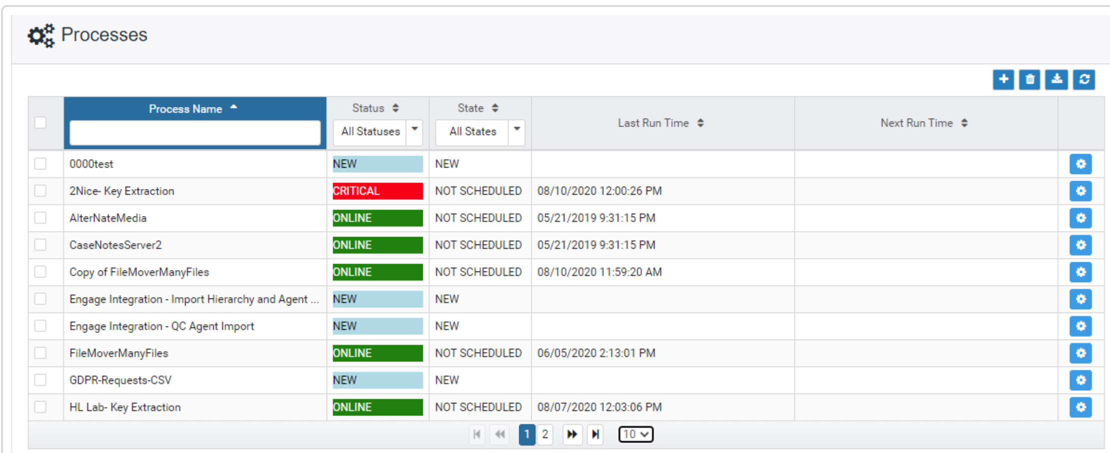
1. Right click on a **NEAT (administration server) server name** to display the Properties tab. The following NEST (worker server) properties are displayed:
   - NEST Host Name
   - NEST Communication Mode
   - NEST User Home
   - NEST Database Host
   - NEST Database Name

- NEST Database Port
- NEST Shutdown Port
- NEST Version
- NEST Service Name
- Patches
- Actions
- (Optional) Cluster Information (this only displays for clustered environments)
- (Optional) Compute Node Information (if there are multiple compute nodes attached to the NEST Server, they are all listed)

PROCESSES

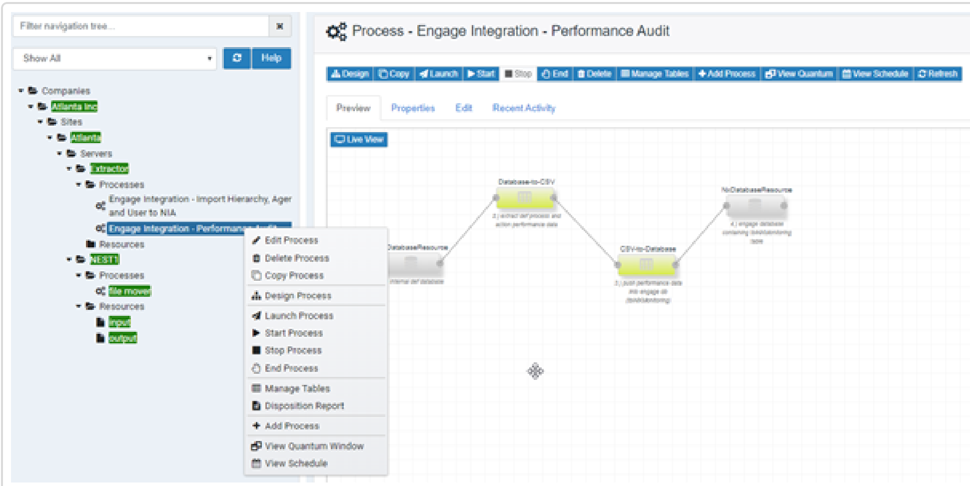
Each server has defined processes. These processes are tasks such as moving files or filtering files. Access the Processes List window by selecting the Processes node in the tree view. When you select the Processes node in the tree view, a list of processes displays with the current status, state, last run time, and next run time details.

The processes list page also has a drop-down box where you can select the list size to display. The options are 10, 15, 20 and 50 and the number of processes shown changes based on the selection.



When you select a process, the configured properties and most recent activity are loaded for the server. You can perform these actions for existing processes: design, copy, launch, start, stop, end, delete, export, manage tables, add process, view quantum, view schedule details, reset quantum and refresh. When you copy a process, the application prompts you to enter a new process name and a process group name.

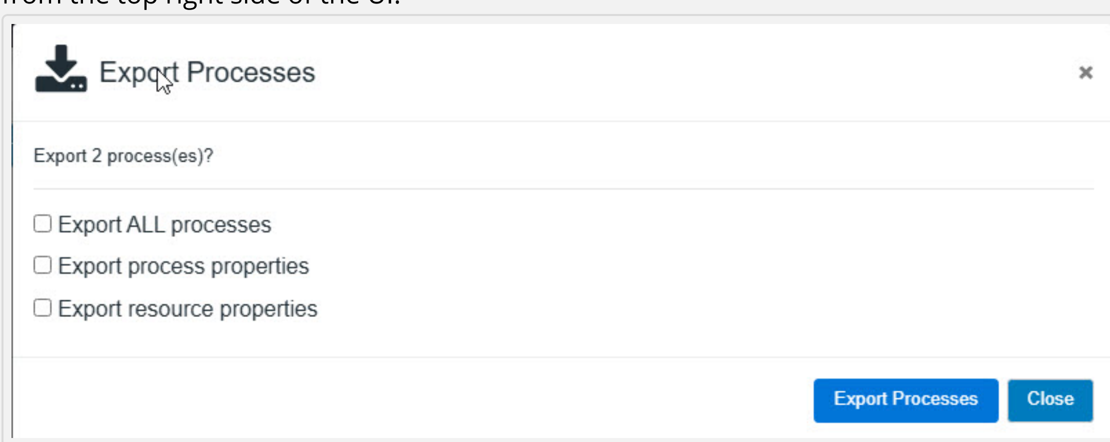
After copying a process, you need to publish it manually.



➔ To export selected processes:

You have the option to export selected published processes or to export all published processes for the current server. The exported processes are added to the exported_processes.zip and placed in the users Downloads folder.

1. In the tree view, click **Processes.**
2. Select the processes you want to export and click **Export selected processes** button from the top right side of the UI.



3. In the Export Processes window:

- Click the **Export Processes** button to export the selected process.
- Select the **Export All Processes** checkbox and then click the **Export Processes** button to export all the processes for the current server.
- Select the **Export process properties** checkbox to export process properties to the process JSON file. Process properties include a subset of the available properties on the process edit tab.
- Select the **Export resource properties** checkbox to export resource properties to the process JSON file. Resource properties include selected resource properties for each resource specified in a process design. You need to click each resource and click **Submit** to complete resource configuration before saving and publishing the new process.

Process Live View

When you select a process from the tree view, a preview of the process as it is currently defined displays. You can optionally select the **Live View** button to see the following information which is refreshed automatically. When live view is enabled, the LIVE VIEW ON message flashes next to the Live View button. Hover over this message to display the date and time stamp for the live view results.

- A live view of action state(s) based on events.
- A working indicator that shows the process work state.
- Summing of file counts for each action.

If the compute node availability is enabled, a drop-down is provided where you can select which compute node whose live results you want to see. The results change based on your selection.

Working in the Process Designer:

When you are working in the Process Designer, you can use these short cut keys as you are making modifications:

ACTION	DESCRIPTION
Shift-Click	Shift click and the drag mouse to select multiple components.
Right-Click	Right-click to display the floating tool palette.
Click	Click on a component to edit the selected component.
Right-Click a Component	Right-click on a component to display the component context menu.
Control-Click	Control-click to select one component at a time. Control click on a component to select that one component. Control-Click the component again to de-select the component.
Shift-Click on Multiple Components	Shift-click to select a range of components. Shift click on a component to start the selection of a range of components. Shift-Click on another component to complete the range, and then select all of the components in between.
Left-Click - Control-A - Add Annotation	Left-Click - Control-A to creates an annotation/note at the location of the last mouse left click.

To add a process to a server:

1. Click **Add a Process**. The Add Process window opens.
   The integration templates are pre-configured process definitions used to simplify DEF process design. When you add a process to a server from the Processes window, you can select one of these templates. The actions in the templates are pre-configured, but you can change them. You will configure the resources for your environment. There are placeholders for the resources in the templates.
   The predefined process templates have been removed from the user interface and are now available on the Data Exchange Framework Confluence site for those who have access to that site (https://tlvconfluence01.nice.com/display/NXDEF/Process+Library). They are also available for download as a complete package via the NiCE Software Download Center (https://nice.subscribenet.com/control/nice/login?nextURL=%2Fcontrol%2Fnice%2Fpurchases). You can also import a process by clicking **Select a Process Template (*.json)** and then navigating to the process that you want to import. The process and resource properties are also imported if they are present in the exported process JSON file. This reduces the amount of configuration needed for a newly imported process. Before saving and publishing the new process, open each resource and click **Submit** to complete the setup.
2. In the **Name**, enter a name for the process.
3. In the **Description**, enter a description for the process.
4. In the **Start Time**, enter or select the time the process should begin.
5. In the **Interval**, specify the interval. Use the up and down arrows to adjust the interval.
6. In the **Interval Type** drop-down, select the interval type. The valid values are ***Second***, ***Minute***, ***Hour***, or ***Day***.
7. In the **Process retention (days)**, enter the number of days you want to keep data for this process. This overrides the server-level retention value. If you leave this field empty, the system will use the server-level retention value.
8. In the **Blackout Start Time**, enter the beginning time for which this process cannot be scheduled to start.

9. In the **Blackout End Time**, enter the ending time for which this process cannot be scheduled to start.
10. In the **Quantum (minutes; 0= none)** select a quantum value in minutes. This value indicates the number of minutes to collect a group of items starting from that point and moving forward. A value of 0 indicates "none". You can also enter negative values in this field to pull historical calls, starting with the latest data and moving backward.
11. In the **Quantum Lower Bound Date**, select a quantum lower date bound. This date is used if the system has no "last execution time" for the action component using it. This is used as a starting point for pulling data. Changing this value is the trigger for resetting the last execution times for any action components within the process that are using it.
12. In the **Quantum Upper Bound Date**, select a quantum upper date bound. This is used when you do not want to allow the quantum window to advance beyond the specified date.
13. In the **Quantum Time-zone Offset (minutes)**, select a quantum time zone offset in minutes to shift the quantum window start and end times to match the time zone of a target system.
14. In the **Quantum Start Time Shift**, enter an offset in "+/dd:hh:mm" format to shift the quantum window start time.
15. In the **Quantum Stop Time Shift**, enter an offset in "+/dd:hh:mm" format to shift the quantum window stop time.
16. In the **Process Group**, specify a process group for this process. This is used for logical separation of incoming data. Each process group will have a separate working table where it tracks files it works on within. Data can be joined across process groups using the Metadata Join action. Alphanumeric characters only are allowed, and all others are removed on save. The maximum amount of characters allowed is 18.
17. In the **Notification email on completion**, specify if you want an e-mail notification to be sent when the process completes. Select ***false*** to *only* receive email notifications for process failures, or ***true*** to *also* receive email notifications for successful process runs.

18. In the **Pct variance notification threshold**, specify a percent variance. If non-zero, it compares the number of records processed in any given run against either a fixed number ('Expected volume,' below), or against the number of records processed seven days prior. If that number does not fall within a range determined by this percentage, a notification email is sent to all defined administrators.

    Example

    If 900 records were processed 7 days ago, and this week 1000 are processed with a variance threshold of 20%, the check would be to see if 900 is between 800 and 1200. Since it is, no notifications would go out. With a variance threshold of 5%, though, the application would be checking to see if last week's 900 is between 950 and 1050. Since it's not, it would send out the notification. The notification also breaks down the number of records processed by each action component that comprises the process.

19. (Optional) In the **Expected volume for variance check**, leave empty to test variance a run seven days prior, or enter a single number to be used for every process run. You can also enter a comma-delimited group of numbers for the days of the week, with the first number starting on Sunday.

    Example

    "1000,5000,6000,5000,3000,3000,1000". If you leave this field blank, the % variance notification threshold that you specified is checked against the number of records processed seven days ago. If you enter a value, the application checks to see if the current run's processed records fall within that number +/- the % variance.

20. In the **Auto-tune thread counts**, specify if you want the application to auto-tune thread counts. When you select ***true***, the application attempts to auto-tune the thread counts used by each action that supports multi-threading, provided those actions don't provide their own non-zero advanced "Thread count" overrides. Select ***false*** to accept the default behavior of using the Server-level "Threads per action" setting as the default, with action-level overrides. This feature works by starting with a single thread per action, then will turn up or down by one in each subsequent run depending on performance comparisons in previous runs.

21. In the **Include concurrently-running actions** field, select "false' to have auto-tuning only consider each action's own performance on previous runs when deciding how many to allot on the next one. Select "true" to have it also consider other DEF actions running concurrently in its decision-making. The differences become apparent as more concurrently-running processes are added to the same server, and that as the number of possible concurrently-running actions grows, auto-tuning will change values much more conservatively.
22. In the **Administrator email**, enter an email account to send process engine notifications to. If nothing is defined here, the application checks the server, site, company and administrator in that order.
23. In the **Maximum days to attempt self-reconciliation** set a maximum number of days back to be used by actions when attempting self-reconciliations. This can be overridden at the Action level.
24. In the **Runtime alert threshold (minutes)**, specify the run time alert threshold. If you specify a non-zero value, an email notification is sent if the process runs for longer than the configured time in minutes during a single run. The default value is 120 minutes.
25. In the **Terminate process at runtime threshold**, specify if you want to force termination of a process that has exceeded its runtime threshold.
26. In the **De-dupe rollover table**, specify if you want to delete any existing rollover table rows with File Names matching incoming working table rows during the end-of-process rollover of working table data in the De-dupe rollover table field.
27. In the **Disable self-cleaning**, specify if you want to disable the default, automatic nightly cleanup, based on the server retention setting, of the working table backing this processes group. If you disable the nightly cleanup for any process in a process group, it disables it for all processes in that group.
28. In the **Audit results**, specify if you want the application to audit process run results. The default value is ***true***. When ***true***, audit results are written out in standard Log4J audit log format to {DEF_DATA}/audit_logs/process_run_results/pid-{process id} (with .yyyymmdd appended with each day's rollover). The following items are captured for each action in CSV format.
29. In the **Compute node availability** drop-down, specify if you want the process to use computed nodes. When this option enabled, and you are in the Live View, you can select which compute node whose live results you want to see.

30. In the **Enable metadata processing**, set to ***true*** to enable metadata processing. When enabled, previously seen metadata skips all but key data delivery steps like Inbox Publish. Two new fields display below this field where you configure which dispositions to not pull back out of the rollover table, with a few common, recommended ones set at the default, and additional metadata that should also be pulled up. Using those settings, imported metadata is tested against the rollover table to see if a matching OriginalFileName exists. If it does, configured disposition column data and metadata are copied into incoming metadata preventing processing by unnecessary actions when not required (for example: media operations). Set to ***false*** to process as usual, skipping nothing. Enabling reprocessing incurs a slight performance hit.
31. In the **Order of processing** drop-down, specify the order (ascending or descending) for file processing. The default is ***no order***.
32. In the **Enable enhanced logging** drop-down, specify how the process activity is logged. The Process level log file is always at the DEBUG level and it is only generated only when you set this configuration to ***true***. If the process runs on a Compute Node, then the process level log file is created on both the Compute Node and the Worker server (NEST). They each log the files that they process to their respective files. The Server log level applies to the main Data Exchange Framework log file. If it is DEBUG, then the process level logs are included in both the main log file and the process log file. If multiple processes have it enabled and are running simultaneously, the specific process logs are included in their specific log files. Any errors encountered in the process will be included in both the process log file and the main log file.
    - ***True*** - Logs are broken out into separate files for each action's logging. These logs are limited to 10MB maximum size for each, with 20 rollovers (each rollover file will be named *.1, *.2, etc.); so a maximum of 200MB will be captured at any time. Note that standard logging is best for a complete picture of all process and Server Tool activity, and enhanced logging is best for troubleshooting, especially when standard logs grow too large to effectively review process or action specific failures.
    - ***False*** - All process activity is logged to the DEF's standard log files as usual.
33. In the **Awaited media failure threshold (%)**, specify a percent variance. This variance is for files with failing status against the files that are in new or non failing status.

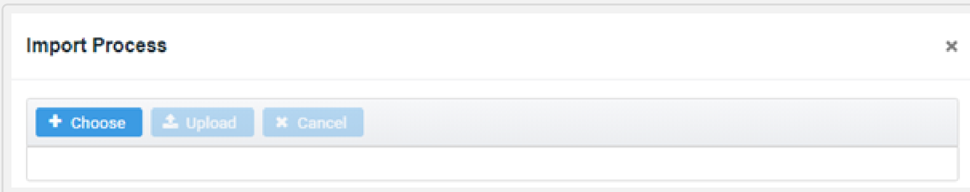
You can specify how many files to process that are in the working table in a failing status. You can use this for backlog mitigation of media files. The files are processed from input resources and metadata.

34. In the **Backlog resource maximum scan limit**, specify the total number of files that you want to scan. These files are not processed from metadata.
35. In the **Check dispositions while scanning**, set to ***true*** to prevent rediscovering of the files that are in the correct disposition. Set to ***false*** to rediscover files.
36. In the **NIA Database Host For SYNC**, set to ***true*** if you want to sync NIA database and import metadata for grouping.
37. In the **Enable Reconciliation Data-gathering**, set to ***true*** to enable reconciliation data. The process must run at least one time to capture metadata from the working table. The data will be displayed in the NIA database in the UTC format.
38. In the **Enable retry process after failure**, set to ***true*** if you want the system to retry as per the schedule if the process fails.
39. In the **Process Type**, select ***production*** to monitor processes even when they are offline or unscheduled. This is the default setting. Select ***testing*** to exclude them from monitoring.
40. Click **Submit**. The process is added to the grid with a status of N. After you add a process, you will design it in the Process Designer. See Designing a Process to learn more.

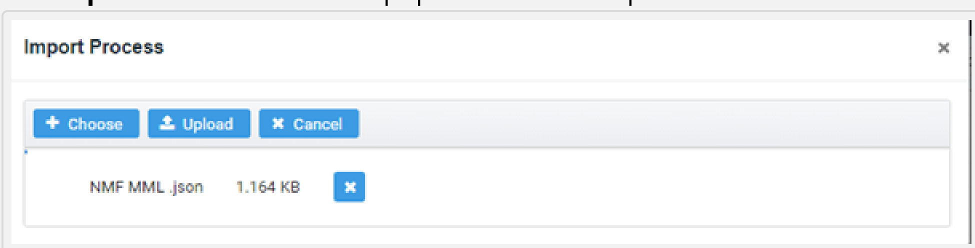
To update first-class fields, you need to modify the ***ndef.properties*** file. This file is located in the NEAT>Data Exchange Framework>etc folder. In the ***ndef.properties*** file, make sure to specify the configuration path for the NIA, where the first-class fields are stored.

To import a process into a process that does not contain an existing process definition:

1. In the tree view, select a process and click **Design**.
2. Click **Import**. The Import Process window opens.



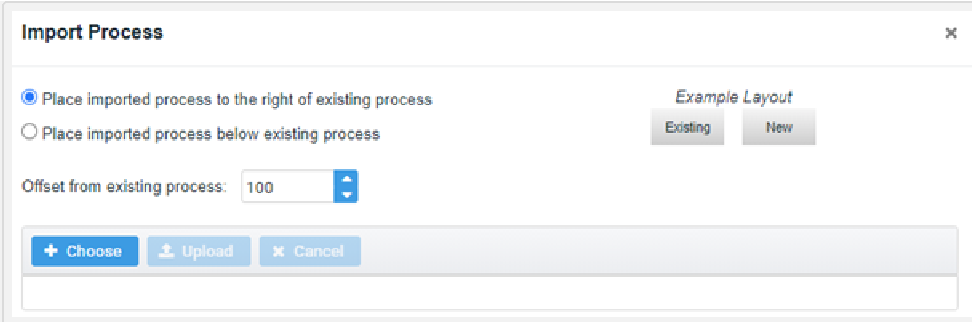
3. Click **Choose**, and in the Open window select a .json file to upload.
4. Click **Open**. The selected file is populated in the Import Process window.



5. Click **Upload**. The process uploads in the designer.

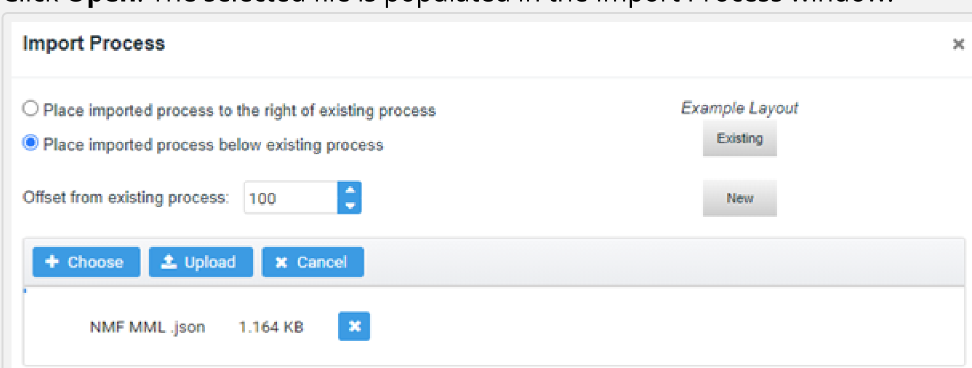
To import a process into a process that contains an existing process definition:

1. In the tree view, select a process and click **Design Process**.
2. In the Process Designer, click **Import**. The Import Process window opens.



3. Select one of the following import options:
   - **Place imported process to the right of existing process.**
   - **Place imported process below existing process.**

   The process and resource properties are also imported if they are present in the exported process JSON file. This reduces the amount of configuration needed for a newly imported process. Before saving and publishing the new process, open each resource and click **Submit** to complete the setup.
4. Specify the offset distance between the existing process definition and the newly imported process definition. The default offset is 100 pixels.
5. Click **Choose**, and in the Open window select a .json file to upload.
6. Click **Open**. The selected file is populated in the Import Process window.



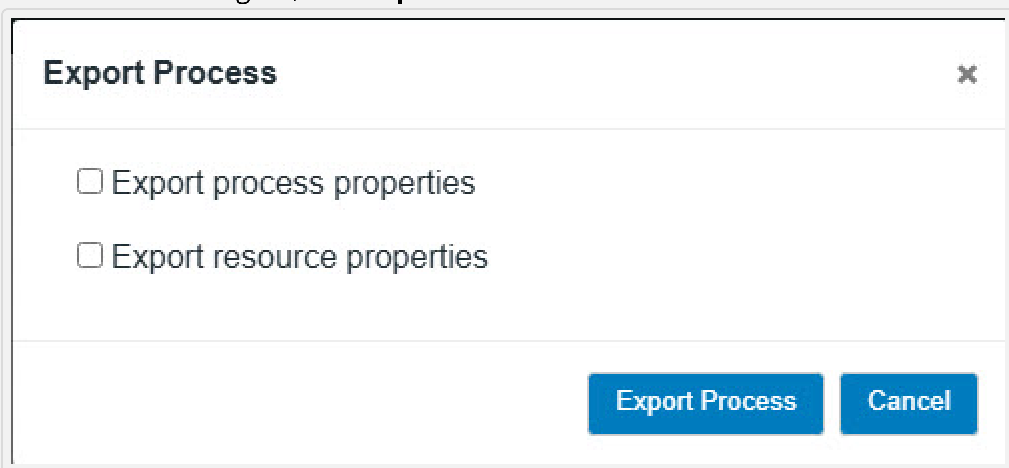
7. Click **Upload**. The process uploads in the designer.

To export processes:

You have the option to export selected published processes or to export all published processes for the current server. The exported processes are added to the exported_processes.zip and placed in the user's Downloads folder.

The process designer Export button is always enabled. An error displays if you try to export a process that is not yet published.

1. In the tree view, select one or more processes from the Process list and click **Design Process**.
2. In the Process Designer, click **Export**.



3. In the Export Process window:
   - Select the **Export process properties** checkbox to export process properties to the process JSON file. Process properties include a subset of the available properties on the process edit tab.
   - Select the **Export resource properties** checkbox to export resource properties to the process JSON file. Resource properties include selected resource properties for each resource specified in a process design. You need to click each resource and click **Submit** to complete resource configuration before saving and publishing the new process.
4. Click the **Export Process** button to export the selected process.

To delete a process:

You cannot delete processes that are currently scheduled or running.

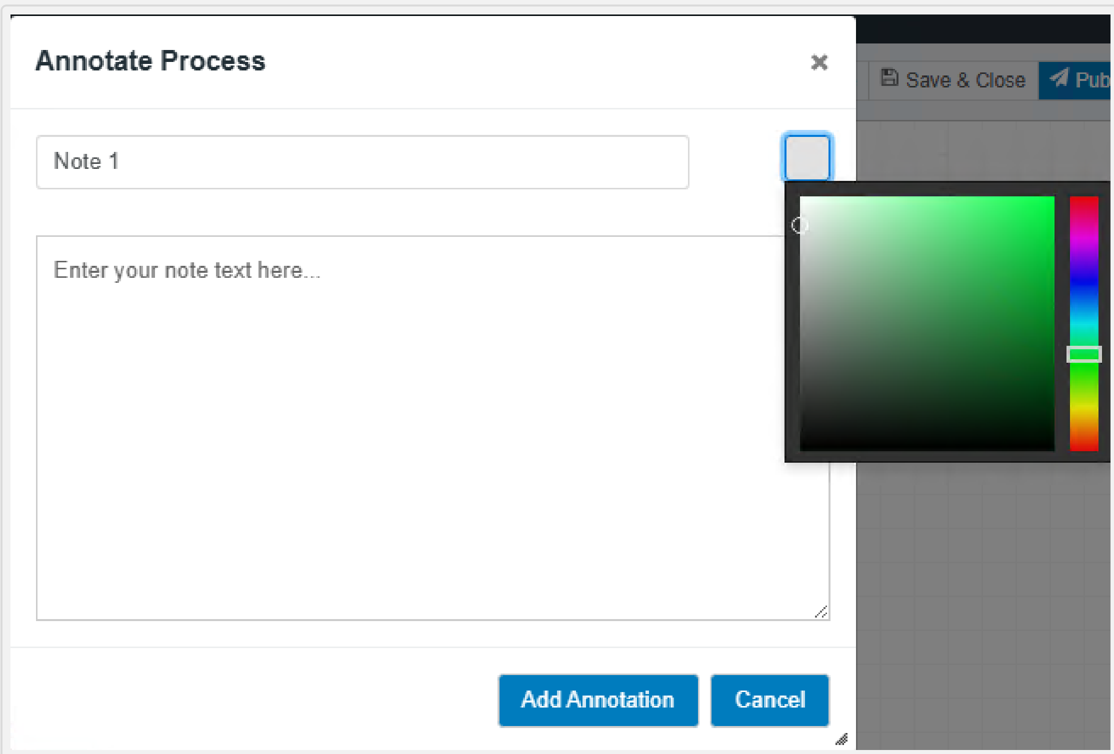
1. In the tree view, select a process and select **Delete Process**. A confirmation message displays.
2. Click **Yes**. A success message displays, and the process is removed from the grid. When you delete a single process and it is the only process in a group, the working table is deleted. When you delete a process, the corresponding logs from nx_process_log and nx_process_block_log are deleted and the NxDatabaseTable and the NxDatabaseColumn are deleted for the working table. When more than one process exists in the group and you delete one process, only the disposition columns are deleted for it.

To add an annotation to a process:

An annotation comprises an annotation title and the annotation text.

1. Left-click on the process designer canvas where you want to place the annotation and press **Ctrl+A**.
2. By default, the annotation title is Note X, where X is a number from 1 to the maximum number of annotations. You can rename the title if needed.

   View Image

3. Click to display the color palette and choose a color for the annotation title bar. The default color is light gray.
4. Add the note in the text box provided. The note can contain text or HTML.
5. Click **Add Annotation**. Annotations are only shown in the process designer, not on the preview page. You can resize, edit, delete, or drag the annotation to another location.

Managing working tables

The Working Table Manager provides administrator control over process group working tables. See Manage Working Table Window to learn more.

Running the Disposition report

Data Exchange Framework administrators can generate a process disposition report from within the user interface. See Process Disposition Report to learn more.

Setting the quantum

The Process Quantum Window provides a graphical interface where you can set the quantum and its associated properties. See Process Quantum Window to learn more.

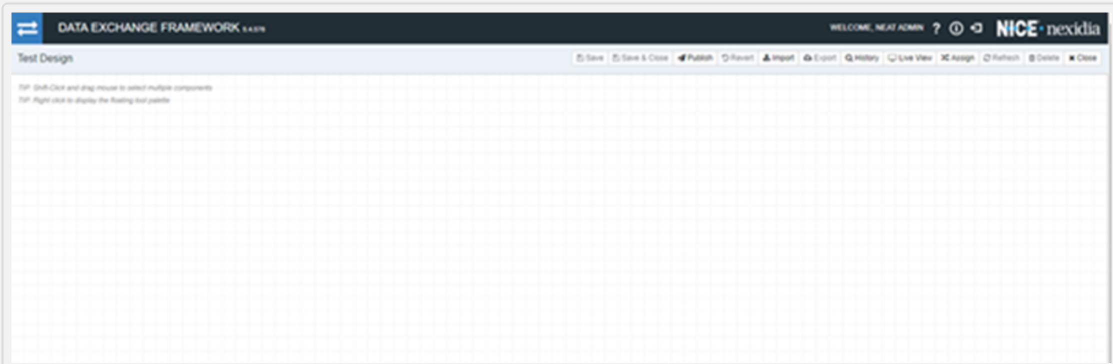
Setting the process schedule

The Process Edit Process Schedule Window provides a graphical interface where you can set the process schedule. See Edit Process Schedule Window to learn more.

Design a Process

To design a process:

1. In the tree view, right click on the process that you want to design and select **Design Process**. The Process Designer opens for the selected process.

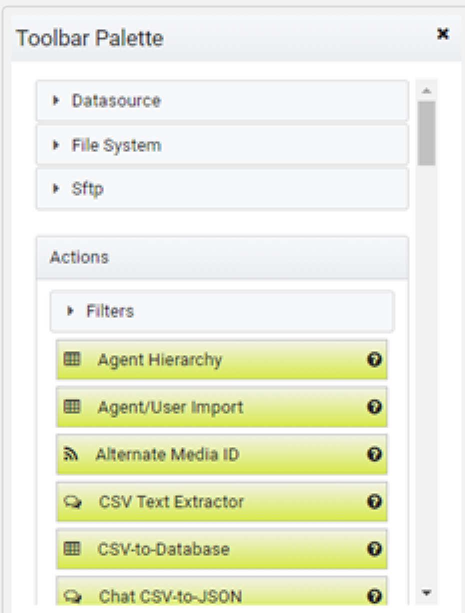


Available Designer Options

- **Save** - Allows you to save the current process as a working draft.
- **Save & Close** - Allows you to save the process as draft and returns you to the grid.
- **Publish** - Allows you to publish a process definition including validation and making the process available to be scheduled for execution (using the Start button in the grid), or to be executed manually.
- **Revert** - Allows you to roll back to the last successfully published definition, and discard any current changes or previously saved drafts.
- **Import** - Allows you to import a process definition JSON file.
- **Export** - Allows you to export a process definition JSON file.
- **History** - Allows you to view all older published versions, and allows you to roll back to them.
- **Compare** - Allows you to compare the process revision history so you can see the differences between previous versions of the process.
- **Live View** - Shows the current processing progress.

- **Assign Resources** - Allows you to assign all resources for an imported process at one time.
- **Refresh** - Reloads the process definition and refreshes the status of the last run. This is useful when you are monitoring processing because when you initially load a process, the status for the last run is displayed. This button is only available if you are not in draft state, and you are not actively editing a fully published process.
- **Delete** - Deletes the process design.
- **Close** - Closes the application.

2. Right-click in the Workflow Designer to display the floating tool bar palette which contains the available actions and resources.



You can also find a specific action quickly using **Quick Filter**. To open the Quick Filter window, click anywhere inside the Worflow designer.

3. Click on the desired action. The Configuration window for the action opens. You can also filter the list by name or groups to narrow the list.
4. Configure the action. See Available Actions for a list of actions and links to their configuration options to learn more.

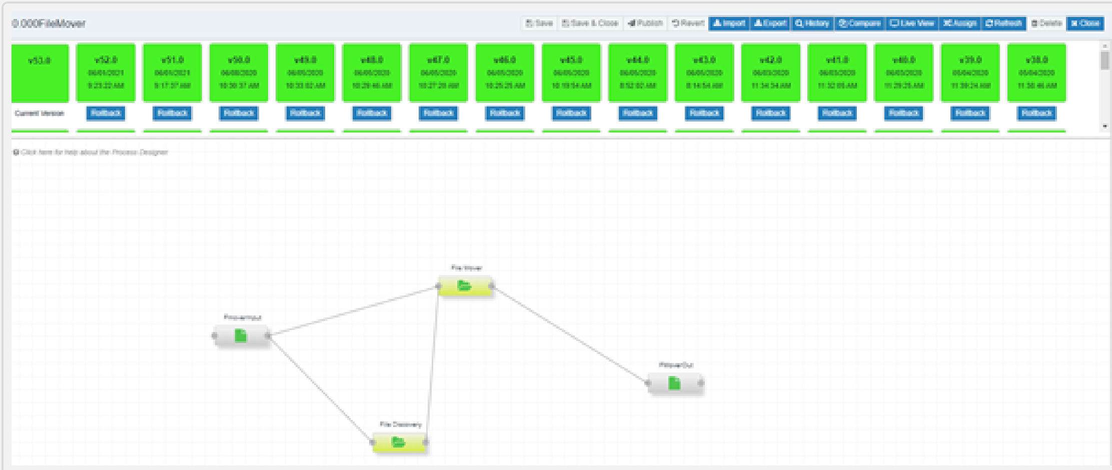
5. Select the resource type to add for the action, and then select a resource from the list that displays. The resource is added to the Workflow Designer.
6. Click on the resource to open its Component Editor. You can do any of the following:
   - Click **Assign to open** the Assign Resource Component list to change the resource assignment. You can enter text in the **Filter resource component list...** field to filter the list.
   - Click **Delete** to delete resource and associated links.
   - Click **Close** to close the Component Editor.
   - Click the **Edit** tab to change the configuration options.
   - Click the **Properties** tab to view the properties.
   - Click the **Labels** tab to add descriptive labels for this resource.
7. Optional. Right click on an action to open a context menu where you can **Edit**, **Disable** or **Delete** the action.
8. Continue adding actions and resources required for this process and verify that the connections are correct.
9. Click **Save**.
10. Click **Publish**. The completed process definition is published. If you make changes to a process but do not publish it, the application prompts you to select one of these actions:
    - Publish and Close Process Designer - Publishes the process and closes the process designer.
    - Return to Process Designer - Returns to the process designer.
    - Continue with Close - Closes the process designer without publishing.

View Process History

➔ To view process history:

1. In the tree view, select a process and then click **Design**.
2. In the process designer, click **History**.

The current version and all previous versions with their last updated date and time displayed.

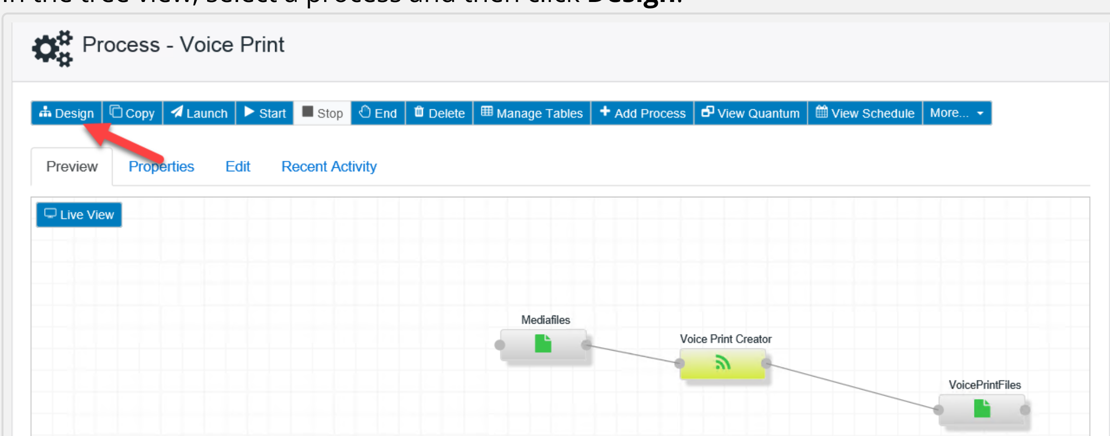


3. (Optional) Click **Rollback** under the version of the process that you want to roll back to.

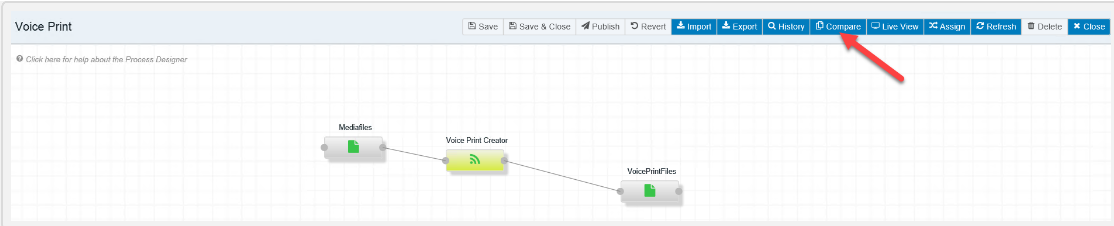
Compare Process History

To compare process history:

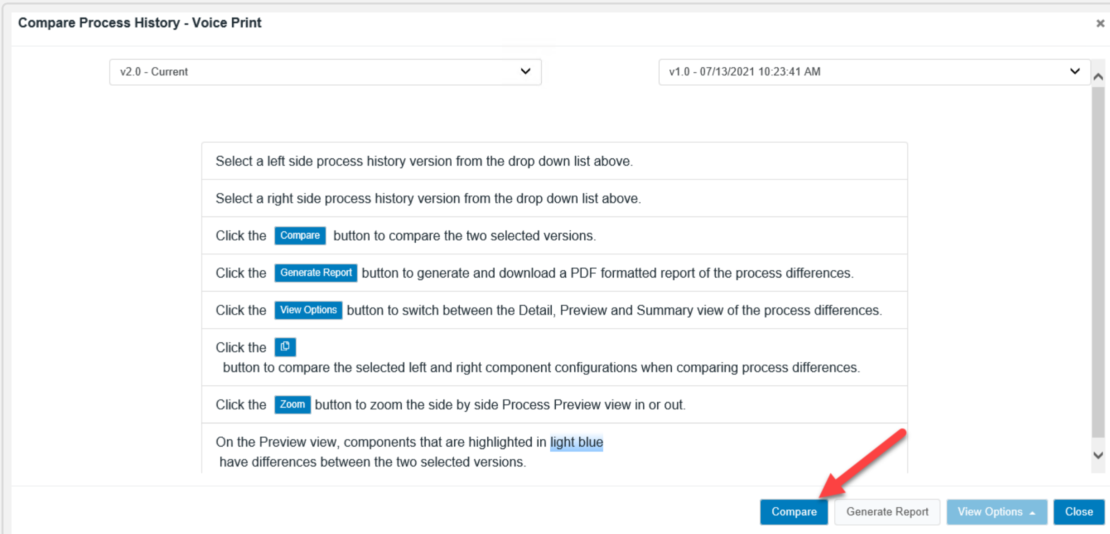
1. In the tree view, select a process and then click **Design**.



2. In the process designer, click **Compare** to compare the process revision history.

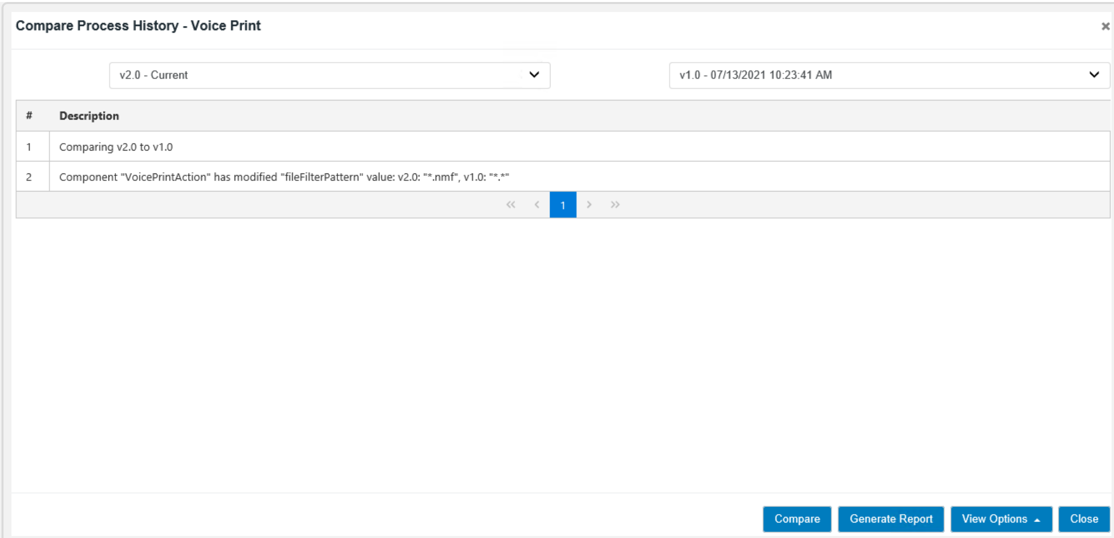


3. In the Compare Process History window, click **Compare**.

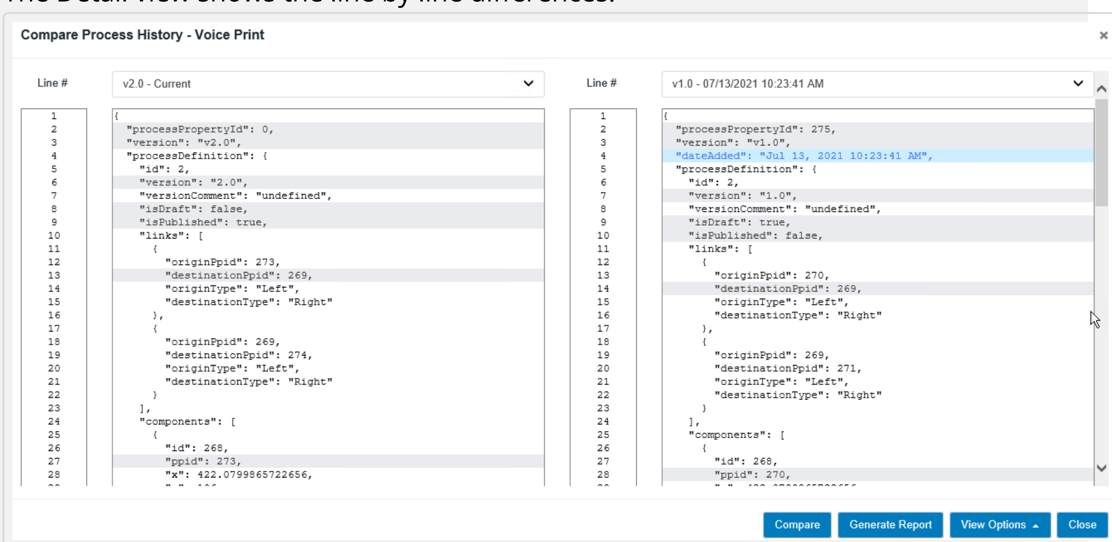


4. Do one of the following:

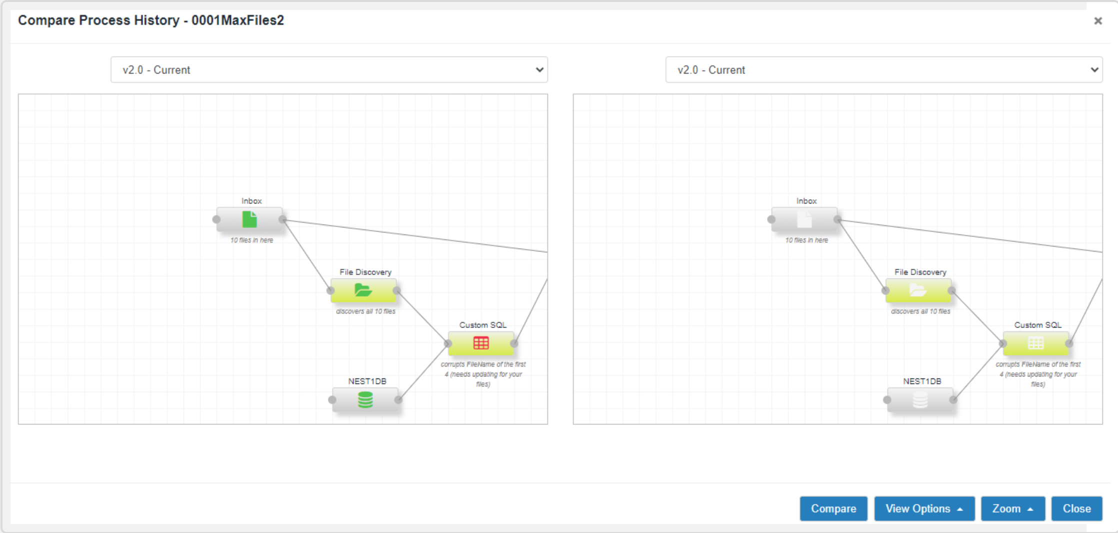
   - If there is only one version of the selected process, a prompt displays indicating that this process only has one version and cannot be compared. Click **OK**.
   - If there are only two versions of the selected process, a prompt displays asking if you want to compare version 2.0 to version 1.0. Click **Yes**. The Summary view of the differences displays.
   - If there are more than two versions of the selected process, in the left and right drop down lists select the process history versions that you want to compare. A prompt displays asking if you want to compare the selected versions. Click **Yes**. The Summary view of the differences displays.



5. (Optional) Click **Generate Report** to generate and download a report of the process differences in PDF format.
6. (Optional) Click **View Options** to switch between the Detail, Preview, and Summary view of the process differences.
   - The Summary view lists the changes that were made.
   - The Detail view shows the line by line differences.



- The Preview view shows the visual differences. You can drag and move the display to see the different actions, and you can zoom the display in and out as needed. In this view, components that are highlighted in light blue have differences between the two selected versions.

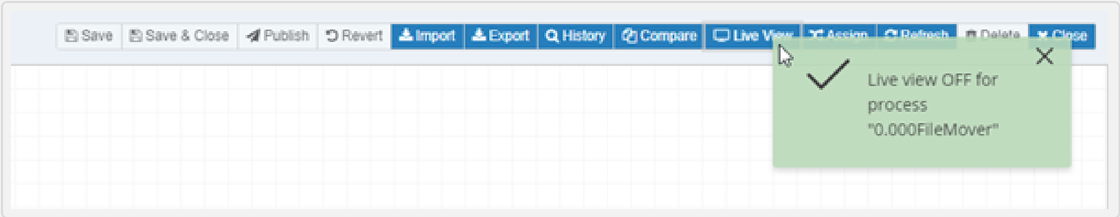


7. Click **Close** to return to the Process Designer.

Show Process Live View

To show the process live view:

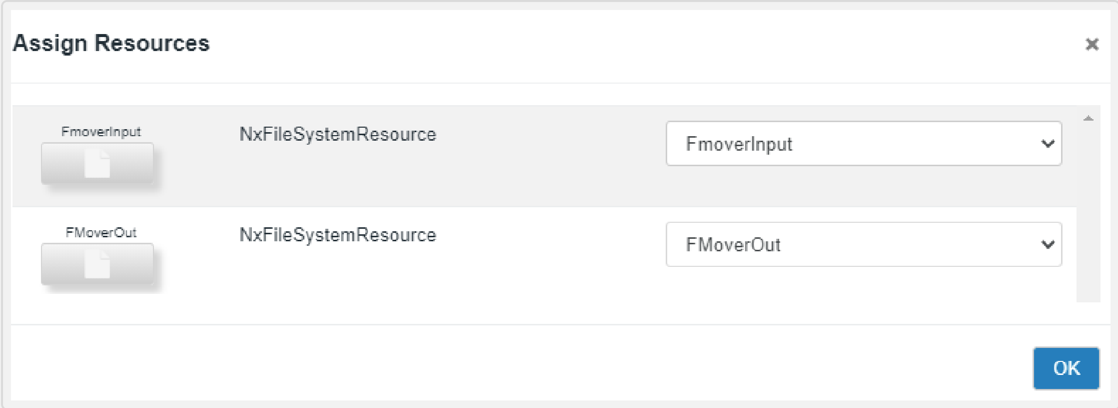
1. In the tree view, select a process and then click **Design**.
2. In the process designer, click the **Live View**.
   A message displays with the current processing progress.



Assign Resource to a Process

To assign a resource to a process:

1. In the tree view, select a process and then click **Design**.
2. In the process designer, click **Assign**.



3. Specify the resources to assign to the process.
4. Click **OK**.

Configure Actions

Once you have added an action to a process in the designer, you need to configure it. See Available Actions for a list of actions and links to their configuration options.

To configure an action:

1. In the process designer, click on the graphic for the action that you added. The Configuration window for the action opens.
2. Specify the configurations for the action in the Edit tab.
3. (Optional) Click the Properties tab to view the current properties for the action.
4. (Optional) Click the Labels link to view and edit the action label.
5. In the **Disable action** , specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This options is useful for troubleshooting the process.
6. Click **Submit**.

Working Table Manager

The Working Table Manager provides administrator control over process group working tables.

- See Working Tables to learn more.
- See Resources and Working Table File Keys to learn more.

To manage working tables:

1. Select a process, then right click to display the context menu and select **Manage Tables**. The Manage Working Table window opens.
2. (Optional) Select the Drop Tab to drop a working a working table. If you choose to drop the current working table, all columns and data are deleted from the working table.
3. (Optional) Select the Query Tab to fill out the select statement and where sub-clause of a query (of either the rollover table or the working table) to be executed against the process group's working table. In a distributed environment, results display after the query has been sent to the Nexidia Worker server, and results have been gathered and transmitted back. Columns available after the last successful run of any process in this process group's working table are listed in the drop-down list, and can be added by selecting an item from the list and then clicking the Add to Where or Add to Select buttons. To find rows in a given disposition, that portion of the query should be in the following form: '..."{PreviousDisposition}" is not null and "{DispositionToCheck} is null...'
4. (Optional) Select the manage Queries tab to manage the saved queries.

Working Tables

Working tables are database tables used by the DEF processes to record metadata and track the record flow. Many working tables could be used when processes are defined within individual process groups within a single Worker server. Process groups are configured at the process level and allow for logical separation of data moving through the framework. They also provide a mechanism for joining metadata arriving from different sources.

By default, a process with no defined process group uses the "working_table." Processes with a defined process group use the "working_table_{process_group_name}." Multiple processes in the same process group use the same working table.

One benefit of process groups is that data flows can be logically separated and optionally joined with data from other process groups. For example, in a system where agent metadata is delivered daily, one process in an "agents" process group might be to monitor for that data and update its contents as the agent metadata arrives. Another process in process group "main" might be to run constantly, pulling in metadata, moving media, etc. When the main process needs agent metadata attached, a Metadata Join action can be used in the main process group that will join against the "working_table_agents" table, pulling appropriate metadata into working_table_main for further processing.

Not all actions used in processes integrate directly with a working table. During process design, you should pay particular attention to ensure that what's being captured matches the business requirements for what's being tracked and how files move from one action to the next. The basic idea with working tables is that some action serves as an entry point for the process and pulls in the basic information on the files to process. Subsequent actions will append additional metadata to each row as they process the files moving through. At a minimum, any time an action touches a working table file, it sets its own "disposition column" for that file to the current timestamp. Some actions may add additional metadata of their own.

Using this kind of working table structure provides the Data Exchange Framework with an intrinsic or self-reconciling nature since any file that fails to process on one run is picked up and attempted again on subsequent runs until either the file is successfully processed, or it falls out of the configured maximum number of days to attempt self-reconciliation.

The number of days an action will attempt self-reconciliation for individual metadata rows moving through the working table is set in the "Maximum days to attempt self-reconciliation" advanced configuration option at the Process level. By default, there is a 10-day self-reconciliation limitation, so actions continue to attempt to act on files in their dispositions until that number of days has passed.

Most actions integrate with the working tables by default; a few will optionally integrate if configured to do so. Additionally, some advanced actions, like Custom SQL, can integrate with the working table but require customized parameters to do so correctly, so you must use them carefully. Others, like Database-to-CSV, are intended as preparatory actions to be followed by Metadata Import from that CSV and offer no working table tracking.

Working table entries that have successfully passed through 'terminal' actions — actions that both integrate with the working table and that have no other such actions downstream of them — are moved into a corresponding 'rollover' table that has the same name as that working table, but with "$X" appended to the end.

Working table and rollover table growth is checked by a nightly Cleanup Monitor that fires every midnight and deletes anything in either table older than the number of days defined under “Server retention (days)” at the Server level. Additionally, the Cleanup Monitor migrates any entries from the working table that are older than the self-reconciliation days into the rollover table and flags them with an ‘NxQuarantined’column that is set to ‘true’. The Cleanup Monitor also handles internal DEF log and file system cleanup.

Processing from Metadata

One important consideration in process design is the use of the “Process files from metadata” configuration option that’s available in most actions. When set to “false,” actions will first look into any input resources attached to them, pull files to process from there, and reconcile them against the disposition columns as described above. When set to “true,” the working table is queried first for files in the correct disposition and the action acts directly on that list, pulling from input resources as necessary without scanning them first.

This approach grants a considerable performance gain by reducing the I/O involved in scanning input resources, so the best practice is generally to have some “entry” action that populates the working table initially from input resources, with subsequent actions all processing from metadata. Processing from metadata also supports cataloging of incoming metadata and files in that initial Metadata Import of metadata from an inbox, and then files can be moved to the catalog, and subsequent actions can be processed from the metadata, pulling directly from the catalog without scanning its collected contents.

If “Process files from metadata” isn’t a configuration option for an action, it will either do so by default or have no working table integration at all.

Other Considerations

By default, the Data Exchange Framework automatically determines how to query the working table based on the process structure as described above. In some exceptional cases, it might be desirable to short-circuit part of the process by having an action in the flow look into different disposition columns than the defaults leading into them or to link up with another action in a separate process in the same process group, and to act on those files instead of the defaults.

In those cases, an advanced configuration option called “Input Disposition(s)” allows that type of override in most actions. This option requires comma-delimited disposition column names in the form that actions are tracked in the working tables (“Disposition-ActionName-UID#”), and these will override the defaults. The default behavior is to look for non-null values in ALL listed disposition column names, but an “OR” condition can be used instead by separating values with the “#” sign.

A more complex condition might be something like, “Disposition-SomeAction-1234,Disposition-AnotherAction-1235#Disposition-OneMoreAction-1236,Disposition-AFinalAction-1237.” In this case, when configured on an action with disposition column “Disposition-MainAction-1238,” the working table query would be: '...where “Disposition-SomeAction-1234” is not null and (“Disposition-AnotherAction-1235” is not null OR “Disposition-OneMoreAction-1236” is not null) and “Disposition-AFinalAction-1237” is not null and “Disposition-MainAction-1238” is null.

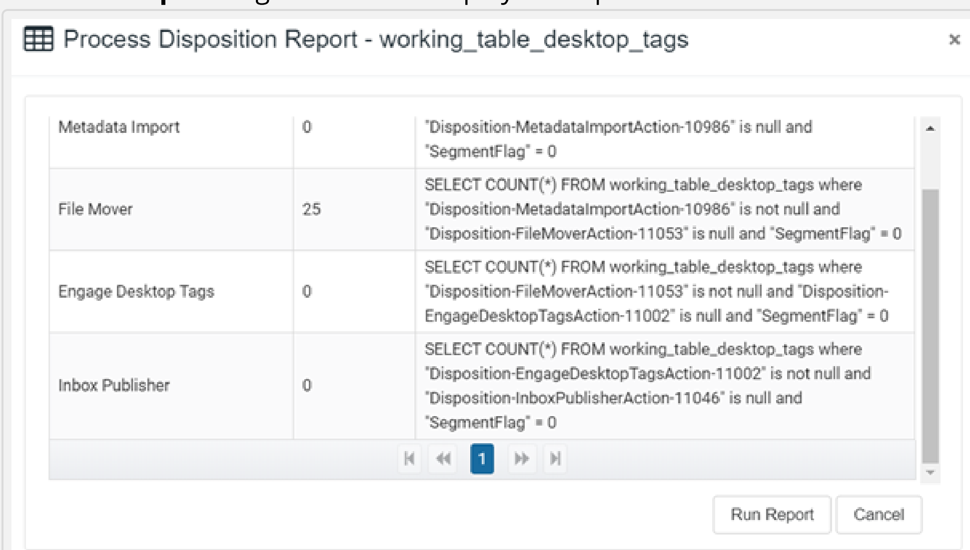
Disposition column names can be found directly in the database or at the bottom of the “CURRENT CONFIGURATION” box that opens when clicking on an action in the process builder. If all that’s desired is an OR condition on incoming actions, an advanced “Default input disposition handling” option is available to control a simple AND/OR toggle on that action to check the dispositions of incoming actions.

While the Cleanup Monitor handles working table cleanup, it may be desirable in some cases to turn off that cleaning and use a custom Cleanup action that can be configured to delete from the working table and age out corresponding file system data simultaneously. In that case, the advanced “Disable self-cleaning” configuration option at the Process level can be used to disable self-cleaning for the entire process group, and a Cleanup action can be configured in its own process as part of the working table process group to be cleaned, to handle cleanup in a more customized way.

Process Disposition Report

Data Exchange Framework administrators can generate a process disposition report from within the user interface. This report shows the counts waiting to be processed at each action at the current time. Note that these counts are different from the Live View counts, which are the files processed by each action at that time.

1. Select a process, then right click to display the context menu and select **Disposition Report**. The Process Disposition Report window opens.
2. Click **Run Report** to generate and display the report.



3. (Optional) Click **Run Report** again to regenerate the results. For multi-server environments "Working... (1/20)" displays, and the application increments the number of retry attempts until a response is received.

Process Quantum Window

The Process Quantum Window provides a graphical interface where you can set the quantum and its associated properties.

To set the quantum:

1. Select a process, then right click to display the context menu and select **View Quantum Window**. The Process Quantum window opens.
2. (Optional) Select from the buttons at the top of the window to perform any of these actions:
   - Clear Quantum
   - Clear Quantum Shift
   - Restore Quantum
   - Reset Quantum
   - Focus Quantum
   - Fit Quantum
3. Select the **Quantum Lower Date Bound**.
4. Specify the **Start Time Shift**.
5. Specify the **Quantum Time-zone Offset (mins)**.
6. Specify the **Quantum (mins)**.
7. Specify the **Quantum Stop Time Shift**.
8. Specify the **Quantum Window Timeline**.
9. (Optional) Make adjustments in the quantum time line to preview different scenarios.
10. (Optional) Hover over the timeline to briefly display the quantum start and end times.
11. Click **OK**.

The Quantum

The quantum is defined at the process level. It is used, along with its corresponding "quantum lower bound," to define time-based windows for processing files and data. Click the View Quantum button in the Process window to open the Process Quantum window.

The quantum defines the interval to use (in minutes), and the quantum lower bound is the starting date and time to use the first time that the action applies the quantum. After an action has run successfully, the following window starts at the end of the previous window rather than at the lower bound and continues along the defined interval. Change the lower bound date to reset the quantum window.

The actions that use the quantum are tracked individually. Typically, only one action in a process uses the quantum, and the rest are processed from metadata based on what it pulls in.

The quantum is integrated into the Database-to-CSV, File Mover, and File Discovery actions, and it is optional. The implementation of File Mover and File Discovery actions is straightforward. In the action configuration, setting the "Apply process quantum" value to true causes files to be picked from input resources within those time windows. For the Database-to-CSV action, you must construct your SQL using embedded windowing parameters that hook into the process quantum. For example, "{WINDOW_START}" and "{WINDOW_END}" parameters can be used in queries like this:

...WHERE disposition = 'COMPLETE|OK'

AND calldate >= '{WINDOW_START}'

and calldate < '{WINDOW_END}'

The Database-to-CSV action substitutes those parameters and adjusts the window after completing the run. You can also apply a maximum discovery bound to the actions. When used, rather than setting the next window start at the current window end + the quantum interval, the next start is the highest date value found during processing. For example, if the window start is at 4:00 today, and the interval is one hour, but the most recent file found is time stamped 4:45, the following window start is 4:45. Without the maximum discovery bound, the following window start is 5:00.

This is a configuration setting for the File Mover and File Discovery actions. For the Database-to-CSV action, you must append a date-based column name to the WINDOW_END parameter. In most cases the column name used is the same one used in the query.

...WHERE disposition = 'COMPLETE|OK'

AND calldate >= '{WINDOW_START}'

and calldate < '{WINDOW_END:calldate}'

When the Database-to-CSV action sees that column name attached to “WINDOW_END,” then it uses the maximum date found at the end of processing while pushing the data to CSV as the next window start.

You can use the quantum in combination with the process run interval to define many types of rolling windows. For example, To have the File Mover pick files from the previous day and leave files from the current day until the next run, set a process start time to run daily at 10:00 a.m. and a quantum with a lower bound 24 hours prior to the start time with a 24-hour quantum interval.

Edit Process Schedule Window

The Process Edit Process Schedule Window provides a graphical interface where you can set the process schedule.

To set the process schedule:

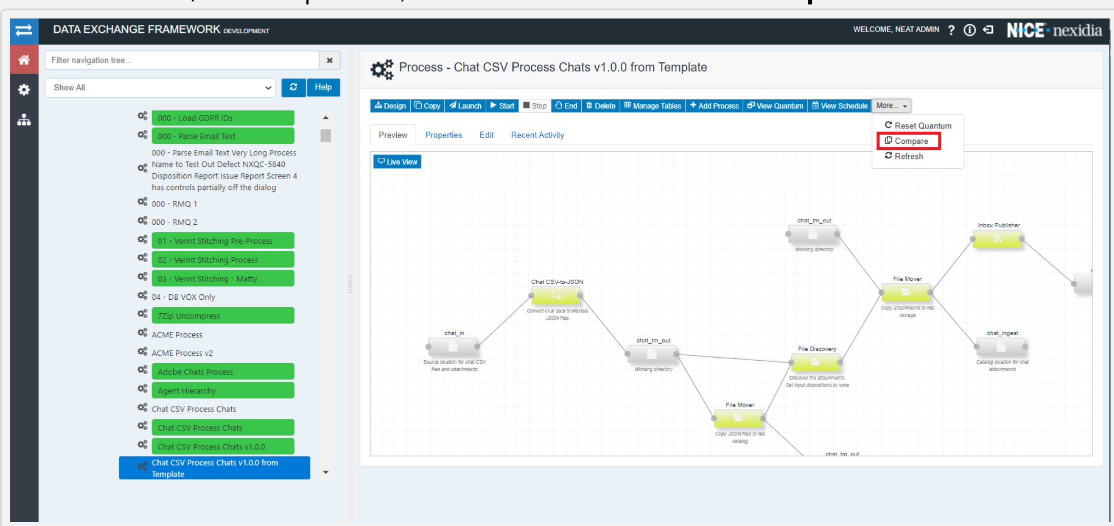
1. Select a process, then right click to display the context menu and select **View Schedule**. The Edit Process Schedule window opens.
2. (Optional) Select from the buttons at the top of the window to perform any of these actions:
   - Clear Schedule
   - Restore Schedule
   - Focus Schedule
   - Fit Schedule
   - Clear Blackout
3. Specify the **Schedule Type**. You can choose different types of Quantum windows based on your use case and the types of interactions you want to ingest in the NIA. There are two main options:
   - If you select ***Interval*** (default), the process will run on a regular interval schedule, such as twice a day (every 12 hours).
   - If you select ***Fixed***, the process will run at the desired schedule specified in the Schedule field. Enter the schedule in the HH24:MM:SS format. You can add multiple entries in a comma-delimited HH24:MM:SS format, for example, 03:00:00, 05:30:00, 18:45:00. The system calculates the next run time based on the schedule specified.
4. Specify the process **start time**.
5. Specify the **Interval**.
6. Specify the **Blackout Start Time**.
7. Specify the number of example **intervals**.
8. Specify the **interval type**.

9. Specify the **blackout end type**.
10. (Optional) Make adjustments in the **Schedule Time line** to preview different scenarios.
11. (Optional) Hover over the time line to briefly display the process start and end times
12. Click **OK**.

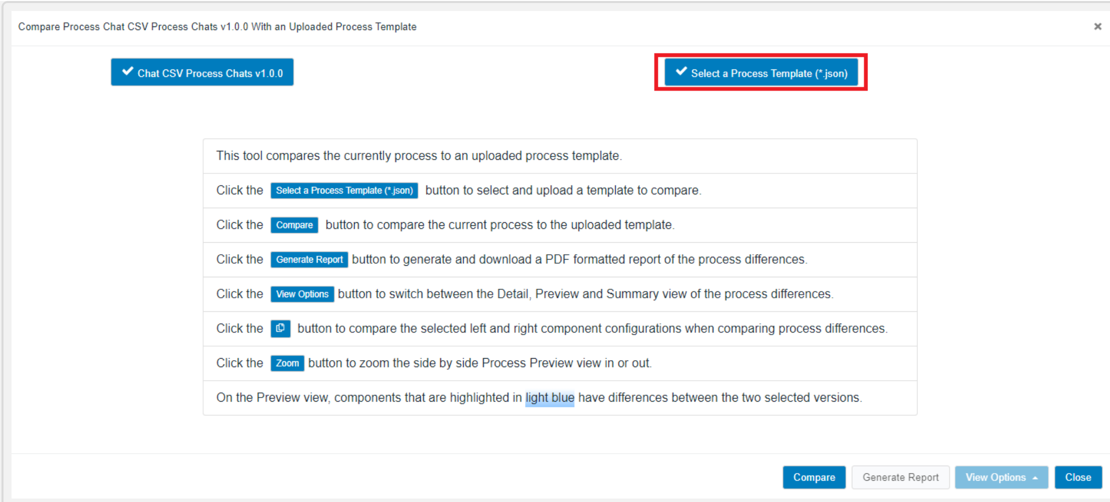
Process comparison with a template

You can compare a current process with a specific version of the template. This helps to find the differences between the process and the template. You can make required changes to the current process to keep it up-to-date.

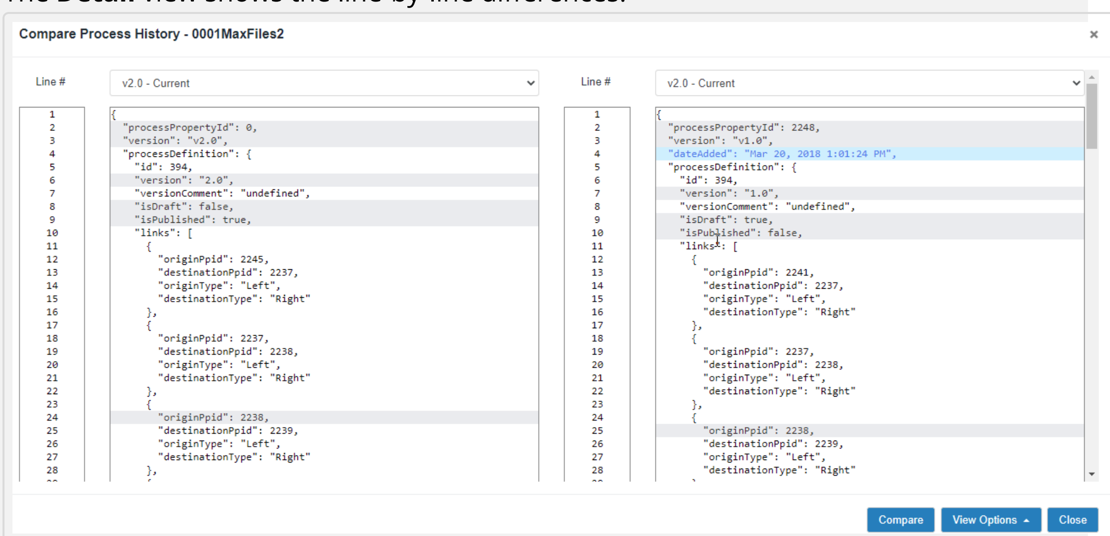
1. In the tree view, select a process, click **More** and then select **Compare**.



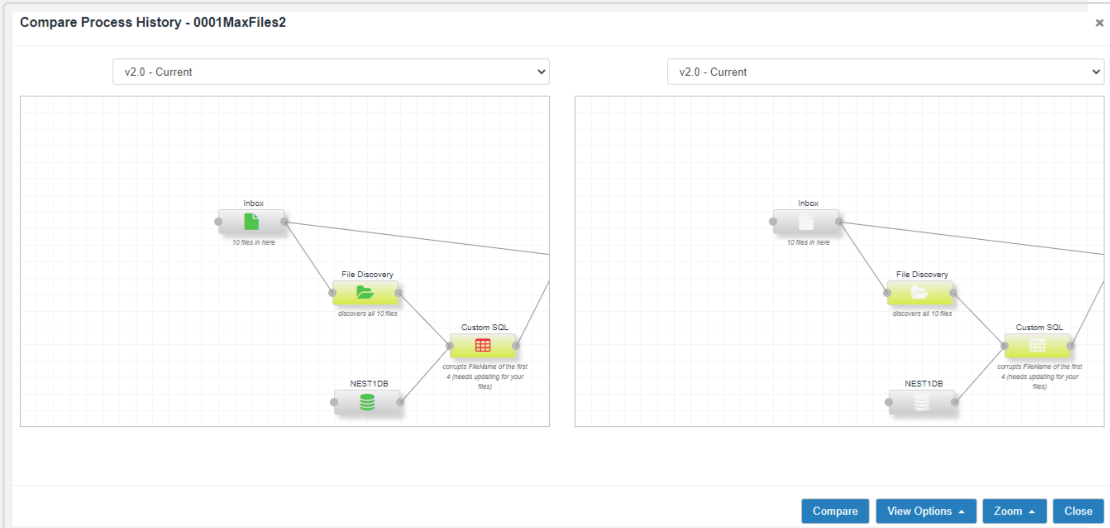
2. In the **Compare Process** window, click the **Select a Process Template (*json)** button and then browse to the required template.



3. Click **Compare**.
4. (Optional) Click the **Generate Report** button to generate and download the report.
5. (Optional) Click the **View Options** button to switch between the **Detail**, **Preview**, and **Summary** view of the report.
   - The **Summary** view lists the changes.
   - The **Detail** view shows the line-by-line differences.



- The **Preview** view shows the visual differences. You can drag and move the display to see the different actions, and you can zoom the display in and out as needed. The highlighted components in light blue indicates differences between the two selected versions.

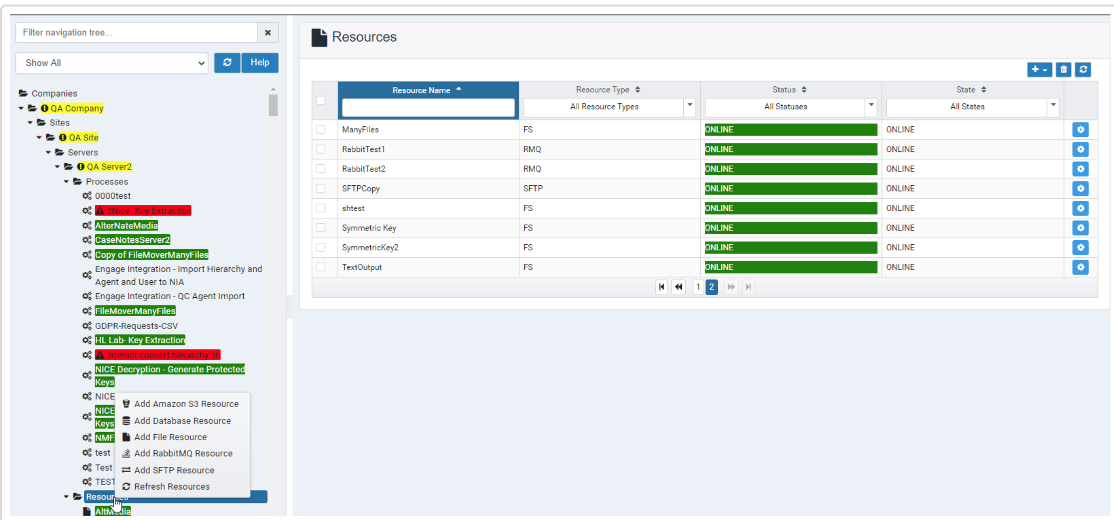


6. Click **Close** to return to the process designer.

RESOURCES

The processes you defined for your servers require resources you can add to your existing servers. Access the Resources page by selecting the Resources node in the tree view. When you right-click and select a resource name, the application loads its configured properties and most recent activity. You can toggle between the Properties, Edit, and Recent Activity tabs to review the resource details.

You can perform the following actions for existing resources: edit, delete, and copy. When you copy a resource, the application prompts you to enter a new resource name. When you delete a resource and a connection exists between the resource and actions, this action also deletes the connection. If a connection exists between resources and actions, you can delete the connection by clicking on the connection and right-click to open a context menu and perform a delete action.



Create all physical file resources outside of the DEF system. This process provides a better planning process and allows for permission validations as the resources are created. The DEF service account needs full control over all file resources used in the DEF processes.

It is best practice to create file resources on network shares whenever it is possible. When you create file resources within DEF, it is convenient to preface the resource with the two-digit preface that identifies the process. This operation keeps all the resources for a single process grouped together.

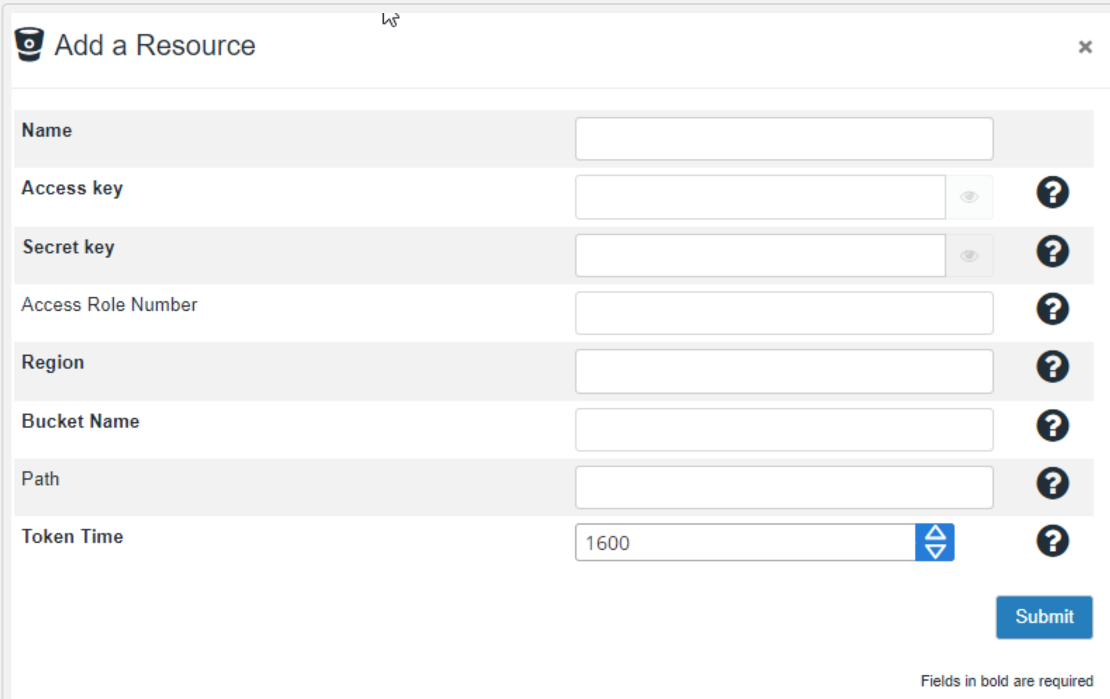
You can define the following resource types:

- Amazon S3
- Database

- File
- RabbitMQ
- SFTP
- Artemis.
- Snowflake
- AWS SQS resource.

To add an Amazon S3 resource:

1. Right-click the Resources node in the navigation tree and select **Add Amazon S3 Resource**.



2. In the **Name**, enter a name for the resource.
3. In the **Access Key**, enter the AWS access key.
4. In the **Secret key**, enter the AWS secret key.
5. In the **Amazon Assume Role**, enter the Amazon Resource Name if needed.
6. In the **Region**, enter the AWS region.
7. In the **Bucket Name**, enter the AWS bucket name.
8. In the **Path**, enter the AWS path under the bucket name.
9. In the **Token Time** drop-down, specify the AWS token time.
10. In the **Use Multi Part** drop-down, select ***true*** to run files in multiple parts.

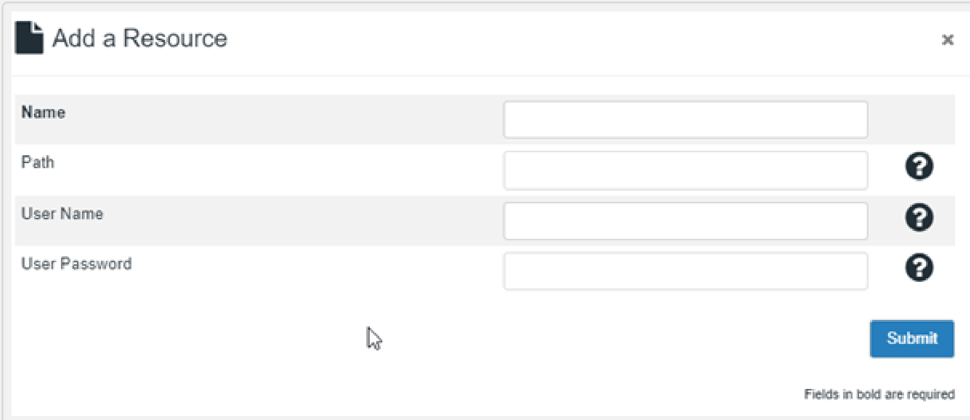
11. In the **Rotate Keys** drop-down, select ***true*** to use rotated keys. Select ***false*** to use a regular AWS connection.
    When this option is set to ***true***, you can set the following options:
    - In the **Rotation Type** drop-down, select ***AWS*** to generate new key and write to AWS credential file path and select ***File*** to read the key information from AWS credential file path.
    - In the **Path to Credential File** drop-down, select the location for the credential file. In the **Path to Credential File** drop-down, select the location for the credential file. This field is mandatory if you have selected ***True*** in the **Rotate Keys** drop-down and this is optional if you have selected ***False*** in the **Rotate Keys** drop-down.
    - The **Rotated Key Handling** field is visible only when you select AWS in the **Rotation Type** field. In this drop-down:
        - Select ***Delete*** to delete the old key after the days specified in the **Days Before Delete/Deactivate** field.
        - Select ***Deactivate*** to deactivate the old key based on the days specified in the **Days Before Delete/Deactivate** field.
12. The **Multi Part Size** field is enabled only when **Use Multi Part** is set to ***true***. In this field, specify the size of files in each part.
13. In the **Multi Part Thread Pool**, specify the number of threads to be run in multi part.
14. In the **KMS Key ID**, specify the AWS Key Management Service (KMS) key ID if your S3 bucket uses a customer-managed encryption key so Amazon can use it . This field is optional and only needed when the customer manages its own keys to encrypt and decrypt data.
15. In the **Number of Days Rotate Key** drop-down, specify how often to rotate keys by generating a new one and marking the old key for deletion or deactivation. Minimum is 1 day and default is 90 days.
16. In the **Number of Days Before Delete/Deactivate** field, specify the number of days to retain deleted or deactivated keys before they are permanently removed or fully deactivated.
17. Click **Submit**.

To add a Database resource:

1. Right-click the Resources node in the navigation tree and select **Add Database Resource**.
2. In the **Name** field, enter the name of the data source.
3. Do one of the following:
   - In the **Connection URL**, enter the URL for the data source. This is a JDBC database connection string.
   - Click to the right of the Connection URL field to open the JDBC Connection Wizard.
     a. In the **Driver Type** drop-down, select one of these driver types: ***jTDS***, ***MySQL***, ***PostgreSQL***, or ***SQL*** Server and click **Next**.
     b. In the **Host Name**, enter the host name of the database server (without the leading //).
     c. Click Next.
     d. In the **Instance Name**, enter an optional instance name and click Next.
     e. In the **Port Number**, enter an optional port number and click **Next**.
     f. In the **Database Name** field, enter a database name and click Next.
     g. Select the **Integrated Security** box if the database connection is using integrated security
     h. Click **Finish**.
4. In the **User ID**, enter the user identification for the data source. Leave this field blank for Windows Integrated Authentication.
5. In the **Password**, enter a password. Leave this field blank for Windows Integrated Authentication.
6. In the **Append DEF Certificate**, set to ***true*** (default value) to set up encrypt and trust certification based on the configuration values set in the ndef properties and append to the JDBC database connection string. Set this field to ***false*** to not apply the ndef configuration values to the JDBC database connection string.
7. Click **Submit**.

To add a file resource:

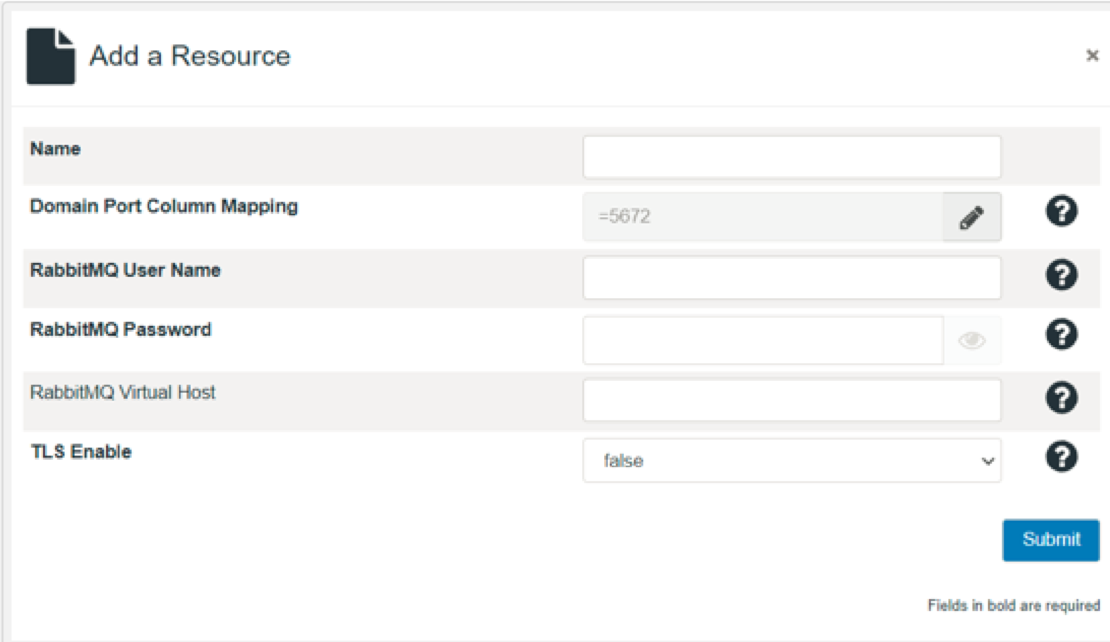
1. Right-click the Resources node in the navigation tree and select **Add a File Resource**.



2. In the **Name**, enter the name of the file source.
3. In the **Path**, enter the path for the file source in the Path field.
4. In the **User Name**, enter the user name (Required for windows authentication only).
5. In the **User Password**, enter the User Password. (Required for windows authentication only).
6. Click **Submit**.

To add a RabbitMQ resource:

1. Right-click the Resources node in the navigation tree and select **Add RabbitMQ Resource**.



2. In the **Name**, enter the RabbitMQ host name.
3. In the **Domain Port Column Mapping**, click the icon to enter the domain and port number. The default port number is 5672.
4. In the **RabbitMQ User Name**, enter the RabbitMQ user name.
5. In the **RabbitMQ Password**, enter the RabbitMQ password.
6. In the **RabbitMQ Virtual Host**, enter the RabbitMQ virtual host.
7. In the **TLS Enable** drop-down, select ***true*** to enable TLS.
8. Click **Submit**.

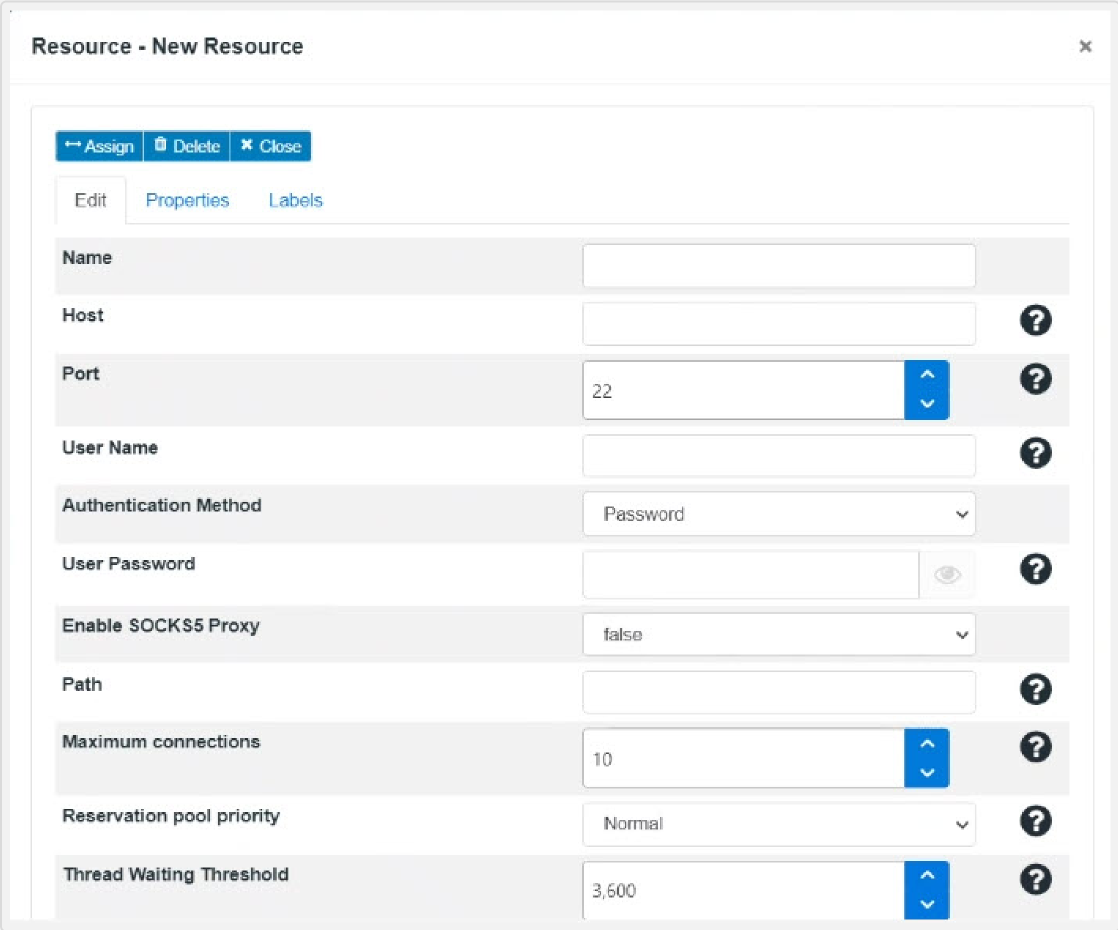
To add a SFTP resource:

FTP accounts normally limit the number of simultaneous connections. If a Data Exchange Framework Network process exceeds this number of connections, the account can be disabled. This causes all processes using that account to fail. In version 5.6 SFTP Connection Pooling was added to the Data Exchange Framework. The number of simultaneous connections is controlled by the Maximum Connections configuration in the SFTP Resource. The default maximum connections to SFTP resource is 10.

- The maximum SFTP Resource connections are tied to the SFTP Login utilized by the resource.
- It is best practice to limit the number of connections across all processes that use an SFTP resource to not exceed the maximum allowed connections.
- The SFTP connection pooling do not allow the number of connections per SFTP login to exceed the maximum connections.

For example, there are two DEF SFTP Resources configured for 10 connections for a total of 20 possible connections. The DEF resources are connected to a single SFTP login. The connection pooling mechanism only allows 10 connection to the SFTP login.

1. Right-click the Resources node in the navigation tree and select **Add SFTP Resource**.



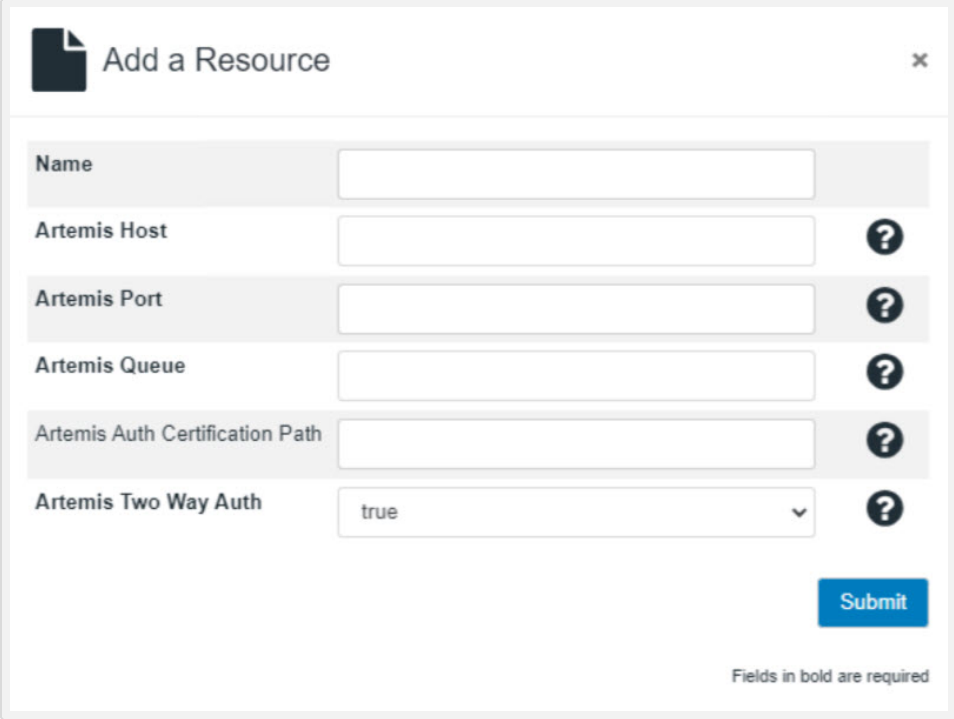
2. In the **Name**, enter the SFTP site name.
3. In the **Host**, enter SFTP host name for login.
4. In the **Port**, specify the SFTP port for login. The default value is 22.
5. In the **User Name**, enter the SFTP user name for login.
6. In the **Authentication Method** drop-down, select one of the following:
   - ***Password***
   - ***File System Key***
   - ***Pasted Key***

The remaining configuration options that are presented to you will vary based on the authentication method you selected in Step 6. Specify values for the options that display.

7. In the **User Password**, enter the SFTP user password for login.
8. In the **Path to user key**, enter the SFTP path.
9. In the **User Key**, enter the SFTP user key for login.
10. In the **Key Passphrase**, enter the SFTP key passphrase for login.
11. In the **Enable SOCKS5 Proxy,** set the value to ***true*** to support the SOCKS5 protocol for enhanced security. This adds an extra layer of protection by routing all SFTP connections through a SOCKS5 proxy. The default value is ***false***. When you set this to true, the following configuration options will be displayed:
    - In the **SOCKS5 Proxy Host**, enter the proxy host for the SFTP SOCKS5 protocol.
    - In the **SOCKS5 Proxy Port**, enter the proxy port for SFTP SOCKS5 protocol. The default value is 1080.
12. In the **Path**, enter the SFTP path.
13. In the **Maximum Connections** drop-down, set this value to a non-zero value greater than two to limit the maximum number of connections any process can take out for this SFTP resource. Keep in mind that on Worker Servers (NEST), one of these connections are reserved for heartbeat resource checks.
14. In the **Reservation Pool Priority** drop-down, select ***high*** if you want to process the new SFTP connection for the same host and username connection to have higher priority than a normal connection.
15. In the **Thread Waiting Threshold** drop-down, specify the waiting time for the SFTP pool connection thread. Default is 3600 seconds.
16. Click **Submit**.

To add Artemis resource:

1. Right-click the Resources node in the navigation tree and select **Add Artemis Resource**.

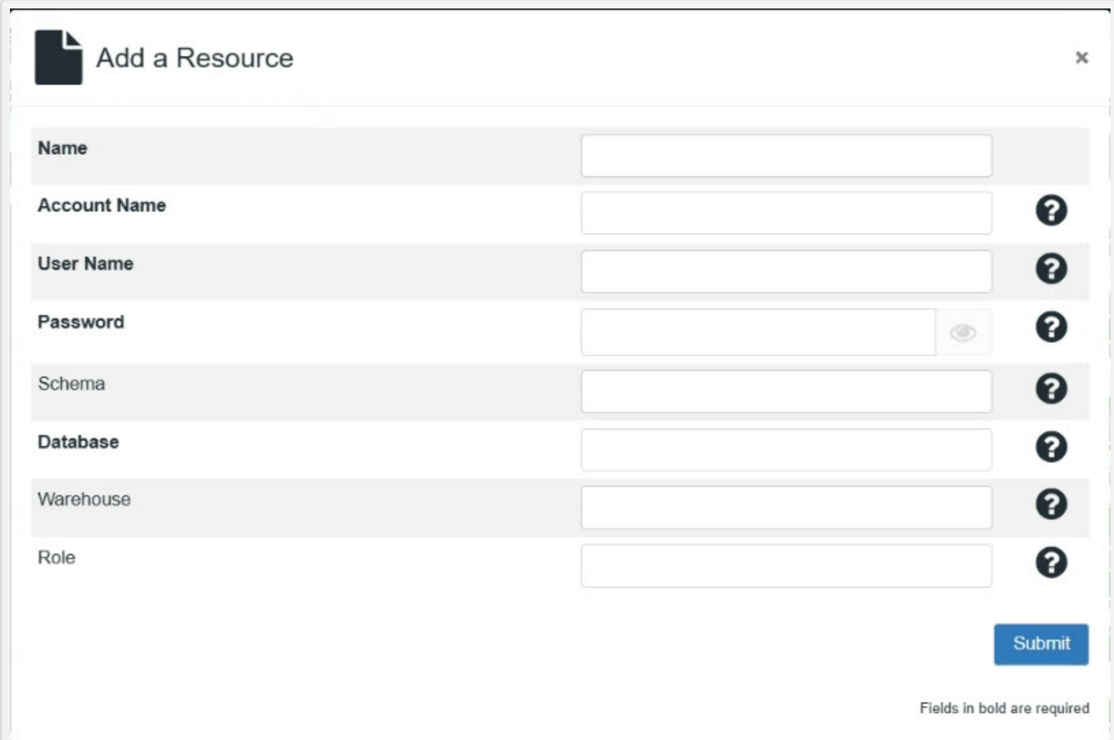


2. In the **Name**, enter a name for the resource.
3. In the **Artemis Host**, enter the Artemis host name or address of the server where data will be sent.
4. In the **Artemis Port**, specify the Artemis port number on the server used for sending data.
5. In the **Artemis Queue**, enter the Artemis queue name on the server where data will be placed.
6. In the **Artemis Auth Certification Path**, specify the location of the Artemis authentication certificate to be added to the DEF cacerts file.
7. In the **Artemis Two Way Auth**, specify whether both the client and server need to authenticate each other. Select ***true*** to use two way client/server authentication.
8. Click **Submit**.

You can use this resource to send and receive messages as part of CXone/Snowflake access.

To add Snowflake resource:

1. Right-click the Resources node in the navigation tree and select **Add Snowflake Resource**.



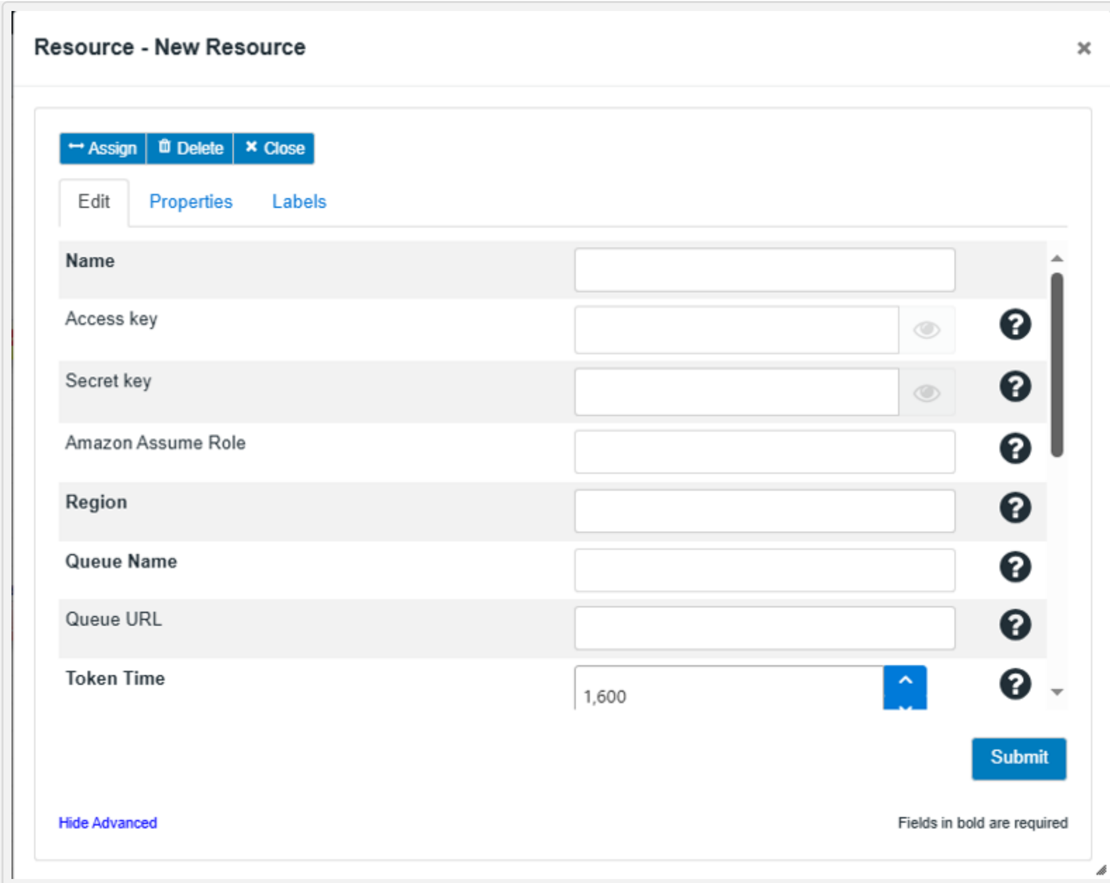
2. In the **Name**, enter a name for the resource.
3. In the **Account Name**, enter the account name to use in Snowflake jdbc.
4. In the **User Name**, enter the user name to connect to Snowflake database.
5. In the **Password**, enter the password to connect to Snowflake database.
6. In the **Schema**, enter the schema name to connect to Snowflake database.
7. In the **Database**, enter the database name to connect to Snowflake database.

8. In the **Warehouse**, enter the warehouse name to connect to Snowflake database.
9. In the **Role**, enter the role to connect to Snowflake database.
10. Click **Submit**.
    You can use this resource to access Artemis queues, which will help you send and receive messages as part of CXone/Snowflake integration.

AWS SQS Resource - You can use this resource to access Artemis queues, which will help you send and receive messages as part of CXone/Snowflake integration.

To add an Amazon SQS resource:

1. Right-click the Resources node in the navigation tree and select **Add Amazon SQS Resource**.



2. In the **Name**, enter a name for the resource.
3. In the **Access Key**, enter the AWS access key.
4. In the **Secret key**, enter the AWS secret key.
5. In the **Amazon Assume Role**, enter the Amazon Resource Name if needed.
6. In the **Region**, enter the AWS region.
7. In the **Queue Name**, enter the AWS SQS Queue Name.
8. In the **Queue URL**, enter the AWS SQS queue URL. This field is optional. It will be derived from queue name if not provided.
9. In the **Token Time** field, enter the AWS Token Time.
10. In the **Message Retention Period (seconds)**, enter how long SQS retains messages (1-1209600 seconds). The default value is 14 days.

11. In the **Visibility Timeout (seconds)** field, enter how long messages are invisible after being received (0-43200 seconds).
12. In the **Max Receive Count** field, enter the maximum number of times a message can be received before moving to DLQ.
13. In the **Rotate Keys** drop-down, select ***true*** to use rotated keys. Select ***false*** to use a regular AWS connection.
    When this option is set to ***true***, you can set the following options:
    - In the **Rotation Type** drop-down, select ***AWS*** to generate new key and write to AWS credential file path and select ***File*** to read the key information from AWS credential file path.
    - In the **Path to Credential File** drop-down, select the location for the credential file. In the **Path to Credential File** drop-down, select the location for the credential file. This field is mandatory if you have selected ***True*** in the **Rotate Keys** drop-down and this is optional if you have selected ***False*** in the **Rotate Keys** drop-down.
    - The **Rotated Key Handling** field is visible only when you select AWS in the **Rotation Type** field. In this drop-down:
      - Select ***Delete*** to delete the old key after the days specified in the **Days Before Delete/Deactivate** field.
      - Select ***Deactivate*** to deactivate the old key based on the days specified in the **Days Before Delete/Deactivate** field.
    - In the **Number of Days Rotate Key** drop-down, specify how often to rotate keys by generating a new one and marking the old key for deletion or deactivation. Minimum is 1 day and default is 90 days.
    - In the **Number of Days Before Delete/Deactivate** field, specify the number of days to retain deleted or deactivated keys before they are permanently removed or fully deactivated.
14. Click **Submit**.

Resources and Working Table File Keys

The “FileName” column in any working table is the unique file key for that row. In many cases that might be a straightforward file name (in a flattened directory structure scanned by File Discovery, e.g.), or in the case of nested directory structures found during discovery, they might indicate paths relative to the path of the resource input they were discovered under; so if an input resource had a path of C:\path\to\media,” and that folder contained “folder1,” “folder2,” and “folder3,” each with a set of media files, the file keys going into the working table during File Discovery would be “folder1\mediafile1.wav,” folder1\mediafile2.wav,” etc.

When subsequent actions use input and output resources, it’s important to remember that the working table file keys must be considered relative to the paths defined in the input and output resources attached to them.

Parametrized Output Paths

Resources allow parameterized paths where the date-based "yyyy," "mm," and "dd" are available, and any other metadata for the row being processed can be substituted in the same way "agentId" is in this example: c:\output\media\path\{agentId}\{yyyy}{mm}{dd}. For input resources, only the non-parameterized part of the path is used (c:\output\media\path).

For outputs, most actions substitute appropriate metadata, build out folder structures where they don't exist, and then apply the file key to the end. Based on the above example, that might output “folder1\mediafile1.wav,” recorded from agent 93, to “c:\output\media\path\93\20160701\folder1\mediafile1.wav.”

The File Mover allows for manipulating the file key during processing by providing an option to rename the file key when it performs a task like moving files from one location to another but flattening the directory structure.

File System Resource Path Validation

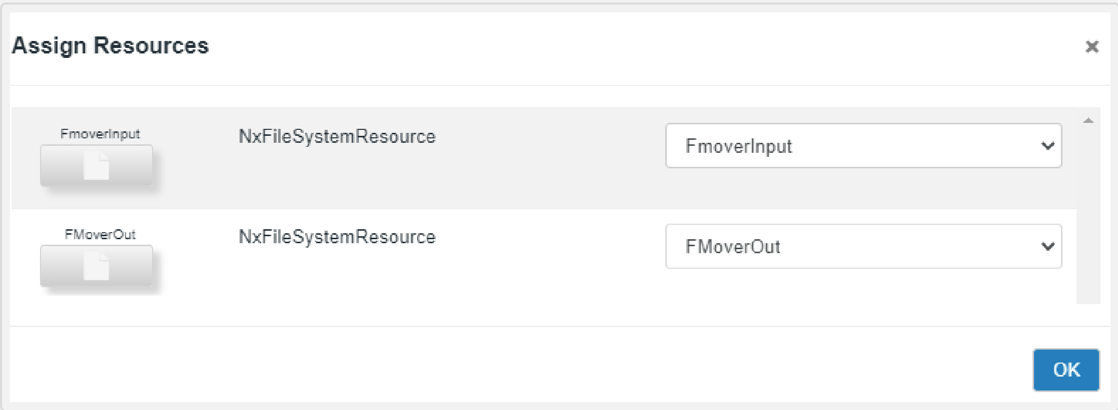
- If the parametrized file system resource is the final output of the process, it publishes successfully.
- If the parametrized file system is an input to another action where the following action is wired to the File Mover action (and not to the file system resource itself):
  - If the File Mover action is set to true for the updated file key, it will publish fine.
  - The publish fails if the file mover is set to false for the update file key.

- If the following action is wired to the parametrized file system resource, the publish fails irrespective of whether the update file key is set to true or false.

Assign Resource to a Process

To assign a resource to a process:

1. In the tree view, select a process and then click **Design**.
2. In the process designer, click **Assign**.



3. Specify the resources to assign to the process.
4. Click **OK**.

ACTIONS

A Data Exchange Framework Action component represents a specific processing algorithm performed on provided data. For example an Action may get data from one area, perform some kind of transform on the data, and then put the resultant data in another area.

In general, DEF Actions process data and DEF Resources can provide data into Actions and/or consume (processed) data output from Actions.

See Available Actions for a list of actions with descriptions, and links to their configuration options.

Available Actions

This table provides a list and description of the available actions and their required input and output resources. Click on the links to display the generic configuration options. The required fields for each action are displayed in bold font in the user interface. You can show or hide the advanced configuration options as needed. To improve user experience and optimize screen space usage, you can resize the action window screen horizontally or vertically by dragging its lower-right corner with your mouse.

ACTION	INPUT	OUTPUT	DESCRIPTION
Agent Hierarchy	File System or SFTP resource	N/A	Feeds the *Interaction Analytics*-friendly Hierarchy CSV to the *Interaction Analytics* Hierarchy Web Service. The *Interaction Analytics* Hierarchy Web Service updates the Agent Hierarchy Data.
Alternate Media ID	File System or SFTP resource	File System resource	Identifies alternate media by looking either for specific metadata field values, or by testing against the file system (either by input resource scanning or processing from metadata). The HasAlternateMedia, AlternateMediaSourcePath, and AlternateMediaExpirationDateUTC metadata fields are populated when the application finds alternate media.
Analytics Hub Response	SQS resource and AWS S3	File System resource	Consumes messages from Amazon SQS (Analytics Hub), retrieves transcripts stored in AWS S3, converts them to VoiceGrid ASR format, and writes the processed output to the file system. Supports batch processing, multi-threading, and cleanup operations.

ACTION	INPUT	OUTPUT	DESCRIPTION
Analytics Hub Request	N/A	SQS resource	Allows seamless integration with Analytics Hub for transcript generation. It sends metadata to Analytics Hub through AWS SQS after a media file is uploaded to an S3 bucket using the File Mover action. The request references the uploaded file and includes language metadata for processing.
AppLink Metadata Retriever	Database resource	N/A	Retrieves Engage metadata for CXone. If no output resource is provided, one will either be inferred from adjacent action I/Os, or automatically provisioned in the {DEF DATA}/fs/folder.
AppLink SAP Business Data Upload	Database resource	N/A	Uploads business data from Engage to CXone.
AppLink SAP Upload	Database resource	N/A	Moves metadata from working table and NMF files from input file system or SFTP resource to CXone. If no input resource is provided, it will either be inferred from adjacent action I/Os, or automatically provisioned in the {DEF_DATA}/fs/ folder.
AppLink SAP Generic Upload	Database resource	N/A	Uploads User Data of reconciliation report and user information to CXone environment from input resource. If no input resource is provided, it will either be inferred from adjacent action I/Os, or automatically provisioned in the {DEF_DATA}/fs/ folder.

ACTION	INPUT	OUTPUT	DESCRIPTION
AppLink SAP User Upload	File System or SFTP resource	N/A	Uploads User Data CSV to CXone from input file system or SFTP resource. If no input resource is provided, it will either be inferred from adjacent action I/Os, or automatically provisioned in the {DEF_DATA}/fs/ folder.
Auto Summary LLM V2	File System or SFTP resource, and LLM File	File System or SFTP resource, and LLM File	Creates JSON files and it displays call summarization outcomes. If no input resource is provided, it will either be inferred from adjacent action I/Os, or automatically provisioned in the {DEF_DATA}/fs/ folder. Provides a concise summary of the contact center conversation for the subsequent agent,, highlighting key details based on the custom prompt. This action was created by combining both the Auto Summary LLM and Auto Summary JSON Creator actions.
Call Summarization Request	File System or SFTP resource	N/A	Sends the media files information to RabbitMQ for Grid service. If no input resource is provided, they will either be inferred from adjacent action I/Os, or automatically provisioned in the {DEF_ DATA}/fs/folder.
Call Summarization Response	File System or SFTP resource	N/A	Call Summarization Response action gets the response back from Grid and saves the information which includes segments list into DEF database. There is no input or output resources.

ACTION	INPUT	OUTPUT	DESCRIPTION
CSV Reformatter	File System resource	File System resource	Converts or re-formats incoming files (for example WFM files) into CSV files that can be imported by the Metadata Import action and other core DEF actions that read CSV files.
CSV Text Extractor	File System resource	File System resource	Captures general text from incoming CSV files, writing each row's corresponding text column data out to a standalone .txt file for the Text Grid to consume. Optionally captures additional metadata from other columns in the CSV file, which can then be mapped to expected NIA fields when publishing to the inbox.
CSV to Database	File System or SFTP resource	Database	Imports data from CSV files into an already defined database table. Provides insert, update, and delete support for keeping tables in sync. Appropriate indexing should be done externally for optimal performance.
CXone Agents Retriever V2	N/A	N/A	Retrieves agent metadata from CXone systems. The agent metadata is recorded as working table metadata for each file retrieved. The API configured in the action is the data source.
CXone Billing Data Submit V1	DEF database resource	AWS Kinesis stream	Retrieves billing data from NIA and submits it to a Kinesis stream. The submitted billing data is a single interaction count for the previous day.

ACTION	INPUT	OUTPUT	DESCRIPTION
CXone Completed Contacts Metadata Retriever V2 Action	REST API		Retrieves metadata from completed contacts based on the channel type. It retrieves DFO emails and chats from the CXone API and sends them to *Interaction Analytics* for further analysis.
CXone Custom Metadata Retriever V2	N/A	N/A	Retrieves custom metadata from CXone systems. Custom metadata is recorded as working table metadata for each file retrieved. The API configured in the action is the data source.
CXone Generic Metadata Retriever Action V2	N/A	N/A	Retrieves generic metadata from CXone systems. Generic metadata is recorded as working table metadata for each file retrieved. The API configured in the action is the data source.
CXone Media Retriever V2	Configured Working Table	Configured file folder where the retrieved media files are written to	Receives audio from CXone systems. The API configured in the action is the data source.
CXone Media Retriever V3	Configured Working Table	Configured file folder where the retrieved media files are written to	Receives audio from CXone systems. The API configured in the action is the data source.
CXone Metadata Retriever V2	N/A	N/A	Retrieves metadata from CXone systems. The API configured in the action is the data source.

ACTION	INPUT	OUTPUT	DESCRIPTION
CXone Report Retriever V2	N/A	N/A	Submits the specified data download or custom report ID and loads the returned metadata into the specified DEF working table. Use of a quantum window is optional. The API configured in the action is the data source.
CXone Snowflake Extractor	Artemis and Snowflake Resources	Database	Extracts the query from the Artemis queue and queries the Snowflake database, then loads the results into the specified database tables.
CXone Snowflake Query	N/A	Artemis Resource	Submits Snowflake commands to the Artemis request queue for the CXone Snowflake Extractor to execute. This action retrieves metadata about interactions and agent/users.
CXone Supplemental Metadata V2	N/A	N/A	Retrieves supplemental metadata from CXone systems. Supplemental metadata is recorded as working table metadata for each file retrieved. The API configured in the action is the data source.
CXone Team Retriever V2	N/A	N/A	Retrieves teams metadata from CXone systems. Teams metadata is recorded as working table metadata for each file retrieved. No input resource is required, since the API configured in the action will be the data source. No output resource is required.

ACTION	INPUT	OUTPUT	DESCRIPTION
CXone Users Retriever V2	N/A	N/A	Retrieves user metadata from CXone systems via the Users Search API. The API, accessed via a POST request, accepts a configurable JSON filter payload. The retrieved user data is 'flattened' and recorded in the DEF working table.
Calabrio User Metadata Retriever	N/A	Working table	Retrieves employee information from the Calabrio API.
Calabrio Media Retriever Action	N/A	N/A	Retrieves media from the Calabrio API. No input resource is required, since the API configured in the action will be the data source. A file system or SFTP output resource is required. NOTE: An initial request to transcode all media to the requested format is required. Downloading will begin after each batch of media is trans-coded, which may take some time.
Calabrio Metadata Retriever Action	N/A	N/A	Retrieves Contact metadata from the Calabrio API and writes the data to the database.
Channel0 Mono Detector	Non-Credentialed File System resource with media	N/A	Analyzes media files through the process. If only a single channel is found, set Channel0=Mono in the metadata.

ACTION	INPUT	OUTPUT	DESCRIPTION
Chat CSV to JSON	File System or SFTP resource	File System or SFTP resource	Converts chat-based CSV files to Text Grid's expected JSON format CSV files are expected to ordered in such a way that chat IDs come in unbroken groups of rows, and turns within them are in temporal order. If that's not the case, previous actions can coerce the data into the right structure by first importing the data into DEF, then using a DB-to-CSV action write out the consumable file, ordering by appropriate fields.
Cleanup	File System or SFTP resource	File System resource	Handles cleanup of DEF working tables, DEF internal logs, and input resources based on a configurable number of hours, in many different configurations. Cleanup is absolutely necessary in all systems to insure that the database and/or file system doesn't grow unchecked.
Compress	Non-Credentialed File System resource	Non-Credentialed File System resource	Creates compressed files from input resources, and writes them to output resources. Supports output of both ZIP and TAR files in various structures with optional GZIP compression of TAR files, as well as MDF has file creation for compressed files and password protection.

ACTION	INPUT	OUTPUT	DESCRIPTION
Convert Engage Hierarchy V2	Database	File System or SFTP resource	Extracts User and Hierarchy data from Engage to NIA friendly CSV formats for Hierarchy, Agent and User imports. The validations performed by the Convert Engage Hierarchy and Agent and User Import actions are logged in the process working table and you can access them via SQL Server Reporting Services (SSRS).
Convert JSON to XML	File System Resource	File System Resource	Converts JSON file to an XML file.
Convert Transcript to ASR	file system or SFTP	file system or SFTP	The ASR format is used to convert the *Interaction Analytics* Standard transcript format into the VoiceGrid ASR.
Convert Transcript To NxPr	file system or SFTP	file system or SFTP	Convert Transcripts to NxPr. The NXPR\NxPr format is used to represent transcripts in a physical format suitable for interchange within *Interaction Analytics*applications.
Custom SQL V2	Database (when not processing against the working table)	N/A	Allows upload of single SQL files instead of manually entering queries, reducing human error. It validates SQL syntax and ensures that only single statements are processed.

ACTION	INPUT	OUTPUT	DESCRIPTION
DEU to Engage	Non-Credentialed File System resource	A database resource	Takes single-hub DEU output files, or the individual zips extracted by the DEU Zip Re packager in DEU multi-hub output scenarios, and pushes relevant CSV data to appropriate Engage tables. Note that the database resource needs to point to an Engage database holding the tbINXTargetMediaSetDefinition and other related tables.
DEU Zip Repackager	Non-Credentialed File System resource	Non-credentialed file system resource	Repackages DEU zip output files whose contents are separated by an Engage Hub into individual zips for each Engage Hub to consume (via a NIA-to Engage process or the newer DEU-to-Engage action).
Database to CSV	A single database resource	File System or SFTP resource	Selects arbitrary data from an attached database resource, and writes results out to a CSV file. Allows for pre- and post-select SQL execution to prep for select and/or clean up data after selection, and integrates with the process-level quantum for pulling data in time-bound windows. All SQL should be optimized externally with the database for best performance.
Duplicate File Detector	File System or SFTP resource	N/A	Detects the files previously processed in the file system or SFTP input resource and optionally integrates with the process quantum. Provides many options for finding the files, either directly from input resources or indirectly by working table metadata.

ACTION	INPUT	OUTPUT	DESCRIPTION
Duplicate Spotting	File System resource	N/A	Identify duplicate segments across incoming media files, and append that information to the metadata about those files. Metadata will be structured as XML, and will identify locations of duplicate segments in a file by timestamps. The action uses the {DEF_ DATA}\workbench\duplicateSpotter.catalog that comes with the DEF installation, but it can be replaced with a more appropriate one after stopping the DEF service.
ElevateAI Near Real-Time Retriever	Artemis Resource connecting to and consuming messages (transcripts) from an Artemis MQ queue	Nx File System resource where VoiceGrid ASR transcript JSON files will be written. Output files follow naming pattern as {callMetaData->interactionIdentifier}.nmf.json Files are written in pretty-printed JSON format and it supports both local and network file system paths	Retrieve ElevateAI real-time punctuated transcript format from Artemis message queue and converts it into VoiceGrid ASR transcript format in a JSON file for downstream processing. A later process inbox publishes (sends) the metadata and transcript to NIA.
ElevateAI Response	File System resource	N/A	Retrieve ElevateAI transcripts and metadata. This action also calculates the Enlighten scores in the JSON files created.

ACTION	INPUT	OUTPUT	DESCRIPTION
ElevateAI Submit	File System resource	N/A	Declare and upload media files to ElevateAI.
EML Parse Text	File System or SFTP resource	File System or SFTP resource	Parses text from email attachments including Microsoft office documents and PDFs and appends parsed text to the email body, reading .eml files from the input resource, and writing the parsed text to the output resource.
EML Text Remove	File System resource	File System resource	Removes specified text from email bodies, reading .eml files from the input resource, and writing the revised .eml to the output resource. Text can be either raw, or Java-style regular expressions. A Java regex testing tool with corresponding javadoc can be found at http/www.regexplanetcom/advanced/java/index.html.
Engage Audit	Varies based on the selected method	Varies based on the selected method	You can attach this action to the end of any process to collect action run-time information, and insert it into the tbINXMonitoring table of an Engage database for monitoring. In cases where the action doesn't have direct access to the Engage database, it can also be used to dump that same data to a CSV file, which can then be picked up by an Engage Audit action in another process on another server with access.

ACTION	INPUT	OUTPUT	DESCRIPTION
Engage Desktop Tags	Database input (Engage database target)	N/A	Gathers Engage Desktop Tag Data from the specified Engage database resource. That data matches interactions in the incoming metadata into a JSON structure expected by *Interaction Analytics*. Inserts this JSON into the 'InteractionTags' column in the working table for this process.
Engage Key Extraction	File System or SFTP resource	File System or SFTP resource	Extracts keys from the NiCE recorder using the process-defined quantum, and is assumed to be the only NMF decrypt-related process running on the key extraction server. On each run, the action queries the recorder for encryption GUIDs generated within the quantum window, extracts corresponding GUID/key/IV triples from the recorder's keystore manager, and uses the latest symmetric key generated by the Generated Protected Key process to encrypt and deliver that in a CSV file for pickup by the Load Keys action running on the decryption server.
Engage Load Keys	File System or SFTP resource	File System or SFTP resource	Load keys from encrypted CSV files generated by the Key Extraction action into the nx_nmf_keystore table for the NMF Decrypt action to use when decrypting NMF files. Ages out old keys with a default of 60 days that should be considered in any new environment. See Encryption Recorder Integration for details on this and other NMF Decrypt-related actions.

ACTION	INPUT	OUTPUT	DESCRIPTION
Engage Media Extract V3	Input Database Resource and optionally an Input ESM Device	File System or SFTP Output Resource	Extracts media from Engage storage for interactions rolling through in current metadata, stitching where appropriate.
Engage NMF Decrypt	File System or SFTP resource	File System or SFTP resource	Decrypts NMF media files using keys loaded into the nx_nmf_keystore table by the Load Keys action. Requires further configuration with the Generated Protected Key, Key Extraction, and Load Keys actions to get those keys into place. See Encryption Recorder Integration to learn more.
Engage NMF MML Decrypt	File System resource	File System or SFTP resource	Decrypts NMF MML media files using keys loaded into the nx_mmf_keystore table by the Load Keys action. Requires further configuration with the Generated Protected Key, Key Extraction, and Load Keys actions. See Encryption Recorder Integration to learn more.
Engage Split Packet Fixer	File system or SFTP	File System or SFTP	Trim audio packets if the interaction/recording start time is between the packet start and end time or packet end time is between the interaction/recording start and end time. This action only works on decrypted files. If no output resource is provided, one will be automatically provisioned in the {DEF_DATA}/fs/ folder.

ACTION	INPUT	OUTPUT	DESCRIPTION
Enlighten Response	N/A	N/A	Gets the response back from the Grid and saves the information (which includes the segments list) into the Data Exchange Frameworkdatabase.
Enlighten Submit	File System resource	N/A	Sends the media files information to RabbitMQ for the Grid service.
Enlighten Transcript Transform	File Input resource	File Output resource	Transforms customer transcript data into files that are consumed by the external Workbench Enlighten executable.
Executable Wrapper	File System resource	File System resource	Runs any executable accessible to the Data Exchange Frameworkservice user account, either standalone or integrated into a regular process flow. For the latter, various parameters are available to use in constructing the call to the executable, and these may or may not require attached input or output file system resources.
ExternalMediaID Capture	Single File System resource	N/A	Using the path of an attached file system resource input, adds an 'ExtemalMediald' column value to the working table for all rows passing through in the current disposition that will be a concatenation of the path of the input resource and the row's 'FileName'.
File Discovery	File System or SFTP resource	N/A	Discovers files in file system or SFTP resources, and adds them to the DEF working tables, optionally integrating with the process quantum.

ACTION	INPUT	OUTPUT	DESCRIPTION
File Filter	File System or SFTP resource		Used as a pre-processing step, deletes any files that fall outside of an expected size range. Passing files are logged in the working table, so actions downstream can process them directly from metadata without an additional scan of the file system.
File Maker	File System or SFTP resource	File System or SFTP resource	Generates files from working table file key metadata using the specified file filter pattern, substituting file key into the asterisk. This generates either a zero-byte file in the specified format, or can optionally copy a specified file into the target location, renaming to file key + file filter pattern.
File Mover	File System of SFTP resource	File System or SFTP resource	Moves or copies tiles between input and output file system and/or SFTP resources, optionally integrating with the process quantum. Provides many different options for moving files, either directly from input resources, or indirectly by way of working table metadata.
File Rename	File System or SFTP resource	N/A	Rename files in an input resource using Java regular expressions, renaming any corresponding working table 'FileName' column entries as appropriate. A Java regex testing tool with corresponding javadoc can be found at http://www.regexpianetcom/advanced/)ava/index.html.

ACTION	INPUT	OUTPUT	DESCRIPTION
GDPR ID Creator	N/A	N/A	Creates GdprID# columns based on configured incoming metadata columns. If the selected metadata column is a phone number, then a phone number normalization function can be applied to the phone number to remove all non-numeric characters.
GDPR Requests Handler	Compliance Service Configuration or File System resource	File System resource	Honors GDPR and other data protection requirements by allowing either direct interaction with *Interaction Analytics*, or using file system resources to process the same requests out of delivered CSV files.
Generate Desktop Tags	CSV and Database	Nexidia database	Imports evaluation data from CSV files or evaluation databases, then stores the results as Desktop Tags on existing interactions. It supports evaluation data from NiCE Quality Central (QC), third-party systems, or other external sources.
Genesys Cloud Job-Metadata Retriever	N/A	N/A	Retrieves the conversation metadata from Genesys Cloud API and saves them in the DEF process working table.

ACTION	INPUT	OUTPUT	DESCRIPTION
Genesys Cloud Transcoded Media Retriever Action	N/A	N/A	Retrieves transcoded media from Genesys cloud systems. No input resource is required, since the API configured in the action will be the data source. A file system or SFTP output resource is required. NOTE: An initial request to transcode the media is necessary. Downloading should succeed if transcoding is complete.
Genesys Cloud Transcoding Requester Action	N/A	N/A	Makes requests of the Genesys Cloud system for the transcoding of media. No input resource is required, since the media requiring transcoding . No file system or SFTP output resource is required. NOTE: Without this transcoding step, the media on the API exists only in a proprietary format, and is not retrievable by the client.
Genesys Cloud User Metadata Retriever Action	N/A	N/A	Retrieves user-metadata from Genesys Cloud API and writes the data to the database.
Genesys IR Media Retriever Action	N/A	N/A	Retrieves media from Genesys recorder systems. No input resource is required, since the API configured in the action will be the data source. A file system or SFTP output resource is required.
Genesys IR Metadata Retriever Action	N/A	N/A	Retrieves Genesys metadata from Genesys IR API and writes the data to the database.

ACTION	INPUT	OUTPUT	DESCRIPTION
GPG Decrypt **Note:** The Data Exchange Framework supports GPG4Win version 3.1.11 and later.	File System or SFTP resource	File System or SFTP resource	Decrypts files using GPG/PGP keys that have already been installed, and are available to the Data Exchange Framework service user account. This action is more of a wrapper around an external executable (like WinGPG) that must be installed and configured outside of the Data Exchange Framework, and requires a standards-compliant command line 'gpg' executable that it can call using the provided pass phrase.
GPG Encrypt **Note:** The Data Exchange Framework supports GPG4Win version 3.1.11 and later.	File System or SFTP resource	File System or SFTP resource	Encrypts files using GPG/PGP keys that have already been installed, and are available to the Data Exchange Framework service user account. This action is more of a wrapper around an external executable (like WinGPG) that must be installed and configured outside of the Data Exchange Framework, and requires a standards-compliant command line 'gpg' executable that it can call using the provided pass phrase.
Generate Protected Keys	File System or SFTP resource	File System or SFTP resource	Generates a symmetric key that's been encrypted against the certificate of the remote key extraction server, and stores it in both the output resource and in a local, configurable keystore. See Encryption Recorder Integration to learn more.

ACTION	INPUT	OUTPUT	DESCRIPTION
Google Metadata Media Retriever Action	N/A	File Folder	Connects to Google Cloud Storage (GCS), retrieves interaction metadata including media file locations from the API and publishes it to Interaction Analytics for further processing, based on a customer-provided API name and authentication method.
Hash Generate/Validate	File System or SFTP resource	N/A	Generates or validates hash files to guard against file corruption during transmission. When generating, the action creates a new file for each incoming file using the original file name plus an extension appropriate to the hash type (*.md5 or *.sha1, e.g.). When validating, it expects the same: Each incoming file should have a corresponding hash file in the form {fileName}. hashTypeExt} that will be used to validate against. Only files passing validation move farther downstream of this action in the process.
Hitachi Retriever	N/A	Single File System or SFTP resource	Retrieves media from Hitachi appliances based on keys included in incoming metadata.
Import JSON	Nested JSON File containing a root array	Individual working table rows created from each JSON object in the array	The Import JSON action uses the working table defined for the process group. It creates one working table row for each object when the input file has a root-level array. If the array contains many objects, the action separates them and adds each one as a separate row for downstream processing.

ACTION	INPUT	OUTPUT	DESCRIPTION
InContact Interaction Retriever	N/A	File System or SFTP resource	Retrieves contact, agent, and custom metadata, plus corresponding audio, email, and chat interactions from InContact systems using the process-defined quantum. Contact, agent, and other configured custom information is recorded as working tale metadata for each file retrieved, or alone if the action is configured to exclude media.
Inbox Publisher	File System or SFTP resource (required when using ESMs only)	A *Interaction Analytics* Database	Pushes collected metadata about processed files to the *Interaction Analytics* Rich Inbox, and optionally remaps some of those fields to target Interaction Analytics fields. Incoming media is expected to contain and ExternalMediaID column that contains the full path to the media files. It should be auto-populated by any single File Mover action earlier in the process when it moves media to its final ingest location.
IVR Redaction	N/A	N/A	Replaces the first few seconds or milliseconds of audio with silence. The IVR redaction accepts input media in WAV format and produces redacted output media as WAV format.

ACTION	INPUT	OUTPUT	DESCRIPTION
Language ID This is a *Interaction Analytics* Workbench function.	Non-credentialed File System resource	N/A	Identifies languages used in incoming media files, and adds that information to the working table's LanguageScore, LanguageName, LanguageId, and LanguageIDFullResults (when Fast processing=0) columns metadata for each file. LanguageScore, LanguageName, and LanguageId are based on the top-scoring language. Language identification is done using the default language pack in the DEF data folder's workbench/languagePacks/LanguagePackALS file. This file can be overwritten by a more appropriate language pack, if needed, after stopping the DEF service account. The new language pack will be picked up on service start.
Language ID for Text Interactions	File System resource	N/A	Identifies languages used in incoming files and adds that information to the working table's (LanguageIDFullResults and also the LanguageName and LanguageScore (for the highest score) columns metadata for each file.
Media Statistics Calculator	File System resource	N/A	Processes media files to calculate various statistics. It can process files either from metadata or from attached input resources. This action is useful for analyzing media files, such as audio and chat transcripts.

ACTION	INPUT	OUTPUT	DESCRIPTION
Media Conversion	File System resource	File System resource	Converts incoming media to the configured format and encoding using codecs available to *Interaction Analytics* Workbench on this system. The configuration options may vary.
Media URL Download	File System resource	File System resource	Downloads media files from specified URLs.
Merge	File System resource	File System resource	Merges media channel files in input resources based on metadata about them.
Metadata Append	N/A	N/A	Allows for the creation of new metadata columns in the working table for use downstream, either empty or created based on existing metadata coming from upstream.
Metadata Filter	N/A	N/A	Filters metadata records moving through this process' working table by performing either non-null checks for specific columns, or by applying a configured SQL 'where' clause. Works by testing against all working table records in the appropriate disposition for this action, and by clearing this action's disposition column for those records that pass the condition. Records that fail will still be re-tested on subsequent runs in case of metadata updates, etc, up to the process' maximum number of self reconciliation days.
Metadata Import	File System or SFTP resource	N/A	Imports metadata from CSV and/or XML files. This is one of the main entry points for metadata collection and processing, and has a variety of options for working with the metadata.

ACTION	INPUT	OUTPUT	DESCRIPTION
Metadata Join	N/A	N/A	Joins against the working table of another process group, and conditionally pulls that metadata into the process working table. This action requires appropriate external indexing of referenced columns for optimal performance.
Metadata Mapping and Export	N/A	File System or SFTP resource	Exports collected metadata for files processed up to this point in the process in either XML or CSV format, optionally remapping some of those fields to known *Interaction Analytics* fields. Also creates corresponding 0-byte *esm media marker files alongside the XML if configured for it.
Metadata Summary	N/A	N/A	Provides summary information on data processed, in the form of groupings of metadata columns and their corresponding counts, which have reached this point in process execution. Results are e-mailed to the appropriate server, site, company, or admin-level administrator.
NIA File Creator	File System resource	File System resource	Creates *.nia files from incoming media files that will identify, in XML form, the time ranges where speech is detected in each channel. File names will match media, but with '.nia' appended.

ACTION	INPUT	OUTPUT	DESCRIPTION
NIA Reprocessing	N/A	An *Interaction Analytics* Database	Attaches to processes during re-runs to let NIA know that incoming media files need to be re-processed. In most cases, this should be added before the Inbox Publish or Metadata Mapping and Export action so that NIA will clear the old results before publishing what would otherwise be repeats.
NTR Centera Retriever	N/A	A file System or SFTP resource	Retrieves media files from a Centera pool. Processes metadata by default, and assumes that a column already exists in working table metadata that contains the Centera clip ID (column name can be configured in the action). Also requires a .pea file for authentication provided by the pool owner.
NTR Decrypt	One non-credentialed file system resource containing files to decrypt and one database pointing to the NTR database holding the key management table	One non-credentialed file System resource	Decrypts media files encrypted by an NTR recorder. Requires the unlimited JCE extensions for 256-bit decryption.

ACTION	INPUT	OUTPUT	DESCRIPTION
NTR Stitch	File System resource	File System resource	Stitches incoming media files together based on metadata about those files in an NTR system. Handles different mixes of VOX and CDR records, and provides stitching of silence to get media to match metadata.
OneAnalytics Publisher	File System or SFTP resource	OneAnalytics AWS S3 bucket	Publishes metadata and media to an Amazon S3 bucket before declaring it to SQS for OneAnalytics to pick up and process. Authentication is handled using Cognito.
OneAnalytics Query Retriever	*Interaction Analytics* Database	OneAnalytics AWS S3 bucket	Retrieves queries from NIA and pushes the queries to an Amazon S3 bucket before declaring it to SQS for OneAanlytics to pick up and process. Handles authentication using Cognito.
Parse Email Text to JSON	N/A	N/A	Parse email text from files including *.eml and *.msg files from the input resource, and writing the parsed text to the output resource. Requires a file system or SFTP resource for both input and output resources. If no input or output resources are provided, they will either be inferred from adjacent action I/Os, or automatically provisioned in the {DEF_DATA}/fs/folder.

ACTION	INPUT	OUTPUT	DESCRIPTION
Parse General Text to JSON	N/A	N/A	Parse general text from files including Microsoft office documents and PDF reading files from the input resource, and writing the parsed text to the output resource. Requires a file system or SFTP resource for both input and output resources. If no input or output resources are provided, they will either be inferred from adjacent action I/Os, or automatically provisioned in the ÍDEF DATAJ/fs/ folder.
PST to EML	Non-credentialed File System resource containing PSTs	Non-credentialed File System	Extracts emails from .pst file inputs, and saves them as .eml files to outputs, tracking either the PSTs or the EMLs in the working table. EML file names will be derived from a combination of the PST file's descriptor node ID, a hash of the subject, and the email's message delivery time.
Process Chain	N/A	N/A	Chains processes in the same process group together, allowing one process to launch another one when it completes. When used as a process exit point, this action triggers execution of the target process.
Process Launch	N/A	N/A	Launches an arbitrary process running on this server. See the Process Chain action for an alternative to launching processes within the same process group whose dispositions should be chained together.
Production Media Folders Path	N/A	File System or SFTP resource	Creates the folder structure in the output resource without moving the media there. *Interaction Analytics* uses this path in the ingest path property.

ACTION	INPUT	OUTPUT	DESCRIPTION
QC Interactions Audit	Varies	Varies	Attach to the end of any process to collect action run-time information, and insert it into the *Interaction Analytics* Evaluations database for monitoring.
Redaction	File System resource	File System resource	Takes the segment information from the response in the DEF database and applies it to the media files by calling the Workbench API.
Redaction Response	N/A	N/A	Gets the response back from Search Grid and saves the information, which includes the segments list into the DEF database.
Redaction Submit	File System resource	N/A	Sends the media files information to RabbitMQ for the Search Grid service. See the *Interaction Analytics Voice Grid Media Worker Administration* Guide for instructions for RabbitMQ Installation and Workflow Model Management.
Remove Duplicate Metadata	File System resource	N/A	Remove duplicate metadata based on the original file name. Requires no input or output resources, as it works directly against the working table for this process.
Replace CSV Header	File System or SFTP	N/A	Replaces the header row in incoming CSV files with a specified header row. Reads *.csv files, rewrites them with the new header as *.csx files, deletes the original *.csv, and tracks the renamed file in the working table. Probably best used as an entry point for incoming CSV metadata.

ACTION	INPUT	OUTPUT	DESCRIPTION
Reporting	N/A	File System or SFTP resource	Generates a report that tells how many files are awaiting processing at each disposition across this processes process group, optionally grouped by other metadata fields, and places it into a CSV file.
RTG Agent-mapping Action	N/A	N/A	Allows mapping the agent data to the Real-time Grid (RTG) system. This RTG agent-mapping action accepts CSV-files containing agent-mapping data to upload to RTG. It tracks that data using a dedicated database-table.
Salesforce Metadata Parser	N/A	EML file for email and CSV fie for chat	Parses the retrieved metadata in the working table. The parsed metadata will be written in the output file. The output file could be EML file for email and CSV file for Chat based on the input provided.
Salesforce Metadata Retriever	Optional	Optional	Connects with the Salesforce API to retrieve metadata (such as chat, email, etc) based on customer requests. This metadata is then analyzed within the *Interaction Analytics* platform. After retrieval, the metadata is inserted into the working table for further processing.
SalesLoft Media Retriever Action	N/A	N/A	Retrieves audio from SalesLoft systems. No input resource is required, since the API configured in the action will be the data source. A file system or SFTP output resource is required.

ACTION	INPUT	OUTPUT	DESCRIPTION
SalesLoft Metadata Retriever Action	N/A	N/A	Retrieves SalesLoft call data records, using the process-defined quantum. Contact information is recorded as working table metadata for each file retrieved. No input resource is required, since the API configured in the action will be the data source. No output resource is required.
SalesLoft Users Retriever Action	N/A	N/A	Retrieves SalesLoft users. Contact information is recorded as working table metadata for each user retrieved. No input resource is required, since the API configured in the action will be the data source. No output resource is required.
Stitch	File System resource	File System resource	Stitches incoming media files together using available metadata. Can stitch segments of a call together based on a sequence column, or on two columns that chain the current segment ID with the next one, optionally padding the call with silence o match the metadata. Also provides an approach to handling multiple recordings of the same call, when a segment ID column is available that can identify duplicates.
Text Redaction	File System or SFTP resource	File System or SFTP resource	Used to enter arbitrary Java-style regular expressions to be applied to incoming text files. See http://www.vogella.com/tutorials/JavaRegularExpressions/article.html for more details.

ACTION	INPUT	OUTPUT	DESCRIPTION
Topic AI Request	Transcript JSON	CSV transcript	Retrieves the correct and active version of the IPM model from the Topic AI editor and sends the media files information to RabbitMQ for processing by the Voice Grid service. If no input resource is provided, resources are inferred from adjacent action inputs/outputs or automatically provisioned in the {DEF_DATA}/fs/ folder.
Topic AI Response	N/A	N/A	Retrieves the response back from the Voice Grid, processes the information, and sends this information to NIA. It then saves the information, which includes segments list into DEF database.
Twilio Calls Metadata Retriever Action	File System or SFTP resource	File System or SFTP resource	Extracts calls metadata from the Twilio.
Twilio Media Retriever Action	File System or SFTP resource	File System or SFTP resource	Extracts media from Twilio.
Twilio Recordings Metadata Retriever Action	File System or SFTP resource	File System or SFTP resource	Extracts recordings metadata from Twilio.

ACTION	INPUT	OUTPUT	DESCRIPTION
Uncompress	Non-credentialed File System resource	Non-credentialed File System resource	Uncompresses ZIP and TAR files from input resources, extracting their contents to output resources. Supports GZIP compression of TAR files, MD5 hash checking, and password protected input files.
User Review	N/A	File System or SFTP resource	Exports specific metadata fields to a CSV file, optionally adding empty ones.
Verint Sitch	Non-credentialed File System resource	Non-credentialed File System resource	Takes a full day of Verint.zip file exports, pre-extracted by a previous Uncompress action in the process, then analyzes the .xml files for 'Phone'-type calls, extracts the metadata, and stitches the media. It is critical that the full day's calls be available at once, since the stitching maps only join together correctly with a complete picture of how all the calls join together.
Voice Print Creator	Non-credentialed File System resource containing media	Non-credentialed File System resource to write voice prints to	Creates voice prints from incoming media file for each channel. The file names will match media, but with '.vpv' appended.
Vonage Media Retriever Action	N/A	File system or FTP resource.	Retrieves call recording from Vonage systems.

ACTION	INPUT	OUTPUT	DESCRIPTION
Vonage Metadata Retriever Action	N/A	N/A	Retrieves call recording metadata from Vonage systems, using the process-defined quantum. Call recording information is recorded as working table metadata for each file retrieved.
Workflow Enlighten	N/A	N/A	This action reads the enlighten scores data from response in the DEF database and ingests to working table. It requires no input and no output file systems. This action calculates the index scores for the autosummarization tag.
Workflow Request	File System resource	N/A	Sends the media files information to RabbitMQ for the Search Grid service. See the *Interaction Analytics* Voice Grid Media Worker Administration Guide for instructions for RabbitMQ Installation and Workflow Model Management.
Worflow Response	N/A	N/A	Gets the response back from Search Grid and saves the information, which includes the segments list into the DEF database.
Workflow Redaction	File System resource	File System resource	Takes the segment information from the response in the DEF database and applies it to the media files by calling the Workbench API.
XSL Transform	File System or SFTP resource	File System or SFTP resource	Transforms incoming XML files to new XML files in output resources using a specified XSL stylesheet.

Agent Hierarchy Action

1. In the **Process files from metadata**, specify if you want to process files from metadata. When you select ***true*** the application processes files from the process groups existing working table. When you specify false, the application processes files from attached input resources. Processing from metadata reduces I/O and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **File filter**, enter the file filter type. For example: CSV.
3. In the **Delimeter**, enter the delimiter character. This is used for non-XML input, and is a single character such as ",". You can also use \t for tab.
4. In the **Host Name**, enter the host name for the Webservice Call.
5. In the **Fail if nothing is imported within # minutes**, specify the number of minutes to fail the import if nothing is imported within that time.
6. Leave the **Force file encoding** blank to auto-detect.
7. In the **Delete files after processing** drop-down, specify if you want the application to delete files after processing.
8. In the **Reservation Duration Minutes**, specify the total minutes that the process can finish the import.
9. In the **Default quote character**, specify the character used to quote column values (for example **"**) in the Default quote character field.
10. In the **Default escape character**, specify the character used to escape column values (for example ****) in the Default escape character field.
11. In the **Failure Threshold**, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
12. In the **Batch Execute Size** drop-down, specify the batch execute size.
13. In the **Disable Email Sending** drop-down, select ***true*** to not send an email with the CSV error validation as an attachment.
14. In the **Send Hierarchy Checker PDF** drop-down, select ***true*** if you want the PDF to contain the description for all possible errors from the Data Exchange Frameworkand Interaction Analytics and how to resolve them.

15. In the **Delete Validation Error CSV** drop-down, select ***true*** to delete the validation error CSV file after sending the email.
16. In the **Hierarchy Validation Contact Email** , enter an email address to send notification emails to for all administrative server messages, as well as worker servers when no other administrative contacts are assigned.
17. In the **SMTP Server**, enter the SMTP server.
18. In the **SMTP user (optional)**, enter the SMTP user.
19. In the **SMTP password (optional)**, enter the SMTP password.
20. In the **Hierarchy validation from Email**, enter the Email address from notifications for hierarchy validation errors. Default value is 'def-heirarchy@niceondemand.com'.
21. In the **Transfer protocol**, configure the transfer protocol for communicating agent/user information to NIA. The default protocol is HTTP.
22. In the **Reconciliation Identifier**, enter the name of the action if there is a reporting action in the process group.
23. In the **Default input disposition handling** drop-down list, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
    - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
24. In the **Input Disposition(s)**, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
25. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
26. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.

27. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
28. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
29. Click **Submit**.

Agent/User Import V2 Action

In the Data Exchange Framework 5.5 SP2 version, the Agent and User Import action was updated to use the new the API endpoints from *Interaction Analytics* 12.3 SP4. In order to use this functionality, you must have Nexidia Interaction Analytics version 12.3 SP4.

1. In the **Process files from metadata**, specify if you want to process files from metadata. When you select ***true*** the application processes files from the process groups existing working table. When you specify ***false***, the application processes files from attached input resources. Processing from metadata reduces I/O and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **File filter**, enter a filter to limit the files processed. For example: CSV.
3. In the **Delimiter**, enter the delimiter character. This is used for non-XML input, and is a single character such as ",". You can also use \t for tab.
4. In the **Flow**, specify the flow. The valid options are Agent or User.
5. In the **Host Name**, enter the host name for the web service call.
6. In the **Default Integrated Authentication in NIA**, select ***true*** to set NIA integrated authentication to true and ignore the integrated authentication value from Engage CSV. Select ***false*** to not set NIA integrated authentication.
7. In the **Integrated Authentication**, select ***true*** to read and import the integrated authentication flag from Engage CSV to NIA. Select ***false*** to not set integrated authentication in NIA.
8. In the **Fail if nothing is imported within # minutes**, specify the number of minutes to fail the import if nothing is imported within that time.
9. In the **Force file encoding**, leave blank to auto-detect. If auto-detect is not working, use a valid Java InputStream encoding.
10. In the **Delete files after processing** drop-down, specify if you want the application to delete files after processing.
11. In the **Default quote character**, enter the character used to quote column values in the output CSV file, for example ".
12. In the **Default escape character**, enter the character used to escape column values in the output CSV file, for example \.

13. In the **Failure Threshold**, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
14. In the **Batch Execute Size** drop-down, specify the Batch Execute Size.
15. In the **Defaut Agent/User Password**, enter the default password that can be used for the new agent/user if it is required. The default value is ***P@ssw0rd1*** and it can be modified as needed.
16. In the **Override NIA fields with Null Value** drop-down, select ***true*** to override NIA fields while passing null for the user/agent metadata: ExternalUserId, ScanAgentMonitor, ScanAgentGroup, MediaPlaybackAuthorizationId, and ProfilesList.
17. In the **Disable Email Sending** drop-down, select ***true*** to not send an email with the CSV error validation as an attachment.
18. In the **Send Hierarchy Checker PDF** drop-down, select ***true*** to send a pdf that contains descriptions of all possible errors from DEF and NIA, along with details on how to resolve them.
19. In the **Delete Validation Error CSV** drop-down, select ***true*** to delete the validation error CSV file after sending the email.
20. In the **Hierarchy Validation Contact Email**, enter an email address to send notification emails to for all administrative server messages, as well as worker servers when no other administrative contacts are assigned.
21. In the **SMTP Server**, enter the SMTP server.
22. In the **SMTP user (optional)**, enter the SMTP user.
23. In the **SMTP password (optional)**, enter the SMTP password.
24. In the **Hierarchy validation from Email**, enter the Email address from notifications for hierarchy validation errors. Default value is ***def-heirarchy@niceondemand.com***.
25. In the **Transfer Protocol**, configure the transfer protocol for communicating agent/user information to NIA. The default protocol is HTTP.
26. In **Reconciliation Identifier**, enter the name of the action if there is a reporting action in the process group.
27. In the **Default input disposition handling** drop-down, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.

- ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.

28. In the **Input Disposition(s)** , override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
29. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
30. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
31. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
32. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
33. Click **Submit**.

Alternate Media ID Action

1. In the **Process files from metadata** drop-down, specify if you want to process files from metadata. When you select ***true*** the application processes files from the process groups existing working table. When you specify false, the application processes files from attached input resources. Processing from metadata reduces I/O and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **File Filter**, enter the file filter type, for example CSV. Java-style regular expressions can also be used if prepended with "regex". Note that when processing files from resources, this acts as a filter, but when processing from metadata, the metadata's file key is inserted in any wildcards, expanding the processing of the one file to all listed variants. For example: (*,*.xml' would process both a file matching the working tale file key '1234567.wav' and its '1234567.wav.xml' counterpart.
3. In the **Identify alternate media from** drop-down, specify where you want to identify the alternate media from. The valid options are ***Metadata Field Test*** or ***File System Test***.
4. In the **Metadata field name**, enter a metadata field name. The name that you enter is used to determine of alternate media is available.
5. In the **Metadata field value**, enter a metadata field value to test for. Use ***{null}*** or ***{not null}*** without quotes to test for those conditions, or enter a literal value to test for.
6. In the **Media expiration, in days** drop-down, specify the number of days after which the media should expire.
7. In the **Expire media from metadata field**, add x days to the value of this metadata field and use that as the media expiration. Leave blank to expire the media x days after this action discovers them.
8. In the **Failure Threshold**, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
9. In the **Reconciliation Identifier**, enter the name of the action if there is a reporting action in the process group.
10. In the **Default input disposition handling** drop-down, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.

- ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.

11. In the **Input Disposition(s)**, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
12. In the **Thread Count** , enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
13. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
14. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
15. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
16. Click **Submit**.

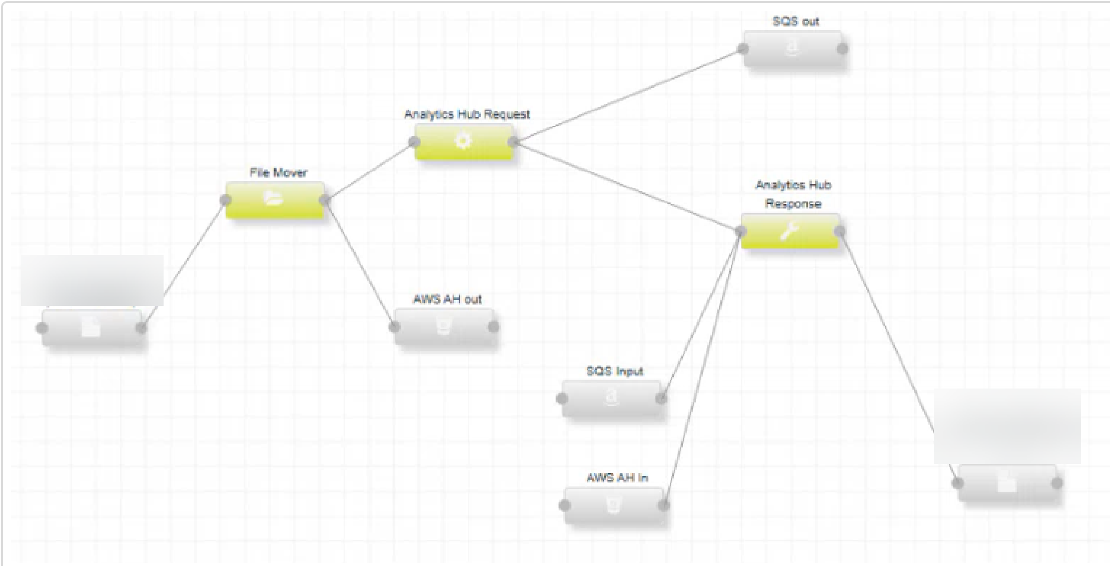
Analytics Hub Response

1. In the **File Name Column** field, enter the column name that stores the processed file names in metadata.
2. In the **Batch Size** field, enter the number of SQS messages to process in each batch. You can enter the values from 1-10000.

3. In the **SQS Wait Time (Seconds)** field, enter how many seconds the system should wait for new messages from the queue. You can choose a value between 0 and 20.
4. In the **Failure Threshold**, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
5. In the **Correlation ID column**, specify the column name for correlation ID in metadata.
6. In the **Cleanup Transcript Files** field, select ***true*** to delete the transcript files from S3 after successful processing.
7. In the **Cleanup Error Messages** field, select ***true*** to delete error messages from SQS after logging them.
8. In the **Delete SQS Messages** field, select ***true*** to delete messages from SQS queue after successful processing and file download.
9. In the **Treat Missing Data as Errors** field, select ***true*** to delete messages from SQS queue after successful processing and file download.
10. The **Remove Messages for Missing Data** field is set to ***true*** by default to remove messages from SQS for missing data that are not in the working table. Change to ***false*** to keep messages in SQS.
11. In the **Retry Submit to Analytics Hub** field, set to **true** to resend the request to Analytics Hub for reprocessing in case of an error in the response message indicating missing files.
12. In the **Max Transcript Size (MB)** field, enter the maximum allowed transcript size in MB (1-500). Files exceeding this limit will be skipped with error logging.
13. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.
14. In the **Default input disposition handling** drop-down list select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all the immediate inputs.
    - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.

15. Optionally enter the input disposition. The Data Exchange Framework automatically chains action component dispositions together as javascript:void()part of its processing, so this field is usually left blank. If you are using split processes, enter a value in this field In the Input Disposition(s) field, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type 'none' to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
16. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
17. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
18. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
19. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
20. Click **Submit**.

You can refer to the following sample process



How It Works?

1. Listens to and consumes messages from the SQS queue.
2. For each message, fetches the corresponding transcript from S3.
3. Converts the transcript into VG ASR format.
4. Writes the converted transcript to the designated location in the file system.

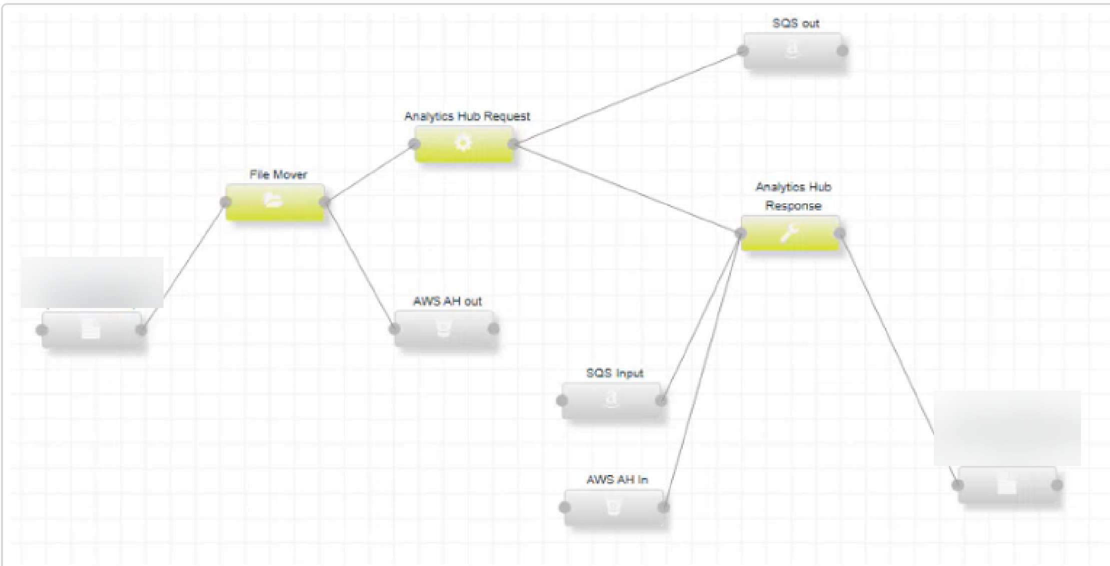
Analytics Hub Request

1. In the **Tenant ID** field, enter the GUID that identifies the tenant. This ID is sent to the SQS queue.
2. In the **Language** field, select the language code for processing. Make sure it matches one of the approved codes. The default value is en-US.
3. In the **S3 Media Upload Path** field, enter the environment prefix for the S3 path. For example, dev, test, prod.
4. In the **File Name Column** field, enter the column name that contains the file name in the metadata.
5. In the **Correlation ID column**, specify the column name for correlation ID in metadata.

6. In the **Failure Threshold**, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
7. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.
8. In the **Default input disposition handling** drop-down list select one of the following:
   - ***AND*** - Allow files to flow through this step that have passed all the immediate inputs.
   - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
9. Optionally enter the input disposition. The Data Exchange Framework automatically chains action component dispositions together as javascript:void()part of its processing, so this field is usually left blank. If you are using split processes, enter a value in this field In the Input Disposition(s) field, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type 'none' to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
10. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
11. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
12. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
13. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.

14. Click **Submit**.

You can refer to the following sample process



How It Works?

1. A media file is uploaded to an S3 bucket using the File Mover action.
2. The Analytics Hub Request action sends metadata to an SQS queue, referencing the uploaded file.
3. A default language code (en-US) is used if none is provided. Language validation is supported for standard SQS queues.
4. Records are loaded into memory and processed one at a time. All records are processed individually to ensure accuracy and validation.

AppLink Metadata Retriever

1. In the **Pre-select SQL** area, optionally, enter SQL to execute prior to the select that will be part of the same transaction (temp table creation, for example). Use {WINDOW_START} and {WINDOW_END} to apply the quantum defined at the process level to the generated SQL. {WINDOW_END} can optionally include a date-based column name in the form {WINDOW_END:dateColumnToTest} that, if used, is tested at the end of processing to determine the max date processed, and is used as the next run's {WINDOW_START}. If not used, the next {WINDOW_START} will be the current run's {WINDOW_END}.
2. In the **Select SQL** area, enter the SQL information. Use {WINDOW_START} and {WINDOW_END} to apply the quantum defined at the process level to the generated SQL. {WINDOW_END} can optionally include a date-based column name in the form {WINDOW_END:dateColumnToTest} that, if used, is tested at the end of processing to determine the max date processed, and is used as the next run's {WINDOW_START}. If not used, the next {WINDOW_START} will be the current run's {WINDOW_END}.
3. In the **Post-select SQL** area, optionally, enter SQL to execute after the select that will be part of the same transaction (data cleanup from the pre-select SQL, for example). Use {WINDOW_START} and {WINDOW_END} to apply the quantum defined at the process level to the generated SQL. {WINDOW_END} can optionally include a date-based column name in the form {WINDOW_END:dateColumnToTest} that, if used, is tested at the end of processing to determine the max date processed, and is used as the next run's {WINDOW_START}. If not used, the next {WINDOW_START} will be the current run's {WINDOW_END}. For items 1-3 above, you can click the **Click here to edit in a larger window** link to open a larger text editing area.
4. In the **Reconciliation Identifier**, enter the name of the action if there is a reporting action in the process group.
5. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.

6. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
7. Click **Submit**.

AppLink SAP Business Data Upload

1. In the **Access key**, enter the AppLink web service access key for the specific customer/ tenant.
2. In the **Secret key**, enter the AppLink web service secret key for the specific customer/ tenant.
3. In the **Business Data Manager API URL**, enter the business data manager API URL.
4. In the **Fail if nothing is imported within # minutes**, specify the number of minutes to fail the import if nothing is imported within that time. The default value of zero indicates no failure.
5. In the **Required column(s)** drop-down, optionally enter the required column or columns. If you enter more than one column, they should be comma-separated. Specify ***true*** to import only required columns, or ***false*** to import all columns in the CSV file.
6. In the **Discovery OpenID configuration URL**, enter the URL. Defaults URL is ***https://cxone.niceincontact.com/.well-known/openid-configuration.***
7. In the **Discovery CXone configuration URL**, enter the CXone configuration URL. Default is ***https://cxone.niceincontact.com/.well-known/cxone-configuration?tenantId=%7BtenantId%7D***.
8. In the **Failure Threshold**, specify the number of failures that causes processing to stop. A value of 0 indicates that the process does not fail.
9. In the **Session credentials expiration (minutes)** drop-down, specify the time to wait before re-authenticating the AppLink session. The default is ***10 minutes***. The session credentials expiration time must be between 1 and 10 minutes.

10. In the **Pre-select SQL** area, optionally, enter SQL to execute prior to the select that will be part of the same transaction (temp table creation, for example). Use {WINDOW_START} and {WINDOW_END} to apply the quantum defined at the process level to the generated SQL. {WINDOW_END} can optionally include a date-based column name in the form {WINDOW_END:dateColumnToTest} that, if used, is tested at the end of processing to determine the max date processed, and is used as the next run's {WINDOW_START}. If not used, the next {WINDOW_START} will be the current run's {WINDOW_END}.
11. In the **Select SQL** area, enter the SQL information. Use {WINDOW_START} and {WINDOW_END} to apply the quantum defined at the process level to the generated SQL. {WINDOW_END} can optionally include a date-based column name in the form {WINDOW_END:dateColumnToTest} that, if used, is tested at the end of processing to determine the max date processed, and is used as the next run's {WINDOW_START}. If not used, the next {WINDOW_START} will be the current run's {WINDOW_END}.
12. In the **Post-select SQL** area, optionally, enter SQL to execute after the select that will be part of the same transaction (data cleanup from the pre-select SQL, for example). Use {WINDOW_START} and {WINDOW_END} to apply the quantum defined at the process level to the generated SQL. {WINDOW_END} can optionally include a date-based column name in the form {WINDOW_END:dateColumnToTest} that, if used, is tested at the end of processing to determine the max date processed, and is used as the next run's {WINDOW_START}. If not used, the next {WINDOW_START} will be the current run's {WINDOW_END}. For items 1-3 above, you can click the **Click here to edit in a larger window** link to open a larger text editing area.
13. In the **Nexidia client ID**, enter the Nexidia client ID to retrieve the CXone access token.
14. In the **Nexidia client secret**, enter the Nexidia secret ID to retrieve the CXone access token.
15. In the **Reconciliation Identifier**, enter the name of the action if there is a reporting action in the process group.
16. In the **Default input disposition handling** drop-down list, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
    - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.

17. In the **Input Disposition(s)**, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type 'none' to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
18. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
19. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
20. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
21. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
22. Click ***Submit***.

AppLink SAP Generic Upload

1. In the **Process files from metadata** drop-down, specify if you want to process files from metadata. Select ***false*** to process files from attached input resources or ***true*** to process them from this process groups existing working table. Processing from metadata reduces input/output and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **File filter**, enter the file filter type. For example: CSV.

3. In the **Delimiter**, enter the delimiter character. This is used for non-XML input, and is a single character such as "**,**". You can use ***\t*** for tab.
4. In the **Access key** field, enter the AppLink web service access key for the specific customer/ tenant.
5. In the **Secret key** field, enter the AppLink web service secret key for the specific customer/ tenant.
6. In the **Storage Access Provider API URL** field, enter the API URL for storage access.
7. From the **Fail if nothing is imported within # minutes** drop-down, specify the number of minutes after which this action will fail if nothing is imported. A value of 0 indicates that the action will not fail.
8. In the **Delete files after processing** field, select ***true*** if you want to delete the files after processing.
9. In the **Discovery OpenID configuration URL**, enter the URL. Defaults URL is ***https://cxone.niceincontact.com/.well-known/openid-configuration.***
10. In the **Discovery CXone configuration URL**, enter the CXone configuration URL. Default is ***https://cxone.niceincontact.com/.well-known/cxone-configuration?tenantId=%7BtenantId%7D***.
11. In the **File Type** drop-down, specify the file type to be uploaded to AppLink.
12. In the **Failure Threshold** field, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
13. From the **Session credentials expiration (minutes)** drop-down, specify the time to wait before re-authenticating the CXone session. The default is ***10 minutes***. The session credentials expiration time must be between 1 and 10 minutes.
14. In the **Nexidia client ID**, enter the Nexidia client ID to retrieve the CXone access token.
15. In the **Nexidia client secret**, enter the Nexidia secret ID to retrieve the CXone access token.
16. In the **Reconciliation Identifier** field, enter the name of the action if there is a reporting action in the process group.
17. From the **Default input disposition handling** drop-down list, select one of the following:

- ***AND***: Allow files to flow through this step that have passed all of the immediate inputs.
- ***OR***: Allow files to flow through this step that have passed any of the immediate inputs.

18. In the **Input Disposition(s)** field, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none***\ to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
19. In the **Thread Count** field, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
20. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
21. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
22. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
23. Click **Submit**.

AppLink SAP Upload

1. In the **AppLink Login URL** field, enter the AppLink web service login URL.
2. In the **Access key** field, enter the AppLink web service access key for the specific customer/ tenant.

3. In the **Secret key** field, enter the AppLink web service secret key for the specific customer/ tenant.
4. In the **Storage Access Provider API URL** field, enter the API URL for storage access.
5. In the **Fail if nothing is imported within # minutes** drop-down, specify the number of minutes after which this action will fail if nothing is imported. A value of 0 indicates that the action will not fail.
6. In the **Delete files after processing** field, select ***true*** if you want to delete the files after processing.
7. In the **Discovery OpenID configuration URL**, enter the URL. Defaults URL is ***https://cxone.niceincontact.com/.well-known/openid-configuration.***
8. In the **Discovery CXone configuration URL**, enter the CXone configuration URL. Default is ***https://cxone.niceincontact.com/.well-known/cxone-configuration?tenantId=%7BtenantId%7D***.
9. In the **Failure Threshold** field, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
10. From the **Session credentials expiration (minutes)** drop-down, specify the time to wait before re-authenticating the CXone session. The default is ***10 minutes***. The session credentials expiration time must be between 1 and 10 minutes.
11. From the **Batch execute size** drop-down, specify the batch execute size. When processing large files, this indicates how often to execute batches against the database.
12. In the **Number of Retries on JSON Failure**, specify the number of times system retries if JSON fails to upload.
13. In the **Number of Seconds To Wait Between Each Retry**, specify the amount of time (in seconds) the system waits for next retry if JSON fails to upload.
14. In the **Default HTTP client timeout (in seconds)**, enter the Apache HTTP client default timeout value. The default is 60.
15. In the **Nexidia client ID**, enter the Nexidia client ID to retrieve the CXone access token.
16. In the **Nexidia client secret**, enter the Nexidia secret ID to retrieve the CXone access token.
17. In the **Reconciliation Identifier** field, enter the name of the action if there is a reporting action in the process group.

18. From the **Default input disposition handling** drop-down list, select one of the following:
    - ***AND***: Allow files to flow through this step that have passed all of the immediate inputs.
    - ***OR***: Allow files to flow through this step that have passed any of the immediate inputs.
19. In the **Input Disposition(s)** field, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none***\ to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
20. In the **Thread Count** field, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
21. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
22. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
23. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
24. Click **Submit**.

AppLink SAP User Upload

1. In the **File filter**, enter the file filter type. For example: CSV.
2. In the **Delimiter**, enter the delimiter character. This is used for non-XML input, and is a single character such as "**,**". You can use ***\t*** for tab.
3. In the **Access key**, enter the AppLink web service access key for the specific customer/tenant.
4. In the **Secret key**, enter the AppLink web service secret key for the specific customer/tenant.
5. In the **Storage Access Provider API URL**, enter the API URL for storage access.
6. In the **Fail if nothing is imported within # minutes** drop-down, specify the number of minutes after which this action fails if nothing is imported. A value of 0 indicates that the action does not fail.
7. In the **Delete files after processing**, select ***true*** if you want to delete the files after processing.
8. In the **Discovery OpenID configuration URL**, enter the URL. Defaults URL is ***https://cxone.niceincontact.com/.well-known/openid-configuration.***
9. In the **Discovery CXone configuration URL**, enter the CXone configuration URL. Default is ***https://cxone.niceincontact.com/.well-known/cxone-configuration?tenantId=%7BtenantId%7D***.
10. In the **Failure Threshold**, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
11. In the **Session credentials expiration (minutes)** drop-down, specify the time to wait before re-authenticating the AppLink session. The default is ***10 minutes***. The session credentials expiration time must be between 1 and 10 minutes.
12. In the **Nexidia client ID**, enter the Nexidia client ID to retrieve the CXone access token.
13. In the **Nexidia client secret**, enter the Nexidia secret ID to retrieve the CXone access token.
14. In the **Reconciliation Identifier** field, enter the name of the action if there is a reporting action in the process group.
15. From the **Default input disposition handling** drop-down list, select one of the following:

- ***AND***: Allow files to flow through this step that have passed all of the immediate inputs.
- ***OR***: Allow files to flow through this step that have passed any of the immediate inputs.

16. In the **Input Disposition(s)** field, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none***\ to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
17. In the **Thread Count** field, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
18. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
19. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
20. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
21. Click ***Submit***.

Auto Summary LLM V2

1. In the **Process files from metadata**, specify if you want to process files from metadata. When you select ***true***, the application processes files from the process groups existing working table. When you specify ***false***, the application processes files from attached input resources. Processing from metadata reduces I/O and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **File filter**, enter the file filter type. For example: *.xml, *.csv, etc. Java-style regular expressions can also be used if prefixed with regex:

   When processing files from resources, this acts as a filter, but when processing from metadata, the metadata's file key is inserted into any wildcards, expanding the processing of the one file to all listed variants. For example, *, *.xml would process both a file matching the working table file key 1234567.wav and its 1234567.wav.xml counterpart.
3. In the **Custom Prompt** field, provide a prompt. You will receive an LLM response based on the custom prompt provided. By default, the custom prompt is: Provide a concise summary of the contact center conversation for the subsequent agent, highlighting the most relevant details for efficient customer assistance.
4. In the **Max Words Response** field, specify the maximum number of words that the LLM will return in its response. The default value is 80.
5. In the **Host Name** field, enter the host name for the LLM API call.
6. In the **Port** field, enter the port number for the LLM API call. The default port is 47400.
7. In **Search directories recursively** field, specify if directories should be searched recursively.
8. In the **Treat missing files as errors** field, by default, missing files are considered acceptable, as systems largely import metadata and then wait for files to arrive, in which case we don't need to capture and track errors when they are not aligned. Change to ***true*** if missing files should be considered errors to be logged and tracked against the action's failure threshold.
9. In **Overwrite existing output files** field, select ***true*** to overwrite files that already exist in the output location and ***false*** to leave existing files untouched. The default value is ***true***.
10. In **Delete inputs process** field, set to ***true*** to delete the input files after they are processed. The default value is ***false***.

11. In the **Failure Threshold**, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
12. In the **Temperature** field, specify the level of temperature to set the level of creativity for the AI's responses in the LLM service call. Set to 1 for AI to be quite creative and varied in its responses, giving you summaries that are imaginative but still make sense. Inserting higher number here will result in more creative responses.
13. In the **Max Token Response** field, specify the maximum number of tokens to be used for the LLM service API call. The default value is 1500.
14. In the **Number of Seconds To Wait Between Each Retry**, specify the amount of time (in seconds) the system waits for next retry if LLM service fails.
15. In the **Retry LLM Service Call** field, specify the number of times to retry submitting the prompt to LLM Service on failure.
16. In the **Transfer Protocol** field, configure the transfer protocol for communicating agent/user information to NIA. The default protocol is HTTP.
17. In the **Reconciliation Identifier** field, enter the name of the action if there is a reporting action in the process group.
18. From the **Default input disposition handling** drop-down list, select one of the following:

    AND: Allow files to flow through this step that have passed all of the immediate inputs.

    OR: Allow files to flow through this step that have passed any of the immediate inputs.
19. In the **Input Disposition(s)** field, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none***\ to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
20. In the **Thread Count** field, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.

21. In the **Maximum files to process** field, specify the maximum number of files this action should process in a given run. This number may not be exact and can fluctuate based on the number of threads being processed concurrently when the maximum is reached. To guarantee a specific count, lock the thread count at 1, but be aware of the performance implications of not multi-threading. Setting the value to 0 uses the DEF internal limit (50,000), which is configurable in **ndef.properties** using the **hard.max.files.to.process** property.
22. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
23. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
24. Click ***Submit***.

Calabrio Media Retriever Action

1. In the **User ID**, enter the ID for the account that you use to access the Calabrio API.
2. In the **Password**, enter the password of the Calabrio account that you want to use.
3. In the **Base URL**, enter the base-URL for the Calabrio API.
4. In the **Batch Size** drop-down, specify the number of media to be transcoded and retrieved at a time. The minimum is ***25*** and maximum is ***100,000***. The default value is ***100***.
5. From the **Failure Threshold** drop-down, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
6. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.
7. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.

8. From the **Terminal action** drop-down, select if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
9. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
10. Click **Submit**.

Calabrio Metadata Retriever Action

1. In the **User ID**, enter the ID for the account accessing the Calabrio API.
2. In the **Password**, enter the password of the Calabrio account being used.
3. In the **Base URL**, enter the base-URL for the Calabrio API.
4. In the **Page Size**, specify the number of rows that you want to retrieve from the API when querying for contact metadata. The number of rows ranges from ***25*** to ***100,000***.
5. In the **Skip unrecorded calls**?, specify whether or not to skip the calls. The default value is ***true***.
6. From the **Audio Type** drop-down, specify the desired format of the audio media when requesting media transcoding prior to downloading. Supported types are ***WAV*** and ***OPUS***. The default format is ***WAV***.
7. From the **Video Type** drop-down, specify desired format for related video media when requesting media transcoding prior to downloading. Supported types are ***NONE***, ***WEBM***, ***WEBM_VP8***, and ***WEBM_VP9***. The default selection is ***NONE***.
8. From the **Failure Threshold** drop-down, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
9. From the **Expand metadata** drop-down, set to ***true*** to expand metadata to include queue name key, etc
10. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.

11. From the **Terminal action** drop-down, select if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
12. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
13. Click **Submit**.

Calabrio User Metadata Retriever Action

1. In the **User ID**, enter the ID for the account accessing the Calabrio API.
2. In the **Password**, enter the password of the Calabrio account being used.
3. In the **Base URL**, enter the base-URL for the Calabrio API.
4. In the **Basic information only** dropdown, set to ***true*** to retrieve the agent information including ID, first name, last name, and employee number.
5. From the **Failure Threshold** drop-down, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
6. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.
7. From the **Terminal action** drop-down, select if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
8. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
9. Click **Submit**.

Call Summarization Request

1. In the **Process files from metadata** drop-down, specify if you want to process files from metadata. Select ***false*** to process files from attached input resources or ***true*** to process them from this process groups existing working table. Processing from metadata reduces input/output and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **File filter** field, enter the file filter type. For example: CSV.
3. In the **File path column name**, enter the name of the metadata column in the process group working table to determine the files to move. Default is ***FileName***.
4. In the **RabbitMQ Host Name**, enter the RabbitMQ host name for login.
5. In the **RabbitMQ Port**, enter the RabbitMQ port number for login. The default value is ***5672***.
6. In the **RabbitMQ Queue Name**, enter the RabbitMQ Queue Name to submit.
7. In the **RabbitMQ User Name**, enter the RabbitMQ user name for login.
8. In the **RabbitMQ Password**, enter the RabbitMQ password.
9. In the **RabbitMQ Virtual Host**, enter the RabbitMQ virtual host for login.
10. In the **Default Text Analysis Model Uri**, enter the path to the location of model that Summarization Grid uses in summarization.
11. In the **Default Scoring Model Uri**, enter the path to the location of scoring model that Summarization Grid uses in summarization.
12. In the **Text Analysis Model Metadata Mapping**, enter the mapping of selected metadata field to the model that Summarization Grid should use in summarization.
    You can also add a suffix for the IPM model. The suffix will be appended to the RabbitMQ Response Queue Name. This will help you understand which IPM model was used while processing in RabbitMQ.
13. In the **Scoring Model Metadata Mapping**, enter the mapping of selected metadata field to the scoring model that Summarization Grid should use in summarization.
14. In the **RabbitMQ Response Queue Name**, enter the RabbitMQ Response Queue Name to submit.

15. In the **Skip Auto Summary**, you can define mappings of metadata fields to values for better call summarization request filtering. When all configured metadata fields match, the system will skip files for auto-summarization. For example, if English is provided as a value for the Language filter for File1, File1 will be skipped for auto summarization. Files that do not match the filter criteria will be processed to the NIA database.
    When you click , the Skip Auto Summary window opens. You can use this window to edit the filter values. Click **OK**.
    The default values are
    - **LanguageId**: English
    - **CallDirection**: Inbound
    - **MediaStatistics:duration_in_milliseconds**: Greater than or equal to 60
    - **MediaStatistics: NonSpeech percent**: Less than 80
    - **NumberOfTurns**: 0.
16. In the **Delete inputs after move** drop-down, set to ***true*** to delete input files after move. Set to ***false*** to copy, leaving input files intact. Default value is ***true***.
17. From the **Send Satellite Logs** drop-down, select ***true*** to diagnose the Voice Grid issues for not being able to process the files. Set to ***false*** to track and log as errors. Default value is ***false***.
18. In the **Failure Threshold**, enter the number of non-fatal processing errors (e.g. inability to find a requested file) to tolerate before failing and stopping the rest of the process from continuing. The default value is 0, which means never fail the process, but continue to track errors in logs, and as warnings in the UI.
19. In **Transcript Type**, set to VoiceGrid ASR to process ASR json file generated by VoiceGrid and set to NxPr to process NxPr file generated by NIA to create CSV file.
20. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.
21. In the **Default input disposition handling** drop-down, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
    - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.

22. In the **Input Disposition(s)**, enter the input disposition. The Data Exchange Framework automatically chains action component dispositions together as javascript:void()part of its processing, so this field is usually left blank. If you are using split processes, enter a value in this field.
23. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
24. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
25. From the **Terminal action** drop-down, select if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
26. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
27. Click **Submit**.

Call Summarization Response V2

1. In the **RabbitMQ Host Name**, enter the RabbitMQ host name for login.
2. In the **RabbitMQ Port**, enter the RabbitMQ port number for login. The default value is ***5672***.
3. In the **RabbitMQ User Name**, enter the RabbitMQ user name for login.
4. In the **RabbitMQ Password**, enter the RabbitMQ password.
5. In the **RabbitMQ Virtual Host**, enter the RabbitMQ virtual host for login.
6. In the **Select top 'n' events**, enter the number of top 'n' combined total of non-attensity and attensity events to accept (agent tasks and scored summaries) from metadata result. Value cannot be negative. The default value is 8. Set the value to 0 to get all events.

7. In the **Select top 'n' attensity events**, enter the number of top 'n' attensity events (scored summaries) to accept from metadata result. The default value is 4. Value cannot be negative. Set the value to 0 to get all events.
8. In the **Failure Threshold**, enter the number of non-fatal processing errors (e.g. inability to find a requested file) to tolerate before failing and stopping the rest of the process from continuing. The default value is 0, which means never fail the process, but continue to track errors in logs, and as warnings in the UI.
9. In the **Enlightened Complaint Enabled**, set to ***true*** to enable complaints data extraction from the Workflow model. The complaint data is then sent to NIA user defined fields (UDFs) to make it available across NIA applications, including Reporting, Search, and Explore. By default, it is set to ***false***.
   After setting this parameter to true, the following parameters are not available for configuration:
   - Select top 'n' events
   - Select top 'n' attensity events
   - Call Summary Collector
   - Include Desktop Tags
   - Include Auto Summary Tags
   - Generate Formatted Transcript File
   - Include Only Binary Outcome
   - Include Only Binary Enlighten
   - Add Transcript For LLM.
10. In the **Max Number of Topics**, enter the maximum number of top-scored topics to include in the working table. Topics are automatically sorted by score (highest first) and the top N topics are stored as indexed columns (Topic1_*, Topic2_*, etc). Set to 0 or leave empty to include all topics. For example, set to 4 to store only the top 4 highest-scored topics.
11. In the **TLS Enable**, set to ***true*** to enable TLS encryption for RabbitMQ connection.
12. In the **Delete Transcript CSV**, specify if you want to delete transcripts after they have successfully processed. The default value is ***true***.

13. In **Call Summary Collector** drop-down, select ***true*** to generate JSON based on metadata, enlighten scores, transcript, and call summarization data.
14. In the **Enlighten Model Names**, specify enlighten score names separated by comma that you want to included in the Call Summarization JSON. Leave empty if you want to include all available enlighten scores.
15. In the **Include Desktop Tags**, select ***true*** to create Desktop Tags.
16. In the **Include Auto Summary Tags**, select ***true*** to include Auto Summary Tags in the working table to generate First-Class AutoSummary Field. By default, the value is ***false***, which means Auto Summary Tags is not included. Start and end offsets are added to the intents First-Class Fields in the JSON output. This ensures the intents and events are displayed on the player's waveform at specific times.
17. In the **Treat Missing Data as Errors**, the default value is ***true***. This means you can analyze and capture more information about why the data is missing. Change to ***false*** if missing data should be considered errors. Errors will be logged and tracked against the action's **failure threshold.**
18. In the **Remove messages from RMQ**, select ***true*** to remove messages from RMQ for the missing data that are not in the working table. This is the default setting. Select ***false*** to keep all messages in RMQ.
19. In the **Generate Formatted Transcript File**, select ***true*** to generate a formatted transcript JSON file. Select ***false*** to not generate a formatted transcript JSON file.
20. In the **Include Only Binary Outcome**, select ***true*** to include only binary Contact Outcome to JSON output. The default value is ***false***.
21. In the **Include Only Binary Enlighten**, select ***true*** to include only binary Enlighten to JSON output. The default value is ***false***.
22. In the **Add Transcript For LLM**, select ***true*** to add a transcript to JSON File for LLM. The default value is ***false***.
23. In the **Retry Submit to VG**, select ***true*** to resend the request to VG for reprocessing in case of an error in the response message indicating missing files. The default value is ***false***.
24. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.
25. In the **Default input disposition handling** drop-down, select one of the following:

- ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
- ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.

26. In the **Input Disposition(s)**, specify how input dispositions should be handled. The default is empty, to automatically chain. Enter ***none*** to disconnect the action from dependencies to prior actions, or enter a comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Use a hash sign instead of a comma to create an OR condition.
27. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
28. In the **Terminal action** drop-down, select if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
29. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
30. Click **Submit**.

AutoSummary JSON File

DEF uses various business rules to generate a summary of the interaction for AutoSummary from the data generated by the machine learning models, such as Intents, Outcomes, Events, and so on. This summary is the heart of the AutoSummary data JSON, which is useful to their consumers for further analysis. In addition, DEF generates a formatted transcript.

If an interaction fails due to missing intent data, it is still sent to the NIA platform. This is because it contains other valuable data for Enlighten, Analytics, and other metrics. This ensures comprehensive analytics can be performed, even without the intent data.

To get more accurate information about sentiment of the interaction, DEF gets the SentimentTransition value from Voice Grid when calculating sentiment scores. It then calculates the following two scores and sends the results to NIA:

- OneHalfSentimentScore
- ThreeFourthsSentimentScore

The default Enlighten model outcome thresholds that appear in the JSON are defined as follows:

- **Sentiment**
  - Very Negative # (-999) – (-0.31)
  - Negative # (-0.31) - 2.489
  - Neutral # 2.489 - 5.538
  - Positive # 5.538 - 8.988
  - Very Positive # 8.988 - 999.0
  - Otherwise invalid
- **Resolution**
  - The interaction is "Resolved" if the score is between # 2 – 999.0
  - The interaction is not Resolved if the score is between (-999.0) - 2
- **Escalation**
  - The interaction is Escalated if the score is between 2.5 - 999.0
  - The interaction is not Escalated for any score outside of this range
- **Confirmed Sale**
  - The interaction has Confirmed Sale if the score is between 6 - 999.0
  - The interaction has not Confirmed Sale for any score outside of this range

CSV Reformatter Action

1. In the **Process files from metadata** drop-down, specify if you want to process files from metadata. Select ***false*** to process files from attached input resources or ***true*** to process them from this process groups existing working table. Processing from metadata reduces input/output and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **File filter**, enter the file filter type. For example: CSV.
3. In the **Force file encoding**, leave blank to auto-detect. If auto-detect is not working, use a valid Java InputStream encoding.
4. In the **Delete inputs after reformat** drop-down, specify if you want to delete the inputs after reformatting is completed.
5. In the **Profile type** drop-down, select a profile to preset specific configuration options required to process the type of input file selected. The custom type sets no configuration options, and you must select all configuration options for it. There are six options: ***Custom (default)***, ***Headerless***, ***Replace Header***, ***Use Existing Header***, ***Predictive Model*** and ***WFM***.
6. In the **comment delimiter**, enter a character to denote a comment line in the transcript file. The default value is #.
7. In the **Lines to skip**, enter the number of lines to skip from the beginning of the file.
8. In the **Use existing header row** drop-down, set to ***true*** to use the existing header row in an incoming file.
9. In the **Custom header row prefix**, enter a string used to identify a header row. For example, set to ***#fields:*** for WFM files.
10. In the **Custom header row**, enter a header row to use if an incoming file has no header row. Column headers must be comma delimited.
11. In the **Default separator character**, enter the character used to separate delimited column values in the input CSV file (for example ,).
12. In the **Default quote character**, enter the character used to quote column values in the output CSV file (for example ").
13. In the **Default escape character**, enter the character used to escape column values in the output CSV file (for example \).

14. In the **Failure Threshold**, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
15. In the **Reconciliation Identifier**, enter the name of the action if there is a reporting action in the process group.
16. In the **Default input disposition handling** drop-down, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
    - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
17. In the **Input Disposition(s)**, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type 'none' to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
18. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
19. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
20. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
21. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
22. Click **Submit**.

CSV Text Extractor Action

1. In the **Process files from metadata** drop-down, specify if you want to process files from metadata. Select ***false*** to process files from attached input resources or ***true*** to process them from this process groups existing working table. Processing from metadata reduces input/output and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **File filter** field, enter the file filter type. For example: CSV.
3. In the **Text Column Name** field, enter the name of the column that holds the general text.
4. In the **ID column(s) for file name** field, enter the name(s) of the column(s) to be used in generating the .txt file names, comma-delimited. File names use the column value(s) in the order entered here, with underscores between each if using more than one in the ID column(s) for the file name field.
5. In the **Delimiter** field, enter the delimiter character. This is used for non-XML input, and is a single character such as "**,**". You can also use ***\t*** for tab.
6. In the **Delete files after processing** drop-down, specify if you want to delete the CSV files after processing.
7. In the **Additional metadata to capture** field, If you want to capture additional column metadata alongside each chat session's JSON record in the working table, enter a comma-delimited list of CSV column headers.
8. In the **Default quote character** field, enter the character used to quote column values in the output CSV file, for example ".
9. In the **Default escape character** field, enter the character used to escape column values in the output CSV file, for example \.
10. In the **Failure Threshold** field, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
11. In the **Force file encoding** field, enter the character encoding of the input file If you want to force file encoding. You will leave this field blank in most cases to auto-detect. If autodetect
is not working, use a valid Java InputStream encoding. See encodingdochtml for more information on supported encodings.

12. In the **Reconciliation Identifier**, enter the name of the action if there is a reporting action in the process group.
13. In the **Default input disposition handling** drop-down list, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
    - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
14. In the **Input Disposition(s)** field, enter the input disposition. The Data Exchange Framework automatically chains action component dispositions together as javascript:void()part of its processing, so this field is usually left blank. If you are using split processes, enter a value in this field.
15. In the **Thread Count** field, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
16. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
17. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
18. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
19. Click **Submit**.

CSV-to-Database Action

1. In the **File filter**, enter the file filter type. For example: CSV.
2. In the **Delimiter**, enter the delimiter character. This is used for non-XML input, and is a single character such as "**,**". You can use ***\t*** for tab.
3. In the **Target Table Name**, enter the target table name.
4. In the **Key column(s)**, enter the key column or columns. If you enter more than one column, they should be comma separated.
5. In the **Required column(s)** drop-down, optionally enter the required column or columns. If you enter more than one column, they should be comma-separated. Specify ***true*** to import only required columns, or ***false*** to import all columns in the CSV file.
6. In the **Import only required columns** drop-down, specify if you want to only import the required columns.
7. In the **Suppress column(s)**, optionally enter the column or columns to suppress. If you enter more than one column, they should be comma separated.
8. In the **How to handle existing data** drop-down, specify how the application should handle existing data the target table. The valid options are:
   - ***UPDATE*** - Update the existing record.
   - ***DELETE*** - Delete the existing record before inserting a fresh one.
   - ***ERROR*** - There should be no existing record with a matching primary key on the table, and the action should fail and send notifications if a primary key violation is encountered. Note that the action will not fail if there is no primary key constraint on the target table. Also note that if you opt to use UPDATE or DELETE, you might consider indexing the columns (if any) in the "Key column(s)" property.
9. In the **Source-to-Target Column Mapping**, optionally enter the source-to-target column mapping. For example:
   mediaFile=ExternalMediaId;startTime=CallDateTime[;...]):
10. In the **fail if nothing is imported within # minutes** drop-down, specify the number of minutes after which this action will fail if nothing is imported. A value of 0 indicates that the action will not fail.
11. In the **Delete Flag** field, optionally enter the delete flag. For example:
    deleted=y[;other=...]):

12. In the **Pre-import delete 'where' sub-clause**, optionally enter the pre-import delete where sub-clause. This specifies which record or records that should be deleted. For example:
 region='NW'.
13. In the **Delete input file(s) after import** drop-down, specify if you want to delete input files after import.
14. In the **Force file encoding**, if you want to force file encoding, enter the character encoding of the input file. Leave this field blank in most cases to auto-detect. If auto-detect isn't working, use a valid Java InputStream encoding.
15. In the **Default quote character**, enter the character used to quote column values in the output CSV file, for example **"**.
16. In the **Default escape character**, enter the character used to escape column values in the output CSV file, for example ****.
17. In the **Failure Threshold**, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
18. In the **Batch execute size** drop-down, specify a value that will execute the batches at a configurable interval within each CSV file in the Batch commit size drop-down list. This option is useful when processing large files. The default value is 1000.
19. In the **Commit after each batch** drop-down, specify if you want to commit to the database after each batch. The default value is ***false*** which will commit to the database only after the .csv file has been fully processes. Select ***true*** to commit to the database after each batch is read from any given .csv file. Selecting ***true*** will take transaction load off of the database if it becomes a problem, but will break intended transactional integrity for processing of each incoming .csv file. ***False*** is the preferred option unless the database cannot handle the transactional load.
20. In the **Reconciliation Identifier**, enter the name of the action if there is a reporting action in the process group.
21. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.

22. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
23. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
24. Click **Submit**.

CXone Agents Retriever V2 Action

1. In the **Discovery OpenID configuration URL**, enter the openID configuration URL to use when authenticating. The default value is ***https://cxone.niceincontact.com/.well-known/openid-configuration***.
2. In the **Discovery CXone configuration URL**, enter the CXone configuration URL. You must enter the token ID when you enter this URL. Defaults to ***https://cxone.niceincontact.com/.well-known/cxone-configuration?tenantId={tenantId}***.
3. In the **Access key**, enter the CXone web service access key.
4. In the **Secret key**, enter the CXone web service secret key.
5. In the **Admin URL**, enter the URL to use when calling the Admin API. The default value is ***https://api-{area}.{domain}/incontactapi/services/v24.0***.
6. From the **CXone web service page size** drop-down, select the number of agents to retrieve per API call. The default is ***100***.
7. In the **Failure threshold**, specify the number of failures that cause processing to stop. ***0*** (default) indicates that the process never fails.
8. From the **Session credentials expiration (minutes)** drop-down, specify the time between ***1*** and ***10*** minutes, to wait before re-authenticating the CXone session. The default is ***10*** minutes.

9. From the **Log CXone service responses** drop-down, specify if you want to log the CXone service responses. The default value is ***False***, so no response logging occurs. If you set the value of this parameter as ***True***, the log file is created in *<user temp folder>/CXone-Response.log*.
10. In the **Nexidia client ID**, enter the Nexidia client ID to retrieve the CXone access token.
11. In the **Nexidia client secret**, enter the Nexidia secret ID to retrieve the CXone access token.
12. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.
13. From the **Default input disposition handling** drop-down list, select one of the following:
    - ***AND***: Allow files to flow through this step that have passed all of the immediate inputs (default).
    - ***OR***: Allow files to flow through this step that have passed any of the immediate inputs.
14. In the **Input Disposition(s)**, specify how input dispositions should be handled. The default is empty, to automatically chain. Enter ***none*** to disconnect the action from dependencies to prior actions, or enter a comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Use a hash sign instead of a comma to create an OR condition.
15. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the **Threads per action** setting at the server level.
16. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. ***0*** (default) indicates that there is no limit.
17. From the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.

18. In the **Disable action**, specify ***true*** if you want to prevent this action from processing any files during its run. Leave the default value ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
19. Click **Submit**.

CXone Billing Data Submit V1 Action

1. In the **Lookup business unit**, set to ***true*** to use the CXone business unit API to look up the business unit. Set to ***false*** to manually specify the business unit.
2. In the **Discovery OpenID configuration URL**, enter the openID configuration URL to use when authenticating. The default value is ***https://cxone.niceincontact.com/.well-known/openid-configuration***.
3. In the **Discovery CXone configuration URL**, enter the CXone configuration URL. You must enter the token ID when you enter this URL. The default value is ***https://cxone.niceincontact.com/.well-known/cxone-configuration?tenantId={tenantId}***.
4. In the **Access key**, enter the CXone web service access key.
5. In the **Secret key**, enter the CXone web service secret key.
6. In the **Business unit URL**, enter the URL to use when calling the Business Unit API. For example, https://api-na1.niceincontact.com/incontactapi/services/v22.0/businessunit.
7. In the **Billing data SQL**, enter the SQL used to retrieve the CXone billing data.
8. In the **Kinesis access key**, enter the Kinesis service access key.
9. In the **Kinesis secret key**, enter the Kinesis service secret key.
10. In the **Kinesis region**, enter the Kinesis service region. The default region is us-west-2.
11. In the **Kinesis stream name**, enter the Kinesis service stream name. The default value is icdev-metering-data-stream.
12. In the **Kinesis payload**, enter the CXone billing Kinesis payload.
    For example,
    {
    "appName": "Nexidia Enlighten XO",
    "eventType": "Meter",

"version": "1.0.0",
"eventId": "{eventId}",
"eventTime": "{eventDate}",
"productName": "Nexidia Enlighten XO",
"featureName": "Nx Bot Automation",
"tenantId": "{businessUnitID}",
"meteredParam": "NEXIDIAENLTNINTERCTN",
"prodTypeId": 562,
"quantity": {interactionCount},
"quantityUnits": "count"
}

The required tokens are used as follows:

- {eventId} - Unique Kinesis identifier in the form of a randomly generated GUID
- {eventDate} - Current date/time of the process run
- {businessUnitId} - Either determined using the Business Unit API or configured by the SA
- {interactionCount} - The previous days interaction count from NIA

The action will fill in the tokens with the actual values.

13. The **Recommended schedule** is a display-only field that shows the recommended time for the process to run to ensure all interactions from the previous day are counted based on the local date, time, and time zone.
14. In the **Failure threshold**, specify the number of failures that cause processing to stop. ***0*** (default) indicates that the process never fails.
15. In the **Nexidia client ID**, enter the Nexidia client ID to retrieve the CXone access token.
16. In the **Nexidia client secret**, enter the Nexidia secret ID to retrieve the CXone access token.
17. From the **Session credentials expiration (minutes)**, specify the time to wait before re-authenticating the CXone session. It must be between ***1*** and ***10*** minutes, the default value is ***10*** minutes.

18. From the **Log CXone service responses**, specify if you want to log the CXone service responses to a log file. The default value is ***False***, so no response logging occurs. If you set the value of this parameter as ***True***, the log file is created in *<user temp folder>/CXone-Billing.log*.
19. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.
20. From the **Default input disposition handling** drop-down list, select one of the following:
    - ***AND***: Allow files to flow through this step that have passed all of the immediate inputs (default).
    - ***OR***: Allow files to flow through this step that have passed any of the immediate inputs.
21. In the **Input Disposition(s)**, specify how input dispositions should be handled. The default is empty, to automatically chain. Enter ***none*** to disconnect the action from dependencies to prior actions, or enter a comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Use a hash sign instead of a comma to create an OR condition.
22. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. ***0*** (default) indicates that there is no limit.
23. From the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
24. In the **Disable action**, specify ***true*** if you want to prevent this action from processing any files during its run. Leave the default value ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
25. Click **Submit**.

CXone Completed Contacts Metadata Retriever V2 Action

1. In the **Discovery OpenID configuration URL** field, enter the openID configuration URL to use when authenticating. The default value is ***https://cxone.niceincontact.com/.well-known/openid-configuration***.
2. In the **Discovery CXone configuration URL** field, enter the CXone configuration URL. You must enter the token ID when you enter this URL. The default value is ***https://cxone.niceincontact.com/.well-known/cxone-configuration?tenantId={tenantId}***.
3. In the **Access key** field, enter the CXone web service access key.
4. In the **Secret key** field, enter the CXone web service secret key.
5. In the **Completed Contacts URL** field, enter the URL used to retrieve completed contacts. For example, ***https://api-{area}.{domain}/incontactapi/services/v24.0/contacts/completed***.
6. In the **Max # of rows to return** field, enter the maximum number of rows of contacts to retrieve at a time. The default value is 100 rows.
7. In the **CXone web service wait time (in milliseconds)** field, enter the amount of time to wait between calls to the metadata search API endpoint. The default value is 50,000 milliseconds.
8. In the **CXone web service number of retries** field, enter the number of retries to the metadata search API endpoint before failing. The default value is 30 retries.
9. In the **Contact ID metadata column name** field, enter the name of the metadata column containing the Contact ID. The default value is contactId.
10. In the **Filter by media type ID** field, enter the media type ID to filter completed contacts. The default value is 9 - Digital.
11. In the **Order by column name** field, enter the name of the metadata column you want to sort the data by. By default, it sorts by the lastUpdateTime column.
12. In the **Failure Threshold** field, specify the number of failures that cause processing to stop. The default value is 0, which indicates that the process never fails.

13. From the **Session credentials expiration (minutes)** drop-down, specify the time between 1 and 10 minutes, to wait before re-authenticating the CXone session. The default value is 10 minutes.
14. In the **Nexidia client ID** field, enter the Nexidia client ID to retrieve the CXone access token.
15. In the **Nexidia client secret** field, enter the Nexidia secret ID to retrieve the CXone access token.
16. In the **Reconciliation Identifier** field, enter a name that will be used in reports for this action.
17. From the **Terminal action** drop-down, specify whether this is a terminal action. Upon process completion, the rows affected by terminal actions are moved to a working table rollover table. By default, actions are considered terminal if they have a corresponding disposition column in the working table, and no downstream actions reference them. This option is available for instances where the default behavior is not desired.
18. In the **Disable action** field, specify ***true*** if you want to prevent this action from processing any files during its run. If you leave it as the default value ***false***, the action will process files as usual. This option is useful for troubleshooting.
19. Click **Submit**.

CXone Custom Metadata Retriever V2 Action

1. In the **Discovery OpenID configuration URL**, enter the openID configuration URL to use when authenticating. The default value is ***https://cxone.niceincontact.com/.well-known/openid-configuration***.
2. In the **Discovery CXone configuration URL**, enter the CXone configuration URL. You must enter the token ID when you enter this URL. Defaults to ***https://cxone.niceincontact.com/.well-known/cxone-configuration?tenantId={tenantId}***.
3. In the **Access key**, enter the CXone web service access key.
4. In the **Secret key**, enter the CXone web service secret key.
5. In the **Reporting URL**, enter the URL to use when calling the Reporting API. The default value is https://api-na1.niceincontact.com/incontactapi/services/v21.0.
6. In the **Failure threshold**, specify the number of failures that cause processing to stop. ***0*** (default) indicates that the process never fails.
7. From the **Session credentials expiration (minutes)** drop-down list, specify the time between ***1*** and ***10*** minutes, to wait before re-authenticating the CXone session. The default is ***10*** minutes.
8. From the **Log CXone service responses** drop-down, specify if you want to log the CXone service responses. The default value is ***False***, so no response logging occurs. If you set the value of this parameter as ***True***, the log file is created in *<user temp folder>/CXone-Response.log*.
9. From the **Handle HTTP Error** drop-down, select how HTTP response code errors should be handled. The available options are ***log warning*** (default), ***log error***, and ***throw exception***.
10. In the **Rename metadata fields**, enter a comma delimited list of metadata field names to rename. The default values are ***startdate,enddate,filename***.
11. In the **Rename metadata fields suffix**, enter the string to be added as a suffix to renamed metadata fields. The default is ***_cm***.
12. In the **Nexidia client ID**, enter the Nexidia client ID to retrieve the CXone access token.
13. In the **Nexidia client secret**, enter the Nexidia secret ID to retrieve the CXone access token.

14. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.
15. From the **Default input disposition handling** drop-down list, select one of the following:
    - ***AND***: Allow files to flow through this step that have passed all of the immediate inputs (default).
    - ***OR***: Allow files to flow through this step that have passed any of the immediate inputs.
16. In the **Input Disposition(s)**, specify how input dispositions should be handled. The default is empty, to automatically chain. Enter ***none*** to disconnect the action from dependencies to prior actions, or enter a comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Use a hash sign instead of a comma to create an OR condition.
17. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the Threads per action setting at the server level.
18. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. ***0*** (default) indicates that there is no limit.
19. From the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
20. In the **Disable action**, specify ***true*** if you want to prevent this action from processing any files during its run. Leave the default value ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
21. Click **Submit**.

CXone Generic Metadata Retriever V2 Action

1. From the **Process files from metadata** drop-down, select ***true*** if the CXone REST API to be called requires a key value such as a Contact ID, and the working table contains the key value to be passed to the CXone REST API. The default value is ***false***
2. From the **Uses quantum** drop-down, select ***true*** if the CXone REST API to be called requires a start and end date. The default value is ***false***.
3. From the **Uses pagination** drop-down, select ***true*** if the CXone REST API to be called requires pagination to return the entire result set. The default value is ***false***.
4. In the **Discovery OpenID configuration URL**, enter the openID configuration URL to use when authenticating. The default value is ***https://cxone.niceincontact.com/.well-known/openid-configuration***.
5. In the **Discovery CXone configuration URL**, enter the CXone configuration URL. You must enter the token ID when you enter this URL. Defaults to ***https://cxone.niceincontact.com/.well-known/cxone-configuration?tenantId={tenantId}***.
6. In the **Access key**, enter the CXone web service access key.
7. In the **Secret key**, enter the CXone web service secret key.
8. In the **Service URL**, enter the CXone service URL to be executed by the action.
9. In the **Service headers**, enter the HTTP headers required by the CXone REST API.
10. In the **Service params**, enter the HTTP query parameters required by the CXone REST API.
11. In the **Response code handlers**, define how to handle the CXone REST API response status code. Currently this configuration option is limited to the following response handlers:
    - “success” response handler
    - “notfound” response handler

    The format of a response handler is:
    <HTTP Response Code>=<response handler>
    The default values are 200, 204 and 400.

12. From the **Response parse type** drop-down, select the type of response returned by the CXone REST API:
    - ***List Type***
    - ***Embedded List Type*** (default)
    - ***Map Type***
13. In the **Embedded list node name**, enter the name of the JSON element that contains the embedded list.
    For example;
    { "data": [{...},
    {...},
    {...}]
    }
    The embedded list node name for this example would be "data".
14. In the **Metadata ID field name**, enter the name of the metadata column in the working table that contains the unique ID for a specific metadata row.
    For example: agentID, userId, teamId.
15. In the **Failure Threshold** drop-down , specify the number of failures that cause processing to stop. A value of 0 indicates that the process never fails.
16. From the **Session credentials expiration (minutes)** drop-down, specify the time between ***1*** and ***10*** minutes, to wait before re-authenticating the CXone session. The default is ***10*** minutes.
17. From the **Log CXone service responses** drop-down, specify if you want to log the CXone service responses. The default value is ***false***, so no response logging occurs. If you set the value of this parameter as ***true***, the log file is created in *<user temp folder>/CXone-Response.log*.
18. In the **Rollover if metadata not found** drop-down, select ***true*** to rollover to the working table if no metadata found. Default is ***true***.
19. In the **Nexidia client ID**, enter the Nexidia client ID to retrieve the CXone access token.
20. In the **Nexidia client secret**, enter the Nexidia secret ID to retrieve the CXone access token.
21. In the **Rename metadata fields**, enter a comma delimited list of metadata field names to rename. The default values are startdate,enddate,filename.

22. In the **Rename metadata fields suffix**, enter the string to be added as a suffix to renamed metadata fields. The default is _gm.
23. In the **Reconciliation Identifier**, enter the name of the action if there is a reporting action in the process group.
24. From the **Default input disposition handling** drop-down, select one of the following:
    - ***AND***: Allow files to flow through this step that have passed all of the immediate inputs (default).
    - ***OR***: Allow files to flow through this step that have passed any of the immediate inputs.
25. In the **Input Disposition(s)**, specify how input dispositions should be handled. The default is empty, to automatically chain. Enter ***none*** to disconnect the action from dependencies to prior actions, or enter a comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Use a hash sign instead of a comma to create an OR condition.
26. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the **Threads per action** setting at the server level.
27. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. ***0*** (default) indicates that there is no limit.
28. From the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
29. In the **Disable action**, specify ***true*** if you want to prevent this action from processing any files during its run. Leave the default value ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
30. Click **Submit**.

CXone Media Retriever V2 Action

1. In the **Discovery OpenID configuration URL**, enter the openID configuration URL to use when authenticating. The default value is ***https://cxone.niceincontact.com/.well-known/openid-configuration***.
2. In the **Discovery CXone configuration URL**, enter the CXone configuration URL. You must enter the token ID when you enter this URL. Defaults to ***https://cxone.niceincontact.com/.well-known/cxone-configuration?tenantId={tenantId}***.
3. In the **Access key**, enter the CXone web service access key.
4. In the **Secret key**, enter the CXone web service secret key.
5. In the **Media playback URL**, enter the data extraction URL to use when calling the data extraction API. The Default value is ***https://{area}.{uhDomain}/media-playback/v1***.
6. In the **Default separator character**, enter the character used to separate delimited column values in the output CSV file. The default is ***,*** (comma).
7. In the **Default quote character**, enter the character used to quote column values in the output CSV file. The default is ***"*** (double quote).
8. In the **Default escape character**, enter the character used to escape column values in the output CSV file. The recommended values for the default escape character is a single backslash: "\" (backslash).
9. In the **Default line end character**, enter the character used to terminate a line in the output CSV file. The default is ***\n***.
10. In the **Failure threshold**, specify the number of failures that cause processing to stop. ***0*** (default) indicates that the process never fails.
11. From the **Session credentials expiration (minutes)** drop-down, specify the time between ***1*** and ***10*** minutes, to wait before re-authenticating the CXone session. The default is ***10*** minutes.
12. **Download voice and screen media** - If you set this option as true and the parse voice and screen option is set to true, then both the voice and screen, and voice only media files will be downloaded.

If this option is set to false and the parse voice and screen option is set to true, then only the voice only media file will be downloaded. The voice and screen media file will not be downloaded in this case.

13. From the **Log CXone service responses** drop-down, specify if you want to log the CXone service responses. The default value is ***false***, so no response logging occurs. If you set the value of this parameter as ***true***, the log file is created in *<user temp folder>/CXone-Response.log*.
14. In the **Default HTTP client timeout (in seconds)**, enter the Apache HTTP client default timeout value. The default is ***60***.
15. From the **Parse voice media type** drop-down, keep the default value ***true*** to parse voice media data.
16. From the **Parse voice and screen media type** drop-down, keep the default value ***true*** to parse voice and screen media data.
17. From the **Parse chat media type** drop-down, select ***true*** to parse chat data.
18. From the **Parse email media type** drop-down, select ***true*** to parse email data.
19. From the **Remove newlines** drop-down, select ***true*** to remove new lines from chat messages. Default is ***false***, which will not remove new lines from chat messages. This option is displayed if either or both "Parse chat media type" and "Parse email media type" are set to true.
20. From the **Parse text from HTML chats** drop-down, parse text from chats that contain HTML. Default is to parse text from chats containing HTML.
    This option is displayed if either or both "Parse chat media type" and "Parse email media type" are set to true.
21. From the **Capture additional chat metadata** drop-down, select ***true*** to capture additional chat metadata.
22. From the **Retrieve DFO chat and email metadata and media** drop-down, select ***true*** if you want to retrieve DFO chat and email metadata and media.
23. In the **DFO messages API URL**, enter the DFO messages API URL used to retrieve DFO chat and email threads. The default is ***https://api-de-au1.niceincontact.com/dfo/3.0/contacts/{caseId}/messages***.

24. In the **DFO custom data API URL**, enter the DFO custom data API URL used to retrieve custom data associated with DFO chat and email threads. The default is ***https://api-de-au1.niceincontact.com/dfo/3.0/contacts/{caseId}***.
25. In the **DFO consumer defined custom metadata fields API URL**, enter the DFO consumer defined custom metadata fields API URL used to retrieve custom metadata fields defined for consumers. The default is ***https://api-de-au1.niceincontact.com/dfo/3.0/consumer-contact-custom-fields***.
26. From the **DFO check for duplicate case IDs** drop-down, select ***true*** if you want to check for duplicate case IDs.
27. From the **DFO generate unique row ID** drop-down, select ***true*** if you want to generate a unique row ID for each row written to the DFO chat CSV output file.
28. In the **DFO unique row ID column name**, enter a name for the DFO unique row ID metadata column. The default name is ***DFORowId***.
29. In the **DFO media type**, enter the name of the DFO media type value. Default is ***BE***.
30. In the **DFO channel type**, enter the name of the DFO channel type. Default is ***Facebook Messenger***, ***Twitter Direct Messages***.
31. In the **Use Contact Media Playback API**, select ***true*** to use the media playback contacts API. Select ***false*** to not use the media playback contacts API.
32. In the **Nexidia client ID**, enter the Nexidia client ID to retrieve the CXone access token.
33. In the **Nexidia client secret**, enter the Nexidia secret ID to retrieve the CXone access token.
34. In the **Parse hold counts**, set to ***true*** to parse hold count metadata from the media playback API and write to the working table. Default is ***false***, which will not to parse hold count metadata.
35. In the **Contacts Media Playback API**, enter the media playback contacts API URL. Defaults to https://na1.nice-incontact.com/media-playback/v1/contacts?acd-call-id={ACDCallIDegMasterContactID}.
36. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.
37. From the **Default input disposition handling** drop-down, select one of the following:
    - ***AND***: Allow files to flow through this step that have passed all of the immediate inputs (default).

- ***OR***: Allow files to flow through this step that have passed any of the immediate inputs.

38. In the **Input Disposition(s)**, specify how input dispositions should be handled. The default is empty, to automatically chain. Enter ***none*** to disconnect the action from dependencies to prior actions, or enter a comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Use a hash sign instead of a comma to create an OR condition.
39. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the **Threads per action** setting at the server level.
40. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. ***0*** (default) indicates that there is no limit.
41. From the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
42. In the **Disable action**, specify ***true*** if you want to prevent this action from processing any files during its run. Leave the default value ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
43. Click **Submit**.

CXone Media Retriever V3 Action

1. In the **Discovery OpenID configuration URL**, enter the openID configuration URL to use when authenticating. The default value is ***https://cxone.niceincontact.com/.well-known/openid-configuration***.

2. In the **Discovery CXone configuration URL**, enter the CXone configuration URL. You must enter the token ID when you enter this URL. The default value is ***https://cxone.niceincontact.com/.well-known/cxone-configuration?tenantId={tenantId}***.
3. In the **Access key**, enter the CXone web service access key.
4. In the **Secret key**, enter the CXone web service secret key.
5. In the **Media playback URL**, enter the data extraction URL to use when calling the data extraction API. The Default value is ***https://{area}.{uhDomain}/media-playback/v1***.
6. In the **Default separator character**, enter the character used to separate delimited column values in the output CSV file. The default is ***,*** (comma).
7. In the **Default quote character**, enter the character used to quote column values in the output CSV file. The default is ***"*** (double quote).
8. In the **Default escape character**, enter the character used to escape column values in the output CSV file. The recommended values for the default escape character is a single backslash: "\" (backslash).
9. In the **Default line end character**, enter the character used to terminate a line in the output CSV file. The default is ***\n***.
10. In the **Failure threshold**, specify the number of failures that cause processing to stop. ***0*** (default) indicates that the process never fails.
11. From the **Session credentials expiration (minutes)** drop-down, specify the time between ***1*** and ***10*** minutes, to wait before re-authenticating the CXone session. The default is ***10*** minutes.
12. **Download voice and screen media** - If you set this option as true and the parse voice and screen option is set to true, then both the voice and screen, and voice only media files will be downloaded.
    If this option is set to false and the parse voice and screen option is set to true, then only the voice only media file will be downloaded. The voice and screen media file will not be downloaded in this case.
13. From the **Log CXone service responses** drop-down, specify if you want to log the CXone service responses. The default value is ***false***, so no response logging occurs. If you set the value of this parameter as ***true***, the log file is created in *<user temp folder>/CXone-Response.log*.

14. In the **Default HTTP client timeout (in seconds)**, enter the Apache HTTP client default timeout value. The default is ***60***.
15. From the **Parse voice media type** drop-down, keep the default value ***true*** to parse voice media data.
16. From the **Parse voice and screen media type** drop-down, keep the default value ***true*** to parse voice and screen media data.
17. From the **Parse chat media type** drop-down, select ***true*** to parse chat data.
18. From the **Parse email media type** drop-down, select ***true*** to parse email data.
19. From the **Parse IVR media type** drop-down, select ***true*** to retrieve IVR media. The default value is ***false***.
    IVR media consists of interactions between customers and IVR (Interactive Voice Response) systems, where customers respond to automated prompts. These interactions are downloaded as voice-only MP4 files. The IVR system is treated as an agent in these interactions. By analyzing IVR media, you can gain valuable insights into customer experiences within the IVR system, helping you improve your overall service.
20. From the **Remove newlines** drop-down, select ***true*** to remove new lines from chat messages. Default is ***false***, which will not remove new lines from chat messages. This option is displayed if either or both "Parse chat media type" and "Parse email media type" are set to true.
21. From the **Parse text from HTML chats** drop-down, parse text from chats that contain HTML. Default is to parse text from chats containing HTML.
    This option is displayed if either or both "Parse chat media type" and "Parse email media type" are set to true.
22. From the **Capture additional chat metadata** drop-down, select ***true*** to capture additional chat metadata.
23. From the **Retrieve DFO chat and email metadata and media** drop-down, select ***true*** if you want to retrieve DFO chat and email metadata and media. Default value is false.
24. In the **DFO messages API URL**, enter the DFO messages API URL used to retrieve DFO chat and email threads. The default value is ***https://api-de-au1.niceincontact.com/dfo/3.0/contacts/{caseId}/messages***.

25. In the **DFO custom data API URL**, enter the DFO custom data API URL used to retrieve custom data associated with DFO chat and email threads. The default is ***https://api-de-au1.niceincontact.com/dfo/3.0/contacts/{caseId}***.
26. In the **DFO consumer defined custom metadata fields API URL**, enter the DFO consumer defined custom metadata fields API URL used to retrieve custom metadata fields defined for consumers. The default is ***https://api-de-au1.niceincontact.com/dfo/3.0/consumer-contact-custom-fields***.
27. In the **DFO check for duplicate case IDs** drop-down, select ***true*** if you want to check for duplicate case IDs.
28. From the **DFO generate unique row ID** drop-down, select ***true*** if you want to generate a unique row ID for each row written to the DFO chat CSV output file.
29. In the **DFO unique row ID column name**, enter a name for the DFO unique row ID metadata column. The default name is ***DFORowId***.
30. In the **DFO media type**, enter the name of the DFO media type value. Default is ***BE***.
31. In the **DFO channel types**, enter the name of the DFO channel type. Default is ***Facebook Messenger***, ***Twitter Direct Messages***.
32. In the **DFO write JSON messages response**, select ***true*** to write out the DFO messages response as a JSON file.
33. In the **DFO strip bots**, select ***true*** to remove chat conversations between the chatbot and the customer from the DFO chat thread. By default, this is set to false.
34. In the **DFO split agents**, select ***true*** to split chat conversations between multiple agents and the customer and save them in separate CSV files. The chat conversation of each agent is saved in a separate file. By default, this is set to true. Chatbot is also considered as an agent.
    - If you set DFO strip bots to false and DFO split agents to false, all chat conversations between chatbot, human agent, and customer are saved in a single CSV file.
    - If you set DFO strip bots to false and DFO split agents to True:
        - Chat conversations between chatbot and the customer are saved in a separate CSV file.

- Chat conversations between multiple agents and the customer are split and saved in separate CSV files.
- If you set DFO strip bots to true and DFO split agents to false:
    - Chat conversations between chatbot and the customer are removed from the DFO chat thread until the next human agent is encountered.
    - Chat conversations between multiple agents and the customer are saved in a single CSV file.
- If you set DFO strip bots to true and DFO split agents to true:
    - Chat conversations between chatbot and the customer are removed from the DFO chat thread until the next human agent is encountered.
    - Chat conversations between multiple agents and the customer are split and saved in separate CSV files.

35. In the **DFO prefix metadata**, select ***true*** to add a prefix to the following parameters: By default, this field is set to false.
    - In the **DFO prefix additional metadata (working table metadata),** enter the prefix to use when adding the additional metadata/working table columns to the output DFO chat CSV file. By default, ***wt_*** is added as a prefix.
    - In the **DFO prefix custom metadata CSV columns**, enter the prefix to use when adding the custom metadata columns to the output DFO chat CSV file. By default, ***cust_*** is added as a prefix.

    For DFO chat survey type chat threads, ***_survey*** is added as a suffix to the CSV output file. For example, ***7023d0f1-23bf-45c0-b864-cf8cd4d26646--be-Chat_survey.csv***.
36. In the **Use Contacts Media Playback API**, select ***true*** to use the media playback contacts API. Select ***false*** to not use the media playback contacts API.
37. In the **Nexidia client ID**, enter the Nexidia client ID to retrieve the CXone access token.
38. In the **Nexidia client secret**, enter the Nexidia secret ID to retrieve the CXone access token.
39. In the **Parse hold counts**, set to ***true*** to parse hold count metadata from the media playback API and write to the working table. Default is ***false***, which will not to parse hold count metadata.
40. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.

41. From the **Default input disposition handling** drop-down, select one of the following:
    - ***AND***: Allow files to flow through this step that have passed all of the immediate inputs (default).
    - ***OR***: Allow files to flow through this step that have passed any of the immediate inputs.
42. In the **Input Disposition(s)**, specify how input dispositions should be handled. The default is empty, to automatically chain. Enter ***none*** to disconnect the action from dependencies to prior actions, or enter a comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Use a hash sign instead of a comma to create an OR condition.
43. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the **Threads per action** setting at the server level.
44. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. ***0*** (default) indicates that there is no limit.
45. From the **Terminal action** drop-down, specify whether this is a terminal action. Upon process completion, the rows affected by terminal actions are moved to a working table rollover table. By default, actions are considered terminal if they have a corresponding disposition column in the working table, and no downstream actions reference them. This option is available for instances where the default behavior is not desired.
46. In the **Disable action**, specify ***true*** if you want to prevent this action from processing any files during its run. Leave the default value ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
47. Click **Submit**.

CXone Metadata Retriever V2 Action

1. In the **Discovery OpenID configuration URL**, enter the openID configuration URL to use when authenticating. The default value is ***https://cxone.niceincontact.com/.well-known/openid-configuration***.
2. In the **Discovery CXone configuration URL**, enter the CXone configuration URL. You must enter the token ID when you enter this URL. Defaults to ***https://cxone.niceincontact.com/.well-known/cxone-configuration?tenantId={tenantId}***.
3. In the **Access key**, enter the CXone web service access key.
4. In the **Secret key**, enter the CXone web service secret key.
5. In the **Data Extraction URL**, enter the URL to use when calling the Data Extraction API. The default value is ***https://{area}.{uhDomain}/data-extraction/v1***.
6. In the **CXone web service wait time (in milliseconds)**, enter the amount of time to wait between calls to the metadata search API endpoint. The default is ***10,000***.
7. In the **CXone web service number of retries**, enter the number of retries to the metadata search API endpoint before failing. The default is ***20***.
8. In the **Segment ID metadata column name**, enter the name of the metadata column containing the Segment Id. The default is ***SegmentID***.
9. In the **ACD Contact ID metadata column name**, enter the name of the metadata column containing the ACD Contact Id. The default is ***ACDCallIDegMasterContactID***.
10. From the **Delete metadata inputs after import?** drop-down, keep the default value ***True*** to delete metadata inputs after they are imported.
11. In the **Failure threshold**, specify the number of failures that cause processing to stop. ***0*** (default) indicates that the process never fails.
12. From the **Session credentials expiration (minutes)** drop-down, specify the time between ***1*** and ***10*** minutes, to wait before re-authenticating the CXone session. The default is ***10*** minutes.
13. In the **Default separator character**, enter the character used to separate delimited column values in the output CSV file. The default is ***,*** (comma).

14. In the **End date metadata column name**, enter the name of the metadata column containing the End date. The default is ***EndDate***.
15. In the **End date format**, enter the end date format defined by the Java SimpleDateFormat class. The default is ***MM/dd/yyyy HH:mm:ss***.
16. In the **Force file encoding**, enter a valid Java InputStream encoding, or leave blank to auto-detect.
17. From the **Log CXone service responses** drop-down, specify if you want to log the CXone service responses. The default value is ***False***, so no response logging occurs. If you set the value of this parameter as ***True***, the system creates a log file in *<user temp folder>/CXone-Response.log*.
18. In the **Default quote character**, enter the character used to quote column values in the output CSV file. The default is **"** (double quote).
19. In the **Default escape character**, enter the character used to escape column values in the output CSV file. The default is **** (backslash).
20. In the **Data extraction API payload version number**, enter the data extraction API payload version number. The default is ***3***.
21. In the **Filter metadata using full date/time**, specify if you want to filter metadata by only date, or by date and time.
    - To filter metadata by only date, use the default ***false*** setting and set the **Data extraction API payload version number** to 5 or below.
    - To filter metadata by date and time, set the value to ***true*** and set the **Data extraction API payload version number** to 7 or higher.
    - The **Data extraction API payload version number**7 supports filtering by date, and by date and time.
22. In the **Data extraction API payload entity name**, enter the data extraction API payload entity name. The default is ***recording-interaction-metadata***.
23. In the **Capture business data metadata**, set to ***true*** to capture business data metadata. The default value is ***false*** to ignore business data metadata. The **Data extraction API payload version number** must be set to version 7 or higher.
24. In the **Business data name column name**, specify the name of the business data name column. The default value is set as ***BusinessDataName***.

25. In the **Business data value column name**, specify the name of the business data value column. The default value is set as ***BusinessDataValue***.
26. In the **Nexidia client ID**, enter the Nexidia client ID to retrieve the CXone access token.
27. In the **Nexidia client secret**, enter the Nexidia secret ID to retrieve the CXone access token.
28. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.
29. From the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
30. In the **Disable action**, specify ***true*** if you want to prevent this action from processing any files during its run. Leave the default value ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
31. Click **Submit**.

CXone Report Retriever V2 Action

1. In the **Discovery OpenID configuration URL**, enter the openID configuration URL to use when authenticating. The default value is ***https://cxone.niceincontact.com/.well-known/openid-configuration***.
2. In the **Discovery CXone configuration URL**, enter the CXone configuration URL. You must enter the token ID when you enter this URL. Defaults to ***https://cxone.niceincontact.com/.well-known/cxone-configuration?tenantId={tenantId}***.
3. In the **Access key**, enter the CXone web service access key.
4. In the **Secret key**, enter the CXone web service secret key.
5. In the **Reporting URL**, enter the URL to use when calling the Reporting API. The default value is ***https://api-{area}.{domain}/incontactapi/services/v24.0***.
6. From the **Report type** drop-down, select the type of report to run. The options are ***Datadownload*** or ***Custom***.
7. In the **Report ID**, enter the report ID of the custom or data download report to run.
8. In the **CXone web service wait time (in milliseconds)** drop-down, specify the amount of time to wait between calls to the metadata search API endpoint.
9. In the **CXone web service number of retries**, specify the number of retries to the metadata search API endpoint before failing.
10. From the **Delete metadata inputs after import?** drop-down, keep the default value ***True*** to delete metadata inputs after they are imported.
11. In the **Failure threshold**, specify the number of failures that cause processing to stop. ***0*** (default) indicates that the process will never fail.
12. From the **Session credentials expiration (minutes)** drop-down, specify the time between ***1*** and ***10*** minutes, to wait before re-authenticating the CXone session. The default is ***10*** minutes.
13. In the **Default separator character**, enter the character used to separate delimited column values in the output CSV file. The default is ***,*** (comma).
14. In the **Force file encoding**, enter a valid Java InputStream encoding, or leave blank to auto-detect.

15. From the **Log CXone service responses** drop-down, specify if you want to log the CXone service responses. The default value is ***false***, so no response logging occurs. If you set the value of this parameter as ***True***, the log file is created in *<user temp folder>/CXone-Response.log*.
16. From the **Use quantum?** drop-down, keep the default value ***true*** to use the Quantum window.
17. In the **Ready state**, enter the report ready state string.
18. From the **Pivot Data** drop-down, select ***true*** to pivot data into a single row on an ID-by-ID row basis when report data comes back as name/value pairs for a specific ID. This action ONLY applies to the data download report type.
19. In the **Pivot data ID field name**, enter the name of the ID field used to pivot the report data.
20. In the **Pivot data name field name**, enter the name of the data name field used to pivot the report data.
21. In the **Pivot data value field name**, enter the name of the data value field used to pivot the report data.
22. In the **Default quote character**, enter the character used to quote column values in the output CSV file. The default is **"** (double quote).
23. In the **Default escape character**, enter the character used to escape column values in the output CSV file. The default is **** (backslash).
24. In the **Nexidia client ID**, enter the Nexidia client ID to retrieve the CXone access token.
25. In the **Nexidia client secret**, enter the Nexidia secret ID to retrieve the CXone access token.
26. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.
27. From the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.

28. In the **Disable action**, specify ***true*** if you want to prevent this action from processing any files during its run. Leave the default value ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
29. Click **Submit**.

CXone Snowflake Extractor Action

1. In the **Failure threshold**, specify the number of failures that cause processing to stop. ***0*** (default) indicates that the process will never fail.
2. In the **Batch Execute Size,** specify the number of records that are processed and submitted to the database at a time until all the data is processed. The default is 100.
3. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.
4. From the **Default input disposition handling** drop-down, select one of the following:
   - ***AND***: Allow files to flow through this step that have passed all of the immediate inputs (default).
   - ***OR***: Allow files to flow through this step that have passed any of the immediate inputs.
5. In the **Input Disposition(s)**, specify how input dispositions should be handled. The default is empty, to automatically chain. Enter ***none*** to disconnect the action from dependencies to prior actions, or enter a comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Use a hash sign instead of a comma to create an OR condition.
6. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. ***0*** (default) indicates that there is no limit.

7. From the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
8. In the **Disable action** field, specify ***true*** if you want to prevent this action from processing any files during its run. Leave the default value ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
9. Click **Submit**.

CXone Snowflake Query Action

1. In the **Query** field, enter the SQL query that will be sent to the Artemis queue to execute.
2. In the **Table Name** field, enter the target table name in the customer database.
3. In the **Request ID** field, select the request ID identifier (Snowflake) that can be sent to Artemis to get data from.
4. In the **Tenant ID** field, enter the tenant ID that can be sent to Artemis to be used in database schema. This field is optional.
5. In the **Service User** field, enter the service user account that can be sent to Artemis to give permission in the database schema. This field is optional.
6. In the **Failure threshold**, specify the number of failures that cause processing to stop. ***0*** (default) indicates that the process will never fail.
7. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.
8. From the **Default input disposition handling** drop-down, select one of the following:
   - ***AND***: Allow files to flow through this step that have passed all of the immediate inputs (default).
   - ***OR***: Allow files to flow through this step that have passed any of the immediate inputs.

9. In the **Input Disposition(s)**, specify how input dispositions should be handled. The default is empty, to automatically chain. Enter ***none*** to disconnect the action from dependencies to prior actions, or enter a comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Use a hash sign instead of a comma to create an OR condition.
10. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. ***0*** (default) indicates that there is no limit.
11. From the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
12. In the **Disable action** field, specify ***true*** if you want to prevent this action from processing any files during its run. Leave the default value ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
13. Click **Submit**.

CXone Users Retriever V2

1. In the **Discovery OpenID configuration UR**L, enter the URL. Defaults URL is ***https://cxone.niceincontact.com/.well-known/openid-configuration.***
2. In the **Discovery CXone configuration URL**, enter the URL. Tenant ID token is required. Default is ***https://cxone.niceincontact.com/.well-known/cxone-configuration?tenantId={tenantId}.***
3. In the **Access key**, enter the CXone web service access key.
4. In the **Secret key**, enter the CXone web service secret key.
5. In the **Admin URL**, enter the Admin URL that is used when calling the Admin API. For example, //{area}.{uhDomain}/user-management/v2.

6. In the **Session credentials expiration (minutes)**, enter the time to wait before re-authenticating the CXone session. The default value is 10 minutes. Session credentials expiration time must be set between 1 and 10 minutes.
7. In the **Log CXone service responses**, specify whether to log the CXone service responses to a log file. The log file will be created in the <user temp folder>/CXone-Response.log. By default, this is not enabled.
8. In the **CXone web service page size**, select the number of users to retrieve per API call. The default is 100 users per call.
9. In the **Failure Threshold**, specify the number of failures that cause processing to stop. A value of 0 indicates that the process can never fail.
10. In the **Session credentials expiration (minutes)** drop-down, specify the time to wait before re-authenticating the AppLink session. The default is 10 minutes. The session credentials expiration time must be between 1 and 10 minutes.
11. In the **Nexidia client ID**, enter the Nexidia client ID to retrieve the CXone access token. This value is the same for all CXone customers.
12. In the **Nexidia client secret**, enter the Nexidia secret ID to retrieve the CXone access token. This value is the same for all CXone customers.
13. In the **User search filter**, enter the user search filter to pass to the CXone users search API.
14. In the **ID property name**, enter the name of the JSON property containing the ID value that is used to generate a unique DEF working table file key. Default value is ***id***.
15. In the **Truncate working table**, Truncate the user list working table before retrieving the latest user list. Default action is to truncate the working table.
16. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.
17. From the **Default input disposition handling** drop-down list, select one of the following:
    - ***AND***: Allow files to flow through this step that have passed all of the immediate inputs (default).
    - ***OR***: Allow files to flow through this step that have passed any of the immediate inputs.

18. In the **Input Disposition(s)**, specify how input dispositions should be handled. The default is empty, to automatically chain. Enter ***none*** to disconnect the action from dependencies to prior actions, or enter a comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Use a hash sign instead of a comma to create an OR condition.
19. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. ***0*** (default) indicates that there is no limit.
20. From the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
21. In the **Disable action**, specify ***true*** if you want to prevent this action from processing any files during its run. Leave the default value ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
22. Click **Submit**.

CXone Suppl Metadata Retriever V2 Action

1. In the **Discovery OpenID configuration URL**, enter the openID configuration URL to use when authenticating. The default value is ***https://cxone.niceincontact.com/.well-known/openid-configuration***.
2. In the **Discovery CXone configuration URL**, enter the CXone configuration URL. You must enter the token ID when you enter this URL. Defaults to ***https://cxone.niceincontact.com/.well-known/cxone-configuration?tenantId={tenantId}***.
3. In the **Access key**, enter the CXone web service access key.
4. In the **Secret key**, enter the CXone web service secret key.
5. In the **Reporting URL**, enter the URL to use when calling the Reporting API. The default value is ***https://api-{area}.{domain}/incontactapi/services/v19.0***.
6. In the **Failure threshold**, specify the number of failures that will cause processing to stop. ***0*** (default) indicates that the process will never fail.
7. From the **Session credentials expiration (minutes)** drop-down list, specify the time between ***1*** and ***10*** minutes, to wait before re-authenticating the CXone session. The default is ***10*** minutes.
8. From the **Log CXone service responses** drop-down, specify if you want to log the CXone service responses. The default value is ***False***, so no response logging occurs. If you set the value of this parameter as ***True***, the log file is created in *<user temp folder>/CXone-Response.log*.
9. From the **Handle HTTP Error** drop-down list, select how HTTP response code errors should be handled. The available options are ***log warning*** (default), ***log error***, and ***throw exception***.
10. In the **Nexidia client ID**, enter the Nexidia client ID to retrieve the CXone access token.
11. In the **Nexidia client secret**, enter the Nexidia secret ID to retrieve the CXone access token.
12. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.
13. From the **Default input disposition handling** drop-down list, select one of the following:

- ***AND***: Allow files to flow through this step that have passed all of the immediate inputs (default).
- ***OR***: Allow files to flow through this step that have passed any of the immediate inputs.

14. In the **Input Disposition(s)**, specify how input dispositions should be handled. The default is empty, to automatically chain. Enter ***none*** to disconnect the action from dependencies to prior actions, or enter a comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Use a hash sign instead of a comma to create an OR condition.
15. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the **Threads per action** setting at the server level.
16. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. ***0*** (default) indicates that there is no limit.
17. From the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
18. In the **Disable action**, specify ***true*** if you want to prevent this action from processing any files during its run. Leave the default value ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
19. Click **Submit**.

CXone Teams Retriever V2 Action

1. In the **Discovery OpenID configuration URL**, enter the openID configuration URL to use when authenticating. The default value is ***https://cxone.niceincontact.com/.well-known/openid-configuration***.

2. In the **Discovery CXone configuration URL**, enter the CXone configuration URL. You must enter the token ID when you enter this URL. Defaults to ***https://cxone.niceincontact.com/.well-known/cxone-configuration?tenantId={tenantId}***.
3. In the **Access key**, enter the CXone web service access key.
4. In the **Secret key**, enter the CXone web service secret key.
5. In the **Admin URL**, enter the user admin URL. The Default value is ***https://api-{area}.{domain}/incontactapi/services/v22.0***.
6. In the **Failure Threshold**, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process never fails.
7. In the **Session credentials expiration (minutes**) drop-down, specify the time to wait before re-authenticating the CXone session. The default is 10 minutes. The session credentials expiration time must be between 1 and 10 minutes.
8. In the **Log CXone service responses** drop-down, specify if you want to log the CXone service responses.
9. In the **Nexidia client ID**, enter the Nexidia client ID to retrieve the CXone access token.
10. In the **Nexidia client secret**, enter the Nexidia secret ID to retrieve the CXone access token.
11. In the **Reconciliation Identifiers**, enter the name of the action if there is a reporting action in the process group.
12. From the **Default input disposition handling** drop-down list, select one of the following:
    - ***AND***: Allow files to flow through this step that have passed all of the immediate inputs (default).
    - ***OR***: Allow files to flow through this step that have passed any of the immediate inputs.

13. In the **Input Disposition(s)**, specify how input dispositions should be handled. The default is empty, to automatically chain. Enter ***none*** to disconnect the action from dependencies to prior actions, or enter a comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Use a hash sign instead of a comma to create an OR condition.
14. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the **Threads per action** setting at the server level.
15. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. ***0*** (default) indicates that there is no limit.
16. From the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
17. In the **Disable action**, specify ***true*** if you want to prevent this action from processing any files during its run. Leave the default value ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
18. Click **Submit**.

Channel0 Mono Detector Action

1. In the **Process files from metadata** drop-down, specify if you want to process files from metadata. When you select ***true*** the application processes files from the process groups existing working table. When you specify false, the application processes files from attached input resources. Processing from metadata reduces I/O and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **Failure Threshold**, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
3. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.
4. In the **Default input disposition handling** drop-down, select one of the following:
   - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
   - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
5. In the **Input Disposition(s)**, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
6. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
7. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.

8. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
9. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
10. Click **Submit**.

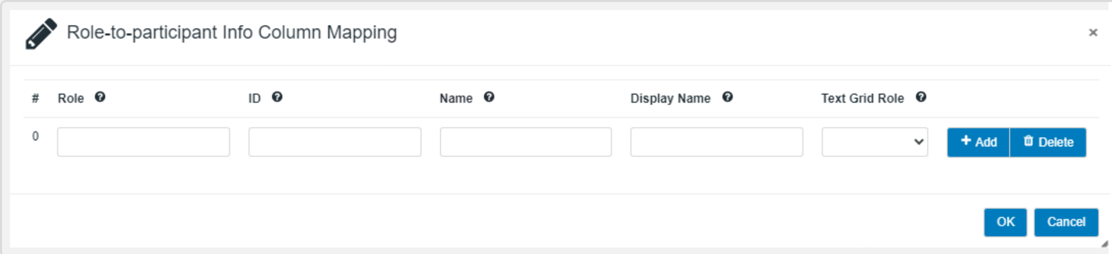
Chat CSV to JSON Action

1. In the **Process files from metadata** drop-down, specify if you want to process files from metadata. When you select ***true*** the application processes files from the process groups existing working table. When you specify false, the application processes files from attached input resources. Processing from metadata reduces I/O and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **File Filter**, enter the file filter type. For example: CSV.
3. In the **Chat ID column name(s)**, enter the name(s) of the column(s) holding the unique ID for each conversation (comma-delimited if more than one). This value is used to generate the .json file name, in the order entered, separated by underscores.
4. In the **Message text column name**, enter the name of the column that contains the chat text.
5. In the **Chat time column name**, enter the name of the column that contains the time stamp of each message in the conversation.
6. (Optional) in the **Message sort column**, enter the optional name of a numeric sequence column to sort messages by. If you do not enter a name, the message time stamp is used instead.
   If the CSV file contains only one message in the chat, 5 milliseconds are added to the end time of the chat. This will ensure the chat duration is not zero and single-message chats can be analyzed. The end time value is configurable.
   In the **Role column name**, enter the name of the column that identifies the role of each participant in the conversation.
7. In the **Role-to-participant info column mapping**, specify a mapping of expected roles to the names of columns holding identifiers (ID, Name, Display Name) for those roles.

   When you click in this field or click , the Role-to-participant Info Column Mapping window opens. Use this window to map chat role entries by specifying these values:

   - **Role** - The role name as defined in the CSV file's role column.
   - **ID** - The CSV column containing the participant identification for the role.
   - **Name** - The CSV column containing the participant name for the role.

- **Display Name** - The CSV column containing the participant display name for the role.
- **Text Grid Role** - The Text Grid role that the CSV role name maps to. The valid options are Unknown, Agent, or Customer.



8. In the **Parse attachments** drop-down, specify if you want to parse attached files. The default value is ***false***.
    a. (Optional) If you selected ***true***" in the **Parse attachments** drop-down, enter the name of the column that holds the location of a file attached to the message. Multiple attached files per chat turn are supported in the attachment column.
    b. (Optional) Specify if you want to append a parse text label to the message body.
    c. (Optional) Click in the **Attachment Types** field and specify the types of attachments that you want to open and extract. The available options are Email, JSON, Microsoft Office Documents, PDF, Plain Text, Rich Text Format, SQL, Visio Documents, Zip, and Other.
9. In the **Delimiter**, enter the delimiter character. This is used for non-XML input, and is a single character such as ***,***. You can also use ***\t*** for tab.
10. In the **Delete files after processing** drop-down, specify if you want to delete .csv files and the corresponding attaching after processing.
11. In the **Additional metadata to capture** field, if you want to capture additional column metadata alongside each chat session's JSON record in working table, enter a comma-delimited list of .csv column headers.
12. In the **Excerpted** field, specify if chats processed by this action are excerpted from a long-lived room or channel.
13. In the **Remove HTML Tags** field, specify whether you want to remove or keep the HTML tags from the .csv file. Select ***true*** if you want to remove the HTML tags.

14. In the **Default quote character**, specify the character used to quote column values (for example **"**) in the Default quote character field.
15. In the **Default escape character**, specify the character used to escape column values (for example ****) in the Default escape character field.
16. In the **Failure Threshold** drop-down list, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
17. In the **IP address column name**, enter the name of the column that contains the IP address of the participant.
18. In the **Force file encoding**, leave blank in most cases to auto-detect. If auto-detect is not working, enter a valid Java InputStream encoding. See Supported Encodings to learn more.
19. In the **Attachment write Limit (KBs)** drop-down, specify the write limit (specified in KBs for example: 1000). The default value is ***-1*** (no write limit).
20. In the **Column header re-mapping** field, This option allows you to use the name of your choice in the CSV file rather than the exact column header names from the working table or the exact tag names from the XML file. You will use the following format: ***incomingHeader1=finalHeader1,iH2=fH2,...***
21. In the **Create new metadata from incoming**, enter a pipe-separated list of new columns to create, using braces to identify the incoming metadata to generate the new data from, for example: 'UCID={iCompoundId}-{iPBXId}|...' Note that it is also possible to indicate the datatype of the column to be created, and that doing so will trigger an automatic indexing of the column. Continuing the previous example, this can be accomplished using this form: 'UCID:navchar(256)={iCompoundId}-{iPBXId}|...'. Note that there are performance implications to adding a new datatype column that will be indexed in an existing and already-large working table.
22. In the **Redact chat text** drop-down, select ***true*** to redact chat text. Select ***false*** to not to redact text.
23. In the **Add Duration Milliseconds** drop-down, add duration milliseconds to end time.
24. In the **Redact chat text** drop-down, select ***true*** to redact chat text to protect sensitive information in your chat before sending data to Analytics.

- In the **Length of redaction replacements** drop-down, enter the value to set the length of each redaction in the text. The minimum value is 3, the maximum is 8, and the default value is 5.
- In the **List of regular expressions**, click the **Edit** icon to enter one or more regular expressions for selecting chat text to redact.
- In the **Redact chat text character**, enter the character used to replace redacted text. The default is X.

In **Output format**, select ***VoiceGrid ASR*** to convert chat JSON files from Text Grid to Voice Grid format for AutoSummary, or ***TextGrid JSON*** to leave the chat JSON files in Text Grid JSON format.

25. In the **Clean new lines** drop-down, select ***true*** if you want to clean the new lines from the chat body which collapses multiple new lines characters into single newline character.
26. In the **Reconciliation Identifier**, enter the name of the action if there is a reporting action in the process group.
27. In the **Default input disposition handling** drop-down, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
    - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
28. In the **Input Disposition(s)**, enter the input disposition. The Data Exchange Framework automatically chains action component dispositions together as javascript:void()part of its processing, so this field is usually left blank. If you are using split processes, enter a value in this field.
29. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
30. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.

31. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
32. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
33. Click **Submit**.

Cleanup Action

There are three cleanup types:

- Files from input resources - Use to scan file system or SFTP resources, and delete anything older than "x" hours. This will be logged against the working table.
- Files from current disposition - Use when plugging into a process workflow, to delete any files that reach this action (this will be all with no action inputs).
- Working table - Use to age out the working table data, and optionally deleting any corresponding files.

To configure the cleanup action (when you select the Files from input resources cleanup type):

1. In the **Cleanup type** drop-down, select the ***Files from input resources*** cleanup type.
2. In the **File filter**, enter the file filter type. For example: CSV.
3. In the **Delete input resource files older than # hours** drop-down, specify the number of hours after which to delete the input resource files. ***1140*** is the default, and one hour is the minimum allowed value.
4. In the **Directory cleanup strategy** drop-down, specify the directory cleanup strategy for emptying directories in input resources after file cleanup. The valid options are ***recursive*** and ***top level***. The default is "***recursive***. When you select ***top level***, the "Top level cleanup depth" option opens. When selected, this determines what sub-folder depth the top-level aging-out process looks to. All higher folders are ignored. 1 = all folders sitting directly in the file system resource, and is the default.
5. In the **Preserve assigned/reserved media** drop-down, specify if you want the application to preserve assigned/reserved media. If you specify ***true***, the action will prevent the deletion of any media that has been assigned or reserved in *Interaction Analytics*. This requires a database resource input to be configured to point to a NexidiaESI database, where the 'Connection URL' field should be in the following form: [\namedinstance];databaseName=NexidiaESI'.

6. In the **Empty folder strategy** drop-down, specify how to handle empty folders in input resources after file cleanup. This setting helps you manage folders based on your needs, allowing for flexible cleanup of input resources.
   - Remove empty folders (default) - The Delete input resource files older than # hours is compared to the last modified dates of top-level folders to determine if they should be deleted, without checking contents.
   - Keep empty folders - Keeps all folders, even if empty.
   - Remove empty folders after # hours - Removes empty folders found in the attached file system resource after the specified time. Enter the number of hours to wait before deleting empty folders.
7. In the **Failure Threshold** drop-down list, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
8. In the **Reconciliation Identifier**, enter the name of the action if there is a reporting action in the process group.
9. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
10. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
11. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
12. Click **Submit**.

To configure the cleanup action (when you select the Files from current disposition cleanup type):

1. In the **Cleanup type** drop-down, select the ***Files from current disposition*** cleanup type.
2. In the **File filter**, enter the file filter type. For example: CSV.
3. In the **Allow multiple input resources** drop-down, specify if you want to allow multiple input resources. Set this value to ***true*** to allow multiple input resources to be attached to the action. This approach is suitable for cleanups where files which may be duplicated across resources, but could cause a lower level of performance otherwise.
4. In the **Directory cleanup strategy** drop-down, specify the directory cleanup strategy for emptying directories in input resources after file cleanup. The valid options are ***recursive*** and ***top level***. The default is ***recursive***. When you select ***top level***, the "Top level cleanup depth" option opens. When selected, this determines what sub-folder depth the top-level aging-out process looks to. All higher folders are ignored. 1 = all folders sitting directly in the file system resource, and is the default.
5. In the **Preserve assigned/reserved media** drop-down, specify if you want the application to preserve assigned/reserved media. If you specify ***true***, the action will prevent the deletion of any media that has been assigned or reserved in *Interaction Analytics*. This requires a database resource input to be configured to point to a NexidiaESI database, where the 'Connection URL' field should be in the following form: [\namedinstance];databaseName=NexidiaESI'.
6. In the **Failure Threshold** drop-down list, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
7. In the **Reconciliation Identifier**, enter the name of the action if there is a reporting action in the process group.
8. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.

9. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
10. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
11. Click **Submit**.

To configure the cleanup action (when you select the Files from working table cleanup type):

1. In the **Cleanup type** drop-down, select the ***Working table*** cleanup type.
2. In the **Delete process group working table records older than # hours** drop-down, specify the number of hours after which to delete the process group working table records. ***1140***is the default, and one hour is the minimum allowed value.
3. In the **Dump old working table records to output location CSV and ZIP** drop-down, specify if you want the application to export the old working database records to a CSV or ZIP file output location.
4. In the **Drop unused working table columns** drop-down, specify if you want the application to drop unused working table columns. Select ***true*** to delete working table columns with no data in them after cleanup.
5. In the **Delete corresponding files from input resources** drop-down, set to true if you want the application to delete files from input resources whose keys match working table records that are being deleted.
6. In the **Directory cleanup strategy**:

- Set to ***Recursive*** to recursively find files to delete and prune empty directories after cleanup. Note that in large trees, using this option may slow down performance. In such cases, a Top-level directory purge may be better. When selected, deletion will be based on the selected Empty folder strategy.
- When set to ***Top Level***, the **Top level cleanup depth** field is displayed. With the default setting of 1, any folder directly in the attached file system resource will be a candidate for cleanup. This means any folder that has not had its contents updated since the cutoff time will be deleted, irrespective of their contents. If desired, changing the cleanup depth will mean only sub-folders at the specified depth will be candidates for cleanup. For example, if set to 2, only sub-folders of top-level folders will be checked for deletion. If set to 3, it will be those sub-folders, and so on.

7. In the **Preserve assigned/reserved media** drop-down, specify if you want the application to preserve assigned/reserved media. If you specify ***true***, the action will prevent the deletion of any media that has been assigned or reserved in *Interaction Analytics*. This requires a database resource input to be configured to point to a NexidiaESI database, where the 'Connection URL' field should be in the following form: [\namedinstance];databaseName=NexidiaESI'.
8. In the **Empty folder strategy** drop-down, specify how to handle empty folders in input resources after file cleanup. This setting helps you manage folders based on your needs, allowing for flexible cleanup of input resources.
   - Remove empty folders (default) - Deletes all empty folders found in the attached file system resource.
   - Keep empty folders - Keeps all folders, even if empty.
   - Remove empty folders after # hours - Removes empty folders found in the attached file system resource after the specified time. Enter the number of hours to wait before deleting empty folders. In the **Delete empty folders older than # hours** drop-down, specify the number of hours after which to delete the empty folders. ***1140 hours*** is the default, and one hour is the minimum allowed value.
9. In the **Failure Threshold** drop-down list, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
10. In the **Reconciliation Identifier**, enter the name of the action if there is a reporting action in the process group.

11. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
12. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
13. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave "false" to allow processing as usual. This option is useful for troubleshooting the process.
14. Click **Submit**.

Compress Action

1. In the **Process files from metadata** drop-down, specify if you want to process files from metadata. Select ***false*** to process files from attached input resources or ***true*** to process them from this process groups existing working table. Processing from metadata reduces input/output and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **File filter**, enter the file filter type. For example: CSV.
3. In the **Compression Type** drop-down, select the compression type. The valid options are ***.zip***, ***.tar*** and ***.7z***.
4. (Optional) If you selected zip as the compression type, in the **Password to secure zipped files with** field, enter the password to use to secure the zipped files.
5. In the **Gzip.tar files** drop-down, specify if tar files can be compressed/uncompressed.
6. In the **7zip.7z files** drop down, specify if 7z files can be compressed/uncompressed.
7. In the **Create file MD5 hash** drop-down, specify if you want the application to create file MD5 hash.
8. In the **Compress each file individually** drop-down, specify if you want the application to compress each file individually. When you specify ***true***, a separate compressed file is created for each file that is encountered. When you specify ***false***, the compressed files that are created will contain multiple files based on the other specifications below.
9. In the **Filename Pattern** drop-down, optionally enter a value in the Filename Pattern field if you want to name the output file something other than the default name of {folderName}.ext (where ext=zip or tar) or {individualFile},ext. You can also use standard resource parameters to reference metadata or dates {yyyy}, {mm}, {dd}, or {MetadataField}.
10. In the **Delete original file(s) after compressing** drop-down, specify if you want the application to delete the files after they are compressed.

11. In the **Checks and sets dispositions** drop-down,specify if you want the application to check and set the dispositions. This option has to do with how the action participates in the process flow: If an action checks and sets dispositions, then when it's looking at items to process, it checks the working table to see if any actions before it in the workflow (actions that also check and set dispositions) have processed the record, and it makes sure that it has not.
12. In the **Failure Threshold** drop-down , specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
13. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.
14. In the **Default input disposition handling** drop-down list, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
    - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
15. In the **Input Disposition(s)**, enter the input disposition. The Data Exchange Framework automatically chains action component dispositions together as javascript:void()part of its processing, so this field is usually left blank. If you are using split processes, enter a value in this field.
16. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
17. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
18. In the **Terminal action** drop-down, specify if this is a terminal action. The working table rows touched by terminal actions are pushed into a working table rollover table on process completion. By default, actions are considered terminal when they have a corresponding disposition column in the working table and no other actions downstream reference them. This option is provided for instances where the default behavior is not desirable.
19. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.

20. Click **Submit**.

Convert Engage Hierarchy V2 Action

This action is only compatible with Engage 7.x.

1. In the **Build Agent** drop-down, specify if you want to build the Agent CSV file. The default value is ***true***.
2. In the **Build User** drop-down, specify if you want to build the User CSV file. The default value is ***true***.
3. In the **Build Agent Supervisor** drop-down, specify if you want to build the Agent Supervisor to the DB table nx_agent_supervisor. The default value is ***true***.
4. In the **Hub ID**, enter the Hub ID to use in conjunction with the User ID for the Unique Agent ID.
5. In the **Fail if nothing is imported within # minutes**, specify the number of minutes to fail the import if nothing is imported within that time. The default value of zero indicates no failure.
6. In the **Output file name**, enter a name to be appended to each of the CSV files generated. You can also optionally include a TIMESTAMP_PARAMETER that will be substituted with the current time in milliseconds from the epoch (1/1/70). If you don't include a timestamp parameter with your output file name the file will be overwritten on each execution of the action. By default, the output CSV file is named using a unique combination of process execution ID and timestamp.
7. In the **Default separator character**, enter the character used to separate delimited column values in the output CSV. For example, ***i.e,*** or **\t** for a tab.
8. In the **Default quote character**, specify the character used to quote column values (for example ***"***) in the Default quote character field.
9. In the **Default escape character**, specify the character used to escape column values (for example ****) in the Default escape character field.
10. In the **Build Agent Supervisor CSV** drop-down, specify if you want to generate the supervisor.csv file. The default value is ***false***.
11. In the **Integrated Authentication**, specify if the target version of Interaction Analytics supports integrated authentication. The default value is "true". When Integrated authentication field is set to ***true***, Get domain from engage user field is available.

12. Set the **Get domain from engage user** to true to get domain from engage user table. If set ***false*** it gets the domain engage active directory.
13. **Domain** field is available when Get domain from engage user field is set to ***false***. In the **Domain**, the Domain is appended to User Name/Login. Leave it as the default value **--** if you want to get the domain from the Engage Database when Integrated Authentication is true; make it empty if you don't want to append anything to User Name/Login; enter domain if you want to append to user Name/Login when Integrated Authentication is ***true***.
14. In the **Supervisor Identification** drop-down, choose the type of identification you want to assign to the supervisor in the Engage. Supervisor Identification is used in Supervisor Column in Agent CSV, Portal Dashboard ID in User CSV and in agent/supervisor mapping. The possible values are: First/Last Name, Email, Unique Supervisor ID or Supervisor ID. Be careful while selecting the value from the drop down because selected value from the drop down should not be kept empty in the Engage. For example if you select Email than make sure the Email filed in the Engage cannot be empty.
15. In **External user ID identification**, select the external user ID column in both Agent and User CSVs. The field can be set to:
    - ***Empty***
    - ***Email***
    - ***UseriD***
    - ***Agent ID***
    - ***OS Login***
16. In the **Exclude nodes from Hierarchy Identification** drop-down, specify if you want to exclude nodes based on the Hierarchy Name or Hierarchy ID.
17. In the **Exclude nodes from Hierarchy-Including All Children**, enter a comma separated hierarchy name list to exclude nodes and all children from the hierarchy.
18. In the **Exclude nodes from Hierarchy** field, enter a comma separated hierarchy name list to exclude them from the hierarchy but keep all children and assign them to the next level up.
19. In the **Site Identification** drop-down, select Site ID to set the Site to the Site ID from Engage, or select Location to set the Site to the Site Location from Engage.

20. In the **Ignore Engage Inactive User Status**, select ***true*** to pass active user status to NIA when the user is not in the Inactive Users group in the Engage hierarchy and the user status is inactive in Engage. Select "***false***" to keep the same user status in NIA and Engage.
21. In the **Append Level to Hierarchy ID** drop-down, select ***true*** to append the level number to the hierarchy ID.
22. In the **Build CSVs W/O Changes in Engage Hierarchy** drop-down, specify ***true*** to import NIA on each run even if there are no changes in the hierarchy or users/agents.
23. (Optional) If you selected ***false*** in the **Build CSVs W/O Changes configuration**, in the **Wait # minutes After Last Change In Engage** drop-down, specify the number of minutes after an Engage data change before generating the hierarchy/agent/user CSV files.
24. In the **Use Engage Validation** drop-down, select ***true*** to disable further hierarchy process including DEF validation if there is an error in Engage validation. select ***false*** to run DEF hierarchy process and validation regardless of Engage validation error.
25. In the **Disable Email Sending** drop-down, select ***true*** to not send an email with the CSV Error Validation as an attachment.
26. In the **Send Hierarchy Checker PDF** drop-down, select ***true*** if you want the PDF to contain the description for all possible errors from the Data Exchange Framework and Interaction Analytics and how to resolve them.
27. In the **Include Profiles** drop-down, select ***true*** if you want to include profiles to user table.
28. In the **Hierarchy Validation Contact Email**, enter an email address to send notification emails to for all administrative server messages, as well as worker servers when no other administrative contacts are assigned.
29. In the **SMTP Server**, enter the SMTP server.
30. In the **SMTP user (optional)**, enter the SMTP user.
31. In the **SMTP password (optional)**, enter the SMTP password.
32. In the **Hierarchy validation from Email**, enter the Email address from notifications for hierarchy validation errors. Default value is ***def-heirarchy@niceondemand.com***.

33. In the **Reconciliation Identifier**, enter the name of the action if there is a reporting action in the process group.
34. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
35. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
36. Click **Submit**.

Convert JSON to XML

1. Set the input and output as File System Resource.
2. In the **Process files from metadata** drop-down, specify if you want to process files from metadata. When you select ***true*** the application processes files from the process groups existing working table. When you specify false, the application processes files from attached input resources. Processing from metadata reduces I/O and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
3. Set the following properties to convert JSON to an XML file:
   a. In the **File Filter** , enter the file type as ***.json***.
   b. In the **Delete Files After Process** drop-down, specify ***true*** if you want to delete input files after processing. The default value is "***false***".
   c. n the **Failure Threshold** drop-down list, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
   d. In the **Reconciliation Identifier**, enter the name of the action if there is a reporting action in the process group.
   e. In the **Default input disposition handling** drop-down, select one of the following:

- ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
- ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.

f. In the **Input Disposition(s)**, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.

g. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.

h. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.

i. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.

j. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.

4. Click **Submit**.

Convert Transcript to ASR

1. In **Process files from metadata**, specify if you want to process files from metadata. Select ***false*** to process files from attached input resources or ***true*** to process them from this process groups existing working table. Processing from metadata reduces input/output and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In **File filter**, '*.xml', '*.xml,*.csv', etc. Java-style regular expressions can also be used if prepended with 'regex:'. Note that when processing files from resources, this acts as a filter, but when processing from metadata, the metadata's file key is inserted into any wildcards, expanding the processing of the one file to all listed variants ('*,*.xml', e.g., would process both a file matching the working table file key '1234567.wav' and its '1234567.wav.xml' counterpart).
3. In **Transcript Type**, select the type of transcript to convert to ASR format. You can select ***Nexidia Standard, NxPr***, or ***Analytics Hub***.
4. If you select Nexidia Standard:
    a. In **Agent channel identifier**, enter the Key column name for agent channel in transcript file. If channel mapping section in the transcript file exist then the channel label will be used instead of this config and if both are empty it will be Unknown.
    b. In **Customer channel identifier**, enter the Key column name for customer channel in transcript file. If channel mapping section in the transcript file exist then the channel label will be used instead of this config and if both are empty it will be Unknown.
5. In the **Search directories recursively** drop-down, specify if you want the application to search the directories recursively (if the files are not being moved based on metadata).
6. In **Overwrite existing output files**, select ***true*** to overwrite files that already exist in the output location and ***false*** to leave existing files untouched. The default value is ***true***.
7. In **Delete inputs after move**, select ***true*** to do a true file move and ***false*** to copy, leaving inputs intact. The default value is ***true***.
8. In **Failure Threshold**, specify the number of failures that will cause processing to stop. A value of ***0*** indicates that the process will never fail.

9. In **Flatten output directory structure**, specify if you want to flatten the output structure of the generated files. Set this option to ***true*** to flatten the output structure of the generated files if the working table key is a nested directory structure; and ***false*** to retain any working table key structure in directory output.
10. In **Reconciliation Identifier**, enter the name of the action if there is a reporting action in the process group.
11. In the **Default input disposition handling** drop-down list, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
    - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
12. In the **Input Disposition(s)**, enter the input disposition. The Data Exchange Framework automatically chains action component dispositions together as javascript:void()part of its processing, so this field is usually left blank. If you are using split processes, enter a value in this field.
13. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
14. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
15. In the **Terminal action** drop-down, specify if this is a terminal action. The working table rows touched by terminal actions are pushed into a working table rollover table on process completion. By default, actions are considered terminal when they have a corresponding disposition column in the working table and no other actions downstream reference them. This option is provided for instances where the default behavior is not desirable.
16. In **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
17. Click **Submit**.

Convert Transcript To NxPr

1. In **Process files from metadata**, specify if you want to process files from metadata. Select ***false*** to process files from attached input resources or ***true*** to process them from this process groups existing working table. Processing from metadata reduces input/output and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In **File filter**, '*.xml', '*.xml,*.csv', etc. Java-style regular expressions can also be used if prepended with 'regex:'. Note that when processing files from resources, this acts as a filter, but when processing from metadata, the metadata's file key is inserted into any wildcards, expanding the processing of the one file to all listed variants ('*,*.xml', e.g., would process both a file matching the working table file key '1234567.wav' and its '1234567.wav.xml' counterpart).
3. In **Transcript Type**, select the type of transcript to convert to NxPr format. You can select ***Nexidia Standard***, ***VoiceGrid Transcript***, ***Chat Transcript***, and ***ElevateAI Transcript***.
4. In **Agent channel identifier**, enter the Key column name for agent channel in transcript file. If channel mapping section in the transcript file exist then the channel label will be used instead of this config and if both are empty it will be Unknown.
5. In **Customer channel identifier**, enter the Key column name for customer channel in transcript file. If channel mapping section in the transcript file exist then the channel label will be used instead of this config and if both are empty it will be Unknown.
6. In **Redacted word identifier**, enter the comma separated value to redact from transcript content. Default value is PII.
7. In **Media Type**, specify the media type that will be published to the inbox. The valid options are ***Audio, Chat, Email***, and ***Text***.
8. In **Language Metadata**, enter the name of the working table column that stores the value for the language in use.
9. In **Language Value Map**, enter the customer-specific language values and map them to the corresponding fixed language identifiers. For example, if the customer’s language column stores English and Spanish, enter the mappings as ***English=en*** and ***Spanish=es-419***.
   The supported fixed language identifiers are en, es-419, fr, de, it, ja, ko, pt-BR, zh-CN.

Examples of valid mappings are:

- English → en=en
- Brazilian Portuguese → pt-BR=pt-BR
- Japanese → ja=ja
- Latin American Spanish → es=es
- European French → fr=fr
- German → de=de
- Italian → it=it
- Korean → ko=ko
- Mandarin Chinese → zh=zh

10. The **Search directories recursively** drop-down is displayed only when the n **Process files from metadata** is set to false. In this field, specify if you want the application to search the directories recursively (if the files are not being moved based on metadata).
11. In **Overwrite existing output files**, select ***true*** to overwrite files that already exist in the output location and ***false*** to leave existing files untouched. The default value is ***true***.
12. In **Delete inputs after move**, select ***true*** to do a true file move and ***false*** to copy, leaving inputs intact. The default value is ***true***.
13. In **Output format**, select ***Deflated*** to compress xml file and to ***Inflated*** not to compress.
14. In **Failure Threshold**, specify the number of failures that will cause processing to stop. A value of ***0*** indicates that the process will never fail.
15. In **Lowest score**, specify the lowest score.
16. In **Highest score**, specify the highest score.
17. In **Flatten output directory structure**, specify if you want to flatten the output structure of the generated files. Set this option to ***true*** to move files to a flat directory structure or leave as ***false*** to maintain the folder structure of files encountered in a recursive move.
18. In **Clean up punctuation**, set to ***true*** to clean up punctuation, for example, V11 transcript and up. Set to ***false*** to keep the original text as is, for example, v10 transcript.

19. In **Flatten output folder structure**, specify if you want to flatten the output structure of the generated files. Set this option to ***true*** to flatten the output structure of the generated files if the working table key is a nested folder structure; and ***false*** to retain any working table key structure in folder output.
20. If the **Flatten output folder structure** is set to true, then the **Update file key on move** field is displayed. In this field, specify if you want the application to update the file key on move. This is necessary for proper working table tracking if the relative path of the file (and working table file key) is likely to change as part of the move. The default value is "true".
21. In **Reconciliation Identifier**, enter the name of the action if there is a reporting action in the process group.
22. In the **Default input disposition handling** drop-down list, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
    - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
23. In the **Input Disposition(s)**, enter the input disposition. The Data Exchange Framework automatically chains action component dispositions together as javascript:void()part of its processing, so this field is usually left blank. If you are using split processes, enter a value in this field.
24. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
25. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
26. In the **Terminal action** drop-down, specify if this is a terminal action. The working table rows touched by terminal actions are pushed into a working table rollover table on process completion. By default, actions are considered terminal when they have a corresponding disposition column in the working table and no other actions downstream reference them. This option is provided for instances where the default behavior is not desirable.

27. In **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
28. Click **submit**.

Custom SQL V2 Action

1. In the **SQL** Mapping field, click , the SQL Mapping window opens. Use this window to add a single SQL statement file and click **OK**. It validates the SQL syntax and will display message if there is any error. You can add multiple SQL statement files.
2. In the **Fail if nothing is updated** drop-down, set to ***true*** if the SQL statement is an update and if the action should fail if nothing is updated.
3. In the **Execute against working table** drop-down, specify if you want the system to execute the SQL against the working table directly instead of a database resource input. If you select ***true***, you will use the "WORKING TABLE" parameter in SQL.
4. The **Resolve SQL** field will appear only if you select ***true*** in the **Execute against working table** drop-down. Click the **Resolve SQL** button. A Resolve SQL pop-up displays. The system executes the SQL against the working table and checks for errors. If the SQL is valid, it will show the success message. If there are issues, it will show an error message. On this pop-up, you can do one of the following:
    a. Click **Copy** to copy the resolved SQL to the clipboard.
    b. Click **Cancel** to exit and return to the custom SQL action configuration.
5. In the **Skip Error Message** drop-down, set to ***true*** to skip error message in validation. Set to ***false*** to show error message.
6. In the **Commit After Each Statement** drop-down, set to ***false*** to commit at the end and rollback if there is any error for each statement. Set to ***true*** to commit for each item in the map with no rollback.
7. In the **Reconciliation Identifier**, enter the name of the action if there is a reporting action in the process group.
8. In the **Default input disposition handling** drop-down, select one of the following:

- ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
- ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.

9. In the **Input Disposition(s)** field, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type 'none' to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
10. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
11. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
12. Click **Submit**.

DEU to Engage Action

1. In the **Process files from metadata** drop-down, specify if you want to process files from metadata. Select ***false*** to process files from attached input resources or ***true*** to process them from this process groups existing working table. Processing from metadata reduces input/output and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **File filter** field, enter the file filter type. For example: CSV.
3. In the **Attempt reprocess w/updates on failure** drop-down, specify if you want to reprocess with updates on failure. If set to "***false***", any failed inserts will count toward the failure threshold, but will otherwise be ignored. When set to "***true***", failed inserts will be re-attempted on a second pass through the failing CSV, deleting first. The intent is to make reprocessing easier.
4. In the **Delete DEU output zip after successful processing** drop-down, select a value to exclude files that are smaller than the selected kb.
5. In the **Exclude files smaller than # kB** drop-down, select a value to exclude files that are less than the selected kb.
6. In the **Exclude files larger than # kB** drop-down, select a value to exclude files that are more than the selected kb.
7. In the **Delete all files that don't pass the filter** drop-down, specify if you want to delete the DEU output zip after successful processing. Select "***false***" to retain files for review purposes.
8. In the **Delete filed with incomplete data** drop-down, specify if you want to delete all files that do not pass the filter. Files that do not pass the filter are logged in the working table as processed (with a "NXFailureMessage" column explaining the reason) so they won't be re-discovered. Set this to "***false***" if you want to retain those files for review.
9. In the **Interaction Metadata NiceInteractionIdcolumn name** field, modify the Interaction metadata NiceInteractionId column name if it is different than the default value of UDFvarchar1.
10. In the **Interaction Metadata ContactId column name** field, modify the Interaction Metadata ContactId column name if it is different than the default value of UDFvarchar2.

11. In the **Interaction Metadata SiteId column name** field, modify the Interaction Metadata SiteId column name if it is different than the default value of UDFint1.
12. In the **Interaction Metadata ContactGMT StartTime column name** field, modify the Interaction Metadata ContactGMT StartTIme column name if is different than the default value of UDFdate1.I
13. In the **Failure Threshold** drop-down list, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
14. In the **Batch execute size** drop-down, specify the batch execute size. When processing large files, this indicates how often to execute batches against the database.
15. In the **Commit after each batch execution** drop-down, specify if you want to commit after each batch execution. Set to "***true***" to commit to the database after each batch is executed; leave "***false***" to commit only after the CSV file has been fully processed. The trade-off here is: "***true***" will take the transaction load off of the database if it becomes a problem, but will break intended transactional integrity for processing of each incoming CSV file. "***False***" should be preferred unless the database simply can't handle the transactional load.
16. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.
17. In the **Default input disposition handling** drop-down list, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
    - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
18. In the **Input Disposition(s)** field, enter the input disposition. The Data Exchange Framework automatically chains action component dispositions together as javascript:void()part of its processing, so this field is usually left blank. If you are using split processes, enter a value in this field.
19. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.

20. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
21. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
22. Click **Submit**.

DEU Zip Repackager Action

1. In the **Process files from metadata** drop-down, specify if you want to process files from metadata. Select ***false*** to process files from attached input resources or ***true*** to process them from this process groups existing working table. Processing from metadata reduces input/output and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **File filter** field, enter the file filter type. For example: ZIP.
3. In the **Delete original DEU output file(s) after processing** drop-down, specify if you want to delete the DEU output zip after successful processing. Select "false" to retain files for review purposes.
4. In the **Failure Threshold** drop-down list, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
5. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.
6. In the **Default input disposition handling** drop-down list, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
    - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
7. In the **Input Disposition(s)** field, enter the input disposition. The Data Exchange Framework automatically chains action component dispositions together as javascript:void()part of its processing, so this field is usually left blank. If you are using split processes, enter a value in this field.
8. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
9. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.

10. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
11. Click **Submit**.

Database-to-CSV Action

1. In the **Pre-select SQL** area, optionally, enter SQL to execute prior to the select that will be part of the same transaction (temp table creation, for example). Use {WINDOW_START} and {WINDOW_END} to apply the quantum defined at the process level to the generated SQL. {WINDOW_END} can optionally include a date-based column name in the form {WINDOW_END:dateColumnToTest} that, if used, is tested at the end of processing to determine the max date processed, and is used as the next run's {WINDOW_START}. If not used, the next {WINDOW_START} will be the current run's {WINDOW_END}.
2. In the **Select SQL** area, enter the SQL information. Use {WINDOW_START} and {WINDOW_END} to apply the quantum defined at the process level to the generated SQL. {WINDOW_END} can optionally include a date-based column name in the form {WINDOW_END:dateColumnToTest} that, if used, is tested at the end of processing to determine the max date processed, and is used as the next run's {WINDOW_START}. If not used, the next {WINDOW_START} will be the current run's {WINDOW_END}.
3. In the **Post-select SQL** area, optionally, enter SQL to execute after the select that will be part of the same transaction (data cleanup from the pre-select SQL, for example). Use {WINDOW_START} and {WINDOW_END} to apply the quantum defined at the process level to the generated SQL. {WINDOW_END} can optionally include a date-based column name in the form {WINDOW_END:dateColumnToTest} that, if used, is tested at the end of processing to determine the max date processed, and is used as the next run's {WINDOW_START}. If not used, the next {WINDOW_START} will be the current run's {WINDOW_END}. For items 1-3 above, you can click the "Click here to edit in a larger window" link to open a larger text editing area.
4. In the **Required column(s)** drop-down, optionally enter the required column or columns. If you enter more than one column, they should be comma-separated. Specify ***true*** to import only required columns, or ***false*** to import all columns in the CSV file.
5. In the **Output file name** field, enter the output file name for the file being output. You can also include the "{TIMESTAMP}" parameter in the name to insert a time stamp. For example: agentRosterFile-{TIMESTAMP}.csv. If you leave this field blank, it is automatically generated.
6. In the **Default separator character** field, enter the character used to separate delimited column values in the output CSV file. The default is ***,*** (comma).

7. In the **Default quote character** field, enter the character used to quote column values in the output CSV file. The default is ***"*** (double quote).
8. In the **Default escape character** field, enter the character used to escape column values in the output CSV file. The default is ****** (backslash).
9. In the **Skip bad CSVs and continue** field:
   - When you select the ***true*** option, the system skips the bad CSV files with formatting errors and logs the error. The processing is continued with the next CSV files. Choose this option to maintain continuous processing when some files may contain errors.
   - When you select the ***false*** (default) option, the system stops processing when it finds a bad CSV file with formatting errors and reports the error. The processing is stopped until the error is fixed manually. Choose this option to ensure immediate attention to formatting issues.
10. In the **Default line end character** field, enter the character used to terminate a line in the output CSV file. The default is ***\n***.
11. In the **Replace new lines** drop-down, specify if you want to replace new line characters with the selected character. The default value is "***Do not replace***".
12. In the **Generate header-only CSV on no result** drop-down, select "***true***" to generate a header-only CSV if there is no result.
13. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.
14. From the **Terminal action** dropdown, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
15. In the **Disable action** field, specify ***true*** if you want to prevent this action from processing any files during its run. Leave the default value ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
16. Click **Submit**. A temporary output file with the extension ***.CSV.NXTMP*** is created first to prevent other processes from accessing incomplete CSV files. After some time, when the file is complete, the action renames the output file to .***CSV***.

Duplicate File Detector Action

1. In the **Process files from metadata** drop-down, specify if you want to process files from metadata. When you select ***true*** the application processes files from the process groups existing working table. When you specify false, the application processes files from attached input resources. Processing from metadata reduces I/O and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **File Filter** field, enter the file filter type, for example CSV. Java-style regular expressions can also be used if prepended with "regex". Note that when processing files from resources, this acts as a filter, but when processing from metadata, the metadata's file key is inserted in any wildcards, expanding the processing of the one file to all listed variants. For example: (*,*.xml' would process both a file matching the working tale file key '1234567.wav' and its '1234567.wav.xml' counterpart.
3. In the **Including File Name to De-Dupe** drop-down, specify how to de-dupe files. Select ***false*** to de-dupe files based on content only, without considering their names or ***true*** to de-dupe files based on a combination of their names and contents. The default value is ***true***.
4. In the **Search directories recursively** drop-down, specify if you want the application to search the directories recursively (if the files are not being moved based on metadata).
5. In the **Apply process quantum** drop-down, specify if you want to apply the quantum that is defined at the process level to this action. If used this will control the time window of files processed. The default value is ***false***.
6. In the **Apply maximum discovery bound to quantum** drop-down:
   - If you set to ***true***: After the quantum is applied, the date of the last file processed will be used as the start of the next quantum window.
   - If set to ***false***: The next quantum window will start immediately after the end of the current window.
7. In the **Delete Duplicate** drop-down, select ***true*** to delete all the files previously processed, or false to leave them in the input resource.
8. In the **Failure threshold** field, specify the number of failures that will cause processing to stop. ***0*** (default) indicates that the process will never fail.

9. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.
10. From the **Default input disposition handling** dropdown list, select one of the following:
    - ***AND***: Allow files to flow through this step that have passed all of the immediate inputs (default).
    - ***OR***: Allow files to flow through this step that have passed any of the immediate inputs.
11. In the **Input Disposition(s)** field, specify how input dispositions should be handled. The default is empty, to automatically chain. Enter ***none*** to disconnect the action from dependencies to prior actions, or enter a comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Use a hash sign instead of a comma to create an OR condition.
12. In the **Thread Count** field, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the **Threads per action** setting at the server level.
13. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. ***0*** (default) indicates that there is no limit.
14. From the **Terminal action** dropdown, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
15. In the **Disable action** field, specify ***true*** if you want to prevent this action from processing any files during its run. Leave the default value ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
16. Click **Submit**.

Duplicate Spotting Action

1. In the **Process files from metadata** drop-down, specify if you want to process files from metadata. When you select "***true***" the application processes files from the process groups existing working table. When you specify ***false***, the application processes files from attached input resources. Processing from metadata reduces input/output and file system scans when some previous action needs to have processed something from input resources first.
2. In the **File filter** field, enter the file filter type. For example: CSV.
3. In the **Maximum hours of media stored in catalog** drop-down, specify the maximum hours of media stored in the catalog. We recommend ***100 hours***.
4. In the **Commit interval** drop-down, specify the interval to commit the files to the server. Use ***0*** or ***1*** to commit after each file. The default value is ***100***.
5. In the **Sensitivity** drop-down, specify the sensitivity level for finding duplication. The valid values are ***1-11***. We recommend a setting of ***6***.
6. In the **Failure threshold** field, specify the number of failures that will cause processing to stop. ***0*** (default) indicates that the process will never fail.
7. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.
8. From the **Default input disposition handling** dropdown list, select one of the following:
   - ***AND***: Allow files to flow through this step that have passed all of the immediate inputs (default).
   - ***OR***: Allow files to flow through this step that have passed any of the immediate inputs.
9. In the **Input Disposition(s)** field, specify how input dispositions should be handled. The default is empty, to automatically chain. Enter ***none*** to disconnect the action from dependencies to prior actions, or enter a comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Use a hash sign instead of a comma to create an OR condition.
10. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. ***0*** (default) indicates that there is no limit.

11. From the **Terminal action** dropdown, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
12. In the **Disable action** field, specify ***true*** if you want to prevent this action from processing any files during its run. Leave the default value ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
13. Click **Submit**.

ElevateAI Near Real-Time Retriever Action

1. In the **Batch Size,** enter the number of transcripts to retrieve and process in each batch from the Artemis message queue at a time. The default value is 100.
2. In the **Agent Participant**, specify which participant is the agent. The default value is ***participantOne***.
   Common values are ***participantOne*** or ***participantTwo'.***
3. In the **Failure threshold**, specify the number of failures that cause processing to stop. ***0*** (default) indicates that the process will never fail.
4. In the **Punctuated transcript property name**, specify the Punctuated transcript JSON property name. The default value is ***punctuatedTranscript***.
5. In the **Summary property name**, specify the JSON property that contains the AI generated summary of the transcript of agent-customer conversation. This summary is used to populate the Autosummary column in the working table and is passed to the Interaction Analytics database. The default value is ***summary***.
6. In the **Interaction ID property name,** specify the JSON property that uniquely identifies each interaction. The default value is ***InteractionID***.
7. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.
8. In the **Default input disposition handling** drop-down list, select one of the following:

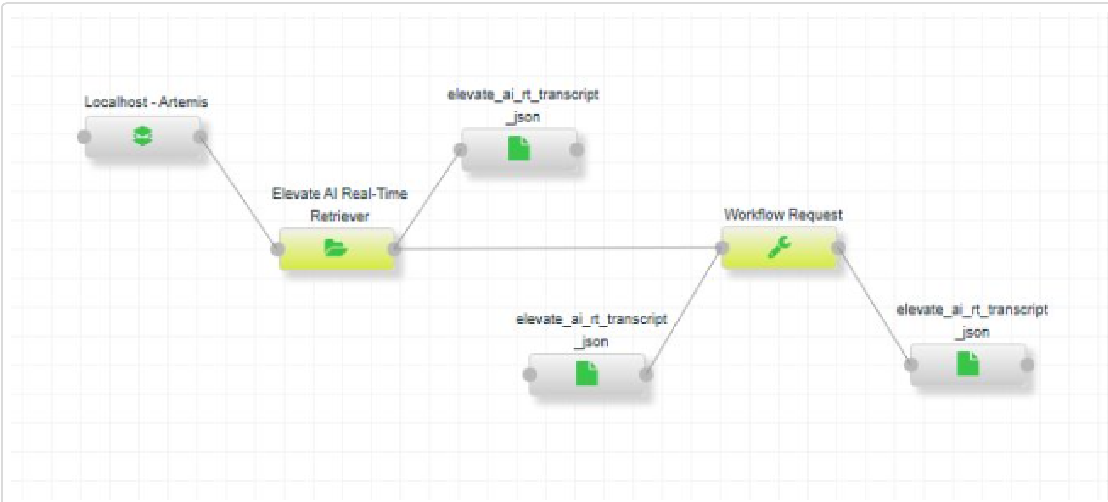
- ***AND*** - Allow files to flow through this step that have passed all the immediate inputs.
- ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.

9. In the **Input Disposition(s)**, override the default input disposition handling of the action by entering a comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
10. In the **Thread Count** , enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
11. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
12. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
13. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
14. Click **Submit**.

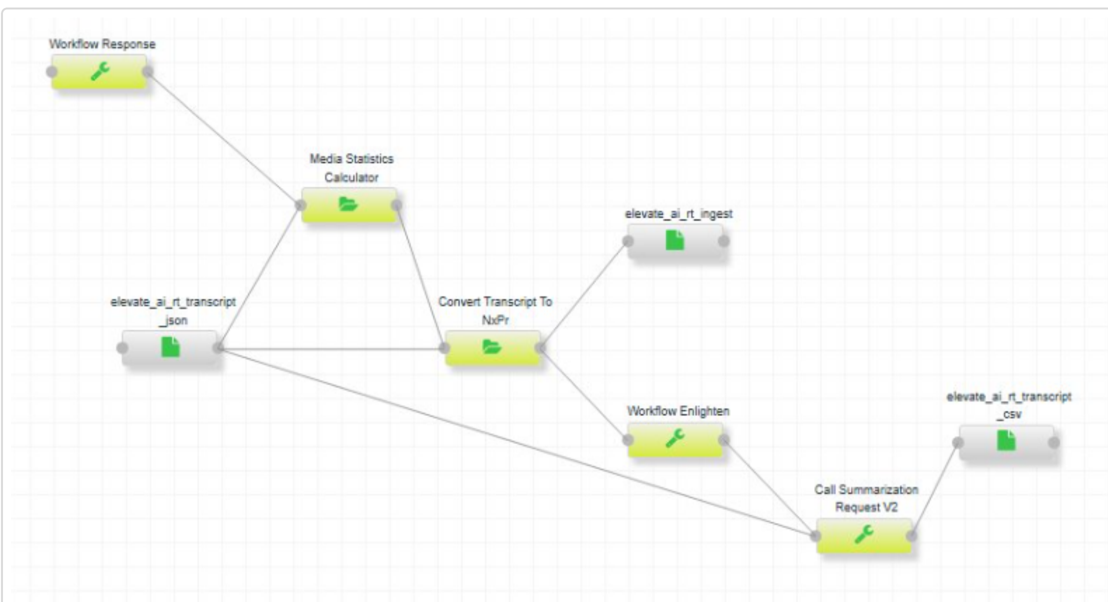
The ElevateAI Near Real-Time Retriever action is used in conjunction with two processes. These two processes run in separate process groups.

- The first process is called the Real-Time process and is comprised of the following three smaller processes:

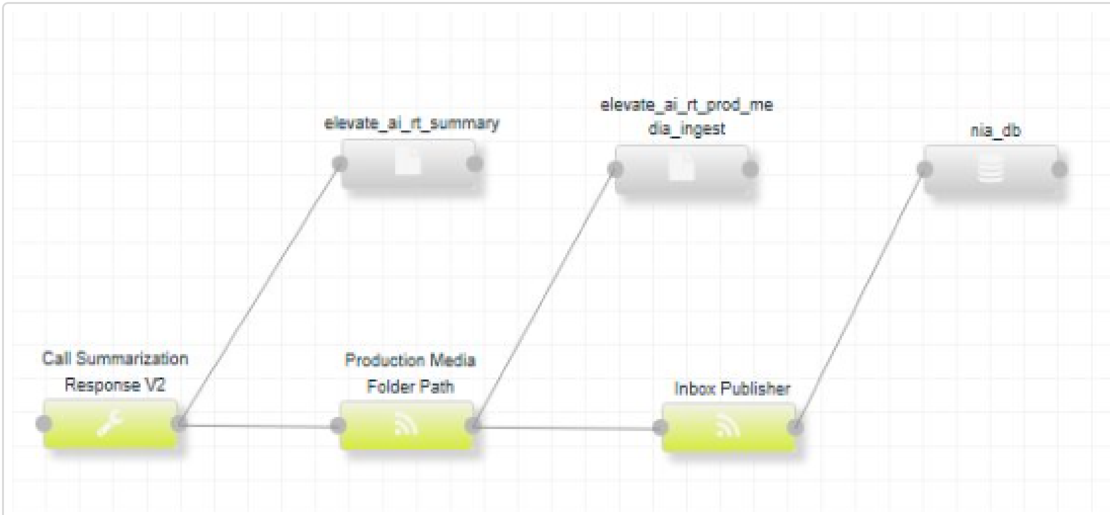
- ElevateAI - RT - Template – 1: Retrieve transcript from Artemis, convert and write transcript to file system, Workflow Request.



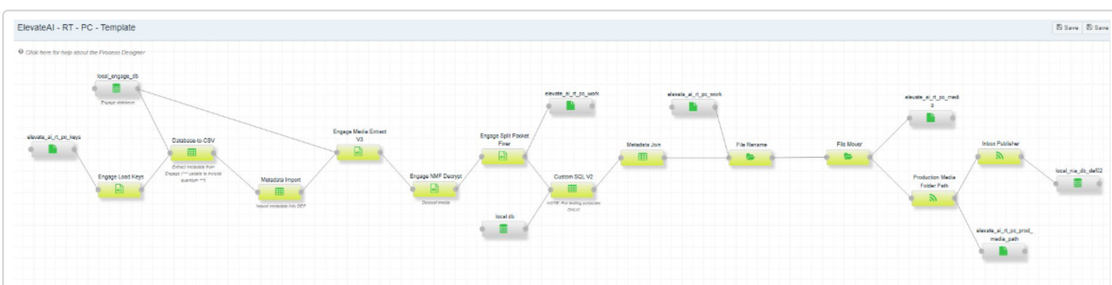
- ElevateAI - RT - Template – 2: Workflow response, Media Statistics Calculator, Convert Transcript to NxPr, Workflow Enlighten, Call Summarization Request



- ElevateAI - RT - Template – 3: Call Summarization Response, Production Media Folder Path, Inbox Publisher.



- The second process is called the Post-Call process: Process name: ElevateAI - RT - PC - Template
  - Extract metadata from Engage, Production Media Folder Path, and Inbox Publisher to Interaction Analytics.
  - The only difference is the updating of the FileName in the working table to match up/line up with the FileName in the Real Time process for a specific NMF file.



Elevate AI Response Action

1. In the **ElevateAI API Key** drop-down, specify if you want to process files from metadata. When you select ***true*** the application processes files from the process groups existing working table. When you specify false, the application processes files from attached input resources. Processing from metadata reduces I/O and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **ElevateAI Check Processing Status URL** field, enter the file filter type, for example CSV, in the File Filter field. Java-style regular expressions can also be used if prefix with "regex". Note that when processing files from resources, this acts as a filter, but when processing from metadata, the file key of the metadata is inserted in any wild-cards, expanding the processing of the one file to all listed variants. For example: (*,*.xml' would process both a file matching the working tale file key '1234567.wav' and its '1234567.wav.xml' counterpart.
3. In the **Retrieve ElevateAI Transcript?** drop-down, select ***true*** if you want to specify the ElevateAI API key.
4. In the **ElevateAI Transcript URL**, specify the ElevateAI API declare interaction payload URL. If this is true, Enlighten scores are calculated. You can see these scores in the ***transcript.json*** file.
5. In the **Retrieve ElevateAI Punctuated Transcript** drop-down, select ***true*** if you want to specify the URL for ElevateAI API declare interaction.
6. In the **ElevateAI Punctuated Transcript URL**, specify the URL for ElevateAI API retrieve punctuated transcript. If this is true, Enlighten scores are calculated. You can see these scores in the ***punctuated_transcript.json*** file.
7. In the **Retrieve ElevateAI CX AI metadata?** drop-down, select ***true*** if you want CX AI metadata to be retrieved.
8. In the **ElevateAI CX AI Metadata URL**, specify the URL for ElevateAI API retrieve CX AI metadata. Defaults to "https://api.elevateai.com/v1/interactions/{interactionIdentifier}/ai". If this is true, Enlighten scores are calculated. You can see these scores in the ***cxai.json*** file.
9. In the **Retrieve ElevateAI CX AI metadata to JSON file?** drop-down, specify ***true***, if you want to retrieve CX AI metadata to a JSON file written to the output resource.

10. In the **Retrieve ElevateAI Retrieve CX AI metadata To Working Table?** drop-down, specify ***true*** if you want to retrieve CX AI metadata to the DEF working table for the specified process.
11. In the **Failure Threshold** field, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
12. In the **Score-to-Model Mapping** drop-down, enter a mapping of expected scores to the names of model holding identifiers (GUID) for those scores. You can optionally click the **Pencil** icon to the right of this field to open the Score-to-Model Mapping window where you can define model name, model GUID, and model thresholds. This adds score names that will be ingested to the working table and the associated model GUID that will be matched to the results back from the Enlighten process and ingested to the working table.
13. In the **ElevateAI Model Location** drop-down, specify the path of the workflow model location.
14. In the **Validate Model Name** drop-down, set to ***true*** to validate Enlighten model name.
15. In the **Generate Auto Summarization tag** drop-down, set to ***true*** to generate auto-summarization tag that includes Enlighten model score index.
16. In the **Success Status List**, enter the list of ElevateAI success status responses returned by the ElevateAI Check Processing Status API separated by comma.
17. In the **Pending Status List**, specify the list of ElevateAI pending status responses returned by the ElevateAI Check Processing Status API separated by comma.
18. In the **Failure Status List**, specify the list of ElevateAI failure status responses returned by the ElevateAI Check Processing Status API by comma.
19. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.
20. In the **Default input disposition handling** drop-down list, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
    - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.

21. In the **Input Disposition(s)**, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
22. In the **Thread Count** , enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
23. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
24. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
25. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
26. Click **Submit**.

Elevate AI Submit Action

1. In the **Process files from metadata** drop-down, specify if you want to process files from metadata. When you select ***true*** the application processes files from the process groups existing working table. When you specify false, the application processes files from attached input resources. Processing from metadata reduces I/O and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.

2. In the **File filter** field, enter the file filter type, for example CSV, in the File Filter field. Java-style regular expressions can also be used if prefix with "regex". Note that when processing files from resources, this acts as a filter, but when processing from metadata, the file key of the metadata is inserted in any wild-cards, expanding the processing of the one file to all listed variants. For example: (*,*.xml' would process both a file matching the working tale file key '1234567.wav' and its '1234567.wav.xml' counterpart.
3. In the **ElevateAI API Key**, specify the ElevateAI API key.
4. In the **ElevateAI Declare Interaction Payload**, specify the ElevateAI API declare interaction payload.
5. In the **ElevateAI Declare Interaction URL**, specify the URL for ElevateAI API declare interaction.
6. In the **ElevateAI Upload Media URL**, specify the URL for ElevateAI API upload media.
7. In the **Failure Threshold** field, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
8. In the **Default HTTP Client Timeout (in seconds)**, specify the Apache HTTP client default timeout value (in seconds). The defaults timeout value is 60 seconds.
9. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.
10. In the **Default input disposition handling** drop-down list, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
    - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
11. In the **Input Disposition(s)**, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
12. In the **Thread Count** , enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.

13. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
14. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
15. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
16. Click **Submit**.

EML Parse Text Action

1. In the **Process files from metadata** drop-down, specify if you want to process files from metadata. When you select "***true***" the application processes files from the process groups existing working table. When you specify ***false***, the application processes files from attached input resources. Processing from metadata reduces input/output and file system scans when some previous action needs to have processed something from input resources first.
2. In the **File filter** field, enter the file filter type. For example: .EML.
3. In the **Save attachments** drop-down, specify if you want to save email attachments.
4. In the **Attachment types** field, click in the Attachment Types field and specify the types of attachments that you want to open and extract. The available options are ***Email***, ***HTML***, ***JSON***, ***Microsoft Office Documents***, ***PDF***, ***Plain Text***, ***Rich Text Format***, ***SQL***, ***Visio Documents***, ***Zip***, and ***Other***.
5. In the **Minimum image size (KBs)** drop-down, specify the minimum image size. Any attached images smaller than this are ignored.
6. In the **Remove embedded attachments** drop-down, specify if you want to remove embedded attachments.
7. In the **Append parse text label to message body** drop-down, specify if you want to append a parse text label to the message body.
8. In the **Delete files after processing** drop-down, specify if you want the application to delete files after processing. When you select "***true***" the application deletes the .eml files after processing.
9. In the **Failure threshold** field, specify the number of failures that will cause processing to stop. ***0*** (default) indicates that the process will never fail.
10. In the **Attachment write Limit (KBs)** drop-down, specify the write limit (specified in KBs for example: 1000). The default value is ***-1*** (no write limit).
11. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.
12. From the **Default input disposition handling** dropdown list, select one of the following:
    - ***AND***: Allow files to flow through this step that have passed all of the immediate inputs (default).

- ***OR***: Allow files to flow through this step that have passed any of the immediate inputs.

13. In the **Input Disposition(s)** field, specify how input dispositions should be handled. The default is empty, to automatically chain. Enter ***none*** to disconnect the action from dependencies to prior actions, or enter a comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Use a hash sign instead of a comma to create an OR condition.
14. In the **Thread Count** field, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the **Threads per action** setting at the server level.
15. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. ***0*** (default) indicates that there is no limit.
16. From the **Terminal action** dropdown, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
17. In the **Disable action** field, specify ***true*** if you want to prevent this action from processing any files during its run. Leave the default value ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
18. Click **Submit**.
    The email media metadata is captured as a First-Class NIA field called "EmailMetadata" in the DEF working table through the Inbox Publisher action and sent to NIA through the NexidiaESI SourceMediaInbox table.

EML Text Remove Action

1. In the **Process files from metadata** field, specify if you want to process files from metadata. When you select "***true***" the application processes files from the process groups existing working table. When you specify ***false***, the application processes files from attached input resources. Processing from metadata reduces I/O and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **File filter** field, enter the file filter type. For example: .EML.
3. In the **Delete files after processing** drop-down, specify if you want the application to delete files after processing. When you select "***true***" the application deletes the .eml files after processing.
4. In the **Remove text/disclaimer** field, enter a list of one or more text strings that should be removed from the email body. Regular expressions are allowed.
5. In the **Failure threshold** field, specify the number of failures that will cause processing to stop. ***0*** (default) indicates that the process will never fail.
6. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.
7. From the **Default input disposition handling** dropdown list, select one of the following:
   - ***AND***: Allow files to flow through this step that have passed all of the immediate inputs (default).
   - ***OR***: Allow files to flow through this step that have passed any of the immediate inputs.
8. In the **Input Disposition(s)** field, specify how input dispositions should be handled. The default is empty, to automatically chain. Enter ***none*** to disconnect the action from dependencies to prior actions, or enter a comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Use a hash sign instead of a comma to create an OR condition.
9. In the **Thread Count** field, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the **Threads per action** setting at the server level.

10. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. ***0*** (default) indicates that there is no limit.
11. From the **Terminal action** dropdown, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
12. In the **Disable action** field, specify ***true*** if you want to prevent this action from processing any files during its run. Leave the default value ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
13. Click **Submit**.

Engage Audit Action

1. In the **Engage Audit** drop-down, specify the engage audit method. The valid options are:
   - ***Direct to DB***
   - ***Direct to CSV***
   - ***CSV to DB***
2. In the **Default quote character** field, enter the character used to quote column values in the output CSV file. The default is **"** (double quote).
3. In the **Default escape character** field, enter the character used to escape column values in the output CSV file. The default is **** (backslash).
4. From the **Terminal action** dropdown, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
5. In the **Disable action** field, specify ***true*** if you want to prevent this action from processing any files during its run. Leave the default value ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
6. Click **Submit**.

Engage Desktop Tags Action

1. In the **Failure Threshold** drop-down, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
2. In the **Internal working batch size** drop-down, specify the internal working batch size. This is the number of desktop tags to resolve in a batch when the total number of files to work on exceeds it. The default value is ***10000***.
3. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.
4. From the **Default input disposition handling** dropdown list, select one of the following:
   - ***AND***: Allow files to flow through this step that have passed all of the immediate inputs (default).
   - ***OR***: Allow files to flow through this step that have passed any of the immediate inputs.
5. In the **Input Disposition(s)** field, specify how input dispositions should be handled. The default is empty, to automatically chain. Enter ***none*** to disconnect the action from dependencies to prior actions, or enter a comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Use a hash sign instead of a comma to create an OR condition.
6. In the **Thread Count** field, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the **Threads per action** setting at the server level.
7. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. ***0*** (default) indicates that there is no limit.
8. From the **Terminal action** dropdown, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.

9. In the **Disable action** field, specify ***true*** if you want to prevent this action from processing any files during its run. Leave the default value ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
10. Click **Submit**.

Engage Key Extraction Action

1. In the **NICE crypto database connection string** field, specify the SQL Server connection string in the following format: ***jdbc:sqlserver://database.server.fqdn;databaseName=nice_crypto;integratedSecurity=true***. Note that the "***database.server.fqdn*** " portion in the default must be changed to match the SQL Server host. You can optionally click to the right of this field to open the JDBC Connection Wizard and specify the following:
    a. In the **Driver Type** drop-down, select one of these driver types: ***jTDS***, ***MySQL***, ***PostgreSQL***, or ***SQL*** Server and click Next.
    b. In the **Host Name** field, enter the host name of the database server (without the leading //) and click **Next**.
    c. In the **Instance Name** field, enter an optional instance name and click **Next**.
    d. In the **Port Number** field, enter an optional port number and click **Next**.
    e. In the **Database Name** field, enter a database name and click **Next**.
    f. Check the **Integrated Security** box if the database connection is using integrated security.
    g. Enter the **Database Name** and **Database Password** and click **Finish**.
2. In the **Keystore location** field, enter the absolute path to the keystore that holds the key that is used to decrypt the symmetric key in the Keystore location field. The symmetric key is found in the input resource.
3. In the **Keystore password** field, specify a password for the keystore in the Keystore password field.
4. In the **Keystore alias** field, specify the alias of the key in the keystore that is used to decrypt the symmetric key in the Keystore alias field.
5. In the **Logger Identifiers** field, enter a comma-separated list of logger identifiers to use when extracting keys.
6. In the **Failure threshold** field, specify the number of failures that will cause processing to stop. ***0*** (default) indicates that the process will never fail.

7. In the **batch execute size** drop-down, specify the maximum number of GUIDs that will be passed to the spDecryptData stored procedure at a time. Note that all GUIDs retrieved in the quantum window will still be processed. This option controls how those GUIDs are partitioned internally for database performance.
8. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.
9. From the **Default input disposition handling** dropdown list, select one of the following:
   - ***AND***: Allow files to flow through this step that have passed all of the immediate inputs (default).
   - ***OR***: Allow files to flow through this step that have passed any of the immediate inputs.
10. In the **Input Disposition(s)** field, specify how input dispositions should be handled. The default is empty, to automatically chain. Enter ***none*** to disconnect the action from dependencies to prior actions, or enter a comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Use a hash sign instead of a comma to create an OR condition.
11. In the **Thread Count** field, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the **Threads per action** setting at the server level.
12. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. ***0*** (default) indicates that there is no limit.
13. From the **Terminal action** dropdown, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
14. In the **Disable action** field, specify ***true*** if you want to prevent this action from processing any files during its run. Leave the default value ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
15. Click **Submit**.

Engage Load Keys Action

1. In the **Process files from metadata** drop-down, specify if you want to process files from metadata. When you select "***true***" the application processes files from the process groups existing working table. When you specify ***false***, the application processes files from attached input resources. Processing from metadata reduces input/output and file system scans when some previous action needs to have processed something from input resources first.
2. In the **Keystore location** field, enter the absolute path to the keystore in the Keystore location field. If the keystore does not exist, it will be created.
3. In the **Keystore password** field, specify a password for the keystore in the Keystore password field.
4. In the **Delete encrypted CSV files after processing** drop-down, specify if you want to delete the encrypted csv files after processing.
5. In the **Delete keys older than # days** drop-down, specify the number of days to have keys older than aged out of the keystore at the start of each run of the action in the Delete keystore keys older than # days field. Since the keystore does not track times, this option depends on the key entry dates in the working table for this process. The default value is zero (don't age out old keys).
6. In the **Failure threshold** field, specify the number of failures that will cause processing to stop. ***0*** (default) indicates that the process will never fail.
7. In the **Crypto key alias** field, change the value to rotate the key being used internally by DEF to encrypt/decrypt the keys pulled from the Key Extract CSV files. This key will be created automatically if it does not already exist.
8. In the **Batch execute size** drop-down, specify the batch execute size. When processing large files, this indicates how often to execute batches against the database.
9. In the **Commit after each batch execution** drop-down, specify if you want to commit after each batch execution. Set to "***true***" to commit to the database after each batch is executed; leave "***false***" to commit only after the CSV file has been fully processed. The trade-off here is: "***true***" will take the transaction load off of the database if it becomes a problem, but will break intended transactional integrity for processing of each incoming CSV file. "***False***" should be preferred unless the database simply can not handle the transactional load.

10. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.
11. From the **Default input disposition handling** dropdown list, select one of the following:
    - ***AND***: Allow files to flow through this step that have passed all of the immediate inputs (default).
    - ***OR***: Allow files to flow through this step that have passed any of the immediate inputs.
12. In the **Input Disposition(s)** field, specify how input dispositions should be handled. The default is empty, to automatically chain. Enter ***none*** to disconnect the action from dependencies to prior actions, or enter a comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Use a hash sign instead of a comma to create an OR condition.
13. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. ***0*** (default) indicates that there is no limit.
14. From the **Terminal action** dropdown, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
15. In the **Disable action** field, specify ***true*** if you want to prevent this action from processing any files during its run. Leave the default value ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
16. Click **Submit**.

Engage Media Extract V3 Action

1. In the **Use CTI data** field, set to ***true*** to attempt to use CIT data for stitching logic. If CTI data is not available for interactions even with this flag set, the action will fall back on non-CTI data. Set to ***false*** to not even attempt the CTI portion (in older systems where those tables aren't available for example, and the CTI table joins fail).
2. In the **Skip Invalid Calls** drop-down, set to ***true*** to skip any call that fails and continue processing the rest of the calls. Set to ***false*** to fail the process.
3. In the **Failure Threshold** drop-down , specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
4. In the **Packet size maximum (kB)** drop-down, adjust the packet size maximum in case of unexpected, but still reasonable, sizes. This guards against trying to process corrupted packets that declare much more and pose potential memory concerns. Packet sizes should not exceed ***1024k***.
5. In the **Extended Storage Type**, select the storage type. Available options are ***None***, ***AWS S3***, ***Centera***, or ***NFS***. Depending on your selection, the following fields display:
   a. If you select ***AWS S3***, **Get Files From ESM Only** displays. Select ***true*** if you want to get the files from ESM only.
   b. If you select ***Centera***, enter the following details in the fields:
      - **Centera pool address** - enter one or more Centera pool IP addresses in the Centera pool address field. Separate multiple pool IP addresses with commas (i.e. 168.159.214.27, 168.159.214.28).
      - **PEA file path** - enter a fully qualified path to an optional PEA file containing Centera authorization credentials (for example, C:\\Users\\Administrator\\NEAT\\etc\\c2armtesting.pea) in the PEA file path field. The PEA file contains center authorization credentials. The Centera file store is password protected, and the PEA file contains the credentials required to access the center file store.
      - **Centera clip ID column name** - enter the name of the column holding the unique clip ID for each file in the Centera clip ID column name field.

- **Centera tag name** - enter the name of the Centera tag name used to identify a file to be extracted (for example, Encrypted) in the Centera tag name. You can apply or tag files with whatever random information you would like. The Centera tag name is used to identify files that have been archived.
- **Number of retries on failure** - select the number of tries the API runs when Centera pool fails.
- **Number of Seconds To Wait Between Each Retry** - select the number of seconds the API waits between each try when Centera pool fails.
- **Get files from ESM Only** - select ***true*** if you want to get the files from ESM only.

c. If you select ***NFS***, **Get Files From ESM Only** displays. Select ***true*** if you want to get the files from ESM only.

6. In the **Include screens**, select:
   - '***true***' to include screens in the extraction
     When you select '***true***', you get additional configuration option, Include H264 Screen. Set '***true***' for including H264 screen else set 'false'.
   - '***false***' to gather only audio
7. In the **Trace packet filtering** drop-down, set to ***true*** to configure output of a per-NMF file under the "def_stitch" subfolder of the default system temp directory that will provide debug information for packets being filtered. Be sure to return to ***false*** after gathering debug information, as this should not be configured by default.
8. In the **Recording duration minimum(s)** drop-down, specify the minimum duration of each individual recording comprising an interaction. The default value is ***1***.
9. In the **Interaction duration minimum(s)** drop-down, specify the minimum duration of an interaction. The default value is ***2***.
10. In the **Set Channel0** field, select ***true*** to display the Channel0 field and set it to agent or customer. Select "***false***" to set Channel0 to unspecified.
11. In the **Flip Channels for Channel0 Agent**t field, set to ***true*** to flip agent and customer stereo channels when call direction from Engage is either 2 or 3. Set to ***false*** to disable flipping.

12. In the **Channel0 For Internal Calls** drop-down, select whose media you want to extract. Available options are ***Customer*** or ***Agent***.
13. In the **Force Mono (Testing Only)** drop-down, set to ***true*** to force the channel to Mono. This option is for troubleshooting only.
14. In the **Allowed Packet Types Audio** field, enter the set of packet/sub-packet combinations that will be stitched. Any packet/sub-packet combinations not listed in this configuration will not be stitched. At least one packet type 4 is required in the combinations of packet/sub-packet. Enter in the format of "{pktType}/{subtype},{pktType2}/{subtype2},...,{pktTypeN}/{subtypeN}".
15. In the **Allowed Packet Types Screen**, enter the combination of packet type and sub-packet type that will be stitched for screen. Any packet/sub-packet combinations not listed in this configuration will not be stitched. At least one packet type 4 is required in the combinations of packet/sub-packet. Should be entered in the format of "{pktType}/{subtype},{pktType2}/{subtype2},...,{pktTypeN}/{subtypeN}"
16. In the **Add Audio Packets To Screen** drop-down, select ***true*** to add the audio packets to screen file.
17. In the **Remove Duplicate Storage Center Paths**, set to ***false*** to point to the first storage center path from the list of available paths. It doesn't check if the pointed path is accessible or not. Set to ***true*** to point to the first accessible storage center path from the list of available paths. This is useful when one of the storage center paths is inaccessible. The media is available in other accessible storage center paths.
18. In the **Remove Duplicate Internal Recordings**, set to ***true*** to de-dupe duplicate audio packets, or ***false*** to leave them as-is. Internal calls between agents may contain two sets of recordings that have duplicate audio packets.
19. In the **Reconciliation Identifiers**, enter the name of the action if there is a reporting action in the process group.
20. In the **Default input disposition handling** drop-down, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
    - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.

21. Optionally enter the **Input dispositions**. The Data Exchange Framework automatically chains action component dispositions together as javascript:void()part of its processing, so this field is usually left blank. If you are using split processes, enter a value in this field.
22. From the **Thread Count** drop-down, select a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
23. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
24. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
25. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
26. Click **Submit**.

Engage NMF Decrypt Action

1. In the **Process files from metadata**, specify if you want to process files from metadata. When you select "***true***" the application processes files from the process groups existing working table. When you specify false, the application processes files from attached input resources. Processing from metadata reduces I/O and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **File filter**, enter the file filter type. For example: .NMF.
3. In the **Keystore location**, enter the absolute path to the keystore containing keys loaded from the recorder by the Load Keys action.
4. In the **Keystore password**, specify a password for the keystore in the Keystore password field.
5. In the **Delete encrypted NMF files after processing** drop-down, specify if you want to delete encrypted NMF files after processing.
6. In the **Flatten output folder structure**, specify if you want to flatten the output structure of the generated files. Set this option to ***true*** to flatten the output structure of the generated files if the working table key is a nested folder structure; and ***false*** to retain any working table key structure in folder output.
7. In the **Failure Threshold**, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
8. In the **Reconciliation Identifiers**, enter the name of the action if there is a reporting action in the process group.
9. In the **Default input disposition handling** drop-down list, select one of the following:
   - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
   - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.

10. In the **Input Disposition(s)**, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type '***none***' to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
11. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
12. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
13. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
14. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave "***false***" to allow processing as usual. This option is useful for troubleshooting the process.
15. Click **Submit**.

Engage NMF MML Decrypt Action

1. In the **Process files from metadata** field, specify if you want to process files from metadata. When you select "***true***" the application processes files from the process groups existing working table. When you specify false, the application processes files from attached input resources. Processing from metadata reduces I/O and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **File filter** field, enter the file filter type. For example: .CSV.
3. In the **Keystore location** field, enter the absolute path to the keystore containing keys loaded from the recorder by the Load Keys action.
4. In the **Keystore password** field, specify a password for the keystore in the Keystore password field.
5. In the **Delete encrypted NMF files after processing** drop-down, specify if you want to delete encrypted NMF files after processing.
6. In the **Failure Threshold** field, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
7. In the **Packet size maximum (kb)** field, specify the packet size. If this action encounters any packets outside of the range specified, the file will fail. The default value is ***16***, and the minimum value is ***4***.
8. In the **Reconciliation Identifiers**, enter the name of the action if there is a reporting action in the process group.
9. In the **Default input disposition handling** drop-down list, select one of the following:
   - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
   - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.

10. In the **Input Disposition(s)** field, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type '***none***' to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
11. In the **Thread Count** field, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
12. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
13. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
14. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave "***false***" to allow processing as usual. This option is useful for troubleshooting the process.
15. Click **Submit**.

Engage Split Packet Fixer Action

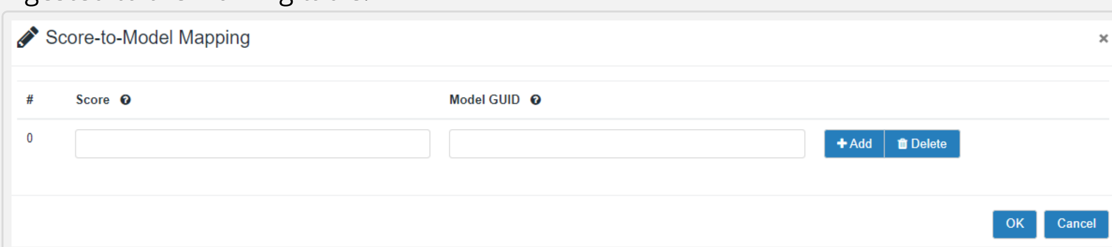
1. In the **File filter field**, set '***.xml**', '*.***xml,*.csv**', etc. Java-style regular expressions can also be used if prepended with 'regex:'. Note that when processing files from resources, this acts as a filter, but when processing from metadata, the metadata's file key is inserted into any wildcards, expanding the processing of the one file to all listed variants ('*,*.xml', e.g., would process both a file matching the working table file key '1234567.wav' and its '1234567.wav.xml' counterpart).
2. In the **Delete NMF files after processing** field , set ***true*** to delete NMF files after processing. Default value is ***false***.
   In the **Failure threshold** field, specify the number of failures that will cause processing to stop. ***0*** (default) indicates that the process will never fail.
3. In the **Packet size maximum (kB)** field, enter the packet size. Packet sizes are expected not to exceed ***256k*** for NMF files, but the maximum can be adjusted here in case of unexpected, but still reasonable, sizes. This value guards the processing of the corrupted packets that declare much more and pose potential memory concerns. Default value is set to ***1024***.
4. In the **Allowed Packet Types Audio**, enter the combinations of packet/sub-packet that will be stitched. Any packet/sub-packet combinations not listed in this configuration will not be stitched. At least one packet type 4 is required in the combinations of packet/sub-packet. Should be entered in the format of "***{pktType}/{subtype}***,***{pktType2}/{subtype2}***, ***{pktTypeN}/{subtypeN}***". Default value is ***4/3***.
5. In the **Trace packet filtering** field, select '***true***' to configure output of a per-NMF file under the 'def_stitch' subfolder of the default system temp directory that will provide debug information for packets being filtered. NOTE that using this option will throttle the action to a single thread, so performance will be significantly worse. Be sure to return to '***false***' after gathering debug information, as this shouldn't be configured by default. Default value is ***false***.
6. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.
7. From the **Default input disposition handling** dropdown list, select one of the following:
   - ***AND***: Allow files to flow through this step that have passed all of the immediate inputs (default).

- ***OR***: Allow files to flow through this step that have passed any of the immediate inputs.

8. In the **Input Disposition(s)** field, specify how input dispositions should be handled. The default is empty, to automatically chain. Enter ***none*** to disconnect the action from dependencies to prior actions, or enter a comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Use a hash sign instead of a comma to create an OR condition.
9. In the **Thread Count** field, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the **Threads per action** setting at the server level.
10. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. ***0*** (default) indicates that there is no limit.
11. From the **Terminal action** dropdown, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
12. In the **Disable action** field, specify ***true*** if you want to prevent this action from processing any files during its run. Leave the default value ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
13. Click **Submit**.

Enlighten Response Action

1. In the **RabbitMQ Host Name**, enter the RabbitMQ host name for login.
2. In the **RabbitMQ Port**, Enter the RabbitMQ port number for login. The default value is ***5672***.
3. In the **RabbitMQ User Name**, Enter the RabbitMQ user name for login.
4. In the **RabbitMQ Password**, Enter the RabbitMQ password.
5. In the **RabbitMQVirtual Host**, Enter the RabbitMQ virtual host for login.
6. In the **Score-to-Model Mapping** drop-down, enter a mapping of expected scores to the names of model holding identifiers (GUID) for those scores. You can optionally click the Pencil icon to the right of this field to open the Score-to-Model Mapping window where you can add score names that will be ingested to the working table and the associated model GUID that will be matched to the results back from the Enlighten process and ingested to the working table.



7. In the **Failure Threshold**, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
8. In the **Reconciliation Identifier**, enter the name of the action if there is a reporting action in the process group.
9. In the **Default input disposition handling** drop-down list, select one of the following:
   - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
   - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.

10. In the **Input Disposition(s)**, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type '***none***' to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
11. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
12. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
13. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave "***false***" to allow processing as usual. This option is useful for troubleshooting the process.
14. Click **Submit**.

Enlighten Submit Action

1. In the **Process files from metadata** field, specify if you want to process files from metadata. When you select "***true***" the application processes files from the process groups existing working table. When you specify false, the application processes files from attached input resources. Processing from metadata reduces I/O and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **file filter** field, enter the file filter type, for example CSV, in the File Filter field. Java-style regular expressions can also be used if prepended with "regex". Note that when processing files from resources, this acts as a filter, but when processing from metadata, the metadata's file key is inserted in any wildcards, expanding the processing of the one file to all listed variants. For example: (*,*.xml' would process both a file matching the working tale file key '1234567.wav' and its '1234567.wav.xml' counterpart.
3. In the **RabbitMQ** Host Name field, enter the RabbitMQ host name for login.
4. In the **RabbitMQ Port** field, Enter the RabbitMQ port number for login. The default value is ***5672***.
5. In the **RabbitMQ User Name** field, Enter the RabbitMQ user name for login.
6. In the **RabbitMQ Password** field, Enter the RabbitMQ password.
7. In the **RabbitMQVirtual Host** field, Enter the RabbitMQ virtual host for login.
8. In the **Workflow ID** field, enter the UUID of the model that Enlighten Grid should use in determining segments scores.
9. In the **Send Workbench Logs** drop-down, specify if you want to send Workbench logs to diagnose the enlighten issues for not being able to return any scores for any files.
10. In the **Failure Threshold** field, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
11. In the **Reconciliation Identifiers**, enter the name of the action if there is a reporting action in the process group.
12. In the **Default input disposition handling** drop-down list, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.

- ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.

13. In the **Input Disposition(s)** field, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type '***none***' to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
14. In the **Thread Count** field, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
15. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
16. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
17. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave "***false***" to allow processing as usual. This option is useful for troubleshooting the process.
18. Click **Submit**.

Enlighten Transcript Transform Action

1. In the **Process files from metadata** drop-down, specify if you want to process files from metadata. When you select ***true*** the application processes files from the process groups existing working table. When you specify false, the application processes files from attached input resources. Processing from metadata reduces I/O and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **File filter** field, enter the file filter type, for example CSV, in the File Filter field. Java-style regular expressions can also be used if prepended with "regex". Note that when processing files from resources, this acts as a filter, but when processing from metadata, the metadata's file key is inserted in any wildcards, expanding the processing of the one file to all listed variants. For example: (*,*.xml' would process both a file matching the working tale file key '1234567.wav' and its '1234567.wav.xml' counterpart.
3. In the **Force file encoding** field, enter the character encoding of the input file If you want to force file encoding. You will leave this field blank in most cases to auto-detect. If auto-detect isn't working, use a valid Java InputStream encoding.
4. In the **Delete transcripts after successfully processed** drop-down, specify if you want to delete transcripts after they have successfully processed. The default value is "true".
5. In the **Default separator character** field, specify the character used to separate delimited column values in the input CSV (for example **,**) in the Default input separator character field.
6. In the **Default quote character** field, enter the character used to quote column values in the output CSV file, for example ".
7. In the **Default escape character** field, enter the character used to escape column values in the output CSV file, for example \.
8. In the **Failure Threshold** field, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
9. In the **Transcript file locale** field, specify the transcript file locale. The default value is "***en_US***".
10. In the **Comment delimiter** field, enter the character used to denote a comment line in the transcript file. The default value is "**#**".

11. In the **Reconciliation Identifiers**, enter the name of the action if there is a reporting action in the process group.
12. In the **Default input disposition handling** drop-down list, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
    - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
13. In the **Input Disposition(s)**, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
14. In the **Thread Count** , enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
15. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
16. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
17. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
18. Click **Submit**.

Executable Wrapper Action

1. In the **Process files from metadata** drop-down, specify if you want to process files from metadata. When you select ***true*** the application processes files from the process groups existing working table. When you specify false, the application processes files from attached input resources. Processing from metadata reduces I/O and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **File filter** field, enter the file filter type. For example: CSV.
3. Enter the full path to the executable. This is the executable to be run.
4. In the **Parameters** field, optionally enter parameters. You can use the %FILE_BATCH% parameter which will be swapped out with an input file listing if one is there. If you used the optional %FILE BATCH% parameter, enter the number of files to pass to it at one time.
5. In the **Number of files to pass to %FILE BATCH% param in batch** drop-down, specify how many files to pass to the %FILE BATCH% parameter in each call. In most cases, this should be the default of "1". If your executable supports multiple file names that can be divided arbitrarily, this number can be increased, and that number of file names will be inserted into any of the available batching parameters, space-delimited.
6. In the **Check exit code for success/failure check** drop-down, specify if you want the application to check the exit code for success or failure.
7. In the **Success exit code** drop-down, specify the Success exit code. Generally, an exit code of "0" indicates success. If your application is different, you can change that value here.
8. Optionally, enter text that should match the output of the executable in the **Executable success response will match regex** field.
9. Optionally, enter text that should match the output of the executable in the **Executable failure response will match regex** field. You can use one or both of the executable response options. If used, a check will be made after each call to the executable to determine if it ran correctly or not.

10. Enter a value in the **Process time out in the Seconds** field to specify how long the application will wait for a process to finish before it terminates it. This typically means that the file being processed will be flagged as an error that counts toward the failure threshold, and processing will continue with the next file.
11. Specify if you want the application to check and set the dispositions (with resource inputs). This option has to do with how the action participates in the process flow: If an action checks and sets dispositions, then when it's looking at items to process, it checks the working table to see if any actions prior to it in the workflow (actions that also check and set dispositions) have processed the record, and it makes sure that it has not.
12. In the **Reconciliation Identifiers**, enter the name of the action if there is a reporting action in the process group.
13. In the **Failure Threshold** field, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
14. In the **Default input disposition handling** drop-down list, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
    - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
15. In the **Input Disposition(s)**, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
16. In the **Thread Count** , enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
17. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.

18. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
19. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
20. Click **Submit**.

ExternalMediaID Capture Action

1. In the **Reconciliation Identifiers**, enter the name of the action if there is a reporting action in the process group.
2. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
3. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave "***false***" to allow processing as usual. This option is useful for troubleshooting the process.
4. Click **Submit**.

File Discovery Action

1. In the **Process files from metadata** drop-down, specify if you want to process files from metadata. When you select ***true*** the application processes files from the process groups existing working table. When you specify false, the application processes files from attached input resources. Processing from metadata reduces I/O and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **File filter** field, enter the file filter type. For example: .XML.
3. In the **Search directories recursively** drop-down, specify if you want the application to search the directories recursively.
4. In the **Apply process quantum** drop-down, specify if you want to apply the process quantum (if the files are not being moved based on metadata). Quantum processing is configured in the Process view.
5. In the **Failure Threshold** field, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
6. In the **Reconciliation Identifiers**, enter the name of the action if there is a reporting action in the process group.
7. In the **Default input disposition handling** drop-down list, select one of the following:
   - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
   - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
8. In the **Input Disposition(s)**, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.

9. In the **Thread Count** , enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
10. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
11. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
12. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
13. Click **Submit**.

File Filter Action

1. In the **Process files from metadata** drop-down, specify if you want to process files from metadata. When you select "***true***" the application processes files from the process groups existing working table. When you specify ***false***, the application processes files from attached input resources. Processing from metadata reduces input/output and file system scans when some previous action needs to have processed something from input resources first.
2. In the **File filter** field, enter the file filter type. For example: CSV.
3. In the **Exclude files smaller than #KB** drop-down, select a value to indicate that files less than this size (in kilobytes) will be excluded from processing.
4. In the **Exclude files larger than #KB**, select a value to indicate that files larger than this size (in kilobytes) will be excluded from processing.
5. In the **Delete all files that don't pass the filter** drop-down, specify if you want the application to delete the files that do not pass the sizing filter specified.
6. In the **Failure threshold** field, specify the number of failures that will cause processing to stop. ***0*** (default) indicates that the process will never fail.
7. In the **Reconciliation Identifiers**, enter the name of the action if there is a reporting action in the process group.
8. From the **Default input disposition handling** dropdown list, select one of the following:
   - ***AND***: Allow files to flow through this step that have passed all of the immediate inputs (default).
   - ***OR***: Allow files to flow through this step that have passed any of the immediate inputs.
9. In the **Input Disposition(s)** field, specify how input dispositions should be handled. The default is empty, to automatically chain. Enter ***none*** to disconnect the action from dependencies to prior actions, or enter a comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Use a hash sign instead of a comma to create an OR condition.

10. In the **Thread Count** field, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the **Threads per action** setting at the server level.
11. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. ***0*** (default) indicates that there is no limit.
12. From the **Terminal action** dropdown, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
13. In the **Disable action** field, specify ***true*** if you want to prevent this action from processing any files during its run. Leave the default value ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
14. Click **Submit**.

File Maker Action

1. In the **File filter**, enter the file filter type. For example: CSV.
2. In the **Type of file to create** drop-down, select the type of file to create.
3. In the **Failure Threshold**, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
4. In the **Flatten output key** drop-down, specify if you want to flatten the output structure of the generated files. Set this option to ***true*** to flatten the output structure of the generated files if the working table key is a nested folder structure; and ***false*** to retain any working table key structure in folder output.
5. In the **Reconciliation Identifiers**, enter the name of the action if there is a reporting action in the process group.
6. In the **Default input disposition handling** drop-down list, select one of the following:
   - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
   - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
7. In the **Input Disposition(s)** field, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
8. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
9. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.

10. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
11. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
12. Click **Submit**.

File Mover Action

1. In the **Process files from metadata** drop-down, specify if you want to process files from metadata. Select ***false*** to process files from attached input resources or ***true*** to process them from this process groups existing working table. Processing from metadata reduces input/output and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **File filter** field, enter the file filter type. For example: CSV. You can use multiple file filter types for processing. You can retrieve media based on specific metadata for all storage types, such as File System, SFTP, and S3.
3. In the **File path column name** drop-down, enter the name of the metadata column in the process group working table that will be used to determine the files to move. Default is FileName. This field is visible if you have set the Process files from metadata parameter to True.
4. In the **Flatten output directory structure** drop-down, specify if you want the application to flatten the output directory structure (if searching inputs recursively). If you set this option to "***false***", the folder structure from the input is reproduced in the output resource. If you set this option to "true", all files from the input will get moved or copied directly into the output resource folder with no nested folder structure.
5. In the **Treat missing files as errors** drop-down, the default value is true to analyze and capture more information about why the data is missing. Change to false if missing files should be considered errors. Errors will be logged and tracked against the action's failure threshold.
6. In the **Overwrite existing output files** drop-down, specify if you want to overwrite existing output files that already exist in the output location.
7. In the **Fail if no files moved** drop-down, specify if this action should fail if no files are moved.
8. In the **Delete inputs after move** drop-down, specify if you want the application to delete the inputs after they are moved.
9. In the **Capture ExternalMediaId metadata during move** drop-down, specify if you want the application to capture the ExternalMediaID metadata during the file move. The valid options are true or false.

10. In the **Gets and sets dispositions** drop-down, specify if you want the application to get and set the dispositions of the file mover. Select "***true***" to track files being moved in the DEF's working table and "***false***' to move files without tracking them.
11. In the **Update file key on move** drop-down, specify if you want the application to update the file key on move. This is necessary for proper working table tracking if the relative path of the file (and working table file key) is likely to change as part of the move. The default value is "***true***".
12. In the **Force to extension** field, optionally enter a file extension to change the working table file key to.
13. In the **Failure Threshold** drop-down list, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
14. In the **Create .nxtmp file and rename during moves** drop-down, specify if you want to create temporary files and rename during moves.
15. In the **File extensions to audit** field, specify the file extensions in a comma-delimited list that are to be audited on copy and delete.
16. In the **Reconciliation Identifier** field, enter the name of the action if there is a reporting action in the process group.
17. In the **Default input disposition handling** drop-down list, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
    - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
18. In the **Input Disposition(s)** field, enter the input disposition. The Data Exchange Framework automatically chains action component dispositions together as javascript:void()part of its processing, so this field is usually left blank. If you are using split processes, enter a value in this field.
19. In the **Thread Count** field, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
20. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.

21. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
22. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
23. Click **Submit**.

File Rename Action

1. In the **Process files from metadata** drop-down, specify if you want to process files from metadata. When you select ***true*** the application processes files from the process groups existing working table. When you specify false, the application processes files from attached input resources. Processing from metadata reduces I/O and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **File filter** field, enter the file filter type. For example: CSV.
3. In the **File name regular expression matcher** field, enter a Java regular expression that identifies groups to extract and use in the "Rename to" field.
4. In the **Rename to** field specify how to rename the field. Use Java regular expression groupings in the format of $1, $2, etc. If you include {timestamp} in the file name, it will be included in the output name.
5. In the **Failure Threshold** field, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
6. In the **Reconciliation Identifiers**, enter the name of the action if there is a reporting action in the process group.
7. In the **Default input disposition handling** drop-down list, select one of the following:
   - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
   - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
8. In the **Input Disposition(s)**, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.

9. In the **Thread Count** field, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
10. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
11. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
12. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
13. Click **Submit**.

GDPR ID Creator Action

1. In the **Request source** drop-down, select either NIA or CSV.
2. In the **Participant ID** field, enter the participant ID used to identify this process to the GDPR management console. This only applies when you selected NIA as the request source.
3. In the **Host name** field, enter the host name used to execute requests on the GDPR management console. This only applies when you selected NIA as the request source.
4. If you selected NIA as the **Request source**, in the Port field enter the port number used to execute requests on the GDPR management console.
5. In the **GdprId 1 Field(s)** field, enter the metadata field or fields that define the GdprId column. For example: PhoneNumber or First Name + ' ' + LastName. This information will be plugged directly into a SQL update statement and must be valid SQL. Click  to the right of the field to select available metadata columns.
6. In the **GdprID 1 Normalization Function** field, select a normalization function to be applied to the GdprId column.
7. Repeat steps 5 and 6 for the GdprId2 -5 field(s) and the GdprId 2-5 Normalization Function.
8. In the **Service Name** field, enter the service name of the GDPR management console services.
9. In the **Delete associated media files?** drop-down, specify if you want to the application to delete the associated media files.
10. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
11. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave "***false***" to allow processing as usual. This option is useful for troubleshooting the process.
12. Click **Submit**.

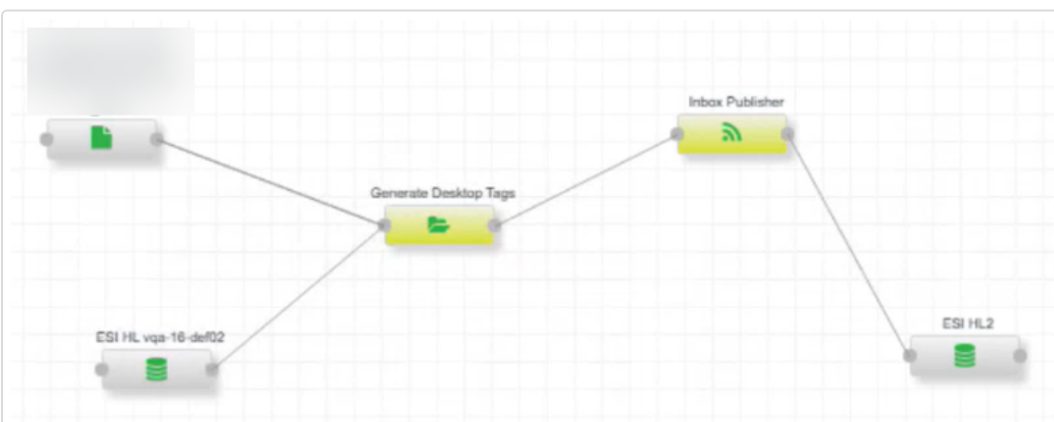
Generate Desktop Tags

1. In the **Source** field, select the evaluation data source.
   - If you select QC, enter a query in the **ESI Query** field. The query must join the temp table and return the IngestFilePath from NexidiaESI.
   - If you select Customer, enter a query in the **ESIDW Query** field. The query must join the temp table and return the IngestFilePath from NexidiaESIDW.
2. In the **Batch Size** field, enter the number of CSV rows to process per batch.
3. In the **Delete CSV after Processing** field, specify if you want to delete the CSV file after processing. Set it to ***true*** to remove the CSV file and set it to ***false*** to retain it.
4. In the **Audit/Evaluator Offset** field, specify the offset value for audits and evaluator reviews. The default value is 5.
5. In the **Default Offset** field, specify the default offset value when no audit/review flag is set. The default value is 0.
6. In the **QC Evaluation Offset** field, specify the offset value for QC evaluations. The default value is 1.
7. In the **Third Party Evaluation Offset** field, specify the offset value for third party evaluations. The default value is 2.
8. In the **CSV Delimiter** field, specify the delimiter used for CSV file parsing. For example, \t = tab.
9. In **Failure Threshold** field, specify the number of failures that will stop processing. A value of 0 indicates that the process will never fail.
10. In the **Quote Character** field, specify the character used to quote column values in the CSV file.
11. In the **Escape Character** field, specify the character used to escape column values in the CSV file.
12. In the **Reconciliation Identifiers** field, enter the name of the action if there is a reporting action in the process group.
13. In **Default input disposition handling** field, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.

- ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.

14. In **Input Disposition(s)** field, you can override the default behavior. Enter a comma-separated list of dispositions the file must pass before this action processes it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
15. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
16. In the **Terminal action** field, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, an action is terminal when it has a matching disposition column in the working table and no other actions depend on it. This option is provided for instances where the default behavior is not desirable.
17. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
18. Click **Submit**.

You can refer to the following sample process:



GDPR Requests Action

1. In the **Request Source** field, select the source type. The possible values are NIA or CSV.
2. In the **Force file encoding** field, leave blank in most cases to auto-detect. If auto-detect is not working, enter a valid Java InputStream encoding. See Supported Encodings to learn more.
3. In the **Participant ID** field, enter the participant ID to identify this process to the GDPR Management Console.
4. In the **Host Name** field, enter the host name used to execute requests on the GDPR Management Console.
5. In the **Port** field, enter the port number used to execute requests on the GDPR Management Console.
6. In the **Age Out Temporary Block Requests** drop-down, select the number of hours to age out temporary block requests older than this value.
7. In the **Service name** field, enter the service name of the GDPR management console services.
8. (Optional) If you selected CSV in the **Request Source** drop-down, .csv is populated in the **File Filter** field.
9. (Optional) If you selected CSV in the **Request Source** drop-down, specify if you want to delete GDPR requests after processing in the **Delete GDPR Requests? after processing** drop-down.
10. (Optional) If you selected CSV in the **Request Source** drop-down, enter the character used to separate delimited column values in the output CSV (for example ,) in the **Default Separator Character** field.
11. (Optional) If you selected CSV in the **Request Source** drop-down, enter the character used to quote column values in the output CSV (for example ") in the **Default Quote Character** field.
12. (Optional) If you selected CSV in the **Request Source** drop-down, enter the character used to escape column values in the output CSV (for example \) in the **Default Escape Character** field.

13. (Optional) If you selected CSV in the **Request Source** drop-down, select the character used to end/terminate a line in the output CSV (for example \n or \r\n) in the **Default Line End Character** drop-down.
14. In the **Delete associated media files?** field specify if you want to delete the associated media files.
15. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
16. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
17. Click **Submit**.

Genesys Cloud Job-Metadata Retriever

1. In the **Client ID**, enter the Client ID to access the API.
2. In the **Client Secret**, enter the Client password or secret to access the API.
3. In the **Base URL**, enter the URL that is the base for all endpoints from which analytics, conversation, and recording metadata can be queried against the Genesys Cloud API.
4. In the **Token Base URL**, enter the URL that is the base for the endpoints from which access or refresh-tokens can be requested for subsequent access to the Genesys Cloud API.
5. In the **Wait Time (in milliseconds) Between Request Retries** drop-down enter the amount of time that you want to wait between requests/retries to the API endpoint.
6. In the **Max Number of Retries for Request** specify the maximum number of retries you want to make for the API endpoint before failing.

7. In the **Page Size**, enter the number of conversations per page to retrieve from the API. There may be several pages of results for each interval (quantum) in a request. The maximum allowed by the API is 10000 conversations per request and the minimum is 25 conversations per request.
8. In the **Failure Threshold**, specify the number of failures that cause processing to stop. A value of 0 indicates that the process can never fail.
9. In the **Job Submit URL**, specify the URL for requesting/submitting job to run in Genesys API.
10. In the **Job Status URL**, specify the URL for requesting job status to the Genesys API.
11. In the **Job Result URL**, specify the URL requesting to retrieve a result (conversations) for a job to Genesys API.
12. In the **Metadata Filters**, specify the filter format which filters the metadata on the API. By default, the data is always filtered by [media-type=voice]. The filter-format must be as follows: '{FILTER}{OR} thisField=value1, theOtherField=value2 {FILTER}{OR} thatField=value3 {FILTER}...'
13. In the **Skip agent-less calls**, specify if you want to skip the recordings for which the agent was absent.
14. In the **Filter Retrieved Metadata (by quantum)?** drop-down, select ***true*** if you want to filter the conversation metadata retrieved from the API by the current quantum (start time of the interval).
15. In the **Start of Day Interval Matching** drop-down, select ***true*** if you want to match the start of day interval matching with the conversations start-time when retrieving metadata.
16. In the **Retrieve Voice Metadata And Media** drop-down, select ***true*** if you want to include voice conversations in the returned data.
17. In the **Retrieve Email Metadata And Media** drop-down, select ***true*** if you want to include email conversations in the returned data.
18. In the **Retrieve Callback Metadata And Media** drop-down, select ***true*** if you want to include callback conversations in the returned data.
19. In the **Retrieve Video Metadata And Media** drop-down, select ***true*** if you want to include video conversations in the returned data.

20. In the **Retrieve Chat Metadata And Media** drop-down, select ***true*** if you want to include chat conversations in the returned data.
21. In the **Reconciliation Identifier**, enter the name of the action if there is a reporting action in the process group.
22. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
23. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
24. Click **Submit**.

Genesys Cloud Transcoded Media Retriever Action

1. In the **Genesys ID**, enter the Genesys account username.
2. In the **Genesys Secret**, enter the Genesys password or secret to access the Cloud.
3. In the **Base URL** field, enter the URL that is the base for all endpoints from which analytics, conversation, and recording metadata can be queried against the Genesys Cloud API.
4. In the **Token Base URL**, enter the URL that is the base for the endpoints from which access or refresh-tokens can be requested for subsequent access to the Genesys Cloud API.
5. In the **Failure Threshold**, specify the number of failures that cause processing to stop. A value of 0 indicates that the process can never fail.
6. In the **Reconciliation Identifiers**, enter the name of the action if there is a reporting action in the process group.

7. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the **Threads per action** setting at the server level.
8. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
9. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
10. Click **Submit**.

Genesys Cloud Transcoding Requester Action

1. In the **Process files from metadata** field, specify if you want to process files from metadata. When you select ***true*** the application processes files from the process groups existing working table. When you specify false, the application processes files from attached input resources. Processing from metadata reduces I/O and file system scans. In order to process data, the previous action must have entered some data into the working table.
2. In the **File filter** field, enter the file filter type, for example CSV. Java-style regular expressions can also be used if prepended with "regex". Note that when processing files from resources, this acts as a filter, but when processing from metadata, the metadata's file key is inserted in any wildcards, expanding the processing of one file to all listed variants.
3. In the **Genesys ID** field, enter the Genesys ID to access the Cloud.
4. In the **Genesys Secret** field, enter the Genesys password or secret to access the Cloud.
5. In the **Base URL** field, enter the URL that is the base for all endpoints from which analytics, conversation, and recording metadata can be queried against the Genesys Cloud API.

6. In the **Token Base URL** field, enter the URL that is the base for the endpoints from which access or refresh-tokens can be requested for subsequent access to the Genesys Cloud API.
7. In the **Failure Threshold** field, specify the number of failures that cause processing to stop. A value of 0 indicates that the process can never fail.
8. In the **Reconciliation Identifiers**, enter the name of the action if there is a reporting action in the process group.
9. In the **Default input disposition handling** drop-down, select one of the following:
   - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
   - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
10. Optionally enter the **Input dispositions**. The Data Exchange Framework automatically chains action component dispositions together as javascript:void()part of its processing, so this field is usually left blank. If you are using split processes, enter a value in this field.
11. From the **Thread Count** drop-down, select a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
12. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
13. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
14. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
15. Click **Submit**.

Genesys Cloud User Metadata Retriever Action

1. In the **Client ID**, enter the Genesys account username.
2. In the **Client Secret**, enter the Genesys password or secret to access the Cloud.
3. In the **Base URL**, enter the base URL Genesys Cloud API.
4. In the **Token Base URL**, enter the URL that is the base for the endpoints from which access or refresh-tokens can be requested for subsequent access to the Genesys Cloud API.
5. In the **Page Size**, enter the number of users that you want to retrieve per request. The minimum value is ***25*** and maximum is ***100***.
6. In the **Status Filter** drop-down, select the status of the users that you want to retrieve. You can select ***None***, ***active***, ***inactive***, or ***deleted***.
7. In the **Failure Threshold**, specify the number of failures that cause processing to stop. A value of 0 indicates that the process can never fail.
8. In the **Reconciliation Identifiers**, enter the name of the action if there is a reporting action in the process group.
9. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the **Threads per action** setting at the server level.
10. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
11. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
12. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
13. Click **Submit**.

Genesys IR Media Retriever Action

1. In the **Genesys ID**, enter the Genesys account username.
2. In the **Genesys Password**, enter the password of the Genesys account.
3. In the **Failure threshold**, enter the number of non-fatal processing errors (inability to find a requested file, e.g.) to tolerate before failing and stopping the rest of the process from continuing. 0=never fail the process, but continue to track errors in logs, and as warnings in the UI. Default value is ***0***.
4. In the **Reconciliation Identifiers**, enter the name of the action if there is a reporting action in the process group.
5. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the **Threads per action** setting at the server level.
6. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
7. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
8. Click **Submit**.

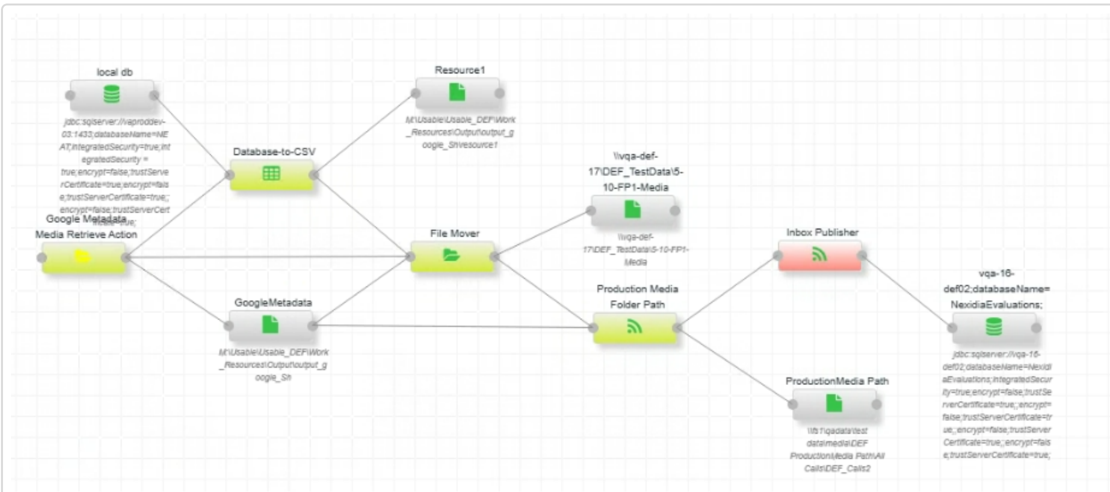
Google Metadata Media Retriever Action

The Google Metadata Media Retriever process workflow performs the following tasks:

1. Connects to a customer-provided Google API endpoint.
2. Authenticates using the customer's method, such as OAuth 2.0.
3. Retrieves interaction metadata, including media file locations from the API.
4. Writes the metadata to the Data Exchange Framework working table.

5. Downloads the media files.
6. Publishes the media and metadata to Interaction Analytics for indexing and analysis.

You can refer to the following sample process:



You can configure the following parameters for the Google Metadata Media Retriever action:

1. In the **Encrypted Service Account JSON**, enter the encrypted JSON content for the Google Service Account.
2. In the **Bucket Name**, enter the Google Cloud storage bucket name used to retrieve data from the Google API.
3. In the **Audio File Extension**, enter the extension of the media files you want to retrieve, for example, ***mp3***.
4. In the **Audio Folder**, enter the folder name where you want to store the media files. You can enter two folder names if you want to organize the files, such as ***voice_recordings, call_recordings***.

5. In the **Batch Size** drop-down, specify the number of objects to fetch per request from the bucket. The default value is ***1000***.
6. In the **Include Properties**, enter a comma-separated list of metadata properties to include from the JSON. If you leave this field blank, all properties are included and only the listed properties are added to the working table. For example, ***transcript,customer_id,agent_id,updateTime.***
7. In the **Failure threshold**, enter how many non-fatal processing errors the system should tolerate before the process stops. For example, a missing file counts as a non-fatal error. A value of 0 means the process never fails but tracks errors in logs and shows them as warnings in the UI. The default value is ***0***.
8. In the **Reconciliation Identifier**, enter the action name if a reporting action exists in the process group.
9. In the **Default input disposition handling** , select one of the following:
   - ***AND*** - Allows files to flow through this step that have passed all immediate inputs.
   - ***OR*** - Allows files to flow through this step that have passed any of the immediate inputs.
10. Optionally enter the **Input dispositions**. The system usually chains dispositions automatically, so you can leave this field blank. If you are using split processes, enter a value in this field.
11. In the **Thread Count**, enter the number of threads this action is allowed to partition its work into. This value overrides the server-level **Threads per action** setting.
12. In the **Terminal action**, specify if this is a terminal action. When the process completes, rows touched by terminal actions move to the working table rollover. By default, an action is terminal when it has a matching disposition column in the working table and no downstream actions reference it. Use this option if you do not want the default behavior.
13. In the **Disable action**, specify if you want to prevent the action from processing files. Leave it set to ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
14. Click **Submit**.

Genesys IR Metadata Retriever Action

1. In the **Genesys ID** field, enter the Genesys ID for the Genesys account.
2. In the **Genesys Password**, enter the password of the Genesys account.
3. In the **Base URL** field, enter the URL that is the base for all endpoints from which analytics, conversation, and recording metadata can be queried against the Genesys Cloud API.
4. In the **Audio type** drop-down, specify the format of the audio media when requesting media transcoding prior to downloading. The allowed values are: ***WAV*** and ***MP3***.
5. In the **File Limit**, enter the number of files to be downloaded per execution.
6. In the **Failure Threshold** field, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
7. In the **Reconciliation Identifiers**, enter the name of the action if there is a reporting action in the process group.
8. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
9. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
10. Click **Submit**.

GPG Decrypt Action

1. In the **Process files from metadata** drop-down, specify if you want to process files from metadata. When you select ***true*** the application processes files from the process groups existing working table. When you specify false, the application processes files from attached input resources. Processing from metadata reduces I/O and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **Passphrase** field, enter the pass phrase.
3. In the **File Input Type** drop-down, specify the file input type. The valid options are ***ASC*** and ***GPG***.
4. In the **Delete inputs after decryption?** drop-down, specify if you want the application to delete the inputs after the decryption is complete.
5. In the **Checks and sets dispositions** drop-down, specify you want the application to check and set the dispositions. Select "***true***" to track files processed in the process group's working table or "***false***" to encrypt without tracking.
6. In the **Failure Threshold** field, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
7. In the **GPG executable process timeout in seconds** drop-down, select a value how long to wait for the call to the gpg executable to complete before timing out and triggering a failure. (0 = no limit).
8. In the **Reconciliation Identifiers**, enter the name of the action if there is a reporting action in the process group.
9. In the **Default input disposition handling** drop-down list, select one of the following:
   - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
   - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.

10. In the **Input Disposition(s)**, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
11. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the **Threads per action** setting at the server level.
12. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
13. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
14. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
15. Click **Submit**.

GPG Encrypt Action

1. In the **Process files from metadata** field, specify if you want to process files from metadata. When you select ***true*** the application processes files from the process groups existing working table. When you specify false, the application processes files from attached input resources. Processing from metadata reduces I/O and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **File filter** field, enter the file filter type. For example: CSV.
3. In the **Recipient** field, enter the public key owner on the DEF service account user's keyring that the files will be encrypted against.
4. In the **Always trust what's in the keystore** drop-down, specify if you want the application to always trust what is in the keystore. When you select ***true*** the application suppresses GPC executable warnings about unsigned keys.
5. In the **File Output Type** drop-down, specify the file output type. The valid options are ***.asc***, ,***gpg***, and ***.png***.
6. In the **Delete inputs after encryption?** drop-down, specify if you want the application to delete the inputs after the encryption is complete.
7. In the **Generate .done file upon completion?** drop-down, specify if you want the application to generate a .done file upon completion. This will create a .done file alongside the .gpg file upon completion.
8. In the **Checks and sets dispositions** drop-down, specify you want the application to check and set the dispositions. Select ***true*** to track files processed in the process group's working table or ***false*** to encrypt without tracking.
9. In the **Failure Threshold** field, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
10. Specify the **GPG executable process timeout in seconds**. A value of "0" indicates there will be no limit.
11. In the **Reconciliation Identifiers**, enter the name of the action if there is a reporting action in the process group.
12. In the **Default input disposition handling** drop-down list, select one of the following:

- ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
- ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.

13. In the **Input Disposition(s)** field, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
14. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the **Threads per action** setting at the server level.
15. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
16. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
17. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
18. Click **Submit**.

Generate Protected Keys Action

1. In the **Process files from metadata** field, specify if you want to process files from metadata. When you select ***true*** the application processes files from the process groups existing working table. When you specify false, the application processes files from attached input resources. Processing from metadata reduces I/O and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **File filter** field, enter the file filter type. For example .CER.
3. In the **Keystore location** field, enter the absolute path to the keystore that will house the symmetric key generated by this action. If the keystore does not exist, it will be created.
4. In the **Keystore password** field, specify a password for the keystore .
5. In the **Delete certificate files after processing** drop-down, specify if you want to delete the certificate files after processing.
6. In the **Failure Threshold**, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
7. In the **Reconciliation Identifiers**, enter the name of the action if there is a reporting action in the process group.
8. In the **Default input disposition handling** drop-down list, select one of the following:
   - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
   - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
9. In the **Input Disposition(s)**, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.

10. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
11. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
12. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
13. Click **Submit**.

Hash Generate/Validate Action

1. In the **Process files from metadata** field, specify if you want to process files from metadata. When you select ***true*** the application processes files from the process groups existing working table. When you specify false, the application processes files from attached input resources. Processing from metadata reduces I/O and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **File filter** field, enter the file filter type. For example: CSV.
3. In the **Hash type** drop-down, select the type of hash to generate.
4. In the **Maximum kb to generate has from** drop-down list. When set to ***zero***, the entire file will be read and hashed. When set to non-zero, only the first "x" kB will be read and used to generate the hash, where "x" is the option's value.
5. In the **Hash operation** drop-down, select "***Generate***" to generate hash files for each incoming file, or "***Validate***" to validate incoming files against their corresponding hash files.
6. In the **Failure Threshold** field, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
7. In the **Reconciliation Identifiers**, enter the name of the action if there is a reporting action in the process group.
8. In the **Default input disposition handling** drop-down list, select one of the following:
   - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
   - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
9. In the **Input Disposition(s)** field, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.

10. In the **Thread Count** field, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
11. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
12. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
13. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
14. Click **Submit**.

Hitachi Retriever Action

1. In the **User Name** field, enter the Hitachi appliance user name.
2. In the **Password** field, enter the Hitachi appliance password.
3. In the **URL Column** field, enter the working table metadata column that holds the full URL to retrieve the corresponding media from.
4. In the **Failure Threshold** field, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
5. In the **Reconciliation Identifiers**, enter the name of the action if there is a reporting action in the process group.
6. In the **Default input disposition handling** drop-down list, select one of the following:
   - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
   - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
7. In the **Input Disposition(s)** field, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
8. In the **Thread Count** field, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
9. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.

10. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
11. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
12. Click **Submit**.

Import JSON Action

1. In **Process files from metadata**, specify whether to process files from metadata. When set to ***true,*** the application processes files from the process group's existing working table. When set to ***false***, the application processes files from attached input resources. Processing from metadata reduces I/O and file system scans, but it requires that a previous action has already added data to the working table
2. In **File Filter**, enter the file type such as ****.json***. *.xml, *.csv, etc. Java-style regular expressions can also be used when prepended with regex:. When processing files from resources, this acts as a filter. When processing from metadata, the system inserts the file key into wildcards, allowing the step to process all matching files. For example, *,*.xml, e.g., would process both a file matching the working table file key 1234567.wav and its 1234567.wav.xml counterpart.
3. In **Include Only Specific Properties**, enter a comma-separated list of properties to ingest. Use exact names for single properties, such as ***AWSAccountId***. For arrays, use formats such as ***RoutingCriteria[n]_Steps[n]_Expiry_DurationInSeconds, RoutingCriteria[n-1]_Steps[n]_Expiry_DurationInSeconds, RoutingCriteria[n+2]_Steps[n]_Expiry_DurationInSeconds***. You can also use prefixes such as ***RoutingCriteria[n]_Steps[n]*** or Agent to ingest all sub-properties.

If you leave this field blank, all properties are ingested. For example, entering ***Tags_aws:connect:instanceId*** results in a database column named ***Tags_aws-connect-instanceId***. Wildcard tokens are preserved exactly as entered.

4. In **Delete After Processing**, specify ***true*** to delete input files after processing. The default value is ***false***.
5. In **Failure Threshold**, specify the number of failures that will stop processing. A value of 0 indicates that the process will never fail.
6. In **Reconciliation Identifier**, enter the name of the action if there is a reporting action in the process group.
7. In **Default input disposition handling**, select one of the following:
   - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
   - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
8. In **Input Disposition(s)**, you can override the default behavior. Enter a comma-separated list of dispositions the file must pass before this action processes it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
9. In **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the server level **Threads per action** setting.
10. In **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates no limit.
11. In the **Terminal action** field, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, an action is terminal when it has a matching disposition column in the working table and no other actions depend on it. This option is provided for instances where the default behavior is not desirable
12. In **Disable action**, specify whether to prevent the action from processing files. Leave this set to ***false*** to allow normal processing. This option is useful for troubleshooting.
13. Click **Submit.**

InContact Interaction Retriever Action

This action is sensitive to the environment and configuration because there needs to be enough RAM allotted to the JVM to handle holding full file contents in memory, Having too many threads working at once could cause an OutOfMemory error that can be corrected by increasing the JVM's memory on startup (in launch.bat).

1. In the **InContact web service base URL** field, enter the InContact web service base URL.
2. In the **Version** drop-down, select the InContact api version from the Version drop-down list. The default version is ***v15.0***. We currently support ***v8.0*** and ***v15.0***.
3. In the **Vendor name** field, enter the vendor name.
4. In the **Application name** field, enter the application name.
5. In the **Business Unit** field, enter the business unit.
6. In the **User name** field, enter the user name.
7. In the **Password** field, enter the password.
8. In the **Contact Metadata** field, enter a comma-delimited list of contact metadata fields to extract. The default contact metadata includes the following:
   - contactId
   - contactStart
   - totalDurationSeconds
   - masterContactId
   - mediaTypeName
   - isOutbound,toAddr
   - fromAddr
   - pointOfContactName
   - campaignName
   - skillName
   - teamName
   - agentId
   - firstName

- lastUpdateTime
- lastName
- agentSeconds
- holdSeconds
- confSeconds
- abandonSeconds
- routingTime
- releaseSeconds
- ACWSeconds
- transferIndicatorName

9. In the **Agent Metadata** field, enter a comma-delimited list of agent metadata fields to extract. The default agent metadata includes the following:
    - userName
    - emailAddress
    - reportToName
    - ntLoginName
    - profileName
    - hireDate
    - terminationDate
    - isActive
    - isSupervisor
    - teamId
    - atHome
    - location
    - city
    - country
    - custom1
    - custom2

- custom3
- custom4
- custom5

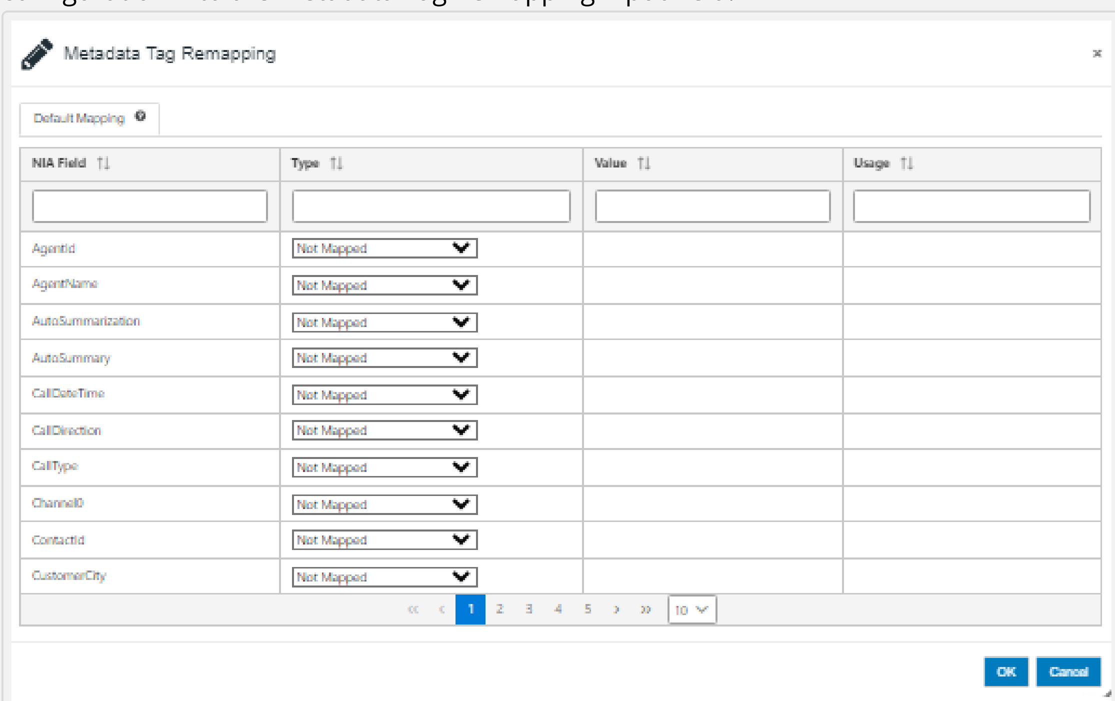
10. In the **Custom Metadata** field, enter a comma-delimited list of custom metadata fields to extract.
11. In the **Exclude Media** drop-down list, specify if you want to exclude media from retrieval through the API. When you select ***true*** only metadata and custom fields are retrieved through the API. This option is unchecked by default.
12. In the **Default separator character** field, enter the character used to separate delimited column values (for example **,**) in the Default separator character field.
13. In the **Default quote character** field, specify the character used to quote column values (for example **"**) in the Default quote character field.
14. In the **Default escape character** field, specify the character used to escape column values (for example ****) in the Default escape character field.
15. In the **Default line end character** drop-down, select the character used to end/terminate a line in the Default line end character drop-down list. The default value is ***\n***.
16. In the **Replace new lines** drop-down, specify if you want to replace new line characters with the selected character in the Replace new lines drop-down list. The default value is ***Do not replace***.
17. In the **Failure Threshold** field, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
18. When not excluding media, the InContact API may still report no media content for some contacts. In the **Exclude contact metadata with no media content** drop-down, select “false” to have that metadata sent downstream without media. This metadata can be identified by the “204” InContactFileType in the working table. Leave ***true*** to ignore metadata for those contacts.
19. In the **Reconciliation Identifiers**, enter the name of the action if there is a reporting action in the process group.

20. In the **Thread Count** field, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
21. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
22. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
23. Click **Submit**.

Inbox Publisher Action

Execution of this action in a process is based on the nia.maintenance.mode setting under the nxdevops user in the nx_user_preferences table. See Maintenance Mode to learn more.

1. The Metadata Tag Remapping configuration allows for renaming columns passing through to columns that *Interaction Analytics* expects. Click in the blank Metadata Tag remapping field or click  to open the Metadata Tag Remapping window and generate this field's contents. The column selections will only be ones the process has been made explicitly aware of, as in the "required metadata" field of the Metadata Import action. You can use other metadata columns if you select a "constant" type and enter 'col:metadataColumnName' as the value. You can also enter constants by directly by leaving the 'col:' portion. You can also click the Export Mapping link to copy the mapping configuration to your clipboard and click the Import Mapping link to paste the mapping configuration into the Metadata Tag Remapping input field.

   Metadata fields that are implicitly mapped are prefixed with ** and marked as Automatically Mapped in the Type column.

The Inbox publisher can map canonical and non-canonical metadata formats with the first-class NIA fields. It can also map Extractor-ingested data format with the first-class NIA fields.

2. In the **Include disposition data in output?** field, specify if you want to include disposition data in the export. The default value is ***false***. Disposition data is internal to the DEF's working table, but can be included in the export. Leave this field as ***false*** since NIA does not have any use for it.
3. In the **NIA target version** drop-down, specify the *Interaction Analytics* target version. The valid options are:
   - ***11.x*** - **Continue to step 8.**
   - ***12.0*** - **Continue to step 4.**
   - ***12.1+ (default)*** - **Continue to step 4.**
4. In the **Media Type** drop-down, specify the media type that will be published to the inbox. The valid options are ***Audio***, ***Chat***, ***Email***, and ***Text***.
5. If you specified a media type of ***chat***, ***email***, or ***text***, in the **Update/Reprocess** drop-down, select ***true*** to declare to *Interaction Analytics* as batch updates, and not as new data to be declared.
6. If you selected version ***12.1+***, specify a value for the **Media present default flag**. The valid options are ***true*** or ***false***. Set to ***true*** to tell *Interaction Analytics* that media exists that matches incoming metadata. Set to ***false*** if no media is present. This setting can be overridden on a per-file basis by including an 'NxMediaPresent' metadata column with a 0/1 bit flag (0=false/ 1=true).
7. If you selected version ***12.0*** or ***12.1+***, specify a value for the **Create index default flag**. The valid options are ***true*** or ***false***. If you select ***true*** , *Interaction Analytics* will index the media for search. Select ***false*** to skip indexing. This setting can be overridden on a per-file basis by including an 'NxCreateIndex' metadata column with a 0/1 bit flag (0=false/ 1=true).
8. In the **Allow Duplicate Mapping** drop-down, set to ***true*** to allow duplicate mapping.
9. In the **Use ESMs** drop-down, select ***true*** to use .esm files as markers for actual media. When you use these, a dummy input resource must be used whose path will serve to build the IngestPath for NIA, and in the working table file key. When not using .esm files, no input resource is allowed.

10. In the **TVP batch size** field, specify the number of records to add to each table-valued parameter object before sending it to SQL Server for processing.
11. In the **PPID for Repository Datetime Action** field, enter the process property ID of the action that can be used for ***RepositoryDatetime*** in reconciliation reports. If you enter 0, the Disposition Date Time value of the following actions will be retrieved.
    Any Media Retriever, ChatCSVtoJSON, CenteraRetriever, HitachiRetriever, InContactInteractionRetriever, ParseEmailTextToJSON, ParseGeneralTextToJSON, PSTToEML, and VerintStitch.
12. In the **Test Ingest Path** field, set to ***true*** to test the ingest path.
    - In the **Output Path for Differences** field, enter the directory path for the output file. This file will include columns for the differing UDFs and their values, showing differences within the ingest path and allowing users to verify the data.
      If no differences are found, the output file will not be created.
    - In the **Ingest Mapping** field, click to set a filter to map XML metadata to the test ingest path.

    This feature allows NIA users to verify DEF-generated data against NIA records. You can create a fake inbox to compare DEF and NIA data. To delete the fake inbox after verification, set the **Test Ingest Path** parameter to ***false*** and the **Delete Test Ingest Tables** parameter to ***true***.
13. If you set the **Test Ingest Path** parameter to ***true***, **Stylesheet Path** field is displayed, where you can enter the path of the Stylesheet. This helps map metadata between old and new versions.
14. In the **Include Monitor Tag** field, set to ***true*** to include monitor tag in the interaction reports.
15. In the **Reconciliation Identifiers**, enter the name of the action if there is a reporting action in the process group.
16. In the **Default input disposition handling** drop-down list, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
    - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.

17. In the **Input Disposition(s)** field, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
18. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
19. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
20. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
21. Click **Submit**. The system checks if the mapped metadata fields exist in the Working Table. If a field is missing, it shows a warning. This helps you see which fields are actually available in the working table. The warning does not stop you from submitting. It just alerts you to possible mismatches. You can then review your mappings and make changes if needed.

Inform Media Retriever Action

1. In the **Inform web service protocol** drop-down, select the protocol for the Inform web service. The options are ***http*** or ***https***.
2. In the **Inform web service hostname** field, enter the host name for the Inform web service.
3. In the **Inform web service port number** field, enter the Inform web service port number if it is different than the default.
4. In the **Inform web service version** drop-down, select the Inform web service API version. Currently, only version 1.0 is supported.
5. In the **User name** field, enter the Inform web service user name.
6. In the **Password** field, enter the Inform web service password.
7. In the **Inform web service wait time (in milliseconds)** drop-down specify the amount of time to wait between calls to the metadata search API endpoint.
8. In the **Inform web service number of retries** field, specify the number of retries to the metadata search API endpoint before failing.
9. In the **Failure Threshold** field, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
10. In the **Media type** drop-down, select the audio media type.
11. In the **Chunk size** field, specify the download audio buffer size.
12. In the **Use basic HTTP client** drop down, select ***false*** to not use the basic HTTP client.
13. In the **Log Inform service responses** drop-down, specify if you want to log Inform service responses to a log file. The log file is created in <user temp folder>/Inform-Response.log. By default, no response logging occurs.
14. In the **Reconciliation Identifier**, enter the name of the action if there is a reporting action in the process group.
15. In the **Default input disposition handling** drop-down list, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
    - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.

16. In the **Input Disposition(s)** field, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
17. In the **Thread Count** field, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
18. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
19. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
20. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
21. Click **Submit**.

Inform Metadata Retriever Action

1. In the **Inform web service protocol** drop-down, select the protocol for the Inform web service. The options are ***http*** or ***https***.
2. In the **Inform web service hostname** field, enter the host name for the Inform web service.
3. In the **Inform web service port number** field, enter the Inform web service port number if it is different than the default.
4. In the **Inform web service version** drop-down, select the Inform web service API version. Currently, only version 1.0 is supported.
5. In the **User name** field, enter the Inform web service user name.
6. In the **Password** field, enter the Inform web service password.
7. In the **Inform web service wait time (in milliseconds)** drop-down specify the amount of time to wait between calls to the metadata search API endpoint.
8. In the **Inform web service number of retries** field, specify the number of retries to the metadata search API endpoint before failing.
9. In the **Inform web service page sizing** field, specify the number of results to bring back per call to the metadata search results API endpoint.
10. In the **Failure Threshold** field, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
11. In the **Resource ID list** field, enter a list of resource IDs to process (for example 2, 4, 5).
12. In the **Use basic HTTP client** drop down, select "***false***" to not use the basic HTTP client.
13. In the **Log Inform service responses** drop-down, specify if you want to log Inform service responses to a log file. The log file is created in <user temp folder>/Inform-Response.log. By default, no response logging occurs.
14. In the **Reconciliation Identifier**, enter the name of the action if there is a reporting action in the process group.

15. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
16. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave "***false***" to allow processing as usual. This option is useful for troubleshooting the process.
17. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave "***false***" to allow processing as usual. This option is useful for troubleshooting the process.
18. Click **Submit**.

IVR Redaction Action

The IVR redaction action replaces the first few seconds or milliseconds of audio with silence. The IVR redaction accepts input media in WAV format and produces redacted output media as WAV format. If the incoming media is not in WAV format, use the Media Convert action to convert MP4 files to WAV format.

1. The **Process files from metadata** always ***Process files from metadata***.
2. In the **File filter**, enter the file filter type. For example: ***.wav***.
3. In the **Metadata column containing time to redact**, specify the name of the metadata column that holds the value of time to redact from the required input media file.
4. In the **Unit of time to redact** drop-down, specify the unit of time to redact. You can select between ***seconds*** or ***milliseconds***. Defaults to ***milliseconds***.
5. In the **Delete inputs media file after processing** drop-down, select ***true*** if you want the application to delete the inputs after they are processed.
6. In the **Failure Threshold** drop-down , specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.

7. In the **Offset buffer (ms)** drop-down, specify the value of time in milliseconds added or subtracted from the required metadata for trimming. Defaults to 0.
8. In the **Reconciliation Identifiers**, enter the name of the action if there is a reporting action in the process group.
9. In the **Default input disposition handling** drop-down list, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
    - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
10. In the **Input Disposition(s)**, enter the input disposition. The Data Exchange Framework automatically chains action component dispositions together as javascript:void()part of its processing, so this field is usually left blank. If you are using split processes, enter a value in this field.
11. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
12. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
13. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
14. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
15. Click **Submit**.

Language ID Action

1. In the **Process files from metadata** field, specify if you want to process files from metadata. When you select ***true*** the application processes files from the process groups existing working table. When you specify false, the application processes files from attached input resources. Processing from metadata reduces I/O and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **File Filter** field, enter the file filter type, for example CSV. Java-style regular expressions can also be used if prepended with "regex". Note that when processing files from resources, this acts as a filter, but when processing from metadata, the metadata's file key is inserted in any wildcards, expanding the processing of the one file to all listed variants. For example: (*,*.xml' would process both a file matching the working tale file key '1234567.wav' and its '1234567.wav.xml' counterpart.
3. In the **Fast processing** drop-down, specify the fast processing option. A value of 0 indicates the entire file will be processed. A value of greater than zero indicates the only the first "x" seconds will be processed. We recommend 240 seconds.
4. In the **Failure Threshold** field, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
5. In the **Reconciliation Identifiers**, enter the name of the action if there is a reporting action in the process group.
6. In the **Default input disposition handling** drop-down list, select one of the following:
   - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
   - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
7. In the **Input Disposition(s)** field, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.

8. In the **Thread Count** field, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
9. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
10. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
11. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
12. Click **Submit**.

Language ID for Text Interactions Action

1. In the **Process files from metadata** field, specify if you want to process files from metadata. When you select ***true*** the application processes files from the process groups existing working table. When you specify false, the application processes files from attached input resources. Processing from metadata reduces I/O and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **File filter** field, enter the file filter type. For example, WAV.
3. In the **Delimiter** field, enter the delimiter character in the Delimiter field. This is used for non-XML input, and is a single character such as ***,***. You can also use ***\t*** for tab.
4. In the **Failure Threshold** field, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail..
5. Leave the **Force file encoding** field blank to auto-detect.
6. In the **Reconciliation Identifiers**, enter the name of the action if there is a reporting action in the process group.
7. In the **Default input disposition handling** drop-down list, select one of the following:
   - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
   - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
8. In the **Input Disposition(s)** field, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a omma to create an OR condition.
9. In the **Thread Count** field, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.

10. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
11. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
12. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave "false" to allow processing as usual. This option is useful for troubleshooting the process.
13. Click **Submit**.

Mattersight BD Retriever Action

1. psIn the **Mattersight web service protocol** drop-down, enter the protocol for Mattresight BD web service with either ***http*** or ***https***. Default is ***https***.
2. In the **Mattersight web service hostname**, enter the host name of the Mattersight web service.
3. In the **Mattersight web service port number**, enter the port number for the Mattersight web service.
4. In the **Service name**, enter the name of the Mattersight service.
5. In the **Mattersight web service version**, enter the API version of the Mattersight web service. Currently only version v1 is supported.
6. In the **Mattersight API username**, enter the access key of the Mattersight web service.
7. In the **Mattersight API password**, specify the secret key of for the Mattersight web service.
8. In the **ANI column name**, enter the name of the name of the metadata column that contains the ANI value.
9. In the **Retrieve behavioral data in batches** drop-down, if you want to retrieve the behavioral data in batches, select ***true***.

10. In the **Failure Threshold** field, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
11. From the **Session credentials expiration (minutes)** drop-down, enter the time to wait before re-authenticating the Vonage session. Session expiration is 120 minutes, however we choose 60 min as a save value, even though the system automatically re-authenticate token/session if expiration time approaches.
12. From the **Default HTTP client timeout(in seconds)** drop-down, select the Apache HTTP client default timeout value (in seconds). Default value is ***60*** seconds.
13. In the **Reconciliation Identifiers field**, enter the name of the action if there is a reporting action in the process group.
14. From the **Default input disposition handling** drop-down, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
    - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
15. In the **Input Disposition(s)**, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type 'none' to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
16. From the **Thread Count** drop-down, select a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
17. From the **Maximum files to process** drop-down, select the maximum number of files to allow this action to process in a given run. The maximum files to process is 10000. A value of 0 indicates that there is no limit.

18. From the **Terminal action** drop-down, select if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
19. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
20. Click **Submit**.

Media Conversion Action

1. In the **Process files from metadata** field, specify if you want to process files from metadata. When you select ***true*** the application processes files from the process groups existing working table. When you specify false, the application processes files from attached input resources. Processing from metadata reduces I/O and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **File filter** field, enter the file filter type. For example: CSV. Java style regular expressions can also be used. When processing files from resources, this acts as a filter. When processing from metadata, the metadata's file key is inserted into any wildcards, expanding processing of the one file to all variants. For example, '*,*.xml' would process both a file matching the working table file key '1234567.wav' and its '1234567.xml.wav' counterpart.
3. In the **Supported Input Formats** field, a read only list of media input format supported by Nexidia Workbench is displayed.
4. In the **Output Format** drop-down, select the output format that you want the media converted into. The valid options are MP4 and WAV.
   - If you select the ***WAV*** output format, a sub-drop down list is displayed that contains a list of the WAV output encoding formats support by the version of Workbench used by the DEF. A file with extension ".wav" will be generated by the conversion process.
   - If you select the ***MP4*** output format, a file with extension ".mp4" will be generated by the conversion process.
5. In the **Delete inputs after conversion** drop-down, specify if you want to delete inputs after the conversion has completed.
6. In the **Failure Threshold** field, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.

7. In the **Set output bitrate (kbps)** field, set transcode target bitrate specified in bits/sec (as 64kbps, 24kbps, 16kbps, etc. Min value is 16kbps). N/A means the system takes the default bitrate for the transcoder in the environment. The output bitrate does not always match the bitrate of the input file. Once the max bitrate is reached for a particular MP4 file, the output bitrate does not exceed even if the bitrate is set to 128 or 256. This Set output bitrate (kbps) is pivot type and is available only when output format field is set to MP4.
8. In the **Set Mix to STREAMALL** field, specify the Workbench Media Accessor API mode. The default value of ***true*** will stream all channels when doing conversions. To flatten the output to mono, change this value to "false".
9. In the **Report-friendly name** field, enter a name to be used in reporting for this action.
10. In the **Default input disposition handling** drop-down list, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
    - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
11. In the **Input Disposition(s)** field, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type 'none' to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
12. In the **Thread Count** field, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
13. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.

14. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
15. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
16. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
17. Click **Submit**.

Media Statistics Calculator Action

1. In **Process files from metadata**, specify if you want to process files from metadata. Select ***false*** to process files from attached input resources or ***true*** to process them from this process groups existing working table. Processing from metadata reduces input/output and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In **File filter**, enter the file filter type. For example: CSV.
3. In the **File Path Column Name**, enter the name of the metadata column in the process group working table that will be used to determine the files to move. The default value is FileName. This field is visible if you have set the **Process files from metadata** parameter to ***true***.
4. In **Transcript Type**, select the type of transcript this action will use to calculate media statistics. The available option is ***VoiceGrid Transcript***.
5. In the **Media Type** drop-down, specify the type of media to calculate media statistics. The options are Audio and Chat.
   The following additional metrics will be calculated from chat turn data and sent to Nexidia to ensure comprehensive chat processing and analysis.

- Start time
- End time
- Turn count
- Average time between turns
- Average response time between customer and agent.
- Participant list
- Chat ID
- NxInteractionUID

6. In **Phrase Calculation Gap**, enter the gap between phrases. The default value is ***300 ms***.
7. In **Delete inputs after process**, specify if you want the application to delete the input files after they are processed.
8. In **Failure Threshold** , specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
9. In **Reconciliation Identifier**, enter a name to be used in reporting for this action.
10. In **Default input disposition handling**, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
    - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
11. In **Input Disposition(s)**, enter the input disposition. The Data Exchange Framework automatically chains action component dispositions together as javascript:void()part of its processing, so this field is usually left blank. If you are using split processes, enter a value in this field.
12. In **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
13. In **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.

14. In **Terminal action**, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
15. In **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
16. Click **Submit**.

Media URL Download

This action downloads the media file based on the URL provided to analyze the interactions further.

A sample process consists of the following:

1. File System Input folder - A folder where the CSV file with URLs is placed.
2. Metadata Import Action - Loads the CSV file into a working table.
3. Media URL Download Action - Downloads the media files based on the URL provided.
4. File System Output folder - A folder where the downloaded media files are saved.

To execute this action, set the following configuration parameters:

1. In the **API Key** field, enter the API key for authentication.
2. In the **API Secret** field, enter the API secret to retrieve the access token.
3. In the **Account ID** field, enter the Account ID if needed. This field is optional.
4. In the **Download URL Column Name** field, enter the name of the column in the metadata that holds the URL required to download the media file.

5. In the **Base Download URL** field, enter the API URL for downloading media. It could be base URL. This field is optional.
6. In the **Metadata Key Column Name** field, enter the metadata key column name used for processing. This is the unique identifier for the record in the WT, and is used for updating the table (action dispositions, etc).
7. In the **Authentication Type** field, select either ***JWT*** or ***OAuth2.0*** for authentication.
    - If you select ***JWT***, specify the Payload Type.
    - If you select ***OAuth***, you need to provide the following details:
        - **Access Token URL** - Enter the API access token URL to obtain the access token for OAuth. This can be a base URL. This field is optional.
        - **Redirect URL** - Enter the redirect URL (Callback URL) for redirection after obtaining the access token. This field is optional. The Redirect URL is included in the request, and the authorization server will redirect the user to this URL after authorization.
8. In the **Payload Type** field, specify how you will provide additional information in the API request.
    - If you select Header, in the **Header Name** field, enter the API/Web Service HTTP headers to include in the API request. Name/value pairs can be any valid HTTP header required by the API/Web Service being called and must be separated by a semi-colon. Click to set the header name/value pairs in the Header Name window. For example, Content-Type=application/json.
    - If you select Query Param, in the **Query Param** field, enter query parameters to add in the URL of the API request. Click to set the parameter name/value pairs in the Query Parameters window. These can be any query parameters required by the Powerboat Service being called. The name=value pairs must be separated by a semi-colon. For example, startDate={startDate}; endDate={endDate}; slippage=100. The values in curly braces { } will be replaced with actual values when the API request is made.
9. In the **Output File Extension** field, enter the file extension for the output file downloaded by the API. If entered, this will append the desired extension to the output file name (or replaces the existing one, forcing an extension). The default value is empty.

10. In the **Failure Threshold** field, specify the number of failures allowed before stopping. A value of 0 means the process will never fail.
11. In the **Update FileName in WT** field, set to ***true*** to rename the file name in the working table to also include the new extension added, if set. Default value is false.
12. In the **Default HTTP Client Timeout (in seconds)** field, enter the timeout value for the HTTP client. The default value is 60 seconds.
13. In the **API/Web Service Wait Time (in milliseconds)** field, enter the amount of time in milliseconds to wait between calls to the API endpoint.
14. In the **API/Web Service Number of Retries** field, enter the number of retries for API calls before failing.
15. In the **Log Service Responses** field, select ***true*** to log service responses. The log file will be created in **<user temp folder>/MediaUrlDownload-Response.log**. The default value is ***false***.
16. In the **Reconciliation Identifier** field, enter a name used for reporting this action.
17. In the **Input Disposition(s)** field, enter a comma-separated list of dispositions that incoming files should have processed through in the working table before this action processes them. This is done to override the default input disposition handling of the action. It is useful for cross-process chaining (within or across servers). Leave blank to automatically chain actions, which is the default behavior. Enter 'none' to disconnect the action from dependencies on prior actions. Use a hash sign instead of a comma to create an OR condition.
18. In the **Thread Count** field, enter the number of threads you want this action to use for partitioning its work. This value overrides the server-level Threads per action setting. If you leave it at the default value of 0, the server-level default is used.
19. In the **Terminal action** drop-down, specify whether this is a terminal action. Upon process completion, the rows affected by terminal actions are moved to a working table rollover table. By default, actions are considered terminal if they have a corresponding disposition column in the working table, and no downstream actions reference them. This option is available for instances where the default behavior is not desired.
20. Click **Submit**.

Merge Action

1. In the **Media File Type** drop-down, specify the media file type. The valid options are ***WAV*** and ***NMF***.
2. In the **Call ID Column Name** field, enter the call ID column name or names. If you enter multiple columns you should identify multi-column uniqueness with a plus mark (+) between the column names. For example: CompoundId+PBXId
3. In the **Interaction ID Column Name** field, enter the interaction ID column name. This is used to determine files to merge from existing working table metadata, and in conjunction with the overall call ID, to name the resulting output file.
4. In the **Recording Side Column Name** field, enter the recording side column name. This is the name of the column that differentiates that channel separation within an interaction.
5. In the **Delete channel files after merging?** drop-down, specify if you want the application to delete channel files after merging.
6. In the **Failure Threshold** field, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
7. In the **Report-friendly name** field, enter a name to be used in reporting for this action.
8. In the **Default input disposition handling** drop-down list, select one of the following:
   - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
   - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
9. In the **Input Disposition(s)** field, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.

10. In the **Thread Count** field, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
11. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
12. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
13. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
14. Click **Submit**.

Metadata Append Action

1. In the **New metadata to create** field, enter a pipe-separated list of new columns to create, using braces to identify the incoming metadata to generate the new data from. For example, 'UCID={iCompoundId}-{iPBXId}|...'. Empty columns can be created by excluding the = portion for example, ('UCID|...'). Note that it is also possible to indicate the datatype of the column to be created, and that doing so will trigger an automatic indexing of the column. Continuing the previous example, this can be accomplished using this form: 'UCOD:nvarchar(256)={iCompoundId}-{iPBXId}|. Note that there are performance implications to adding a new data typed column that will be indexed in an existing and already large working table.
2. In the **Treat missing metadata as errors** drop-down, if new columns are being created using existing metadata, and any of the parameters defined above are empty or null, select here whether to treat those as errors that prevent the row from moving forward. Orphaned rows might wait for metadata arrival on subsequent runs (from a Metadata Join, for example), or they may eventually age out of the working table at the process-level max self-reconciliation days, depending.
3. In the **Replace full value on failure** drop-down, in the case of new metadata columns that are created using a combination of existing metadata set this to "true" to replace the full value if any parameters are missing, or "false" if only the missing parameters should be substituted. For example, with UCID={iCompoundId}-{iPBXId}, if iCompoiundId=1000 for a row but iPBXId is empty, "true" would mean UCID would be set to the substitution string, where "false" would mean that UCID would be set to a concatenation of "1000'" and the substitution string.
4. In the **Missing metadata substitution string** field, leave blank to replace parameters that have no corresponding metadata with empty strings, or enter something here to be substituted in place of missing metadata parameters.
5. In the **Failure Threshold** field, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
6. In the **Report-friendly name** field, enter a name to be used in reporting for this action.
7. In the **Default input disposition handling** drop-down list, select one of the following:
   - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.

- ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.

8. In the **Input Disposition(s)** field, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
9. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
10. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
11. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
12. Click **Submit**.

Metadata Filter Action

1. In the **Required metadata fields**, enter the required metadata fields in a comma separated format.
2. In the **Where clause**, enter the where clause for the metadata filter.
3. In the **Failure Threshold**, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
4. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.
5. In the **Default input disposition handling**, select one of the following:
   - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
   - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
6. In the **Input Disposition(s)**, specify how input dispositions should be handled. The default is empty, to automatically chain. Enter ***none*** to disconnect the action from dependencies to prior actions, or enter a comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Use a hash sign instead of a comma to create an OR condition.
7. In the **Terminal action**, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
8. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave "false" to allow processing as usual. This option is useful for troubleshooting the process.
9. Click **Submit**.

Metadata import Action

1. In the **Process files from metadata** drop-down, specify if you want to process files from metadata. Select ***false*** to process files from attached input resources or ***true*** to process them from this process groups existing working table. Processing from metadata reduces input/output and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **File filter** field, enter the file filter type. For example: CSV.
3. In the **Import Type** drop-down, select the import type. The valid options are primary and secondary. If you select primary, a new row is created in working tables. If you select secondary, additional columns are added for each metadata.
4. In the **Import file key column** field, for primary imports enter the name of the CSV column, XML tag or JSON property that will serve as the working table's 'FIleName' key for the file. This can also be left blank, in which case the XML file name, excluding the extension will be used with XML files, or a key will be auto-generated in the case of CSV files. For secondary imports, this will be the comma-separated list of column(s) to join against, in the form 'matchingColumn', 'incomingColum:workingTableColumn' (for mapping) 'matchingColumn,incomingColumn:workingTableColumn', etc.
5. In the **Required metadata** field, enter the required/expected metadata in a comma-separated format.
6. In the **Import only required metadata** drop-down, specify if only the required metadata should be imported.
7. In the **delete metadata inputs after import** drop-down, specify if the metadata inputs will be deleted after import.
8. If the import type is primary, specify the number of minutes after which this action will fail if nothing is imported. A value of 0 indicates that the action will not fail. If the import type is secondary, then the Failure Threshold specified below is used.
9. Enter the **primary CSV delimiter character**. This is used for non-XML input, and is a single character such as ",". You can use ***\t*** to represent a tab.

10. Specify if you want to prepend the XML file's relative path to the file key. This is only applicable when XML files are being imported. Set to ***true*** to prepend the relative path to the XML file's filename when generating the working table's file key. This is relevant only when XML files may be inside of a nested folder structure, and you need to retain that path information for processing farther down the line.
11. Optionally, enter the force file key to extension for the primary XML. This is only applicable when XML files are being imported. This option lets you enter ".wav" so that when the metadata is imported from "12345.xml", the application puts"12345.wav" into the working table as the file key instead of just the "12345" as with the default behavior. If the import file key column is provided (#2 above), then this field does not apply.
12. Specify if you want to refresh the process group's working table before importing new data, optionally saving "segments" (which include import file names). This is mainly for secondary metadata like agent, VDN, and other data;
13. Specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
14. Specify whether you want the application to strip the full path from the key column, leaving only the file name.
15. In the **Path information to prepend to key column** field, enter a relative path to manually prepend file path information to incoming file keys.
16. In the **Column header re-mapping** field, This option allows you to use the name of your choice in the CSV file rather than the exact column header names from the working table or the exact tag names from the XML file. You will use the following format: 'incomingHeader1=finalHeader1,iH2=fH2,...'
17. In the **Create new metadata from incoming** field, enter a pipe-separated list of new columns to create, using braces to identify the incoming metadata to generate the new data from, for example: 'UCID={iCompoundId}-{iPBXId}|...' Note that it is also possible to indicate the datatype of the column to be created, and that doing so will trigger an automatic indexing of the column. Continuing the previous example, this can be accomplished using this form: 'UCID:navchar(256)={iCompoundId}-{iPBXId}|...'. Note that there are performance implications to adding a new datatype column that will be indexed in an existing and already-large working table.

18. In the **Force file encoding** field, leave blank in most cases to auto-detect. If auto-detect is not working, enter a valid Java InputStream encoding. See Supported Encodings to learn more.
19. In the **Default quote character** field, enter the character used to quote column values in the output CSV file, for example ".
20. In the **Default escape character** field, enter the character used to escape column values in the output CSV file, for example \.
21. In the **Report-friendly name** field, enter a name to be used in reporting for this action.
22. In the **Default input disposition handling** drop-down list, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
    - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
23. In the **Input Disposition(s)** field, enter the input disposition. The Data Exchange Framework automatically chains action component dispositions together as javascript:void()part of its processing, so this field is usually left blank. If you are using split processes, enter a value in this field.
24. In the **Thread Count** field, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
25. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
26. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
27. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
28. Click **Submit**.

Metadata Join Action

1. Enter the **process group** that you want to join against. This allows you to join data from two different processes.
2. Enter the **key column or columns to join against**. If you enter more than one column, they should be comma separated.
3. Enter the **additional join constraints**. Additional join constraints to add additional SQL constraints to the key columns being joined. This should be in the form of a SQL join subclause. For example: "SQL 'where' subclause form; aliases: wt=working table; pg=join table".
4. Enter the **data column(s) to update**. If you enter more than one column, they should be comma separated.
5. In the **Report-friendly name** field, enter a name to be used in reporting for this action.
6. In the **Default input disposition** handling drop-down list, select one of the following:
   - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
   - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
7. In the **Input Disposition(s)** field, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
8. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.

9. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
10. Click **Submit**.

Metadata Mapping and Export Action

1. In the **Export type** drop-down, select the export file type. The valid options are ***XML*** and ***CSV***.
2. In the **Output file name**, enter a name to that needs to be applied to each of the files generated.
   By default, the output file will be named using a unique combination of process execution ID and timestamp. Optionally, you can include following time-stamp:
   - {TIMESTAMP} – This will be substituted with the current time in milliseconds from the epoch (1/1/70).
   - {DATETIME} – This will be substituted with the current date-and-time (including milliseconds) in the format: yyyy-MM-dd HH-mm-ss_SSS.

   Alternatively, you can specify a date-time-format string enclosed in curly brackets ({...}). Invalid date-time formats are removed from the file-name.
   If you do not include a timestamp parameter with your output file name, the file will be overwritten on each execution of the action.
3. If you selected the XML output type, in the **Generate ESM files** drop-down, specify if you want to create an empty .esm file at the same time the XML file is created. This is used when sending to NIA and the media will not sit alongside the generated XML.
4. In the **Include disposition data in output?** drop-down, specify if disposition data should be included in the output.
5. The **Metadata Tag Remapping** configuration allows for renaming columns passing through to columns that *Interaction Analytics* expects. Click in the blank Metadata Tag remapping field or click to open the Metadata Tag Remapping window and have it generate this field's contents. The column selections will only be ones the process has been made explicitly aware of, as in the "required metadata" field of the Metadata Import action. You can use other metadata columns if you select a "constant" type and enter 'col:metadataColumnName' as the value. You can also enter constants by directly by leaving the 'col:' portion. You can also click the Export Mapping link to copy the mapping configuration to your clipboard and click the Import Mapping link to paste the mapping configuration into the Metadata Tag Remapping input field. When a field is being used in another Inbox Publisher on the same NEAT, the In the Usage column shows the process

that is using it, and the metadata field(s) to which it is mapped.

6. In the **Default separator character** field, enter the character used to separate delimited column values (for example **,**) in the Default separator character field.
7. In the **Default quote character** field, specify the character used to quote column values (for example **"**) in the Default quote character field.
8. In the **Default escape character** field, specify the character used to escape column values (for example ****) in the Default escape character field.
9. In the **Default line end character** drop-down, select the character used to end/terminate a line in the Default line end character drop-down list. The default value is ***\n***.
10. In the **Replace new lines** drop-down, specify if you want to replace new line characters with the selected character in the Replace new lines drop-down list. The default value is ***Do not replace***.
11. (Optional) If you selected the CSV output type, in the **Generate header-only CSV on no result** drop-down, select ***true*** if you want to generate a header-only CSV file if there is no result.
12. In the **Generate FileHashKey** drop-down, select ***true*** to generate a numeric FileHashKey value depending on the file key and the UID of the particular process execution run.
13. In the **Reconciliation Identifiers**, enter the name of the action if there is a reporting action in the process group.
14. In the **Default input disposition handling** drop-down list, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
    - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
15. In the **Input Disposition(s)** field, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.

16. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
17. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
18. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
19. Click **Submit**. If you have selected the CSV option in the Export type, a temporary output file with the extension ***.CSV.NXTMP*** is created first to prevent other processes from accessing incomplete CSV files. After some time, when the file is complete, the action renames the output file to ***.CSV***.

If you specify during Admin Server Configuration that you want to use the *Interaction Analytics* user interface parameters for the Metadata Mapping and Export action user interface display, the Interaction Analytics mapping fields in the Configure Metadata Mapping and Export action accept any customized values from *Interaction Analytics*, and change the user interface names of those fields to match the values in *Interaction Analytics* Search and Reporting.

Metadata Summary Action

Use this action within the process anywhere you need to have information gathered and sent to you on what's been processed up to that point, broken down by an arbitrary number of metadata columns.

1. In the **Metadata columns to group by** field, specify the metadata columns to group. To do this, enter a comma-delimited list of metadata columns (that are in the working table) to group counts by for reporting purposes.
2. In the **Report-friendly name** field, enter a name to be used in reporting for this action.
3. In the **Default input disposition handling** drop-down list, select one of the following:
   - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
   - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
4. In the **Input Disposition(s)** field, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
5. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
6. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
7. Click **Submit**.

NIA File Creator Action

1. In the **Process files from metadata** field, specify if you want to process files from metadata. When you select ***true*** the application processes files from the process groups existing working table. When you specify false, the application processes files from attached input resources. Processing from metadata reduces I/O and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **File filter** field, enter the file filter type. For example: CSV. When processing files from resources, this acts as a filter. When processing from metadata, the metadata's file key is inserted into any wildcards, expanding the processing of the one file to all listed variants. For example, *,*.xml would process both a file matching the working table file key 1234567.wav and its 1234567.wav.xml counterpart.
3. In the **Channel names** field, enter a comma-separated list of channels expected in incoming media files. The names will be used in the corresponding sections of the generated .nia file to represent each channel. The default value is customer, agent.
4. In the **Failure Threshold** field, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
5. In the **Flatten output key** drop-down, specify if you want to flatten the output structure of the generated files.
6. In the **Allow mono files** drop-down, specify if you want to allow mono files. Select **true** to allow mono files to pass through the action without error, or "false" to error out mono files and track them as failures.
7. In the **Minimum non-speech duration (ms)** drop-down, specify the minimum duration in milliseconds for identifying non-speech segments. The default value is ***2000***.
8. In the **Minimum speech duration (ms)** drop-down, specify the minimum duration in milliseconds for identifying speech segments. The default value is ***50***.
9. In the **Speech flare (ms)** drop-down, specify the Workbench speech flare setting to use, in milliseconds. The default value is ***50***.
10. In the **Report-friendly name** field, enter a name to be used in reporting for this action.
11. In the **Default input disposition handling** drop-down list, select one of the following:

- ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
- ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.

12. In the **Input Disposition(s)** field, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
13. In the **Thread Count** field, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
14. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
15. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
16. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
17. Click **Submit**.

NIA Reprocessing Action

1. In the **File filter**, enter the file filter type. For example: CSV.
2. In the **Batch size**, enter the batch size that you want to re-process. The maximum batch size is 100,000.
3. In the **Failure Threshold**, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail..
4. In the **Report-friendly name**, enter a name to be used in reporting for this action.
5. In the **Default input disposition handling** drop-down list, select one of the following:
   - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
   - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
6. In the **Input Disposition(s)**, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
7. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
8. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
9. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
10. Click **Submit**.

NTR Centera Retriever Action

1. In the **Centera pool address**, enter one or more Centera pool IP addresses in the Centera pool address field. Separate multiple pool IP addresses with commas (i.e. 168.159.214.27, 168.159.214.28).
2. In the **PEA file path**, enter a fully qualified path to an optional PEA file containing Centera authorization credentials (for example, C:\\Users\\Administrator\\NEAT\\etc\\c2armtesting.pea) in the PEA file path field. The PEA file contains center authorization credentials. The Centera file store is password protected, and the PEA file contains the credentials required to access the center file store.
3. In the **Centera clip ID column name**, enter the name of the column holding the unique clip ID for each file in the Centera clip ID column name field.
4. In the **Centera tag name**, enter the name of the Centera tag name used to identify a file to be extracted (for example, Encrypted) in the Centera tag name. You can apply or tag files with whatever random information you would like. The Centera tag name is used to identify files that have been archived.
5. In the **Failure Threshold**, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
6. In the **Report-friendly name**, enter a name to be used in reporting for this action.
7. In the **Debug Top Tag** drop-down, select ***true*** to add debugging log on top tag.
8. In the **Reconciliation Identifiers**, enter the name of the action if there is a reporting action in the process group.
9. In the **Default input disposition handling** drop-down list, select one of the following:
   - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
   - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.

10. In the **Input Disposition(s)**, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
11. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
12. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
13. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
14. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
15. Click **Submit**.

NTR Decrypt Action

1. In the **File filter**, enter the file filter type. For example, '.ntr'. You can also use '.*xml', '.*csv', etc. Java style regular expressions can also be used. When processing files from resources, this acts as a filter, but when processing from metadata, the metadata file key is inserted into any wildcards, expanding the processing of the one file to all listed variants for example ('*,*.xml', etc. would process both a file matching the working table file key '1234567.wav'' and its ''1234567.wav.xml' counterpart).
2. In the **Call time column**, enter the column name in the imported metadata that contains the time the media file was recorded in the Call time column field. This is used to resolve the decryption key.
3. In the **Token ID column**, enter the column name in the imported metadata that contains the identifier for the source system in the Token ID column field. This is used to resolve the decryption key in systems where they keys and metadata are accumulated from multiple NTR recorders, but decryption only happens on demand (Smart Index, etc.).
4. In the **Delete originals after decrypting** drop-down, specify if you want to delete original files after decrypting.
5. Specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
6. In the **Report-friendly name**, enter a name to be used in reporting for this action.
7. In the **Default input disposition handling** drop-down list, select one of the following:
   - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
   - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
8. In the **Input Disposition(s)**, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.

9. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
10. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
11. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
12. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
13. Click **Submit**.

NTR Stitch Action

1. Enter the **Vox file name** column. If not using the default FileName value, make sure your column is indexed for best performance (either directly in SQL Server or by setting a 'RequiredField(s)' option with a datatype in a preceding Metadata Import action.
2. Enter the **CDR call ID column** name.
3. Enter the **Vox call ID column name**.
4. Enter the **Call start column**.
5. Enter the **Call end column**.
6. Enter the **Call start millisecond fraction column**.
7. Enter the **Call end millisecond fraction column**.
8. Enter the **Vox start column**.
9. Enter the **Vox end column**.
10. Enter the **Vox call start millisecond fraction column**.
11. Enter the **Vox call end millisecond fraction column**.
12. Select the codec time resolution in milliseconds from the drop-down list. The default value is zero for no time correction. 40 milliseconds works for GSM-FR65-byte frames. This option was added for a specific problem correcting time frames on GSM-FR media where a small number of milliseconds were being lost from the ends of VOX records during stitching, and caused mostly inaudible audio gaps.
13. Specify the **stitch silence from the** drop-down list. Set to ***false*** to disable the stitching of silence into calls to match reported media durations in metadata.
14. Specify if you want the application to **delete segments after stitching**.
15. Specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
16. Enter the silence buffer size in bytes. This is the size of the buffer to use when stitching silence into calls. The recommended is multiples of 4096.
17. Enter a name to be used in reporting for this action.
18. In the **Default input disposition handling** drop-down list, select one of the following:

- ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
- ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.

19. Optionally enter the ***input disposition***. The Data Exchange Framework automatically chains action component dispositions together as javascript:void()part of its processing, so this field is usually left blank. If you are using split processes, enter a value in this field.
20. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
21. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
22. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
23. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
24. Click **Submit**.

OneAnalytics Publisher Action

1. In the **Interaction ID column name**, enter the name of the column in the metadata that holds the interaction ID to pass to OneAnalytics. In most cases, this should be an ExternalMediaId captured upstream, either with a File Mover or the ExternalMediaIdCapture action.
2. In the **Interaction Type**, enter the OneAnalytics interaction type.
3. In the **Channel map**, enter the mapping of which speaker (agent or customer) belongs to which channel in stereo media files.
4. In the **Delete incoming media after processing** drop-down, specify if you want to delete the incoming media after processing.
5. Enter the **AWS access key**.
6. Enter the **AWS secret key**.
7. Enter the **AWS region**.
8. Enter the **OneAnalytics tenant ID**.
9. Enter the **service user name**.
10. Enter the **AWS cognito user pool ID**.
11. Enter the **AWS cognito identity pool ID**.
12. Enter the **AWS cognito application client ID**.
13. In the **Destination Location**, enter the AWS S3 bucket that media and metadata will be dropped into.
14. In the **Media queue name**, enter the AWS SQS message queue name.
15. In the **Failure Threshold**, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
16. In the **Language ID**, enter the language identifier to be mapped to a OneAnalytics language pack.
17. In the **Session Credentials expiration (minutes)**, specify how long before session credentials are configured in AWS to expire. Note that this is for long runs of this action, where the AWS access token has expired before the end of the run. Once this number of minutes has passed, a new token will be acquired.

18. In the **Remove the interaction ID extension** drop-down, select ***true*** to remove the file extension from the interaction ID or ***false*** to not remove it. The default value is ***true***.
19. In the **Force interaction ID to lowercase** drop-down, select ***true*** to force the interaction ID to lowercase or false to not force the interaction ID to lowercase. The default value is ***true***.
20. In the **Disable certificate checking** drop-down, select ***true*** to disable certificate checking (recommended only for testing) ***false*** to not disable certificate checking. The default value is ***false***.
21. In the **SSM Endpoint**, enter the AWS Simple Management service endpoint, for example https://10.180.34.36.
22. In the **SQS Endpoint**, enter the AWS Simple Queue System service endpoint, for example https://10.180.34.36.
23. In the **Report-friendly name**, enter a name to be used in reporting for this action. This is only relevant when another process in this process group has a Reporting action, and you want to give this action a special name for it to use.
24. In the Default input disposition handling drop-down list, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
    - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
25. In the **Input Disposition(s)**, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
26. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
27. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.

28. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
29. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
30. Click **Submit**.

OneAnalytics Query Retriever Action

1. Enter the **AWS access key**.
2. Enter the **AWS secret key**.
3. Enter the **AWS region**.
4. Enter the **OneAnalytics tenant ID**.
5. Enter the **service user name**.
6. Enter the **AWS cognito user pool ID**.
7. Enter the **AWS cognito identity pool ID**.
8. Enter the **AWS cognito application client ID**.
9. Enter the **AWS S3 location** that media and metadata will be dropped into.
10. Enter the **AWS SQS message queue name**.
11. In the **Failure Threshold**, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
12. Specify how long before session credentials are configured in AWS to expire. This is for long runs of the action, where the AWS access token has expired before the end of the run. Once this number of minutes has passed a new token is acquired.
13. In the **Report-friendly** name, enter a name to be used in reporting for this action.
14. In the **Default input disposition handling** drop-down list, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
    - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
15. In the **Input Disposition(s)**, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.

16. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
17. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
18. Click **Submit**.

Parse Email Text to JSON

1. In the **Process files from metadata** drop-down, specify if you want to process files from metadata. When you select ***true***, the application processes files from the process groups existing working table. When you specify ***false***, the application processes files from attached input resources. Processing from metadata reduces input/output, and file system scans when some previous action needs to have processed something from input resources first.
2. The **File Filter** is the general text file name filter, and by default, this field shows ****.eml,*.msg***.
3. In the **Delete files after processing** drop-down, select ***true*** if you want to delete input text files after parsing text.
4. In the **Clean text** drop-down, select ***true*** if you want to remove multiple blank lines and carriage returns from the parsed string.
5. In the **Clean symbols** drop-down, select ***true*** if you want to remove special symbol characters like the copyright symbol from the parsed string.
6. In the **Parse text from attachments** drop-down, select ***true*** if you want to parse text from email attachments.
   The system cannot parse the attached zip files that are password protected.

7. In the **Attachment write Limit (KBs)** drop-down, specify the write limit (specified in KBs, for example, 1000). The default value is -1 (no write limit).
8. In the **Output Format** drop-down, select ***TextGrid JSON*** to generate output in the TextGrid JSON format, select ***VoiceGrid ASR*** to generate an output in the VoiceGrid ASR JSON format, and select ***TextGrid JSON & VoiceGrid ASR*** to generate both TextGrid JSON and VoiceGrid ASR JSON formats.
9. In the **Failure Threshold**, specify the number of failures that can cause processing to stop. A value of 0 indicates that the process can never fail.
10. The **Text encoding** is used when parsing text data and is only applicable for the email subject field. Use UTF-8 as default. For Hindi-based languages, use ISO-8859-1.
    The Metadata parse strategy is used when parsing the email metadata. The "tika" option uses Apache Tika, and the "jakarta" option uses Jakarta mail to parse the metadata from the specified email. Note that if the file type is ****.msg***, select the Metadata parse strategy as "tika."
11. In the **Metadata parse strategy**, specify the parsing strategy to use when parsing the email metadata. The tika option uses Apache Tika to parse the metadata from the specified email. The jakata option uses Jakata Mail to parse the metadata from the specified email. tika is the default parsing strategy. If the file type to be processed is *.msg, then select the tika metadata parsing strategy.
12. In the **Unique Output File Name**, use the process execution UID to make the output file name unique.
13. In the **Redact Email Text** drop-down, select ***true*** to redact email text to protect sensitive information in your email before sending data to Analytics. Select ***false*** to not redact.
    - If you select ***true*** In the **Redact Email Text** drop-down:
        - In the **Length of Redaction Replacements** drop-down, enter the value to set the length of each redaction in the text. The minimum value is 3, the maximum is 8, and the default is 5.
        - In the **List of Regular Expressions for redaction**, click the **Edit** icon to enter one or more regular expressions for selecting text to redact.
        - In the **Redact Email Text Character**, enter the character used to replace redacted text. The default is X.

The action applies your defined regular expression patterns to the full email text. Matching content is redacted before conversion to TextGrid format. Redacted emails are then sent to Analytics.

14. In the **Reconciliation Identifier**, enter the name of the action if there is a reporting action in the process group.
15. In the **Default input disposition handling** drop-down list, select one of the following:
    - ***AND*** - Allows files to flow through this step that are passed all of the immediate inputs.
    - ***OR*** - Allows files to flow through this step that are passed any of the immediate inputs.
16. In the **Input Disposition(s)**, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file is already passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type 'none' to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
17. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
18. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
19. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
20. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
21. Click **Submit**.

The email media metadata is captured as a First-Class NIA field called "EmailMetadata" in the DEF working table through the Inbox Publisher action and sent to NIA through the NexidiaESI SourceMediaInbox table.

Parse General Text to JSON

1. In the **Process files from metadata** drop-down, specify if you want to process files from metadata. When you select ***true***, the application processes files from the process groups existing working table. When you specify ***false***, the application processes files from attached input resources. Processing from metadata reduces input/output, and file system scans when some previous action needs to have processed something from input resources first.
2. The **File Filter** is the general text file name filter, and by default, this field shows ****.txt,*.rtf,*.pdf,*.doc,*.xls,*.docx,*.xlsx,*.ppt,*.pptx***.
3. In the **Delete files after processing** import drop-down, select ***true*** if you want to delete input text files after parsing text.
4. In the **Clean text** drop-down, select ***true*** if you want to remove multiple blank lines and carriage returns from the parsed string.
5. In the **Clean symbols** drop-down, select ***true*** if you want to remove special symbol characters like the copyright symbol from the parsed string.
   If the character set is Non-English (i.e. Hindi, Chinese, etc), do not set Clean symbols to ***true***.
6. In the **Output Format** drop-down, select ***TextGrid JSON*** to generate output in the TextGrid JSON format, select ***VoiceGrid ASR*** to generate an output in the VoiceGrid ASR JSON format, and select ***TextGrid JSON & VoiceGrid ASR*** to generate both TextGrid JSON and VoiceGrid ASR JSON formats.
7. In the **Failure Threshold** drop-down list, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process can never fail.
8. In the **Unique Output File Name**, use the process execution UID to make the output file name unique.
9. In the **Redact text** drop-down, select ***true*** to redact email text to protect sensitive information in your email before sending data to Analytics. Select ***false*** to not redact.

- If you select ***true*** In the **Redact text** drop-down:
  - In the **Redact text character**, enter the character used to replace redacted text. The default is X.
  - In the **Length of redaction replacements** drop-down, enter the value to set the length of each redaction in the text. The minimum value is 3, the maximum is 8, and the default is 5.
  - In the **List of regular expressions for redaction**, click the **Edit** icon to enter one or more regular expressions for selecting text to redact.

The action applies your defined regular expression patterns to the full email text. Matching content is redacted before conversion to TextGrid format. Redacted emails are then sent to Analytics.

10. In the **Reconciliation Identifier**, enter the name of the action if there is a reporting action in the process group.
11. In the **Default input disposition handling** drop-down list, select one of the following:
    - ***AND*** - Allows files to flow through this step that are passed all of the immediate inputs.
    - ***OR*** - Allows files to flow through this step that are passed any of the immediate inputs.
12. In the **Input Disposition(s)** field, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file is already passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
13. In the **Thread Count** field, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the Threads per action setting at the server level.
14. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.

15. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
16. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
17. Click **Submit**.

PST to EML Action

1. In the **Process files from metadata** drop-down, specify if you want to process files from metadata. When you select ***true*** the application processes files from the process groups existing working table. When you specify false, the application processes files from attached input resources. Processing from metadata reduces I/O and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **File filter** field, change this only if your PST file will come with a different extension.
3. In the **Delete files after processing** drop-down, specify if you want the application to delete files after processing. When you select ***true*** the application deletes the .pst files after processing.
4. In the **Working table tracking** drop-down, specify the working table tracking. The valid options are ***PST*** or ***EML***. Select ***PST*** to only track .pst files in the working table, or ***EML*** to track those files instead. When you select ***EML***, the .pst files are recorded but are also flagged for no further tracking.
5. In the **Retain PST folder structure** drop-down, specify if you want to retain the PST folder structure. Select ***false*** to flatten the output of the .eml files to the output resource, or ***true*** to retain the folder structure defined in the .pst file.
6. In the **Prepend process execution UID to top-level folder name** drop-down, specify if you want to prepend the process execution UID to top-level folder name. This field displays if you selected the ***true*** option in the Retain PST folder structure field above,
7. In the **Remove text/disclaimer** field, enter a list of one or more text strings that should be removed from the email body in the Remove text/disclaimer field. Regular expressions are allowed. When you click in this field, the Remove Text window opens. You can configure to remove text by doing one of the following:
   - Remove text matching a specific, literal string.
   - Remove text matching a regex. If a regex is used, you must check the **Is Regular Expression** check box for that entry. The text that you specify to remove from the email body will not be included in indexing by Nexidia Text Grid.

8. In the **Failure Threshold** field, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
9. In the **Report-friendly name** field, enter a name to be used in reporting for this action.
10. In the **Default input disposition handling** drop-down list, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
    - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
11. In the **Input Disposition(s)** field, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
12. In the **Thread Count** field, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
13. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
14. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
15. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
16. Click **Submit**.

Process Chain Action

1. In the **Process** field, select another process in this processes same process group to chain to from this list. If both processes belong to the same process group, input dispositions are chained together. If not, the process is launched.
2. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
3. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
4. Click **Submit**.

Process Chaining

The idea of process chaining comes from the need to have processes in the same process group running independently, on their own schedules, but with the input dispositions of one processes actions tied to the output dispositions of the actions in another process. For example, process #1 might constantly discover and move files, and process #2 might take those files and generate NIA files from them. Since the NIA File action would require files to have been completely moved, it would depend on files having been touched by the File Mover in process #1.

Prior to DEF 3.5, processes like these could be chained together by configuring the "Input Disposition(s)" option in the "Advanced" section of the NIA File action to match the output disposition in process #1 ("Disposition-FileMoverAction-####"). Starting in DEF 3.5, the Process Chain action facilitates process chaining in a more portable way. Additionally, it's also made it possible for one process to trigger the launch of another. For example, process #1 can fire process #2, at which point process #1 is free to launch again on its schedule while process #2 handles downstream processing. Furthermore, process #2 could launch another process #3 further downstream, etc.

The Process Chain action can be added to the start or end of a process, and the position determines how it's used. When the action is a process exit point, with no Process Chain entry point in the configured target process, the Process Chain action launches the configured process without chaining any dispositions. If a Process Chain action exists as an entry point in the target process, however, then the input dispositions will be automatically chained together in such a way that the inputs to process #1's Process Chain action will carry over to the action(s) connected to process #2's Process Chain entry point.

There are no restrictions on circular dependencies, so if you launch process #1, and it has a Process Chain that launches process #2, and process #2 has a Process Chain that launches process #1, they'll keep firing each other off in a loop until interrupted with either an "End Process" command or a DEF service restart. You should be careful how you use this process!

Process Launch Action

1. In the **Process** drop-down, select a process for this action to launch in the Process drop-down list. This allows you to launch a process from a process without consideration for the process group.
2. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
3. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
4. Click **Submit**.

Configure Production Media Folder Path Action

1. In the **Production Media Folder Path** field, enter the media folder structure that will be created in the output resource based on the specified metadata. This will also become the ingest file path in the NIA.Example: media\{metadata1}\{metadata2}.
2. In the **Failure Threshold** field, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
3. In the **Report-friendly name** field, enter a name to be used in reporting for this action.
4. In the **Default input disposition handling** drop-down list, select one of the following:
   - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
   - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
5. In the **Input Disposition(s)** field, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type 'none' to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
6. In the **Thread Count** field, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
7. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
8. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.

9. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
10. Click **Submit**.

QC interactions Audit Action

1. In the **Engage Audit Method** field, specify the engage audit method. The valid options are:
   - ***Direct to DB***
   - ***Direct to CSV***
   - ***CSV to DB***
2. In the **Source system** field, enter the user-defined display name for the Source System.
3. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. Actions are considered terminal when they have a corresponding disposition column in the working table, and they are not referenced by other actions downstream. This option is used in cases where the default behavior is not desirable
4. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
5. Click **Submit**.

Redaction Action

1. In the **File filter** field, enter the file filter type. For example: CSV.
2. In the **Redact tone** field, enter an audio tone to use for redaction. If left blank, the tone for redacted segments is silence.
3. In the **Delete inputs after move** drop-down, specify if you want to delete inputs after move. Select "true" for a true file move or "false" for to copy the file and leave inputs intact.
4. In the **Failure Threshold** field, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
5. In the **Report-friendly name** field, enter a name to be used in reporting for this action.
6. In the **Default input disposition handling** drop-down list select one of the following:
   - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
   - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
7. In the **Input Disposition(s)** field, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
8. In the **Thread Count** field, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
9. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.

10. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
11. Click **Submit**.

Redaction Response Action

1. Enter the **RabbitMQ host name** for login.
2. Enter the **RabbitMQ port number** for login. The default value is 5672.
3. Enter the **RabbitMQ user name** for login.
4. Enter the **RabbitMQ password**.
5. Enter the **RabbitMQ virtual host** for login.
6. In the **Failure Threshold** field, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
7. In the **Report-friendly name** field, enter a name to be used in reporting for this action.
8. In the **Default input disposition handling** drop-down list, select one of the following:
   - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
   - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
9. In the **Input Disposition(s)** field, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
10. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
11. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.

12. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave "false" to allow processing as usual. This option is useful for troubleshooting the process.
13. Click **Submit**.

Redaction Submit Action

1. In the **Process files from metadata** drop-down, specify if you want to process files from metadata. When you select ***true*** the application processes files from the process groups existing working table. When you specify false, the application processes files from attached input resources. Processing from metadata reduces I/O and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **File filter** field, enter the file filter type. For example: CSV.
3. Enter the **RabbitMQ host name** for login.
4. Enter the **RabbitMQ port number** for login. The default value is 5672.
5. Enter the **RabbitMQ user name** for login.
6. Enter the **RabbitMQ password**.
7. Enter the **RabbitMQ virtual host** for login.
8. In the **Workflow ID** field, enter the UUID of the model that Redaction Grid should use in determining segments to redact.
9. In the **Send Workbench Logs** drop-down, specify if you want to send workbench logs to diagnose the redaction issues.
10. In the **Generate transcripts** drop-down, specify if you want to generate Voice Grid transcript files. If you select this option, a file output resource must be specified and it must be a network path.
11. In the **Failure Threshold** field, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
12. In the **Report-friendly name** field, enter a name to be used in reporting for this action.
13. In the **Default input disposition** handling drop-down list select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
    - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.

14. Optionally enter the Input Disposition. The Data Exchange Framework automatically chains action component dispositions together as javascript:void()part of its processing, so this field is usually left blank. If you are using split processes, enter a value in this field In the **Input Disposition(s)** field, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type 'none' to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
15. In the **Thread Count** field, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
16. In the **Maximum files to proces**s field, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
17. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
18. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave "false" to allow processing as usual. This option is useful for troubleshooting the process.
19. Click **Submit**.

Remove Duplicate Metadata Action

1. In the **Delete Duplicate Metadata** drop-down, select ***true*** if you want to delete the duplicate calls from the working table. Select ***false*** if you want to flag the duplicate calls as segments to be rolled off to the rollover table.
2. In the **Failure Threshold** drop-down list, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
3. In the **Reconciliation Identifiers**, enter the name of the action if there is a reporting action in the process group.
4. In the **Default input disposition handling** drop-down, select one of the following:
   - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
   - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
5. Optionally enter the **Input dispositions**. The Data Exchange Framework automatically chains action component dispositions together as javascript:void()part of its processing, so this field is usually left blank. If you are using split processes, enter a value in this field.
6. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
7. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
8. Click **Submit**.

Reporting Action

1. In the **Gather report data for previous # days** drop-down, specify the number of previous day's data you want to include for reporting.
2. In the **Date column override**, specify another working table column of timestamp data type in the Data Column Override field. By default, days are broken down by the working table UpdateDate column. The value you enter must be the correct data type and should be indexed for the best performance.
3. In the **Output file name**,specify a name in the Output File Name field. If you do not specify a name here, the default name of extractor-report--{timestamp}.csv is used. The timestamp is in milliseconds since the epoch (1/1/1970). You can use the {yyyy}, {mm}, {dd}, and {processGroup} substitution strings.
4. In the **Metadata columns to group by**, enter the metadata columns to group by in a comma-delimited format By default, the reporting action will break down the number of records in each available disposition for the requested time period. To break down further (for example by agent), enter a comma-delimited list of fields to group by.
5. In the **Exclude dispositions with zero results** drop-down, specify if you want the application to exclude disposition data that contains zero results.
6. In the **Exclude dispositions of actions with no report-friendly names** drop-down, specify if you want the application to exclude disposition data that contains no report friendly names.
7. In the **Query rollover table** drop-down, specify if you want to generate the report based on working table or rollover table. The working table shows in-process items and the rollover table shows completed items. Specify ***true*** to get completed counts.
8. In the **Terminal action** drop-down, specify if this is a terminal action. The working table rows touched by terminal actions are pushed into a working table rollover table on process completion. By default, actions are considered terminal when they have a corresponding disposition column in the working table and no other actions downstream reference them. This option is provided for cases where the default behavior is not desirable..
9. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for

troubleshooting.

10. Click **Submit**.

Replace CSV Header action

1. In the **Process files from metadata**, specify if you want to process files from metadata. When you select "true" the application processes files from the process groups existing working table. When you specify false, the application processes files from attached input resources. Processing from metadata reduces I/O and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **File filter**, enter the file filter type. For example: CSV.
3. In the **Replacement header**, enter the header that will replace the existing header. This should be comma-delimited in a form that matches the file header being replaced.
4. In the **Failure Threshold**, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
5. In the **Force file encoding**, enter the character encoding of the input file. You will leave this field blank in most cases to auto-detect. If auto-detect isn't working, use a valid Java InputStream encoding.
6. In the **Report-friendly name**, enter a name to be used in reporting for this action.
7. In the **Default input disposition handling** drop-down list, select one of the following:
   - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
   - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
8. In the **Input Disposition(s)**, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type 'none' to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR

condition.

9. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
10. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
11. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
12. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave false to allow processing as usual. This option is useful for troubleshooting the process.
13. Click **Submit**.

RTG Agent Mapping Action

The RTG Agent Mapping allows you to map the agent data to the Real-time Grid (RTG) system. This RTG agent-mapping action accepts CSV-files containing agent-mapping data to upload to RTG. It tracks that data using a dedicated database-table.

The action can be configured for the following types of agent-csv source type.:

- Engage
- Mattersight

The action requires the process designer to choose the mapping of the key RTG items to column names in the source CSV file and these choices change depending on the type of the agent-CSV source selected. Also, the mapping data of subsequent process runs replaces the previously uploaded mapping information. If no new information is provided, the previous data reaffirm with the RTG system.

The action requires a file system or SFTP resource input and database output. If you do not provide input resource, one is either inferred from adjacent action I/O or automatically provisioned in the {DEF_DATA}/fs/ folder.

Properties

PARAMETER	SETTING	DESCRIPTION
Agent CSV source type	***Engage***	Options: • Engage • Mattersight This option impacts the availability of the CSV column-mapping options (advanced) described in this table.
Default separator character	***,*** (comma)	Only applicable when CSV files are being imported. \t = tab
Default escape character	**** (back slash)	The character used to escape column values in the output CSV. (i.e. \\).
Default quote character	***“*** (double quote)	The character used to quote column values in the output CSV. (i.e. “)
Delete Agent CSV after processing	***True***	Deletes the agent csv-file after successful processing. This defaults to TRUE.
Disable action	***False***	Set to 'true' to prevent this action from processing any files during its run. Leave 'false' to allow processing as usual.
Engage Agent CSV Agent ID column name	***AgentId***	**Advanced**. When “Engage” CSV-type is selected, this declares the name of the “AgentId” column in the CSV-file.
Engage Agent CSV Domain column name	***Domain***	**Advanced**. When “Engage” CSV-type is selected, this declares the name of the “Domain” column in the CSV-file.

PARAMETER	SETTING	DESCRIPTION
Engage Agent CSV OSLogin column name	***OSLogin***	**Advanced**. When “Engage” CSV-type is selected, this declares the name of the “OS Login” column in the CSV-file.
Engage Agent CSV IUserId column name	***IUserId***	**Advanced**. When “Engage” CSV-type is selected, this declares the name of the “IUserId” column in the CSV-file.
Engage Agent CSV HubId column name	***HubId***	**Advanced**. When “Engage” CSV-type is selected, this declares the name of the “HubId” column in the CSV-file.
File filter	***.csv**	'*.xml', '*.xml,*.csv', etc. You can also use Java-style regular expressions if prepended with 'regex:'. When processing files from resources, this acts as a filter, but when processing from metadata, the metadata's file key is inserted into any wildcards, expanding the processing of the one file to all listed variants ('*,*.xml', e.g., would process both a file matching the working table file key '1234567.wav' and its '1234567.wav.xml' counterpart). Based on the following filter, you can use the same folder to house different CSV source types (see the "agent-csv source type" property in the table). The typical filters are: • *Engage_Agent*.csv (for Engage data) • *Engage_Agent*.csv (for Mattersight data)

PARAMETER	SETTING	DESCRIPTION
Force file encoding	***UTF-8***	Leave empty. In most cases it auto-detect. If auto-detect does not works, you can specify a valid Java InputStream encoding. For more information on supported encoding, see encoding.doc.html
Mattersight Agent CSV Agent ID column name	***AgentId1***	**Advanced**. When "Mattersight" CSV-type is selected, this declares the name of the "Agent Id" column in the CSV-file.
Mattersight Agent CSV Domain column name	***True***	**Advanced**. When "Mattersight" CSV-type is selected, this declares the name of the "Domain" column in the CSV-file.
Mattersight Agent CSV ACDId column name	***ACDId1***	**Advanced**. When "Mattersight" CSV-type is selected, this declares the name of the "ACD Id" column in the CSV-file.
Mattersight Agent CSV HubId column name	***HubId***	**Advanced**. When "Mattersight" CSV-type is selected, this declares the name of the "Hub Id" column in the CSV-file.
NIA web service URL		The URL service that returns the agents slated for observation in the RTG system. The agents returned from this service are typically a subset of the agents included in the source CSV-files. The intersection of these two sets of data are sent to RTG.

Salesforce Metadata Parser Action

1. In the **Metadata ID Field Name**, enter the name of the metadata column in the working table that contains the unique ID for a specific metadata row.
   For example: agentID, userId, teamId.

2. In the **Media Parse Type**, select the type of media to be parsed. The default value is ***chat***.
3. In the **Metadata Message Field Name**, enter the metadata message field name, for example – ***Body*** or ***Message***.
4. In the **Bot/System Name**, specify the Bot/System/Chatbot name. For example, ***SelfServiceBot***.
5. In the **Process All Messages**, specify if you want to process all messages. Set to ***false*** to process only messages that have Agents. Default value is ***true,*** which will process all messages.
6. In the **Default Separator Character**, enter the character used to separate delimited column values in the output CSV.
7. In the **Default Quote Character**, specify the character used to quote column values (for example **"**) in the Default Quote Character field.
8. In the **Default Escape Character**, specify the character used to escape column values (for example ****) in the Default escape character field.
9. In the **Default Line End Character**, enter the character used to terminate a line in the output file. The default is \n.
10. In the **Failure Threshold** drop-down , specify the number of failures that cause processing to stop. A value of 0 indicates that the process never fails.
11. In the **Remove System/Bot Messages** drop-down, set to ***true*** to remove system-generated (bot) messages from the message body. The default value is ***false***, which will not remove system-generated entries (bot) from the message body.
12. In the **Fields To Populate**, enter a comma-delimited list of fields to show the generated output file.
13. In the **Parse Text From HTML Messages**, specify if you want to parse text from chats that contain HTML. The default is ***true*** to parse text from messages containing HTML.
14. In the **Generate Output Per Agent** drop-down, select true to split chat conversations between multiple agents and the customer and save them in separate files. The default value is ***false***, which will generate single output file that contains all chat conversations of agents.

- If you set Remove System/Bot Messages parameter to true and Generate Output Per Agent parameter to true:
  - Chat conversations between system/bot and the customer are removed from the chat thread until the next human agent is encountered.
  - Chat conversations between multiple agents and the customer are split and saved in separate output files.
- If you set Remove System/Bot Messages parameter to false and Generate Output Per Agent parameter to false, all chat conversations are saved in a single output file.
- If you set Remove System/Bot Messages parameter to false and Generate Output Per Agent parameter to True:
  - Chat conversations between system/bot and the customer are saved in a output file.
  - Chat conversations between multiple agents and the customer are split and saved in separate output files.
- If you set Remove System/Bot Messages parameter to true and Generate Output Per Agent parameter to false:
  - Chat conversations between system/bot and the customer are removed from the chat thread until the next human agent is encountered.
  - Chat conversations between multiple agents and the customer are saved in a single output file.

15. In the **Delete Message Text From WT**, delete message text from working table after processing to preserve resources. The default value is ***true***, which will delete message text from working table.
16. In the **Custom Metadata**, enter a comma-delimited list of custom metadata fields to extract.
17. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.
18. In the **Default input disposition handling** drop-down list, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.

- ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.

19. In the **Input Disposition(s)**, specify how input dispositions should be handled. The default is empty, to automatically chain. Enter ***none*** to disconnect the action from dependencies to prior actions, or enter a comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Use a hash sign instead of a comma to create an OR condition.
20. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the **Threads per action** setting at the server level.
21. From the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
22. In the **Disable action**, specify ***true*** if you want to prevent this action from processing any files during its run. Leave the default value ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
23. Click **Submit**.

Salesforce Metadata Retriever Action

1. In the **Process files from metadata**, specify if you want to process files from metadata. When you select ***true*** the application processes files from the process groups existing working table based on the metadata ID field provided. When you specify ***false***, the application will extract media from Salesforce based on a provided URL, query, parameters and quantum. Default is false.
2. From the **Uses quantum** drop-down, select ***true*** to use a quantum window to retrieve data. The default value is false. When **processFromMetadata** is set to ***false*** (default), then **Uses quantum** must be set to ***true***.

3. In the **Salesforce Client ID**, enter the Client ID to access the Salesforce API.
4. In the **Salesforce Client Secret**, enter the Secret ID to access the Salesforce API.
5. In the **Base Endpoint URL**, enter the URL that is the base for all endpoints from which analytics, conversation, and recording metadata can be queried against the Salesforce API. For example, https://zenefits.my.salesforce.com
6. In the **Access Token URL**, enter the access token URL to use for authentication. Example: "{baseUrl}/services/oauth2/token"
7. In the **Data Extraction URL**, enter the extraction URL. For example: /services/data/v56.0/query/?q=
8. In the **SQL Query**, enter the query of metadata fields to extract.
9. From the **Response parse type** drop-down, select the type of response expected to be returned by the Salesforce REST API:
   - List Type
   - Embedded List Type (default)
   - Map Type | Pivot List
10. In the **Metadata ID Field Name**, enter the name of the metadata column in the working table that contains the unique ID for a specific metadata row.
    For example: agentID, userId, teamId.
11. In the **Embedded list node name**, enter the name of the JSON element that contains the embedded list.
    For example;
    { “data”: [{...},
    {...},
    {...}]
    }
    The embedded list node name for this example would be “data”.
12. In the **Metadata Message Field Name**, enter the metadata message field name, for example – ***Body*** or ***Message***.
13. In the **Failure Threshold** drop-down , specify the number of failures that cause processing to stop. A value of 0 indicates that the process never fails.
14. In the **Salesforce Username**, specify the name for Salesforce account.

15. In the **Salesforce Password**, specify the password for Salesforce account.
16. In the **Access Token**, specify the authorization token issued for Salesforce account.
17. In the **Salesforce API version**, specify the Salesforce API version. For example, 50.0.
18. In the **Grant Type,** enter the grant type. A grant type of client credentials indicates that you are using OAuth 2.0 for authorization.
19. In the **Column Header Re-mapping** field, specify the name of your choice for a specific field to be populated in a working table or in CSV. The default format is: incomingHeader1=finalHeader1,iH2=fH2,...
20. In the **Write Output Json File,** set to ***true*** to write API response to JSON file. Default is ***false***.
21. In the **Custom Metadata,** enter a comma-delimited list of custom metadata fields to extract.
22. In the **Rename Metadata Fields**, enter a comma delimited list of metadata field names to rename. The default values are startdate,enddate,filename.
23. In the **Rename Metadata Fields Suffix**, enter the string to be added as a suffix to renamed metadata fields. Default is ***_sf***. For example, enddate would be renamed to enddate_sf.
24. From the **Log Service Responses** drop-down, specify if you want to log the Salesforce service responses. The default value is ***false***, so no response logging occurs. If you set the value of this parameter as ***true***, the log file is created in *<user temp folder>/Salesforce-Response.log*.
25. In the **HTTP Client Timeout**, enter the Apache HTTP client default timeout value. The default is 60 seconds.
26. In the **Retry Wait Time** drop-down, specify the amount of time in milliseconds to wait between calls/retries to the API endpoint. Default is 500.
27. In the **Number of Retries**, enter the number of retries that you want for the API endpoint before failing. Default is 20.
28. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.
29. In the **Default input disposition handling** drop-down list, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.

- ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.

30. In the **Input Disposition(s)**, specify how input dispositions should be handled. The default is empty, to automatically chain. Enter ***none*** to disconnect the action from dependencies to prior actions, or enter a comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Use a hash sign instead of a comma to create an OR condition.
31. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the **Threads per action** setting at the server level.
32. From the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
33. In the **Disable action**, specify ***true*** if you want to prevent this action from processing any files during its run. Leave the default value ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
34. Click **Submit**.

SalesLoft Media Retriever Action

1. In the **Recording URL column name**, specify the name of the recording URL.
2. In the **Recording file extension**, specify the recording file extension. Defaults extension is ***wav***.
3. In the **Throw exception if media not found (404)?** drop-down, select ***true*** if you want to display the media cannot be found (404) message. Otherwise, log "404 NOT FOUND" to the "recording_state" column.
4. In the **Failure Threshold** drop-down list, specify the number of failures that will cause processing to stop. A value of ***0*** indicates that the process will never fail.

5. In the **Default HTTP client timeout** drop-down list, specify the Apache HTTP client default timeout value (in seconds). Defaults value is ***60*** seconds.
6. In the **Report-friendly name**, enter a name to be used in reporting for this action.
7. In the **Default input disposition handling** drop-down list, select one of the following:
   - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
   - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
8. In the **Input Disposition(s)**, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
9. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
10. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
11. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
12. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
13. Click **Submit**.

SalesLoft Metadata Retriever Action

1. In the **Call record URL**, specify the recording URL.
   In the **API key**, specify the Salesoft API key.
2. In the **Quantum date field column name**, specify the name to use for the quantum window start or end date. Options include: ***created_at*** or ***updated_at***.
3. In the **SalesLoft web service wait time (in milliseconds)**, specify the amount of time to wait between calls to the metadata search API endpoint.
4. In the **Retrieve user metadata?** drop-down, select ***true*** if you want to retrieve SalesLoft user metadata associated with the interaction.
5. In the **Retrieve called person metadata?** drop-down, select ***true*** if you want to retrieve SalesLoft called person metadata associated with the interaction.
6. In the **Retrieve additional call metadata?** drop-down, select ***true*** if you want to retrieve SalesLoft call metadata (beyond the call-record) associated with the interaction.
7. In the **Retrieve cadence metadata?** drop-down, select ***true*** if you want to retrieve SalesLoft cadence metadata associated with the interaction.
8. In the **Retrieve step metadata?** drop-down, select ***true*** if you want to retrieve SalesLoft step metadata associated with the interaction.
9. In the **Retrieve account metadata?** drop-down, select ***true*** if you want to retrieve SalesLoft account metadata associated with the interaction.
10. In the **Page size**, specify the number of rows that you want to return per call to the SalesLoft call data records API.
11. In the **Default HTTP client timeout (in seconds)**, specify the Apache HTTP client default timeout value (in seconds). The defaults timeout value is ***60 seconds***.
12. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.

13. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave "false" to allow processing as usual. This option is useful for troubleshooting the process.
14. Click **Submit**.

SalesLoft Users Retriever Action

1. In **the Users URL**, specify the SalesLoft Service URL. Example: https://api.salesloft.com/v2/users.json
2. In the **API key**, specify the Salesoft API key.
3. In the **Page size**, specify the number of rows that you want to return per call to the SalesLoft call data records API.
4. In the **Default HTTP client timeout (in seconds)**, specify the Apache HTTP client default timeout value (in seconds). The defaults timeout value is ***60 seconds***.
5. In the **Report-friendly name**, enter a name to be used in reporting for this action.
6. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
7. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
8. Click **Submit**.

Stitch Action

1. In the **Media file type** drop-down, select the media file type. The valid options are WAV and NAF.
2. In the **Stitching method** drop-down, select a stitching method. The valid options are "Sequence" and "Chain".
3. In the **Call duration correction** drop-down, specify a value in the Call duration correction drop-down list. The valid values are ***None***, ***Duration Column***, and ***Start/End Columns***.
4. In the **Call ID column name**, enter the call ID column name or names. If you enter multiple columns you should identify multi-column uniqueness with a plus mark (+) between the column names.
   For example: CompoundId+PBXId
5. In the **Segment ID column name**, enter the segment ID column name. This is used when one call may have multiple recordings.
6. In the **Segment sequence column name**, enter the segment sequence column name. This required if you are not chaining.
7. In the **Delete segments after stitching** drop-down, specify if you want the application to delete segments after stitching.
8. In the **Failure Threshold**, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
9. In the **Report-friendly name**, enter a name to be used in reporting for this action.
10. In the **Default input disposition handling** drop-down list, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
    - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.

11. In the **Input Disposition(s)**, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
12. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
13. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
14. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
15. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
16. Click **Submit**.

Topic AI Request

1. In the **Process files from metadata** drop-down, specify if you want to process files from metadata. Select ***false*** to process files from attached input resources or ***true*** to process them from this process groups existing working table. Processing from metadata reduces input/output and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **File filter** field, enter the file filter type. For example: CSV.

3. In the **File path column name**, enter the name of the metadata column in the process group working table to determine the files to move. Default is ***FileName***.
4. In the **RabbitMQ Host Name**, enter the RabbitMQ host name for login.
5. In the **RabbitMQ Port**, enter the RabbitMQ port number for login. The default value is ***5672***.
6. In the **RabbitMQ Queue Name**, enter the RabbitMQ Queue Name to submit.
7. In the **RabbitMQ User Name**, enter the RabbitMQ user name for login.
8. In the **RabbitMQ Password**, enter the RabbitMQ password.
9. In the **RabbitMQ Virtual Host**, enter the RabbitMQ virtual host for login.
10. In the **Text Analysis Primary Model URI**, enter the path to the location of the English primary model that Summarization Grid uses in summarization.
    Note that this path should not contain any spaces.
11. In the **Default Text Analysis Model GUID**, enter the IPM model GUID to get the correct latest version of the IPM model.
12. In the **Default Scoring Model URI,** enter the path to the location of scoring model that Summarization Grid uses in summarization. Note that this path should not contain any spaces.
13. In the **NIA API Base URI**, enter the Base URI for Nexidia API. Default value is ***true***.
14. In the **Analytics Hub API Based URI**, enter the base URI for Analytics Hub API. Default value is ***true***.
15. In the **Default Model Sub Type**, enter the Model Sub Type to get based and secondary model.
16. In the **Text Analysis Model Metadata Mapping**, enter the mapping of selected metadata field to the model that Summarization Grid should use in summarization. You can also add a suffix for the IPM model. The suffix will be appended to the RabbitMQ Response Queue Name. This will help you understand which IPM model was used while processing in RabbitMQ.
17. In the **Integrator ID**, enter the integrator ID for Analytics Hub API requests.
18. In the **RabbitMQ Response Queue Name**, enter the RabbitMQ Response Queue Name to submit.

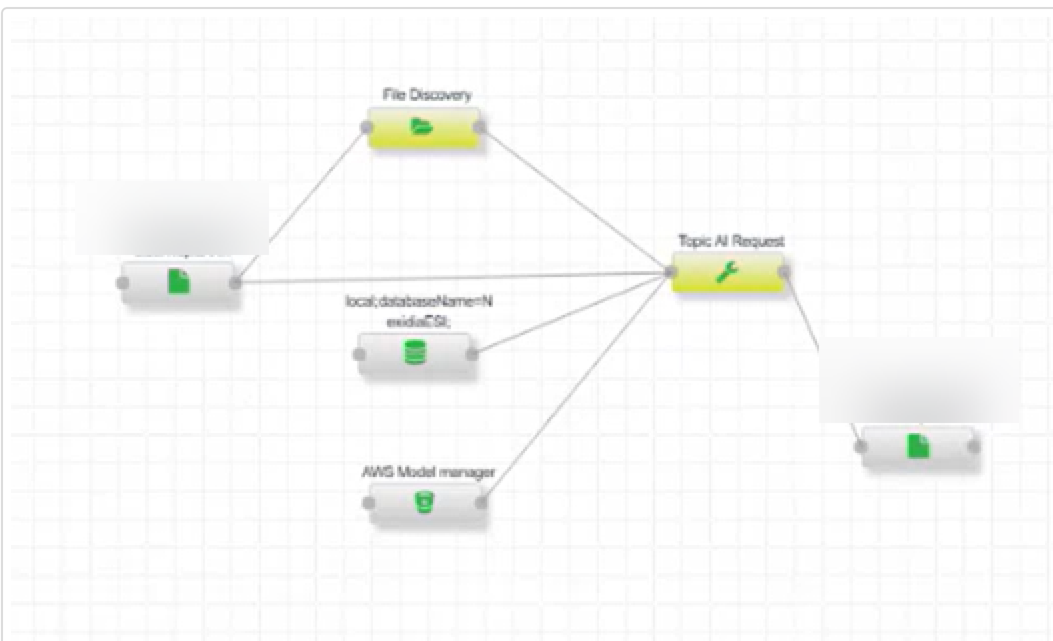
19. In the **Filter Auto Summary**, set filter to include only auto summary that matches these filters.
You can define mappings of metadata fields to values for better call summarization request filtering. When all configured metadata fields match, the system will skip files for auto-summarization. For example, if English is provided as a value for the Language filter for File1, File1 will be skipped for auto summarization.
Files that do not match the filter criteria will be processed to the Nexidia database.
When you click , the Skip Auto Summary window opens. You can use this window to edit the filter values. Click **OK**.
The default values are
    - **LanguageId**: English
    - **CallDirection**: Inbound
    - **MediaStatistics:duration_in_milliseconds**: Greater than or equal to 60
    - **MediaStatistics: NonSpeech percent**: Less than 80
    - **NumberOfTurns**: 0.
20. In the **Text Analysis Path**, enter the Text Analysis Path to executable file.
21. In the **Text Analysis Server Port**, enter the port of the Text Analysis Satellite gRPC server. The default value is 9090.
22. In the **Delete inputs after move** drop-down, set to ***true*** to delete input files after move. Set to ***false*** to copy, leaving input files intact.
23. From the **Send Satellite Logs** drop-down, select ***true*** to diagnose the Voice Grid issues for not being able to process the files. Set to ***false*** to track and log as errors. Default value is ***false***.
24. In the **Wait for or die**, specify the time to wait for all RMQ messages to be confirmed. The default value is 10000 ms.
25. In the **Failure Threshold**, enter the number of non-fatal processing errors (e.g. inability to find a requested file) to tolerate before failing and stopping the rest of the process from continuing. The default value is 0, which means never fail the process, but continue to track errors in logs, and as warnings in the UI.
26. In **Transcript Type**, set to ***VoiceGrid ASR*** to process ASR json file generated by VoiceGrid and set to ***NxPr*** to process NxPr file generated by NIA to create CSV file.

27. In the **User ID**, enter the User ID for authentication endpoint.
28. In the **User Name**, enter the User Name for authentication endpoint. Default value is ***true***.
29. In the **Service Startup Timeout (ms)**, enter the maximum time to wait for service startup validation in milliseconds. Default is 300000ms (5 minutes).
30. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.
31. In the **Default input disposition handling** drop-down, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
    - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
32. In the **Input Disposition(s)**, enter the input disposition. The Data Exchange Framework automatically chains action component dispositions together as javascript:void()part of its processing, so this field is usually left blank. If you are using split processes, enter a value in this field.
33. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
34. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
35. From the **Terminal action** drop-down, select if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
36. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
37. Click **Submit**.

DEF Workflow Steps:

1. Validate IPM Model Version
    - Queries the NIA database for the current IPM model version.
    - Compares it with the DEF environment version:
        - If versions match, the existing IPM model is used.
        - If versions differ:
            - Downloads secondary and base IPM models from Analytics Hub.
            - Combines them to create a comprehensive model.
            - Updates the DEF database with the latest version and GUID.
2. Places the final combined model in a file system location accessible by Voice Grid.
3. Sends media file details to RabbitMQ for Voice Grid processing.

You can refer to the following sample process:



Topic AI Response

1. In the **RabbitMQ Host Name**, enter the RabbitMQ host name for login.
2. In the **RabbitMQ Port**, enter the RabbitMQ port number for login. The default value is ***5672***.
3. In the **RabbitMQ User Name**, enter the RabbitMQ user name for login.
4. In the **RabbitMQ Password**, enter the RabbitMQ password.
5. In the **RabbitMQ Virtual Host**, enter the RabbitMQ virtual host for login.
6. In the **Select top 'n' events**, enter the number of top 'n' combined total of non-attensity and attensity events to accept (agent tasks and scored summaries) from metadata result. Value cannot be negative. The default value is 8. Set the value to 0 to get all events.
7. In the **Select top 'n' attensity events**, enter the number of top 'n' attensity events (scored summaries) to accept from metadata result. The default value is 4. Value cannot be negative. Set the value to 0 to get all events.
8. In the **Failure Threshold**, enter the number of non-fatal processing errors (e.g. inability to find a requested file) to tolerate before failing and stopping the rest of the process from continuing. The default value is 0, which means never fail the process, but continue to track errors in logs, and as warnings in the UI.
9. In the **Delete Transcript CSV**, specify if you want to delete transcripts after they have successfully processed. The default value is ***true***.
10. In **Call Summary Collector** drop-down, select ***true*** to generate JSON based on metadata, enlighten scores, transcript, and call summarization data.
11. In the **Enlighten Model Names**, specify enlighten score names separated by comma that you want to included in the Call Summarization JSON. Leave empty if you want to include all available enlighten scores.
12. In the **Include Desktop Tags**, select ***true*** to create Desktop Tags.
13. In the **Include Auto Summary Tags**, select ***true*** to include Auto Summary Tags in the working table to generate First-Class AutoSummary Field. By default, the value is ***false***, which means Auto Summary Tags is not included. Start and end offsets are added to the intents First-Class Fields in the JSON output. This ensures the intents and events are displayed on the player's waveform at specific times.

14. In the **Treat Missing Data as Errors**, the default value is ***true***. This means you can analyze and capture more information about why the data is missing. Change to ***false*** if missing data should be considered errors. Errors will be logged and tracked against the action's **failure threshold.**
15. In the **Remove messages from RMQ**, select ***true*** to remove messages from RMQ for the missing data that are not in the working table. This is the default setting. Select ***false*** to keep all messages in RMQ.
16. In the **Generate Formatted Transcript File**, select ***true*** to generate a formatted transcript JSON file. Select ***false*** to not generate a formatted transcript JSON file.
17. In the **Include Only Binary Outcome**, select ***true*** to include only binary Contact Outcome to JSON output. The default value is ***false***.
18. In the **Include Only Binary Enlighten**, select ***true*** to include only binary Enlighten to JSON output. The default value is ***false***.
19. In the **Add Transcript For LLM**, select ***true*** to add a transcript to JSON File for LLM. The default value is ***false***.
20. In the **Retry Submit to VG**, select ***true*** to resend the request to VG for reprocessing in case of an error in the response message indicating missing files. The default value is ***false***.
21. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.
22. In the **Default input disposition handling** drop-down, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
    - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
23. In the **Input Disposition(s)**, specify how input dispositions should be handled. The default is empty, to automatically chain. Enter ***none*** to disconnect the action from dependencies to prior actions, or enter a comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Use a hash sign instead of a comma to create an OR condition.
24. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.

25. In the **Terminal action** drop-down, select if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
26. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
27. Click **Submit**.

Twilio Calls Metadata Retriever Action

1. In the **Account SID**, enter the user name for the Twilio account.
2. In the **Authentication token**, enter the password of the Twilio account being used.
3. In the **Base URL**, enter the base-URL for the Twilio.
4. In the **Calls URL**, enter the URL for fetching all calls within the time frame from Twilio API.
5. From the **Audio Type** drop-down, specify the desired format of the audio media when requesting media transcoding prior to downloading. Supported types are ***WAV*** and ***MP3***. The default format is ***WAV***.
6. In the **Page Size**, specify the number of rows that you want to retrieve from the Twilio API when querying for contact metadata. The number of rows ranges from 1 to 1,000. Default is 50.
7. From the **Calls Status** drop-down, specify status of the calls that you want to include. Supported statuses are ***completed***, ***queued***, ***ringing***, ***in-progress***, ***canceled***, ***failed***, ***busy***, or ***no-answer***.
8. In the **Quantum window date format**, specify the quantum window date format. Defaults to yyyy-MM-dd'T'hh:mm:ss'Z'.
9. From the **Log Twilio service responses**, select the service responses to a log file. Default is ***no response logging occurs***.

10. In the **Default HTTP client timeout (in seconds)**, enter the Apache HTTP client default timeout value. The default is ***60***.
11. In the **Twilio web service wait time (in milliseconds)**, enter the amount of time that you want to wait the service between calls to the API endpoint.
12. In the **Twilio web service number of retries**, enter the number of retries that you want for the API endpoint before failing.
13. From the **Failure Threshold** drop-down, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
14. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.
15. From the **Terminal action** drop-down, select if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
16. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
17. Click **Submit**.

Twilio Media Retriever Action

1. In the **Account SID** field, enter the user name for the Twilio account.
2. In the **Authentication Token** field, enter the password of the Twilio account being used.
3. In the **Base URL** field, enter the base URL for the Twilio.
4. In the **Quantum Window Date Format** field, specify the quantum window date format. The default value is yyyy-MM-dd'T'hh:mm:ss'Z'.
5. In the **Failure Threshold** drop-down, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
6. In the **Extract Stereo** field, set to ***true*** to extract stereo recordings. You may need to provide additional parameters for stereo retrieval, such as RequestedChannels=2.

If both stereo and mono recordings are available, DEF will download only the stereo recordings. If no stereo recordings are available, DEF will download only the mono recordings.

7. In the **Add Parameters to Requests** field, set to ***true*** to add extra parameters to the API requests.
8. In the **Service Headers** field, specify the API/Web service HTTP headers to include in the API requests. You can use any valid HTTP header name/value pairs required by the API/Web service being called. For example, Content-Type=application/json; Accept=application/json. Click to open the Service Headers window where you can define the service headers.
9. In the **Service Parameters** field, specify the API/Web service query parameters for the API requests. For example, startDate=2019-05-22; endDate=2019-05-29; version=1. Click to open the Service Parameters window where you can define the service parameters.
10. In the **Default HTTP Client Timeout (in seconds)** field, enter the Apache HTTP client default timeout value. The default is ***60***.
11. In the **Wait Time (in milliseconds)** field, enter the amount of time that you want to wait between calls to the API endpoint.
12. In the **Number of Retries** field, enter the number of retries that you want for the API endpoint before failing.
13. In the **Reconciliation Identifier** field, enter a name to be used in reporting for this action.
14. In the **Thread Count** field, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the Threads per action setting at the server level.
15. In the **Terminal action** drop-down, specify whether this is a terminal action. Upon process completion, the rows affected by terminal actions are moved to a working table rollover table. By default, actions are considered terminal if they have a corresponding disposition column in the working table, and no downstream actions reference them. This option is available for instances where the default behavior is not desired.

16. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
17. Click **Submit**.

Twilio Recordings Metadata Retriever Action

1. In the **Account SID**, enter the user name for the Twilio account.
2. In the **Authentication token**, enter the password of the Twilio account being used.
3. In the **Base URL**, enter the base-URL for the Twilio.
4. From the **Audio Type** drop-down, specify the desired format of the audio media when requesting media transcoding prior to downloading. Supported types are ***WAV*** and ***MP3***. The default format is ***WAV***.
5. In the **Quantum window date format**, specify the quantum window date format. Defaults to yyyy-MM-dd'T'hh:mm:ss'Z'.
6. From the **Log Twilio service responses**, select the service responses to a log file. Default is ***no response logging occurs***.
7. In the **Default HTTP client timeout (in seconds)**, enter the Apache HTTP client default timeout value. The default is ***60***.
8. In the **Twilio web service wait time (in milliseconds)**, enter the amount of time that you want to wait the service between calls to the API endpoint.
9. In the **Twilio web service number of retries**, enter the number of retries that you want for the API endpoint before failing.
10. From the **Failure Threshold** drop-down, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
11. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.

12. From the **Terminal action** drop-down, select if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
13. In the **Thread Count** field, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the Threads per action setting at the server level.
14. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
15. Click **Submit**.
    If both stereo and mono recordings are available, DEF will download only the stereo recordings. If no stereo recordings are available, DEF will download only the mono recordings.

Text Redaction Action

1. In the **Process files from metadata** drop-down, specify if you want to process files from metadata. When you select ***true*** the application processes files from the process groups existing working table. When you specify false, the application processes files from attached input resources. Processing from metadata reduces I/O and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **File filter**, enter the file filter type. For example: CSV. Java style regular expressions can also be used. when processing files from resources, this acts as a filter. When processing from metadata, the metadata's file key is inserted into any wildcards, expanding processing of the one file to all variants. For example, '*,*.xml' would process both a file matching the working table file key '1234567.wav' and its '1234567.xml.wav' counterpart.
3. In the **Regular expression(s)**, enter a list of one or more regular expressions to be used in select text to redact from incoming fields in the Regular expression(s) field.
4. In the **Delete inputs after processing** drop-down, specify if you want to delete inputs after the processing has completed.
5. In the **Failure Threshold**, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
6. If the incoming file encodings are not UTF-8, enter UTF-8 in the Encoding field. This must match a valid Java encoding. See https://docs.oracle.com/javase/8/docs/technotes/guides/intl/encoding.doc.html for more details.
7. In the **Report-friendly name**, enter a name to be used in reporting for this action.
8. In the **Default input disposition handling** drop-down list, select one of the following:
   - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
   - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.

9. In the **Input Disposition(s)**, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type 'none' to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
10. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
11. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
12. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
13. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
14. Click **Submit**.

Uncompress Action

1. In the **Process files from metadata**, specify if you want to process files from metadata. When you select ***true*** the application processes files from the process groups existing working table. When you specify false, the application processes files from attached input resources. Processing from metadata reduces I/O and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **Compression type** drop-down, select the compression type. The valid options are ***.zip*** and ***.tar***.
3. In the **Password to unzip files with**, enter the password to use for unzipping the files.
4. In the **Files are gzipped** drop-down, specify if files are gzipped.
5. In the **Files are 7zipped** drop-down, specify if the files are 7zipped.
6. In the **Check provided MD5 hash** drop-down, specify if you want the application to check the provided MD5 hash.
7. In the **Delete after uncompressing** drop-down, specify if you want the application to delete the compressed file after uncompressing.
8. In the **Checks and sets dispositions** drop-down, specify if you want the application to check and set the dispositions. This option has to do with how the action participates in the process flow: If an action checks and sets dispositions, then when it's looking at items to process, it checks the working table to see if any actions prior to it in the workflow (actions that also check and set dispositions) have processed the record, and it makes sure that it has not.
9. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.
10. In the **Default input disposition handling** drop-down list, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
    - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.

11. In the **Input Disposition(s)**, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
12. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
13. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
14. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
15. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
16. Click **Submit**.

User Review Action

1. In the **Columns to review** field, define which metadata columns will be put into the CSV file in the Columns to review (comma-separated) field. You can enter multiple columns by comma-separating them.
2. In the **empty columns to add** field, specify the empty columns to add to the CSV file in the Empty columns to add (comma-separated) field. You can enter multiple columns by comma-separating them.
3. In the **Report-friendly name** field, enter a name to be used in reporting for this action.
4. In the **Default input disposition handling** drop-down list, select one of the following:
   - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
   - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
5. In the **Input Disposition(s)** field, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
6. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
7. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
8. Click **Submit**.

Verint Stitch Action

1. In the **Process files from metadata**, specify if you want to process files from metadata. When you select "true" the application processes files from the process groups existing working table. When you specify false, the application processes files from attached input resources. Processing
from metadata reduces I/O and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **Session Metadata Fields**, enter the session metadata fields to capture. The session metadata fields are contained on the following Xpath: recordings/contacts parent=truetaggedbycomponent=IFServices/contact/sessions/session.
3. In the **Delete source files after stitching?** drop-down, specify if you want to delete the source .xml and .wav files after stitching.
4. In the **Clean un-processed files?** drop-down, specify if you want to clean up the .xml and .wav files that were not part of the files that were stitched.
5. In the **Failure Threshold** drop-down, specify the number of non-fatal processing errors failures to tolerate before failing and stopping the rest of the process from continuing. A value of 0 indicates that the process will never fail, but continues to track errors in the logs and provides warnings in the user interface.
6. In the **Log Stitching Maps** drop-down, specify if you want to log stitching maps to <JAVA temporary folder> /<time (ms)>_Verint_Stitching_Maps.txt.
7. In the **Report-friendly name**, enter a name to be used in reporting for this action.
8. In the **Default input disposition handling** drop-down list, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
    - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.

9. In the **Input Disposition(s)**, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition. .
10. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
11. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
12. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
13. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
14. Click **Submit**.

Voice Print Creator

1. In the **Process files from metadata**, specify if you want to process files from metadata. When you select "true" the application processes files from the process groups existing working table. When you specify false, the application processes files from attached input resources. Processing from metadata reduces I/O and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **File filter**, enter the file filter type. For example: CSV. Java style regular expressions can also be used. when processing files from resources, this acts as a filter. When processing from metadata, the metadata's file key is inserted into any wildcards, expanding processing of the one file to all variants. For example, '*,*.xml' would process both a file matching the working table file key '1234567.wav' and its '1234567.xml.wav' counterpart.
3. In the **Channel names**, enter a comma-separated list of channels expected in incoming media files. The names will be used in the corresponding sections of the generated .nia file to represent each channel. The default value is customer, agent.
4. In the **Voice Biometrics Model Path** field,
5. In the **Failure Threshold**, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
6. In the **Flatten output key** drop-down, specify if you want to flatten the output structure of the generated files.
7. In the **Allow mono files** drop-down, specify if you want to allow mono files. Select ***true*** to allow mono files to pass through the action without error, or ***false*** to error out mono files and track them as failures.
8. In the **Maximum Duration (ms)** drop-down, specify the maximum duration in milliseconds for identifying voiceprint.
9. In the **Start Duration (ms)** drop-down, specify the start duration in milliseconds for identifying voiceprint.
10. In the **Report-friendly name**, enter a name to be used in reporting for this action.
11. In the **Default input disposition handling** drop-down list, select one of the following:

- ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
- ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.

12. In the **Input Disposition(s)**, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type 'none' to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
13. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
14. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
15. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
16. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
17. Click **Submit**.

Vonage Media Retriever Action

1. In the **Authentication URL**, enter the authentication URL to use when authenticating. For example: https://{region}.newvoicemedia.com/Auth/connect/token.
2. In the **Media extraction URL**, enter the URL to use when calling the media extraction API, where {contentKey} is a chosen **Media Type**. Example: https://{region}.api.newvoicemedia.com/interaction-content/interactions/{guid}/content/{contentKey}.
3. In the **Client ID**, enter the Vonage client ID from your credentials.
4. In the **Client Secret**, enter the Vonage client secret from your credentials.
5. In the **Region**, enter the region where your account is created.
6. From the **Vonage web service wait time (in Milliseconds)** drop-down, select the amount of time to wait when attempting to download the media file using media API endpoint. Default is ***500***.
7. From the **Vonage web service number of retries** drop-down, select the number of retries to the media download API. Default is ***20***.
8. From the **Failure Threshold** drop-down, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
9. In the **Scope**, enter the value of scope that determines the API and endpoints with which the bearer access token can be used.
10. In the **Grant Type**, enter the grant type. A grant type of client credentials indicates that you are using OAuth 2.0 for authorization.
11. In the **Data extraction API payload version number**, enter the data extraction API payload version number. The version number is used when submitting a data extraction request. Defaults is ***3***.
12. In the **Accept Header**, enter the Interaction content API (in JSON format) with API version. Accept header is the content types that the response can return.
13. From the **Media Type** drop-down, select the type of the content. Supported types are ***callRecording***, ***screenRecording voicemail***, ***transcript***, ***chatTranscript***, ***categorizationResult***, and ***attachment***.

14. From the **Session credentials expiration (minutes)** drop-down, enter the time to wait before re-authenticating the Vonage session. Session expiration is 120 minutes, however we choose 60 min as a save value, even though the system automatically re-authenticate token/session if expiration time approaches.
15. From the **Log Vonage service response** drop-down, if you want to log the Vonage service responses, select **true**.
16. From the **Default HTTP client timeout(in seconds)** drop-down, select the Apache HTTP client default timeout value (in seconds). Default value is ***60*** seconds.
17. In the **Reconciliation Identifiers field**, enter the name of the action if there is a reporting action in the process group.
18. From the **Default input disposition handling** drop-down, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
    - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
19. In the **Input Disposition(s)**, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type 'none' to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
20. From the **Thread Count** drop-down, select a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
21. From the **Maximum files to process** drop-down, select the maximum number of files to allow this action to process in a given run. The maximum files to process is 10000. A value of 0 indicates that there is no limit.

22. From the **Terminal action** drop-down, select if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
23. In the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
24. Click **Submit**.

Vonage Metadata Retriever Action

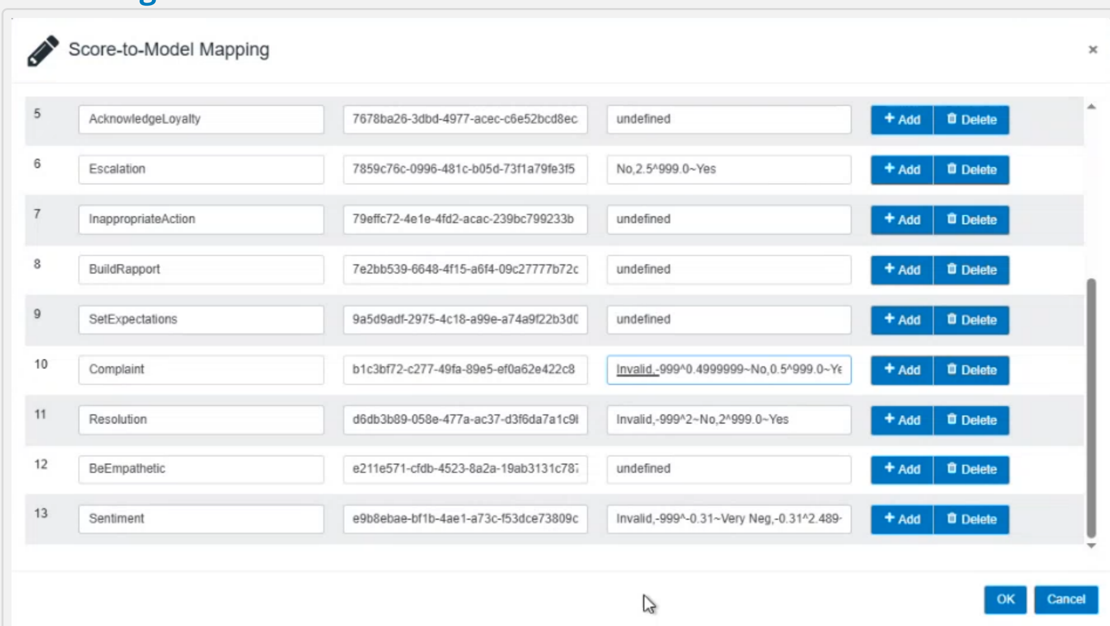
1. In the **Authentication URL**, enter the authentication URL to use when authenticating. For example: https://{region}.newvoicemedia.com/Auth/connect/token.
2. In the **Data extraction URL**, enter the extraction URL. For example: https://{region}.api.newvoicemedia.com/interaction-content/interactions.
3. In the **User-Admin extraction URL**, enter the user admin URL. For example: https://{region}.api.newvoicemedia.com/useradmin/users.
4. In the **Client ID**, enter the Vonage client ID from your credentials.
5. In the **Client Secret**, enter the Vonage client secret from your credentials.
6. From the **Region**, select the region where your account is created.
7. From the **Failure Threshold** drop-down, select the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
8. In the **Scope**, enter the value of scope that determines the API and endpoints with which the bearer access token can be used. The scope defines the data repository that is being used in this particular integration and the type of access to data repository. Multiple data repositories can be listed, separated by comma, the access type is presented follow the colon. Example: interaction-content:read, users:read
9. In the **Grant Type**, enter the grant type. A grant type of client credentials indicates that you are using OAuth 2.0 for authorization.

10. In the **Accept Header**, enter the Interaction content API (in JSON format) with API version. Accept header is the content types that the response can return.
11. In the **Accept Header for User Admin**, enter the User API (in JSON format) with API version.
12. From the **Limit** drop-down, specify the write limit (specified in KBs for example: 1000). The default value is ***1*** (no write limit).
13. From the **Media Type**, select the type of content. Supported types are ***callRecording***, ***screenRecording voicemail***, ***transcript***, ***chatTranscript, categorizationResult***, and ***attachment***.
14. From the **Users Status to Include in Response** drop-down, select the type of users. Available options are ***Active***, ***Archived***, and ***All***. Default is Active.
15. From the **User Request Limit** drop-down, enter the number of items per page returned in response. The minimum is ***1*** and maximum is ***5000***. Default is ***25***. Suggested to have this value set to ***2000*** to avoid multiple calls to API.
16. From the **Session credentials expiration (minutes)** drop-down, enter the time to wait before re-authenticating the Vonage session. Session expiration is 120 minutes, however we choose 60 min as a save value, even though the system automatically re-authenticate token/session if expiration time approaches.
17. From the **Log Vonage service response** drop-down, specify if you want to log the Vonage service responses.
18. In the **Reconciliation Identifiers field**, enter the name of the action if there is a reporting action in the process group.
19. From the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
20. From the **Disable action** field, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
21. Click **Submit**.

Workflow Enlighten

1. In the **Score-to-Model mapping**, enter the path for a mapping of expected scores to the names of model holding identifiers (GUID) for those scores. Click **Edit** to map the scores with the respective model.

   View image

2. In the **Include Sentiment Scores**, set to ***true*** to include sentiment scores as NIA fields. The default value is true.
3. In the **Workflow Model Location**, enter the path to the location of the workflow model.
4. In the **Validate Model Name**, set to ***true*** to validate Enlighten Model name.
5. In the **Generate AutoSummarization Tag**, set to ***true*** to generate a AutoSummarization tag that includes Enlighten model score index.
6. In the **Include Index**, set to ***true*** to include index and behavior scores to the working table. In the **Media Type**, select the type of media that will be used to calculate the index scores- ***Audio, Chat, Email***, and ***Text***.
7. In the **Batch Update Size**, enter the number of database updates to batch together for improved performance. The valid range is 1-1000, and the default value is 50.

8. In the **Reconciliation Identifier**, enter the name of the action if there is a reporting action in the process group.
9. In the **Default input disposition handling** drop-down list select one of the following:
   - **AND** - Allow files to flow through this step that have passed all of the immediate inputs.
   - **OR** - Allow files to flow through this step that have passed any of the immediate inputs.
10. In the **Input Disposition(s)**, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type ***none*** to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
11. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the **Threads per action** setting at the server level.
12. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
13. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
14. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
15. Click **Submit**.

How to Change ENL CSAT Index Thresholds in the DEF Database

During installation, the Data Exchange Framework system creates a lookup table. This table contains the default index score ranges for different Enlighten models.

If a customer requests a different range, you need to:

- Update the values in the NEST database table **EXTScoreIndex_Lookup** and run a query against this table based on the customer's requirements.
- Update the same table in the NEAT database. Processing does not require this second update, but aligning both tables helps avoid confusion.

Workflow Redaction

1. In the **File filter**, enter the file filter type. For example: CSV.
2. In the **Redact tone**, enter an audio tone to use for redaction. If left blank, the tone for redacted segments is silence.
3. In the **Delete inputs after move** drop-down, specify if you want to delete inputs after move. Select "true" for a true file move or "false" for to copy the file and leave inputs intact.
4. In the **Failure Threshold** , specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
5. In the **Report-friendly name**, enter a name to be used in reporting for this action.
6. In the **Default input disposition handling** drop-down list select one of the following:
   - **AND** - Allow files to flow through this step that have passed all of the immediate inputs.
   - **OR** - Allow files to flow through this step that have passed any of the immediate inputs.
7. In the **Input Disposition(s)**, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type 'none' to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
8. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
9. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.

10. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
11. Click **Submit**.

Workflow Request

1. In the **Process files from metadata** drop-down, specify if you want to process files from metadata. When you select ***true*** the application processes files from the process groups existing working table. When you specify false, the application processes files from attached input resources. Processing from metadata reduces I/O and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **File filter**, enter the file filter type. For example: CSV.
3. In the **RabbitMQ queue name**, enter the RabbitMQ Queue Name to submit.
4. In the **RabbitMQ host name,** enter the RabbitMQ host name for login.
5. In the **RabbitMQ port**, enter the RabbitMQ number for login. The default value is 5672.
6. In the **RabbitMQ user name**, enter the RabbitMQ user name for login.
7. In the **RabbitMQ password**, enter the RabbitMQ password.
8. In the **RabbitMQ virtual host,** enter the RabbitMQ virtual host for login.
9. In the **Workflow ID**, enter the UUID of the model that Redaction Grid should use in determining segments to redact.
10. In the **RabbitMQ Response Queue Name**, enter the RabbitMQ Response Queue Name to submit. The Workflow Request Action creates the Response Queue if it does not exist already.
11. In the **Send Workbench Logs** drop-down, specify if you want to send workbench logs to diagnose the redaction issues.
12. In the **Generate transcripts** drop-down, specify if you want to generate Voice Grid transcript files. If you select this option, a file output resource must be specified and it must be a network path.
13. In the **Wait for or die**, enter the time to wait for all RabbitMQ messages to be confirmed. The default value is 10,000 msec (10 sec).
14. In the **Language Code**, select the language code for the workflow request.
15. In the **Failure Threshold**, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
16. In the **TLS Enable**, set to ***true*** to enable TLS encryption for RabbitMQ connection.

17. In the **Reconciliation Identifier**, enter a name to be used in reporting for this action.
18. In the **Default input disposition handling** drop-down list select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
    - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
19. Optionally enter the input disposition. The Data Exchange Framework automatically chains action component dispositions together as javascript:void()part of its processing, so this field is usually left blank. If you are using split processes, enter a value in this field In the Input Disposition(s) field, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type 'none' to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
20. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
21. In the **Maximum files to process** field, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
22. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
23. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
24. Click **Submit**.

Workflow Response

1. Enter a name in the **RabbitMQ Queue name**.
2. Enter the host name in the **RabbitMQ host name** for login.
3. Enter the port number in the **RabbitMQ port** for login. The default value is 5672.
4. Enter a username in the **RabbitMQ user name** for login.
5. Enter a password **RabbitMQ password**.
6. Enter the virtual host name in the **RabbitMQ virtual host** for login.
7. In the **Failure Threshold**, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
8. In the **Channel Flipping** drop-down, select one of the following:
   - ***None*** — To not to flip the channels for agents or customers.
   - ***Direct*** — To flip the channels for all the agents or customers.
   - ***Metadata-Based*** — To flip the channels for the agents or customers if channels matches the specified metadata. A **Metadata Channels Flip Check** shows. Specify the metadata.
     For example, “Direction” can be the metadata field, and “Outgoing” can indicate a need to flip; so **Metadata Channels Flip Check** accepts “Direction=Outgoing” as the flip condition.
9. In the **Report-friendly name**, enter a name to be used in reporting for this action.
10. In the **Default input disposition handling** drop-down, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
    - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
11. In the **Input Disposition(s)**, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type 'none' to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR

condition.

12. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
13. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
14. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
15. Click **Submit**.

XSL Transform Action

1. In the **Process files from metadata**, specify if you want to process files from metadata. When you select ***true*** the application processes files from the process groups existing working table. When you specify ***false***, the application processes files from attached input resources. Processing from metadata reduces I/O and file system scans, but some previous action needs to have entered something into the working table first in order to have data to process from.
2. In the **File filter**, enter the file filter type. For example: CSV.
3. In the **XSL stylesheet**, enter a stylesheet that is used to transform the xml.
4. In the **Default inputs after successful transforms** drop-down, specify if you want the application to delete inputs after successful transforms.
5. In the **Checks and sets dispositions** drop-down, specify if you want the application to check and set the dispositions. This option has to do with how the action participates in the process flow: If an action checks and sets dispositions, then when it's looking at items to process, it checks the working table to see if any actions prior to it in the workflow (actions that also check and set dispositions) have processed the record, and it makes sure that it has not.
6. In the **Force to extension**, enter a value in the "Force to extension" field. When checking and setting dispositions, this can be used to match the media file extension in the working table. It is also useful in situations where the .xml being transformed should map to a media file in the working table, and the transform should be recorded against it. In that case, '.wav' here would force the extension of '1234567.xml' to '1234567.wav' to match the media file when it is tracked.
7. In the **Failure Threshold** field, specify the number of failures that will cause processing to stop. A value of 0 indicates that the process will never fail.
8. In the **Force XML encoding**, if you want to force XML encoding, enter the character encoding of the input file. You will leave this field blank in most cases to auto-detect the character encoding of the XML input files. The valid options are:
   - ***UTF-8***
   - ***UTF-16***
   - ***UTF-32***

- ***ISO-8859-1***
- ***Windows-1252***
- ***ASCII***

9. In the **Reconciliation Identifier** field, enter the name of the action if there is a reporting action in the process group.
10. In the **Default input disposition handling** drop-down list, select one of the following:
    - ***AND*** - Allow files to flow through this step that have passed all of the immediate inputs.
    - ***OR*** - Allow files to flow through this step that have passed any of the immediate inputs.
11. In the **Input Disposition(s)**, override the default input disposition handling of the action by entering comma-separated list of dispositions that any incoming file should already have passed through in the working table before letting this action process it. This is useful for cross-process chaining (within or across servers). Leave blank to automatically chain, which is the default behavior, or type 'none' to disconnect the action from dependencies to prior actions. Use a hash sign instead of a comma to create an OR condition.
12. In the **Thread Count**, enter a value for the number of threads this action is allowed to partition its work into. This value overrides the "Threads per action" setting at the server level.
13. In the **Maximum files to process**, specify the maximum number of files to allow this action to process in a given run. A value of zero indicates that there is no limit.
14. In the **Terminal action** drop-down, specify if this is a terminal action. On process completion, the working table rows touched by terminal actions are pushed into a working table rollover table. By default, actions are considered terminal when they have a corresponding disposition column in the working table, and when no other actions downstream of them reference them. This option is provided for instances where the default behavior is not desirable.
15. In the **Disable action**, specify if you want to prevent this action from processing any files during its run; leave ***false*** to allow processing as usual. This option is useful for troubleshooting the process.
16. Click **Submit**.

MAINTENANCE MODE

You can disable and enable all Inbox Publisher actions from pushing to *Interaction Analytics* by calling a single stored procedure in the Worker server (NEST) database from outside of Data Exchange Framework. The EnterNIAMaintenanceMode and ExitNIAMaintenanceMode stored procedures are located in the Worker server (NEST) databases after start up. You can call these stored procedures through SSMS using the EXEC command.

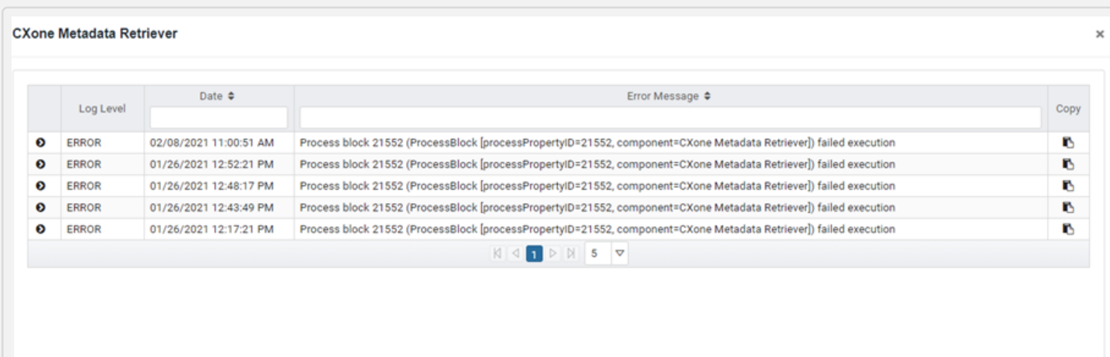
- When you call the EnterNIAMaintenanceMode stored procedure, this prevents all Inbox Publishers from publishing to *Interaction Analytics*(while leaving the rest of the process to keep working as-is and on-schedule).
- When you call the ExitNIAMaintenanceMode stored procedure, this will clear them to start pushing to *Interaction Analytics*again.

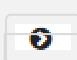
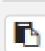
MONITOR AND TROUBLESHOOT THE DATA EXCHANGE FRAMEWORK

The Component View Log Modal window provides a usable view of the logs within the Data Exchange Framework user interface. You can use this window to easily troubleshoot and monitor the Data Exchange Framework activities.

To access and use the Component View Log Modal Window:

1. In the tree view, click on the process that you want to troubleshoot to display it in the Process Preview window.
2. In the Process Preview window, click on an action to open the Component View Log Model for the selected action.



3. View the log details. You can scroll through the logs and select to view 5, 10, or 20 logs at a time and you can also filter and sort the columns.
   - Log Level - This is the level the log issue. The possible options are ERROR, DEBUG, INFO, WARN, ERROR and FATAL.
   - Date - The date of the log entry.
   - Error Message - The error message content.
4. (Optional). Click  to expand the exception content.
5. (Optional). Click  to copy the content to the clipboard.

WORKING WITH THE INTEGRATION TEMPLATES

The integration templates are pre-configured process definitions used to simplify DEF process design. When you add a process to a server from the Processes window, you can select one of these templates. The actions in the templates are pre-configured, but you can change them. You will configure the resources for your environment. There are placeholders for the resources in the templates.

The integration templates are pre-configured process definitions used to simplify DEF process design. When you add a process to a server from the Processes window, you can select one of these templates. The actions in the templates are pre-configured, but you can change them. You will configure the resources for your environment. There are placeholders for the resources in the templates.

The predefined process templates have been removed from the user interface and are now available on the Data Exchange Framework Confluence site for those who have access to that site (https://tlvconfluence01.nice.com/display/NXDEF/Process+Library). They are also available for download as a complete package via the NiCE Software Download Center (https://nice.subscribenet.com/control/nice/login?nextURL=%2Fcontrol%2Fnice%2Fpurchases). You can also import a process by clicking **Select a Process Template (*.json)** and then navigating to the process that you want to import.

5

Other Tools

DEF REST API

REST Authentication

Before any DEF REST endpoint can be executed, you must first execute the login/authenticate REST endpoint. The login/authenticate REST endpoint returns an authentication token. The authentication token must be sent as a header parameter with each request as a "token" parameter. Authentication tokens expire in 30 minutes. After expiration, another login/authentication token must be acquired.

DESCRIPTION	LOGIN/AUTHENTICATE
HTTP Method	GET
URL	http://localhost:8080/NEAT/def/api/v1/login/authenticate
Parameters	username, Header, String, Valid, DEF username password, Header, String, Valid DEF password
Response Type	application/json

DESCRIPTION	LOGIN/AUTHENTICATE
Responses	200 - Success 401 - Unauthorized access attempt 403 - Not authenticated 412 - Precondition failed 500 - Internal server error
Curl Example	curl -X GET "http://vqa-2k12-16:8080/NEAT/def/api/v1/login/authenticate" -H "accept: application/json" -H "username: defuser" -H "password: password"

Response	{
	"token":
	"eyJraWQiOiIxNDk4MTMwNjg1MzgwIiwiYWxnIjoiUlMyNTYifQ.eyJpc3MiOiJuZXhpZGlhLmNvbSIsImV4cCI6M
	TQ5ODEzMjQ4NSwianRpIjoiS29mRWlieUt3N2pieIQ2RW5jZ2xKUSIsImlhdCI6MTQ5ODEzMDY4NSwibm
	JmIjoxNDk4MTMwNTY1LCJzdWIiOiJhZG1pbiIsInJvbGVzIjpbImNsaWVudCJdfQ.ABGdK1UPTGbf4R7Hv RsvSxYNlGHaqWo3Ub3TX212VBS3ECGUIch3KLbZl0TvFgHcifJ-aki8kYTpLXff01bRFWAnVQ1sdYGZ-p 909brnpwMOh8ESQx960_
	pW3nfX3T7SGDmeRYj58dKdeaAxDdXCU4LHgb2xPV5XtYjUwGChEuSa5os1
	pUmnseN3lR0NY9WISMzST6l6tFIZPaV1q_NfMhC42xWFQoaU0i5PNdZ197_
	QW2QYXr809rEBMsm-Z8D Kg_
	rVrJRWThjVgZUP0rO1zRrbb8bKGYDV2ZcE8yVj2siAF7T3yJxPzOuwwpNFdqDHWZ9AuZlJjAOr6I3vcITJOQ"
	}

Automated Password Change REST Endpoints

Change password REST endpoints are provided for these entities: File Resources, SFTP Resources, Database Resources, Keystores, Users, DEF Database. The following tables document each Change Password REST endpoint.

1. Update File Resource Password

DESCRIPTION	UPDATE A FILE RESOURCE PASSWORD
Description	Update a file resource password
HTTP Method	PUT
URL	http://localhost:8080/NEAT/def/api/v1/resources/{file resource id}/fileresource
Parameters	Token, Header, String, Authentication token returned by the authentication service file resource id, Path, Integer, File resource database ID new password Body, String, { "password": "newpassword147" }
Content Type	application/json
Response Type	application/json
Responses	200 - Success 401 - Unauthorized access attempt 403 - Not authenticated 404 - Not found 500 - Internal server error

2. Update SFTP Resource Password

DESCRIPTION	UPDATE A SFTP PASSWORD
HTTP Method	PUT
URL	http://localhost:8080/NEAT/def/api/v1/resources/{sftp resource id}/sftpresource
Parameters	token, Header, String, Authentication token returned by the authentication service SFTP resource id, Path, Integer, SFTP resource database ID new password Body, String, { "password": "newpassword147" }
Content Type	application/json
Response Type	application/json
Responses	200 - Success 401 - Unauthorized access attempt 403 - Not authenticated 404 - Not found 500 - Internal server error
Response	Returns the updated SFTP resource as a JSON response.

3. Update Database Resource Password

DESCRIPTION	UPDATE A DATABASE PASSWORD
HTTP Method	PUT

DESCRIPTION	UPDATE A DATABASE PASSWORD
URL	http://localhost:8080/NEAT/def/api/v1/resources/{data resource id}/dataresource
Parameters	token, Header, String, Authentication token returned by the authentication service database resource id, Path, Integer, database resource database ID new password Body, String, { "password": "newpassword147" }
Content Type	application/json
Response Type	application/json
Responses	200 - Success 401 - Unauthorized access attempt 403 - Not authenticated 404 - Not found 500 - Internal server error
Response	Returns the updated database resource as a JSON response.

4. Update Keystore Password

DESCRIPTION	UPDATE A KEYSTORE PASSWORD
HTTP Method	PUT
URL	http://localhost:8080/NEAT/def/api/v1/servers/{server id}/changekeystorepassword

DESCRIPTION	UPDATE A KEYSTORE PASSWORD
Parameters	Token, Header, String, Authentication token returned by the authentication service server id, Path, Integer, server database ID location, Header, String, Keystore current password new, Header, String, New keystore password
Content Type	application/json
Response Type	application/json
Responses	200 - Success 401 - Unauthorized access attempt 403 - Not authenticated 404 - Not found 500 - Internal server error

5. Update DEF User Password

DESCRIPTION	UPDATE DEF PASSWORD
HTTP Method	PUT
URL	http://localhost:8080/NEAT/def/api/v1/users/{user id}/changepassword
Parameters	token, Header, String, Authentication token returned by the authentication service user id, Path, Integer, User database ID new, Header, String, New user password
Content Type	application/json

DESCRIPTION	UPDATE DEF PASSWORD
Response Type	application/json
Responses	200 - Success 401 - Unauthorized access attempt 403 - Not authenticated 404 - Not found 500 - Internal server error
Response	Returns the updated database resource as a JSON response.

6. Update DEF Database Password

DESCRIPTION	UPDATE DEF DATABASE PASSWORD
HTTP Method	PUT
URL	http://localhost:8080/NEAT/def/api/v1/servers/{server id}/changeserverpassword
Parameters	token, Header, String, Authentication token returned by the authentication service server id, Path, Integer, Server database ID new, Header, String, New DEF Database password
Content Type	application/json
Response Type	application/json

DESCRIPTION	UPDATE DEF DATABASE PASSWORD
Responses	200 - Success 401 - Unauthorized access attempt 403 - Not authenticated 404 - Not found 500 - Internal server error
Response	Returns the updated database resource as a JSON response.

Re-start DEF server REST endpoint

A restart DEF server REST endpoint is also provided. This endpoint must be called to ensure that all password changes take effect. You can either restart all Administration and Worker servers or restart a specified server. The following tables document each Restart REST endpoint.

1. Restart all Administration and Worker Servers

DESCRIPTION	RESTART ALL ADMINISTRATION AND WORKER SERVERS
HTTP Method	PUT
URL	http://localhost:8080/NEAT/def/api/v1/servers/restart
Parameters	forceshutdown, Query Param, Boolean, Force shutdown flag: true - force shutdown
Content Type	application/json
Response Type	application/json
Responses	200 - Success 401 - Unauthorized access attempt 403 - Not authenticated 500 - Internal server error
Response	Returns an HTTP status code.

2. Restart Specified Server

DESCRIPTION	RESTART A SPECIFIED DEF SERVER
HTTP Method	PUT

DESCRIPTION	RESTART A SPECIFIED DEF SERVER
URL	http://localhost:8080/NEAT/def/api/v1/servers/{server id}restart
Parameters	token, Header, String, Authentication token returned by the authentication service server id, Path, Integer, Server database ID forceshutdown, Query Param, Boolean, Force shutdown flag: true - force shutdown
Content Type	application/json
Response Type	application/json
Responses	200 - Success 401 - Unauthorized access attempt 403 - Not authenticated 404 - Not found 500 - Internal server error
Response	Returns an HTTP status code.

Testing REST Endpoints

You can use the following tools to test REST endpoints:

- Postman - This is a Chrome Browser plug-in that allows you to execute REST endpoints through the Chrome browser. Postman can be found at https://www.getpostman.com/.
- Curl - This is a command line tool that allows you to execute REST endpoints from the command line. Curl can be found at https://curl.haxx.se/.

WIZARDS

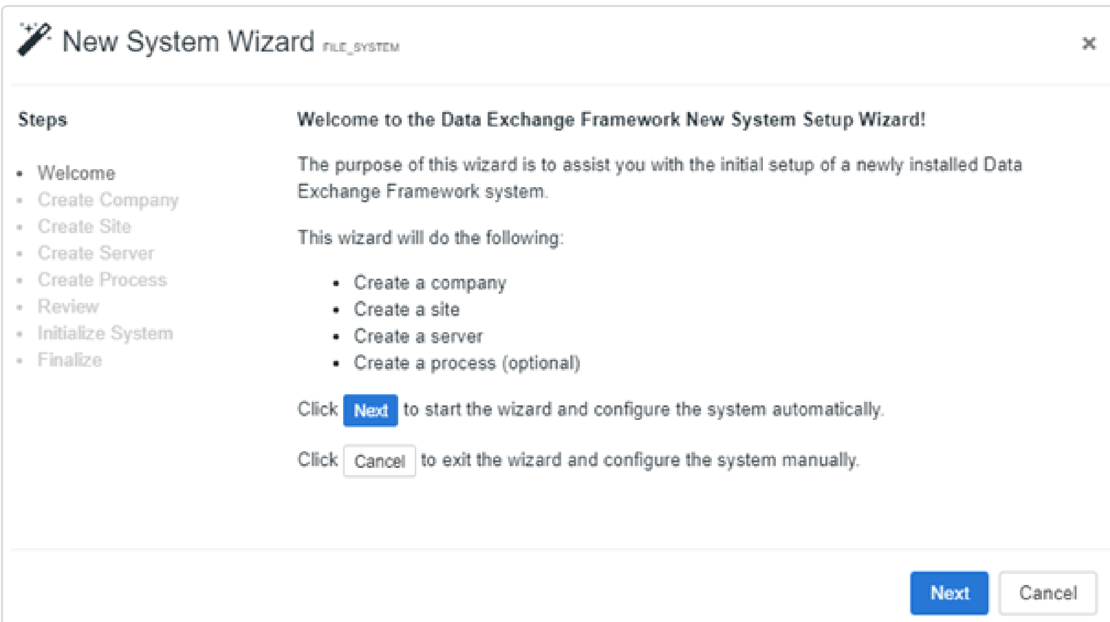
The Data Exchange Framework has the following wizards that you can optionally use to streamline setup and process analysis.

- First Time Setup Wizard
- Disposition Report Wizard

New System Setup Wizard

You can optionally use the New System Wizard to assist you with the initial setup of the Data Exchange Framework. The wizard populates the customer, site, server, and process components from templates.

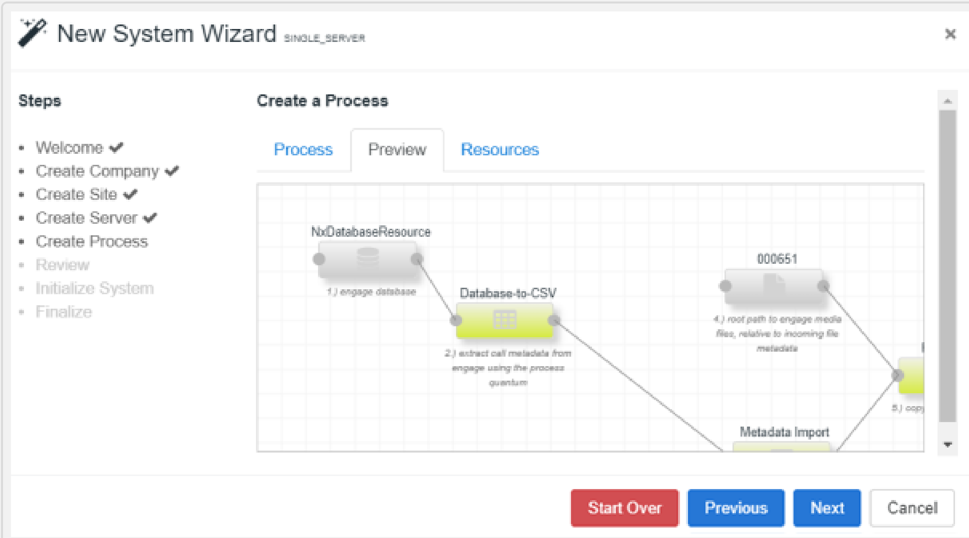
The New System Wizard opens when you log into the Data Exchange Framework after the installation. This window displays each time you log in until you create the first company. You can click the New System Setup Wizard button from the Companies window to access the New System Wizard. You can cancel also out of the New System Wizard and create the hierarchy manually.



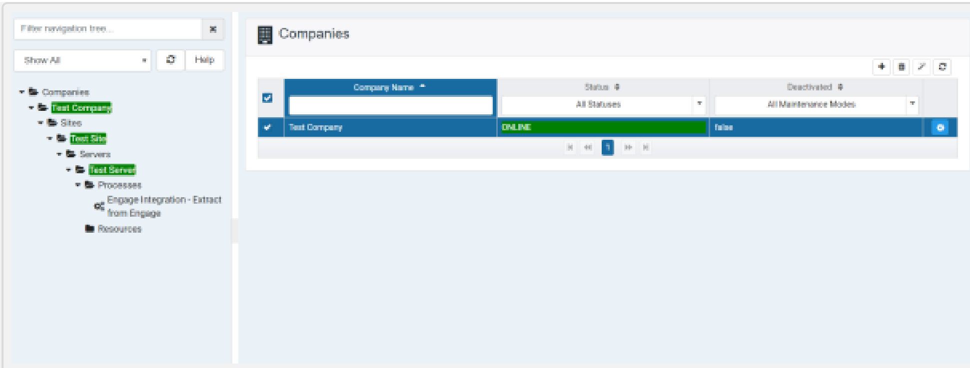
To automatically setup the system:

1. Click **Next** to start the wizard.
2. In the **Company Name** field, enter a name and click **Next**.
3. Enter the **site information** for the server type you are installing and click **Next**.
4. Enter the **server information** for the server type you are installing and click **Next**.

5. Specify the process information and click **Next**. You can select a template or manually enter the required process information. If you select a template, you have the option to preview the process and review the list of resources that it requires.
6. (Optional) Preview the process diagram. Click **Previous** to go back and make modifications, or **Next** to continue.



7. (Optional) Review the required resources for the process.
8. Review the information that you specified for the process. Click **Previous** to go back and make modifications, or **Next** to continue.
9. Review the new system setup status and click **Next.**
10. Review the setup success or failure details.
    - If the setup succeeded, the next steps that you should take are listed.
    - If the setup failed, the appropriate error messages display.
11. Click **Finish** to open the application.



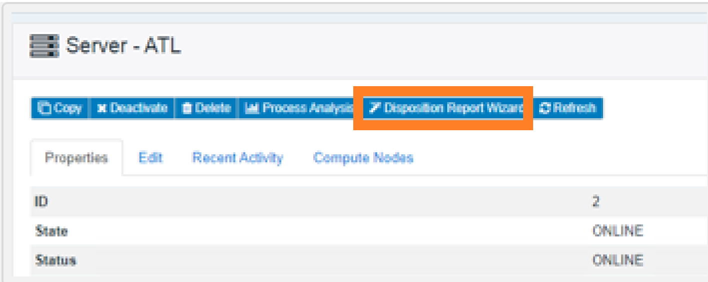
Disposition Report Wizard

You can use the disposition report to analyze the disposition count results for various actions of a specified process. The report provides the status of each interaction as it moves through the Data Exchange Framework into Interaction Analytics. You can:

- Select the process for which you want to generate the disposition report. The report generates disposition count results for various actions in that selected process, such as the number of pending calls (disposition count) for the Metadata Import Action, the stitching disposition count for the Engage Media Extract Action, the Pending Redaction disposition count for the Redaction Submit V2 Action, and the Ingested disposition count for the Inbox Publisher Action.
- Find out the interaction count gaps between various actions and then take the required steps to fix the gaps.

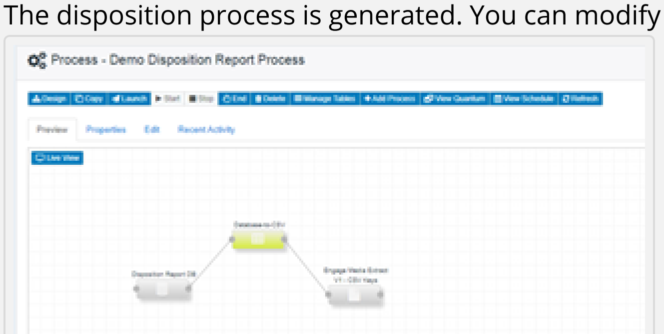
To generate the Disposition Report:

1. In the **Data Exchange Framework** application, select the server and click the **Disposition Report Wizard** button to launch the wizard.



The **Disposition Report Wizard** is launched and lists the items you can configure and the prerequisites.

2. Click **Next** and then specify the disposition report process group, name, and time.
3. Click **Next** and then select one or more processes for which you want to generate the disposition report.
   Some actions cannot be used in the disposition report because they do not report dispositions.
4. Click **Next** and then specify names for each action. These names are used and displayed in the disposition report.
5. Click **Next** and then select the metadata fields for grouping data.
6. Click **Next** and then select how often the process runs.
7. Click **Next** and then select time for historical data.
8. Click **Next** and then select or create a disposition report file or SFTP output resource.
9. Review the disposition generation process and modify various objects such as the action, schedule, and resource.
10. Click **Finish** to finalize and generate the disposition process.
    The disposition process is generated. You can modify it as needed.



You can modify the SQL code for the report in the generated process using the **Design** mode and the **Database to CSV** action.

COMPUTE NODES

Compute nodes provide the ability to horizontally scale a Worker server quickly and easily in response to increased workload or backlog, without the overhead of configuring and balancing load across multiple worker servers. Installation involves pointing the compute node at the install and data directories of the Worker server, then configuring desired processes for compute node availability through the administrative user interface.

Once complete, and with the process running on its configured schedule, the Worker server divides the load between itself and any attached compute nodes for all actions that process from metadata, dynamically balancing load over time until each runs its processes roughly the same time as the Worker server. Between that balancing and the Data Exchange Framework thread count auto-tuning for actions, all available servers or virtual machines can be attached to a Worker server, regardless of specs, and each is utilized to its fullest extent.

Unlike regular worker servers, compute nodes are scheduled to attempt a launch at the top of every minute for any process that is started or scheduled and has “Compute node availability” set to “Enabled” in its configuration (under “Advanced” options). Since compute nodes only execute actions that process from metadata, they are always looking for work they receive from the primary Worker server.

Setup, Shutdown, and Workload Partitioning

When a compute node is attached to a Worker server, it registers itself by UUID in that server's {DEF_DATA}/nodes folder, and will touch its node file every 10 seconds. That folder is constantly monitored by the worker for any nodes that have touched those files within the last minute. As soon as the worker becomes aware of active nodes, any of its actions in processes with 'Compute node availability' set to 'Enabled' that also process from metadata will dynamically partition all rows in the working table in that disposition across all available nodes, using the 'for_node' column in the working table.

Each node then picks up work for its next run as usual from the working table, but only for its own 'for_node' value. (Note that the Worker server's node will always be '00000000-0000-0000-0000-000000000000', and that any rows that have not yet been partitioned will have for_node = null.)

Before partitioning, the action will start by setting 'for_node' to null for any UUIDs that aren't in its active compute nodes list; so as soon as a compute node leaves the cluster, any remaining work that it hasn't completed will be reassigned to another node on the next run of the process.

Dynamic Weighting

Out of the box, load between worker and compute nodes is distributed equally on the first run before any compute node process run time results are available. However, once those results are available, work is weighted based on the performance of the compute node and the worker node over the last ten available runs where the total records processes were greater than 1000. Compute node weighting is tracked in the nx_user_activity table.

You can override the default weighting behavior dynamically by explicitly setting a weight other than 1.0 in any given node's registry file in the Worker server's {DEF_DATA}/node/ folder. You can override it permanently by making the same change to the compute node's {DEF_INSTALL}/etc/ndef.properties file's ndef.cn.load.weight property.

The root node is always weighted at 1.0, and compute nodes are relative to that. A 0.5, for example, would mean that the node can handle about half of what the root node can, so 900

records being partitioned across the two would give 300 to the compute node and 600 to the root.

2.0 would mean the other way around. The compute node can process twice what the root node can, so 600 would go to the compute node and 300 to the root.

Process Design

Because compute nodes execute only those actions that process from metadata, the recommended process design for optimal growth with compute nodes is to have a minimal number of actions that are introducing metadata into the process, with the remainder processing from that metadata. For those very simple processes (like some NMF Decrypt deployments) that currently only have a single action pulling files from the file system and acting on them, a simple adjustment to get them working with compute nodes is to add a File Discovery action before them, then changing them to process from metadata.

This design isn't necessary or even necessarily enforced, but any actions not processing from metadata must be processed by the root node, losing the compute node scalability. In cases where those actions are in the middle of a process design, users might see some "staggering" behavior as the compute node handles actions up to that point; the root node then handles the action that doesn't process from metadata on its next run, then the compute node picks up and continues processing on its next run.

In cases where actions don't provide a "Process from metadata" or "Gets and sets dispositions" option, the compute node will decide how to handle them. Some don't integrate with the working table, others have to process from the file system, and others have to process from metadata. In these cases, no option to change that behavior will be available. The same staggering described above may be observed in those cases. Current Data Exchange Framework templates are already designed with these considerations in mind.

Other Considerations

Many traditional deployments of the Data Exchange Framework involve storing the "DEF Data" and other file system resource folders on a local drive (M:, e.g.). While this is discouraged for high-availability systems in favor of common shares on an external storage device, the fact remains that many deployments will still be local, and since compute nodes rely on the same folder paths as the Worker server, they must be taken into consideration when installing a compute node.

The default behavior of a compute node is to convert any such local, drive letter-based paths to administrative shares under the hood (M:\some\folder --> \\hostname\M$\some\folder, e.g.), making the attachment of the compute node seamless. This does require, however, that the service account user that the Data Exchange Framework servers are installed under must also belong to the Administrators group in Windows. If such permissions aren't allowed at the deployment site, then two options are available: First is to edit any file system resources using local paths, and convert them to shares. This is the recommended approach.

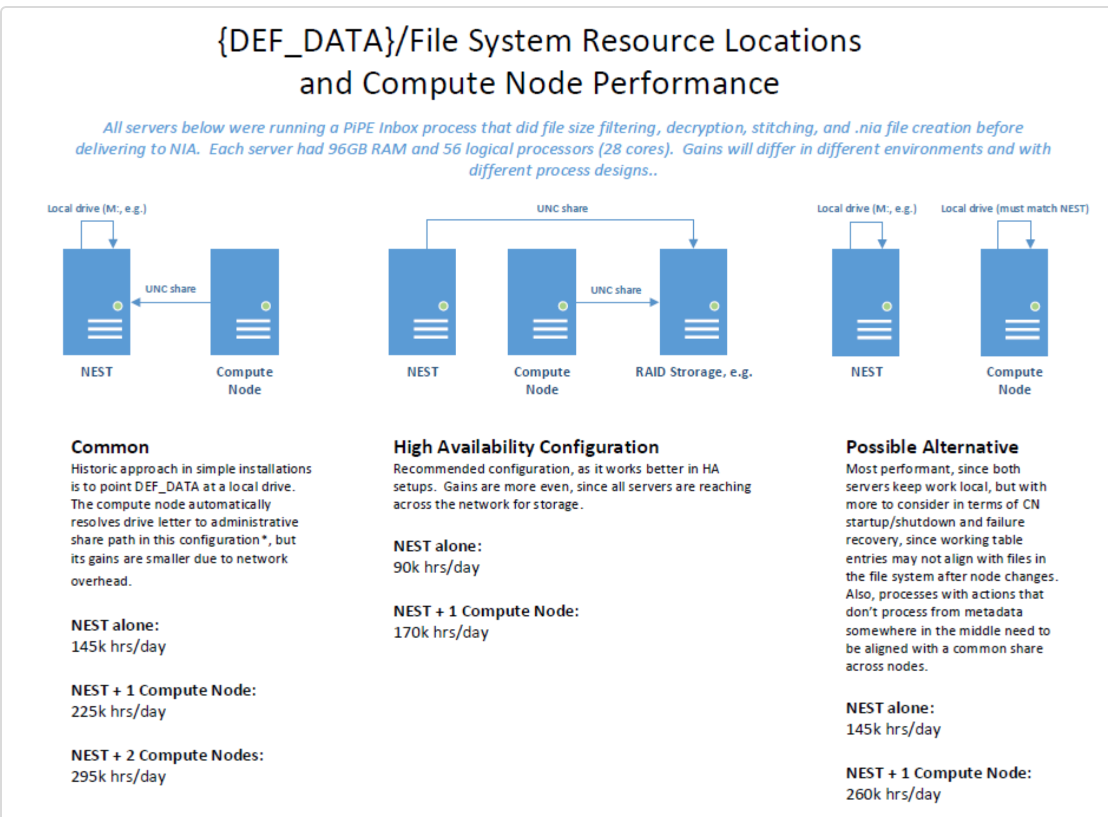
The second option is to leave things as they are, letting the compute node reproduce the same paths locally. One advantage of this approach is that the compute node can do its disk I/O locally without traversing the network, yielding better performance for actions heavy on disk I/O. However, the downside is that care must be taken when disabling the compute node or in failure scenarios to ensure that any files currently in process are migrated to an appropriate server. It also may require changes to process designs, particularly in scenarios where there are actions in mid-process that don't process from metadata and require the outputs of previous actions. In such cases, explicit external shares should be declared between these actions.

This approach also requires that the compute node have matching drive letters available and the permissions to write to their corresponding paths. It also requires adding "ndef.cn.disable.fs.path.adjust=true" To disable the automatic path re-writing before restarting the service, to the etc/ndef.properties file in the Data Exchange Framework installation folder.

You should carefully weigh the performance implications of each of these configurations. Retaining local folders may give the best performance but is the most brittle for failure and the trickiest for process design. Conversely, having all compute nodes share the same file system will hit an upper bound on disk I/O for that server. See Compute Node performance for performance numbers from different configurations in one target performance-testing environment.

Compute Node performance

The following graphic shows the performance numbers from different configurations in one target performance-testing environment.



REDACTION PERFORMANCE

Redaction is accomplished in conjunction with RabbitMQ and a specialized Redaction Grid. The DEF process uses these actions to perform the integration. These three actions are strung together in a single process and can be used in a standalone process.

- The RedactionSubmit action pushes redaction requests to RabbitMQ for Grid to pick up.
- The RedactionResponse action picks up completed requests from RabbitMQ.
- The Redaction action performs the actual file redaction using Nexidia's Workbench.

Redaction scaling occurs at the Redaction Grid level. When more throughput is required, separate Redaction Grid servers can be stood up, each pointing at the same RabbitMQ instance, and the load is seamlessly distributed among them.

Redaction model files are approximately 1GB and are identified in the DEF and Redaction Grid by a configurable "Workflow ID." Each Redaction Grid server requires approximately 1GB of RAM per thread and should be tuned accordingly.

To install RabbitMQ:

1. Install erlang http://www.erlang.org/downloads.
2. Install RabbitMQ https://www.rabbitmq.com/install-windows.html.
3. Change directory to "PATH\RabbitMQ Server\rabbitmq_server-3.7.7\sbin".
4. Run "rabbitmq-plugins enable rabbitmq_management".
5. Start the service.

To add user "guest2" in CMD:

1. "Path\RabbitMQ Server\rabbitmq_server-3.7.7\sbin\rabbitmqctl" add_user guest2 guest2
2. "PATH\RabbitMQ Server\rabbitmq_server-3.7.7\sbin\rabbitmqctl" set_user_tags guest2 administrator
3. "PATH\RabbitMQ Server\rabbitmq_server-3.7.7\sbin\rabbitmqctl" set_permissions -p / guest2 ".*" ".*" ".*"

Search Grid

Search Grid uses port 8080. You can change this in the application file using the tag server.port=8080.

The target machines for a Media Worker installation must have a compliant JVM (version 1.8) pre-installed and the corresponding %JAVA_HOME% system environment variable defined.

If you are using a 28-physical core box for the Grid server, do not use 56-core hyperthreading, and turn off NUMA.

To install the service on a suitable Windows host:

1. Extract the provided deployment zip file (NxVgMediaWorker-1.0.zip) on a local drive.
2. Edit the application.properties file as necessary. The rabbitMQ instance address and user credentials must be changed to their realistic settings (see the RMQ Administration Guide for RMQ installation procedures and user provisioning).
   spring.rabbitmq.addresses=localhost:5672
   spring.rabbitmq.username=guest
   spring.rabbitmq.password=guest
3. Activate the Workbench license. Change directory to /activation, and run the activateWorkbenchLicense.bat script.
4. Install the NxVgMediaWorker Windows Service.
   - In the /bin directory, run the installation script: .\service.bat install [service-user password]
   - If the service user and password are supplied, the installed NxVgMediaWorker Windows service will be running under that user's credential and be given corresponding file access privileges. Depending on the specific client environment conditions, this may be necessary in order for Media Worker to process the requested network media files.
5. To start the Service from the command line: net start NxVgMediaWorker.

To uninstall the service

Uninstalling the service is not required regularly. You will most likely only need to uninstall it in situations such as upgrading Java from version 1.8 to 10. The uninstall procedure uses this script file: .service.bat remove.

ENCRYPTION (NICE) RECORDER INTEGRATION

The NMF Decrypt action decrypts NMF media files. This action requires further configuration with the Generated Protected Key, Key Extraction, and Load Keys actions. The following tasks are performed:

- Reads encrypted NMF from the SFTP server.
- Parses each packet identifying encryption GUIDs.
- Maintains map of GUIDs to ciphers (Rijndael or AES).
- If a GUID not mapped to cipher is encountered, reads the key and IV associated with the GUID from the Keystore.
- Cipher is created and cached.
- Writes the decrypted NMF to the staging area encrypted by Vormetic.

If you are not using the Engage Integration templates to build out the Engage/Nexidia system, the NMF file decryption process requires these actions:

- Generate Protected Key - The Generate Protected Key (GPK) action is responsible for generating the symmetric keys used to encrypt and decrypt key information pulled from the nice_crypto database in the NIM and Engage recorders. This action is launched on demand for initial key creation, and for subsequent key rotation. It takes a self-signed certificate from the Worker server that will run the Key Extraction action as an input, generates a symmetric key encrypted against that certificate, writes it out to the output resource, and adds the unencrypted key to the configured keystore. It also tracks the UUIDs of keys created and the certificates they were generated from in the Data Exchange Framework working table.
- Key Extraction - The Key Extraction action is intended to run as the only action in the only process on an isolated Worker server. It's responsible for extracting keys from the recorder, encrypting them with the symmetric key from the previous action, and writing the encrypted contents to a CSV file for loading by the next action.
- Load Keys - The Load Keys action is responsible for reading the encrypted CSV files generated by the previous action, decrypting them using the appropriate symmetric key from the keystore in the GPK action, and importing key and initialization vector information into the DB table nx_nmf_keystore for use by the NMF Decrypt action.
- NMF Decrypt - The NMF Decrypt action uses the DB table nx_nmf_keystore from the previous action to decrypt incoming NMF files.

The Generate Protected Key, Load Keys, and NMF Decrypt actions are expected to run on the same Worker server, with the Key Extraction action running on a standalone Worker server.

Initial Setup

The Worker server that will run the Key Extraction action must have a self-signed certificate manually created as a preliminary step. To create it, you'll first create the keystore using this command (replace the italicized items with actual values).

keytool -storetype JCEKS -keystore *c:\temp\keyStore.ks* -genkey -keyalg RSA -keysize 2048 -validity 3650 -alias *keyAlias*

The keytool utility can be found in the jre/bin folder of the DEF install directory.The keystore location and key alias must match the previous command.

Follow the prompts, and verify that the key password is the same as the keystore password when prompted. After the keystore has been created, use the following command to export the public key from the certificate into a .cer file that can be consumed by the Generate Protected Key action.

keytool -storetype JCEKS -keystore *c:\temp\keyStore.ks* -export -alias *keyAlias* -file *certName.cer*

Process Design

The first three actions (Generate Protected Key, Key Extraction, and Load Keys) are expected to be individual actions in their own processes, with the Key Extraction action located on its own Worker server. The NMF Decrypt action can plug in to any standard NiCE interaction extraction flow, and is integrated into system templates for easier deployment.

To configure:

1. Generate the self-signed certificate on the Key Extraction Worker server with the Java keytool utility. Make a note the keystore location, password, and alias for use in step #4.
2. On the Decryption Worker server, configure the Generate Protected Key action in its own process, with an input folder that will hold the certificate from step #1, and an output folder where the encrypted symmetric key will be written out. The keystore location points to a keystore file that is created if it doesn't exist. It can be any file path that makes sense for key storage. Copy the certificate file to the input folder, and launch the process.
3. Copy the encrypted symmetric key back to the Key Extraction Worker server.

4. On the Key Extraction Worker server, configure the Key Extraction action in its own process with no input resources, and an output resource for the encrypted CSV files (this output should match the input of the Load Keys action in step #6).
   - The database connection string is in JDBC format. For example: “jdbc:sqlserver://server.fqdn;databaseName=nice_crypto;integratedSecurity=true” (change “server.fqdn” to your SQL Server's fully-qualified domain name).
   - The symmetric key path is the location of the encrypted symmetric key copied over from step #3.
   - The keystore location, password, and alias are the same as those used when you created the self-signed certificate in step #1.
5. On the Manage and Monitor tab configure the Key Extraction process to have a Quantum Lower Date Bound with an appropriate date for beginning key extraction from the recorder database, a Quantum of 0, and schedule it to run on an appropriate interval. This configuration loads all keys from the lower date bound up to the present moment on the first run. On subsequent runs it keeps pulling new keys that have been generated since the previous run.
6. On the Decryption Worker server, configure the Load Keys action in its own process, with an input resource that will contain the encrypted CSVs from the Key Extraction action (this should match the output location of the Key Extraction process), and no output resource. The keystore location and password should match those of the Generate Protected Key action from step #2. This action decrypts the encrypted CSV files from the Key Extraction process, and loads them into the DB table nx_nmf_keystore for the NMF Decrypt action to use in decrypting interaction files.
7. On the Decryption Worker server, you can now add the Decrypt NMF action to any interaction extraction workflow. Configure it with the same keystore location and password as the Generate Protected Key and Load Keys actions.

MULTI-TENANT SUPPORT FOR NICE ENGAGE

The Data Exchange Framework supports the extraction of data from individual tenants from a single instance of NiCE Engage.

The Data Exchange Framework has three templates that perform full extraction from Engage. You can change these templates to pull specific tenant data from a multi-tenant system.

To extract data from individual tenants:

1. When deploying, select one of the Engage Integration Extract templates.
    - Engage Integration - Extract and Prepare Media and Metadata with Redaction
    - Engage Integration - Extract and Prepare Media and Metadata without Redaction
    - Engage Integration - Extract from Engage (On-Prem)
2. Make the following SQL change in the initial DB-to-CSV action of the template you chose.
3. Add the highlighted join condition below to the portion of the DB-to-CSV query that creates the #MediaEngageTemp table and use the desired tenant ID.

```
select distinct
Inter.iInteractionID
,Inter.dtInteractionGMTStartTime
,Inter.dtInteractionGMTStopTime
,dtInteractionLocalStartTime
,dtInteractionLocalStopTime
,nvcDialedNumber
,Inter.iContactID
,Inter.dtContactGMTStartTime
,iInitiatorUserID
,Inter.iMediaTypesId
into #MediaEngageTemp
from nice_interactions..vwInteraction Inter  (nolock)
inner JOIN [nice_storage_center].[dbo].[vwStorageCenter] Store (nolock)
```

```
on Inter.iInteractionID = Store.iInteractionID

inner JOIN [nice_admin].dbo.tblScSamUnit SCUnit (nolock)

on Store.siUnitID = SCUnit.siUnitID

INNER JOIN nice_admin..tblTenantUsers Tenant

ON Tenant.iUserId = Inter.iInitiatorUserID

AND tenant.ItenantID = 4

where Store.tiMediaTypeId = 2 and store.ifsarchiveclass=2

and Store.dtUpdateTime BETWEEN  @windowStartFormatted and @windowEndFormatted
```